MAGNETOM

Replacement of Parts System

Patient Handling

Symphony, A Tim System

Document Version

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This chapter describes the replacement procedure for all patient handling components of the MR system and consists of the following:

- Patient table, **ptab**
- Exchangeable tabletop (option), **trolley**
- Physiological measurement unit (option), **pmu**
- Video monitoring and intercom

CAUTION

Strong magnetic field.

Magnetic parts may cause injury.

» **Ramp down the magnet or maintain a safe distance to the magnetic field.**

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

Switching **off** the patient table via circuit breaker F22 may cause the patient table to reload.

For disassembling/assembling the patient table covers, we recommend to use a mid-size Philips screwdriver.

Adhere to ESD regulations and use the appropriate tool kit: anti-static wristband and grounded tabletop pad (TDK equipment Type 8005 from 3M Deutschland GmbH).

Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance <= 100 mOhm with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. direct position drive) and the protective earth wire connected to the "main component" (e.g. patient table).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

This chapter described all FRUs in the Patient Handling System that can be replaced.

Patient table					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tool**
Guide rails	120 min.	(→ Guide rails / Page 15)	■ n.a.	■ Vertical table movement	Non-magnetic tool kit Electric drill Hammer Plastic block Table transport device Loctite 221
Lifting motor/ vertical	30 min.	(→ Lifting motor / toothed belt of vertical drive / Page 34)	■ n.a.	■ Vertical table movement	Non-magnetic tool kit
Toothed belt - vertical drive	30 min.	(→ Lifting motor / toothed belt of vertical drive / Page 34)	■ n.a.	■ Vertical table movement	Non-magnetic tool kit
Spindle	120 min./60 min.	(→ Spindle / Page 41)	■ n.a.	■ Vertical table movement	Non-magnetic tool kit Table transport device Loctite 221
Brake Disk	60 min.	(→ Brake Disk / Page 75)	■ n.a.	■ Vertical table movement	Non-magnetic tool kit Table transport device Loctite 221
Tabletop plate	60 min.	(→ Tabletop / Page 85)	■ n.a.	■ Horizontal table movement	Non-magnetic tool kit
Energy chain	30/45 min.	(→ Body of the energy chain / Page 93)	■ n.a.	■ Horizontal table movement	Non-magnetic tool kit
Guide rollers	30/45 min.	(→ Guide rollers / Page 99)	■ n.a.	■ Horizontal table movement	Non-magnetic tool kit Loctite 221

Patient table					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tool**
MDSD device	120 min.	(→ MDSD device / Page 103)	■ n.a.	■ Horizontal table movement	Non-magnetic tool kit Non-magnetic torque wrench Philips screwdriver, medium size Flat blade screwdriver, 4mm Flat wrench, 22/27 mm
Horizontal drive unit	120 min.	(→ Horizontal drive unit / Page 132)	■ n.a.	■ Horizontal table movement	Non-magnetic tool kit Non-magnetic torque wrench Flat wrench, 22 mm Non-magnetic torque wrench Philips screwdriver, medium size Flat blade screwdriver, 4mm Loctite 221
Toothed belt of horizontal drive	60 min.	(→ Toothed belt of horizontal drive / Page 164)	■ n.a.	■ Horizontal table movement	Non-magnetic tool kit Philips screwdriver, medium size
Sound transducer	60 min.	(→ Sound transducer / Page 178)	■ n.a.	■ Table functions, patient communication	Non-magnetic tool kit Flat blade screwdriver, 4mm Philips screwdriver, medium size
Circuit boards of patient table electronics, A4120, A4140, A4110	30 to 90 min.)	n.a.	■ Check status of DIP switches	■ Table functions	Non-magnetic tool kit

Patient table					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tool**
RF coil plugs	30 min. for each side	(→ RF coil plugs / Page 186)	■ n.a.	■ SeSo RCCS plug test 1-10 or Quality Assurance / Coil check	Non-magnetic tool kit
Switch S30	60 min.	(→ Control switch S30 / Page 200)	■ n.a.	■ Switch test in Service mode	Non-magnetic tool kit
Cable	180 min.	(→ Cable replacement / Page 212)	■ n.a.	■ Function test of replaced cable	Non-magnetic tool kit
Transformer A4114	30 min + Magnet ramp time	■ n.a.	■ n.a.	■ Function test of the transformer - Check table movement/ horizontal and vertical	**Caution:** Magnetic part. Ramp down the magnet.
Control panel	20 min.	(→ Control panel / Page 221)	■ n.a.	■ Function test of control panel	Non-magnetic tool kit
Light localizer/ display	25 min./ 20 min.	(→ Light localizer/ Display / Page 223)	■ Light localizer	■ Function test of light localizer / display	Non-magnetic tool kit

Trolley					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tool**
Locking hooks	120 min.	(→ Locking Hooks (Mechanical) / Page 227)	■ n.a.	■ Check for smooth and safe trolley operation	Non-magnetic tool kit
Keypads/PCB	20 min.	(→ Keypads / PCBs / Page 238)	■ n.a.	■ Check smooth and save trolley operation	Non-magnetic tool kit

PMU					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tool**
PDAU	20 min.	(→ PDAU / Page 252)	▪ n.a.	▪ SeSo PMU Test	Non-magnetic tool kit
PMU Antenna	25 min.	(→ PMU antenna / micro-phone / Page 255)	▪ n.a.	▪ SeSo PMU Test	Non-magnetic tool kit
PERU	2 min.	n.a.	▪ n.a.	▪ SeSo PMU Test	n.a.
PPU	2 min.	n.a.	▪ n.a.	▪ SeSo PMU Test	n.a.
Sensor charger	5 min.	n.a.	▪ n.a.	▪ SeSo PMU Test	n.a.
Cable kit PMU Display	15 min.	n.a.	▪ n.a.	▪ SeSo PMU Test	n.a.

Video monitoring / intercom					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tool**
Microphone	25 min.	(→ PMU antenna / micro-phone / Page 255)	▪ n.a.	▪ Acoustic test	Non-magnetic tool kit
Fan !Consists mag. parts, maintain safe distance!	30 min.	n.a	▪ n.a.	▪ Functional test	Non-magnetic tool kit
Intercom central unit	10 min.	n.a.	▪ n.a.	▪ Bi-directional acoustic test, table stop button	Non-magnetic tool kit
Control panel intercom	5 min.	n.a.	▪ n.a.	▪ Bi-directional acoustic test, table stop button	Non-magnetic tool kit

Video monitoring / intercom					
FRU	**Time**	**Instructions**	**Adjustments**	**Final checks**	**Tool**
Patient monitor	5 min.	n.a.	■ n.a.	■ Visual test	Non-magnetic tool kit
CCD camera	15 min.	(➔ CCD camera / Page 259)	■ n.a.	■ Visual test	Non-magnetic tool kit

3.1 Guide rails

3.1.1 General

The table is moved in a vertical direction via four preassembled bearing systems above two guide rails. The guide rails and the bearings must be replaced together.

3.1.2 Tools and Time required

3.1.2.1 Time

The estimated time does **not include** ramping the magnet.

Approx. 120 min

3.1.2.2 Tools

- Loctite 221
- Non-magnetic tool kit

Fig. 1: Allen key

- 5 mm Allen key delivered with the guide rail kit.

- Table transport device.

Fig. 2: Table transport device

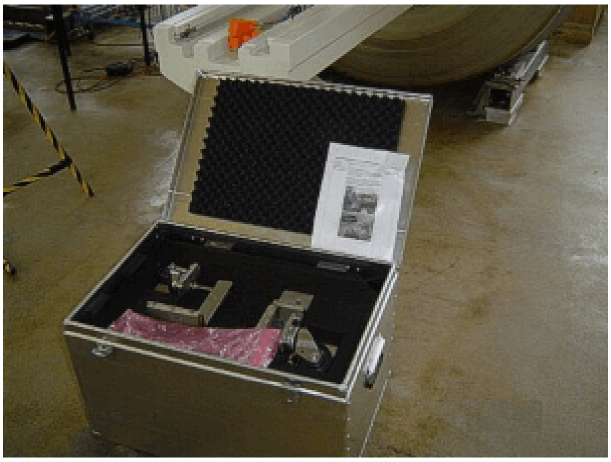
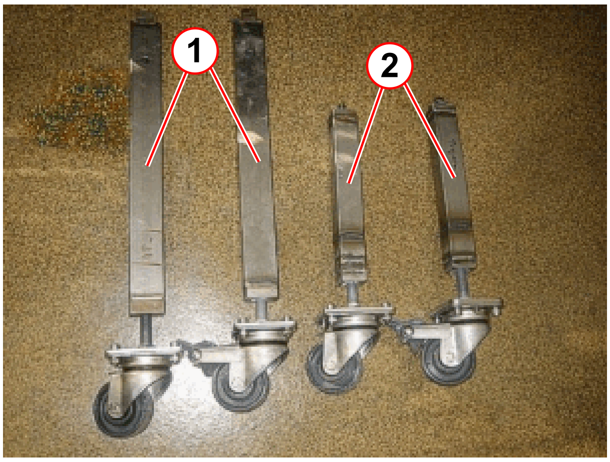


Fig. 3: Table transport device



(1) Transport roller 1
(2) Transport roller 2

3.1.3 Preliminary

- Ramp down the magnet.
- Switch OFF F22 (RF-room) in the ACC.

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Ramp down the magnet**

3.1.4 Disassembly

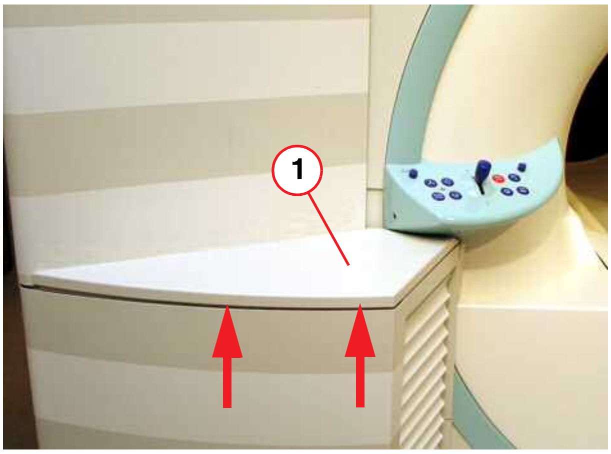
3.1.4.1 Covers

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 4: Service access cover removed

- Remove the service access cover.

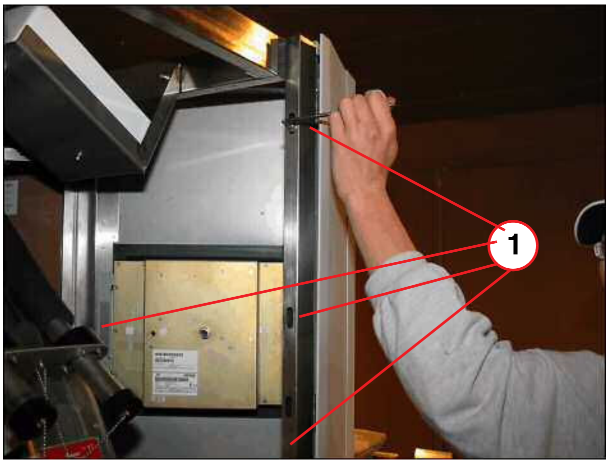
Fig. 5: Lifting column, top cover



(1) Top cover

- Remove top cover of lifting column.

Fig. 6: Top front cover above lifting column



(1) Screws to be removed

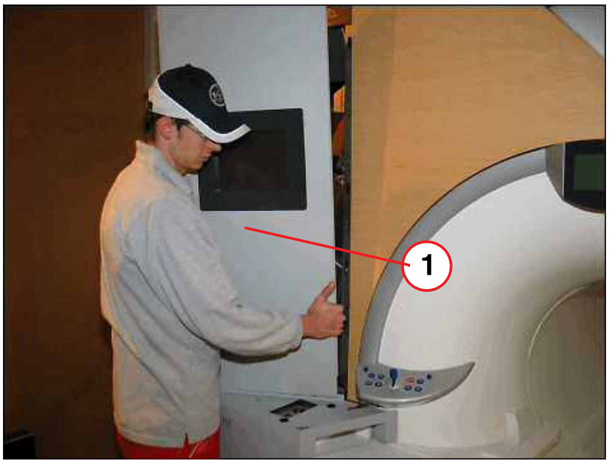
- Remove all screws from the top front cover.

Fig. 7: Top front cover above lifting column

(1) Screws to be removed

- Remove all screws inside the top front cover and disconnect cables, if necessary.

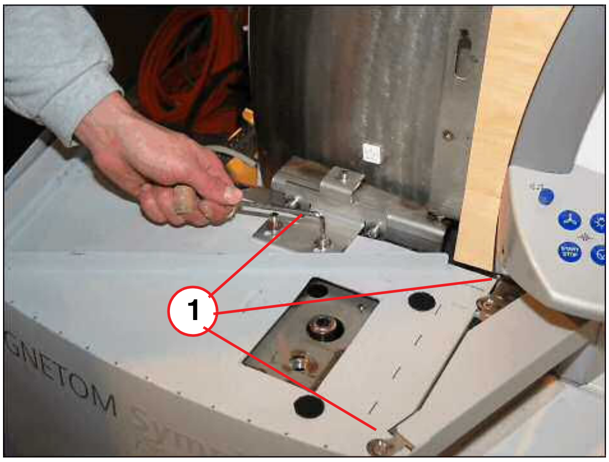
Fig. 8: Top front cover above lifting column



(1) Remove cover

- Remove top front cover.

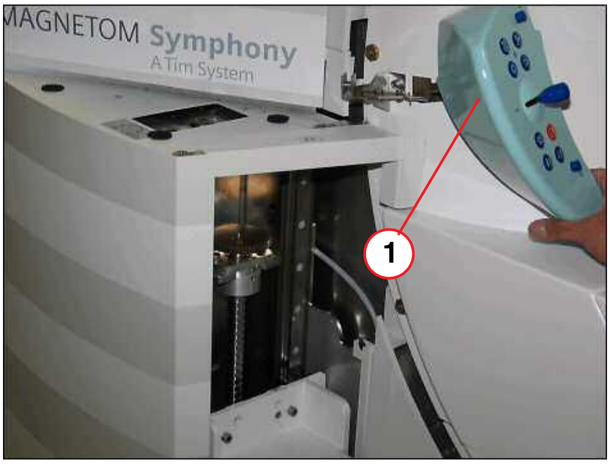
Fig. 9: Lifting column cover



(1) Screws to be removed

- Remove screws from the lifting column and bracket.

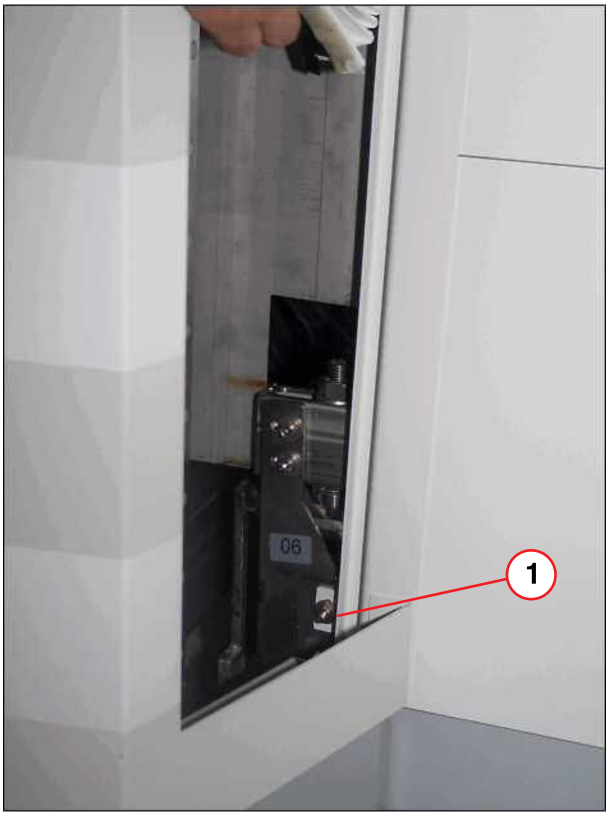
Fig. 10: Control panel



(1) Control panel removed

- Separate the bellows from the adhesive tape and remove upper bellows.
- Loosen screw of operating panel, disconnect cable if necessary and remove it.
- Remove the screw behind the control panel to disassembly the lifting column cover.

Fig. 11: Lower part of lifting column



(1) Bellows removed, screw to be removed

- Separate the bellows from the adhesive tape and remove lower bellows.
- Remove screw on the lower part of the lifting column.

Fig. 12: Lifting column cover

- Remove lifting column cover.

3.1.4.2 Support Beam

Fig. 13: Remove M8 Allen screw

- Remove the four M8 Allen screws, two above and two below.

Fig. 14: Remove two Allen screws inside the support beam

- Remove two Allen screws inside the support beam.
- Move the table up by hand, use the crank and mount the table service device.

- Mount the table transport device (service aid) (➔ Fig. 15 Page 22).
  - Rollers must be evenly positioned/must be level on the floor so that you can return to the guide attachment position (➔ Fig. 16 Page 22).

⚠ CAUTION

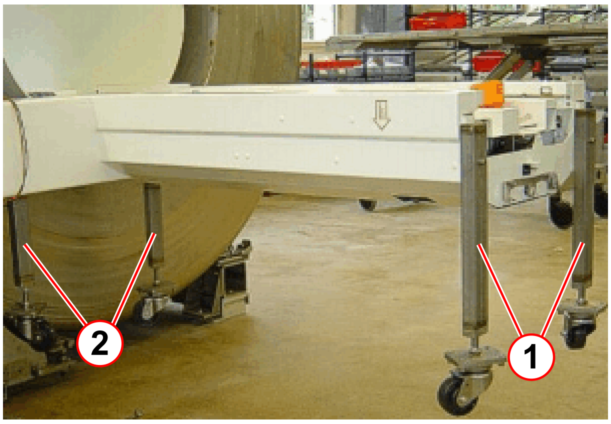
Possible damage of the spindle nut.

» **After the four transport rollers reached the floor and the load is removed from the spindle nut, do not continue to lower the table with the manual crank**

- Lower the table (using the crank) (➔ Fig. 17 Page 22) until the patient table is resting on the table service device. After the four transport rollers reached the floor and the load is removed from the spindle nut, do not continue to lower the table with the crank.

- It takes 5 to 10 turns of the crank to remove the load from the spindle nut located behind the lifting carriage plate!

Fig. 15: Table transport device installation



(1) Transport roller 1
(2) Transport roller 2

Fig. 16: Transport roller

Fig. 17: Lower the table with the crank until the load is removed from the spindle nut.

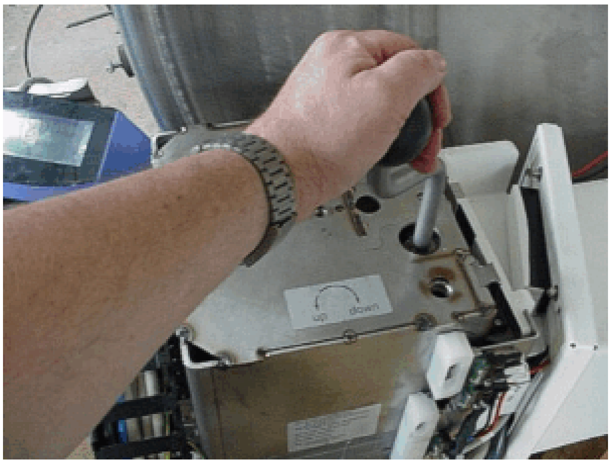


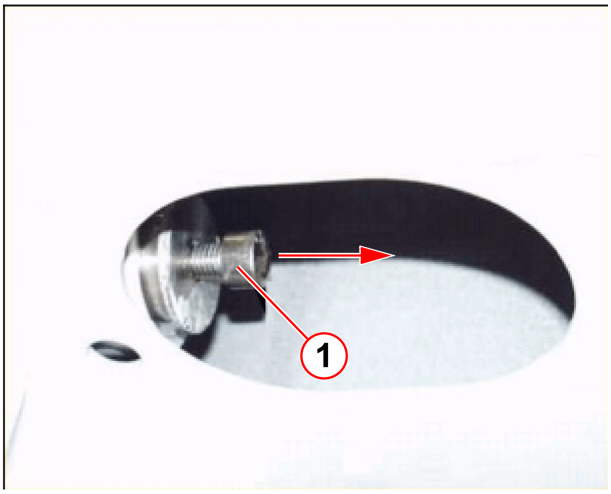
Fig. 18: Energy chain for lifting column

- Dismount the energy chain.

Fig. 19: Remove the M6 Allen screw

- Remove the M6 Allen screw of the spindle nut driver.

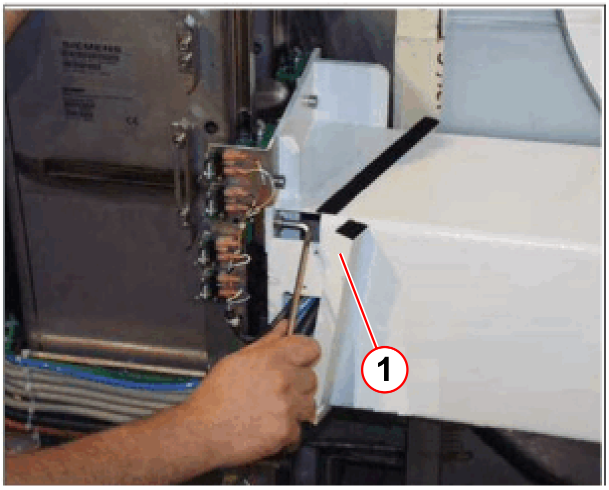
Fig. 20: M10 Allen screw, inside the support beam



() M10 Allen screw

- Pull back the M10 Allen screw (inside the support beam) to release the spindle nut from the spindle nut driver. The table with the lifting column is now free mechanically!

Fig. 21: M8 Allen screw of support beam



(1) Bellow support

- Remove the bellows support and remove the remaining M8 Allen screws.

Fig. 22: Remove the rest of all (16 in total) other M8 Allen screws

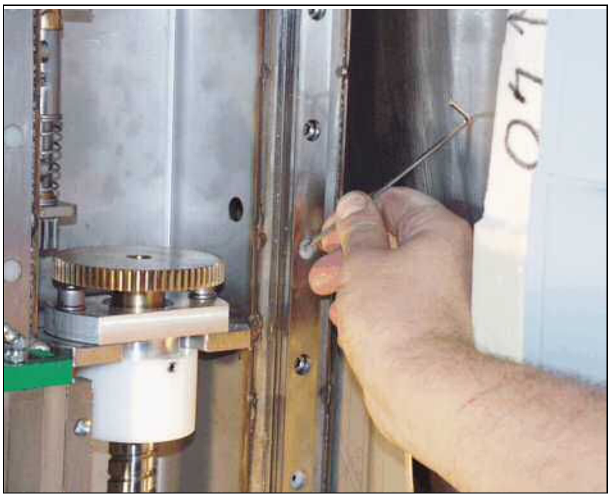
- Remove all other M8 Allen screws and carefully remove the switch panel.
- To access the guide rails, carefully push aside the table.

3.1.4.3 Guide rails

Fig. 23: Plastic covers from the guide rails

- Remove closing plugs if necessary (only used until 06/2008) by using an electric drill with a 5 mm bit and bore up the plastic closing plugs.

Fig. 24: Clinch rings



- Remove plastic clinch rings if necessary using a small screwdriver.

- Remove all M6 Allen screws and remove guide rail.

Fig. 25: Allen screws of guide rails

Fig. 26: Guide rail, removed

- Repeat the disassembly procedure on the other guide rail.

3.1.5 Assembly

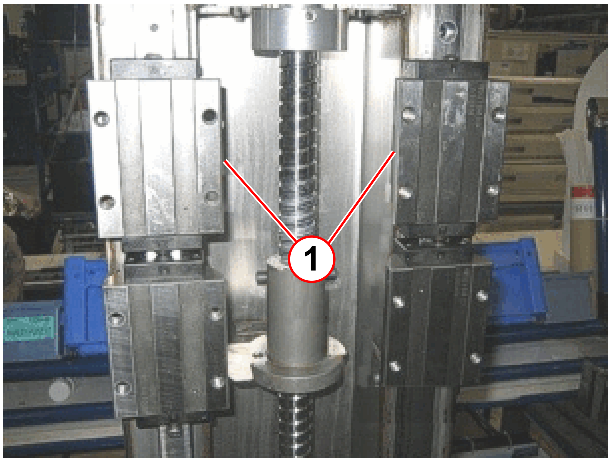
3.1.5.1 Guide rails

> Take into consideration the right and left side of the guide rail when mounting the new ones (ground edges of guide cassettes facing inward (→ Fig. 28 Page 28)).

- The two bearings remain at the bottom of the guide rail.

Fig. 27: Tighten the replaced guide rail from the middle up

Fig. 28: Ground edges of guide cassettes

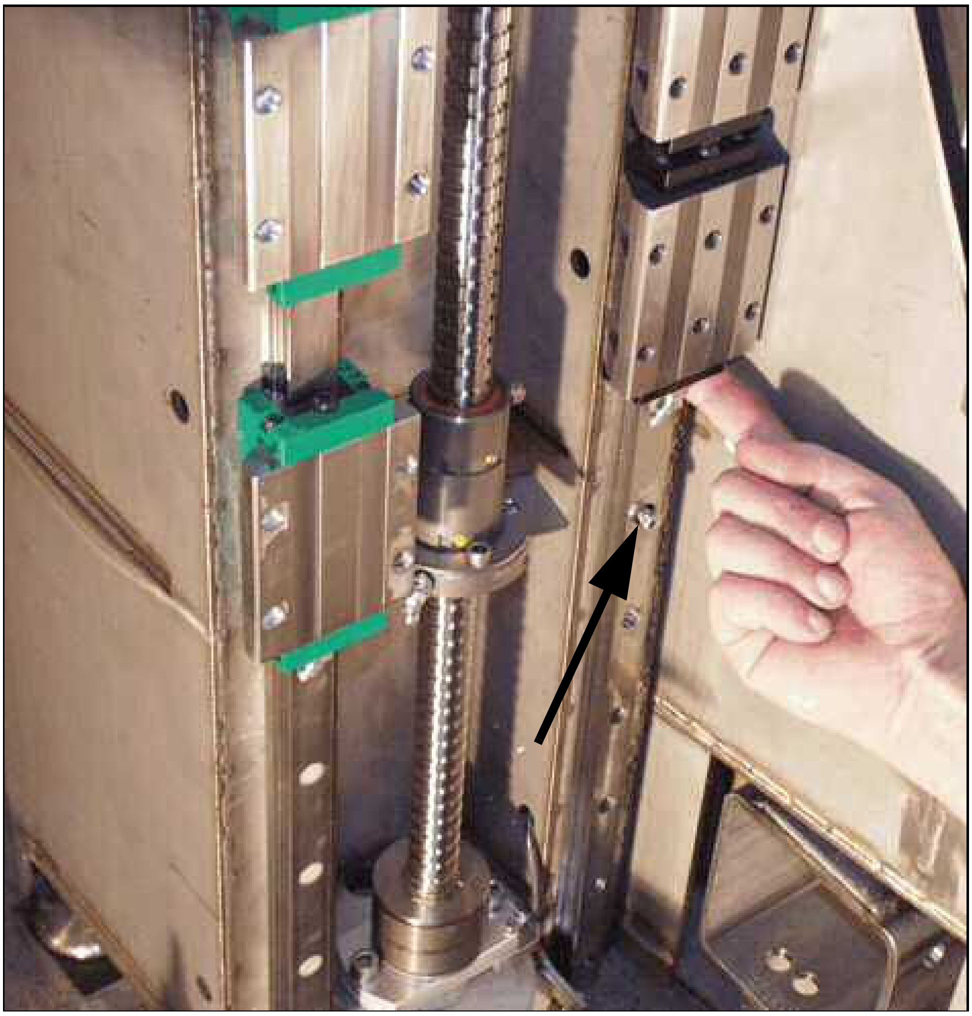


(1) Ground contact surface

- Reinsert Allen screws (apply Loctite 221) into counterbores from the upper part of the guide rail and tighten them. The bearings remain at the bottom of the guide rail.

- Move the bearings up. Insert one Allen screw as far as the bearings can remain on it.

Fig. 29: Move bearings up, insert screw, to bear on the screw



- Reinsert the rest of the Allen screws (apply Loctite 221) into counterbores from the lower part of the guide rail and tighten them.

> The bearings must not drop down with force, as this would break the plastic end covers. Always be careful when moving the bearings.

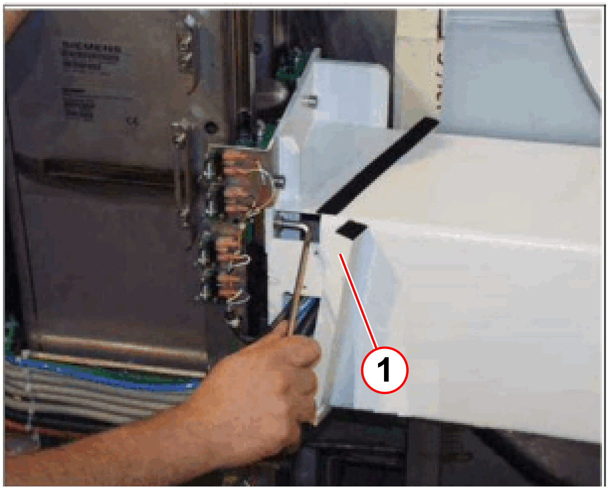
- Repeat the procedure with the other guide rail.
- Before beginning with the reassembly, make sure the bearings are mounted correctly, the guide rails are clean, and no objects are present.

3.1.5.2 Support beam

Fig. 30: Move table back

- Carefully move the table back to the lifting column.
- Move the bearings up and install them on the support beam with one screw on each corner.
- The bearings must be seated accurately to the support beam.
- There are 4 fixing holes each for correct mounting.

Fig. 31: M8 Allen screw of support beam



(1) Bellow support

- Reassemble the switch bracket and insert all other Allen screws (apply Loctite 221) and tighten them.

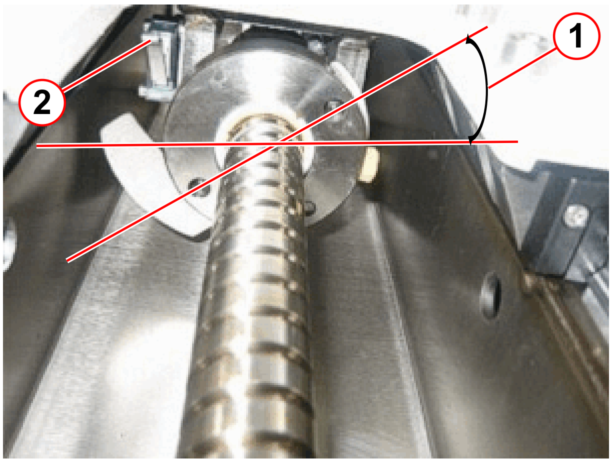
Once the support beam has been installed, the spindle nut in the column must be threaded back through the nut driver.

- Turn the spindle nut by hand so that it moves up underneath the nut driver and hold it in a diagonal/slanting position (approx. 45°). Collision switch S30 visible from underneath (➔ Fig. 32 Page 31), (➔ Fig. 33 Page 31).
- Use the crank to continue moving the spindle nut up so that it is inserted far enough into the nut driver.
- Turn the spindle nut in the nut driver back to the correct position (➔ Fig. 34 Page 31). This ensures that the semicircular switching plate is in the right position to be able to actuate the cam switch of collision switch S30.

- Hold the spindle nut in the correct position and push the nut driver back on the M10 (➔ Fig. 20 Page 24).

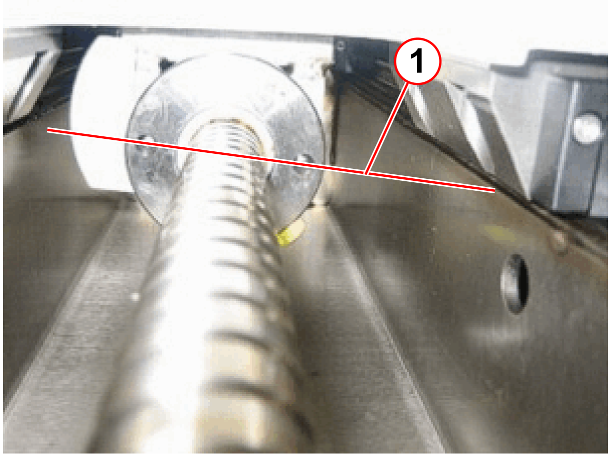
Fig. 32: Spindle nut

Fig. 33: Positioning



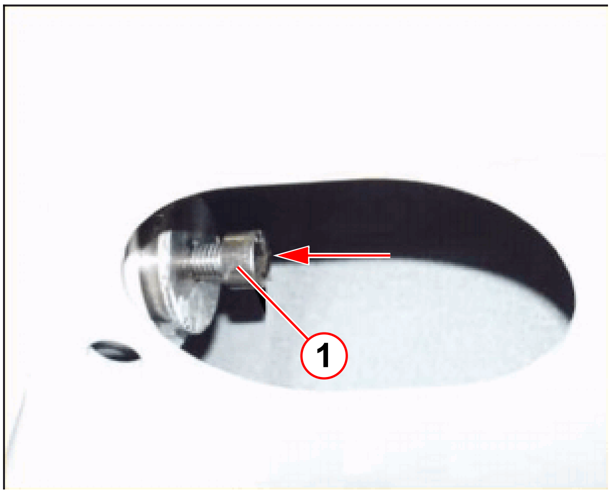
(1) Approx. 45°
(2) Collision switch S30

Fig. 34: Correct position



(1) Correct position

Fig. 35: M10 Allen screw, inside the support beam



() M10 Allen screw

- Pull back the M10 Allen screw (inside the support beam) to connect the spindle nut to the spindle nut driver.

- Tighten the M6 Allen screw of the spindle nut driver.

Fig. 36: M6 Allen screw of spindle nut driver

Before applying the load again, the nut driver should display a small amount of end play and rotational play.

The table load is applied again by continuing to crank upwards.

This relieves the load on the transport device.

3.1.6 Concluding steps

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 37: Energy chain for lifting column

- Reassemble the energy chain.
- Move the table up by hand, use the crank and remove the table service device.
- Install all covers in reverse order.
- Switch ON F22 (RF-room) in the ACC.

3.1.7 Adjustments

- n.a.

3.1.8 Final checks

- Switch on the patient table and check for proper function.
- Perform the safety test of the S30 spindle nut safety switch
- Ramp up the magnet and do a check of the magnetic field.

3.2 Lifting motor / toothed belt of vertical drive

The vertical lifting motor consists of magnetic objects. When replacing the motor, use caution around the magnetic field. Maintain a safe distance to the magnetic field during this procedure.

> **CAUTION**
>
> **The vertical motor consists of magnetic objects.**
>
> **Magnetic objects may cause injury.**
>
> » **Maintain a safe distance to the magnetic field.**

3.2.1 Tools and time required

3.2.1.1 Time

Approx. 30 min.

3.2.1.2 Tools

Non-magnetic tool kit.

3.2.2 Preliminary

- Switch OFF F22 (RF-room).

3.2.3 Disassembly

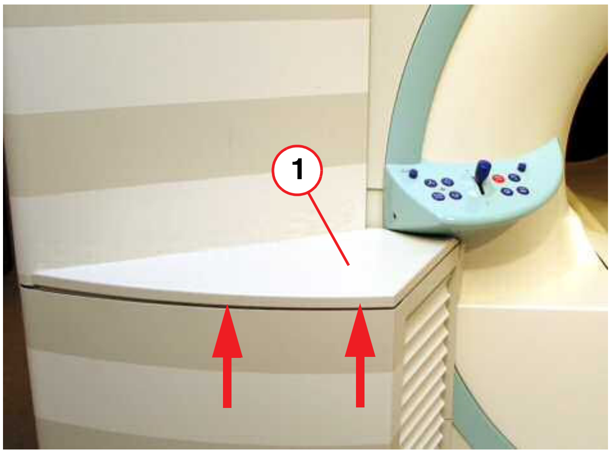
3.2.3.1 Covers

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 38: Service access cover removed

- Remove the service access cover.

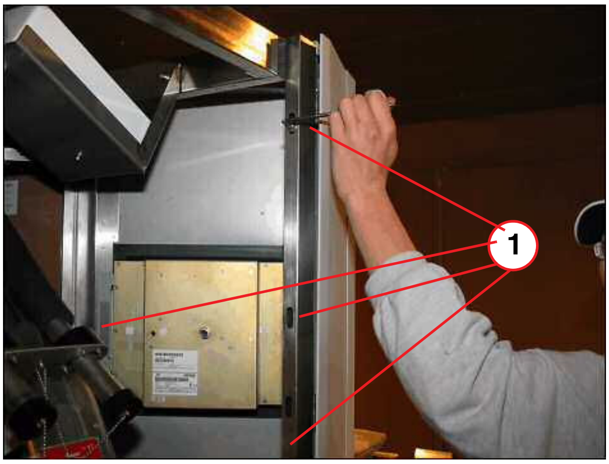
Fig. 39: Lifting column, top cover



(1) Top cover

- Remove top cover of lifting column.

Fig. 40: Top front cover above lifting column



(1) Screws to be removed

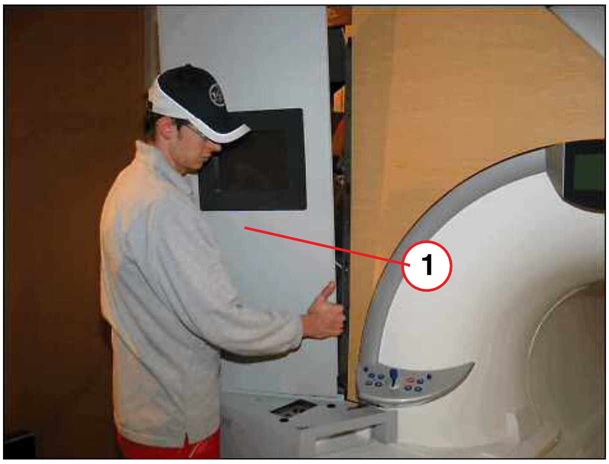
- Remove all screws of top front cover.

Fig. 41: Top front cover above lifting column

(1) Screws to be removed

- Remove all screws inside the top front cover and disconnect cables, if necessary.

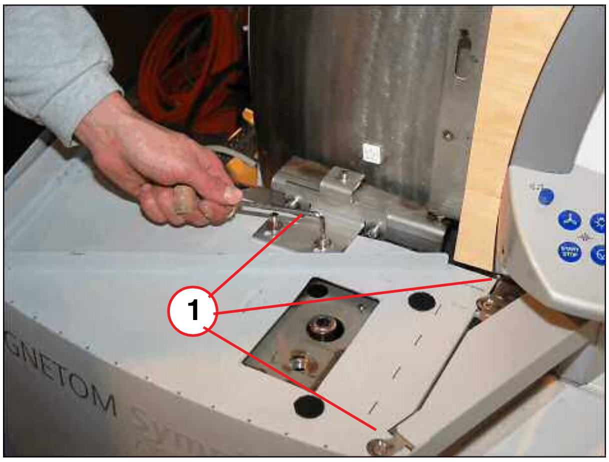
Fig. 42: Top front cover above lifting column



(1) Remove cover

- Remove top front cover.

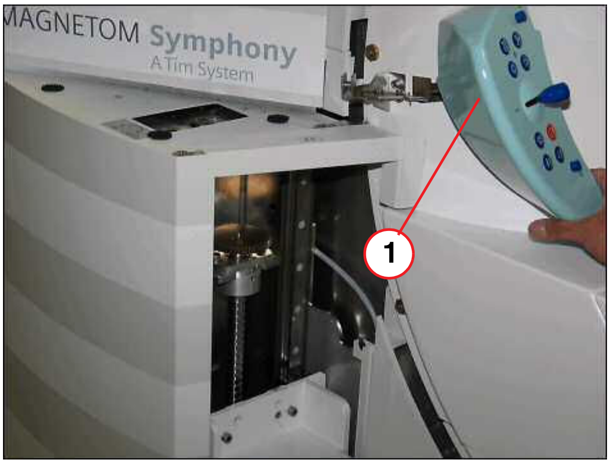
Fig. 43: Lifting column cover



(1) Screws to be removed

- Remove screws from the lifting column and bracket.

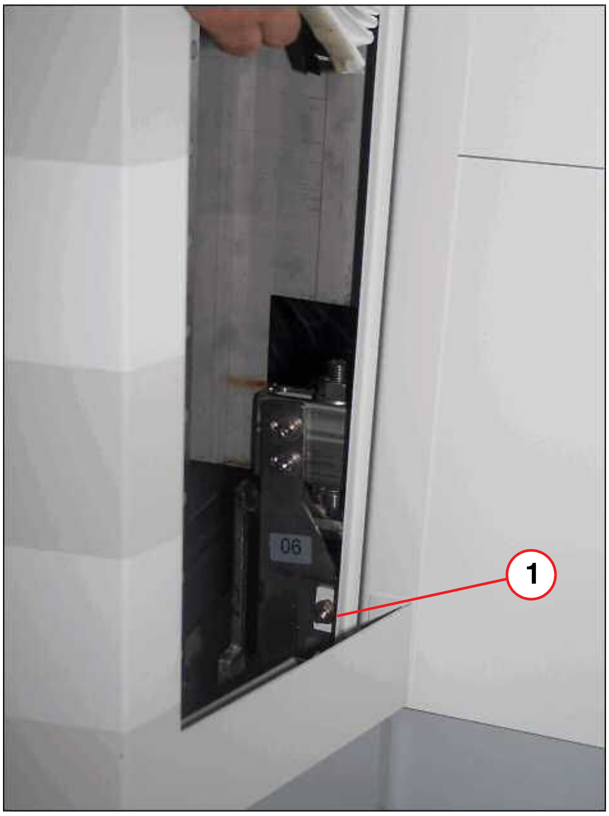
Fig. 44: Control panel



(1) Control panel removed

- Separate the bellows from the adhesive tape and remove upper bellows.
- Loose screw of operating panel, disconnect cable if necessary and remove it.
- Remove the screw behind the control panel to disassembly the lifting column cover.

Fig. 45: Lower part of lifting column



(1) Bellows removed, screw to be removed

- Separate the bellows from the adhesive tape and remove lower bellows.
- Remove screw on the lower part of the lifting column.

Fig. 46: Lifting column cover

- Remove lifting column cover.

3.2.3.2 Vertical motor

Drive the table to the lowest position before replacing the motor or the toothed belt.

Fig. 47: Remove covers and disconnect cables

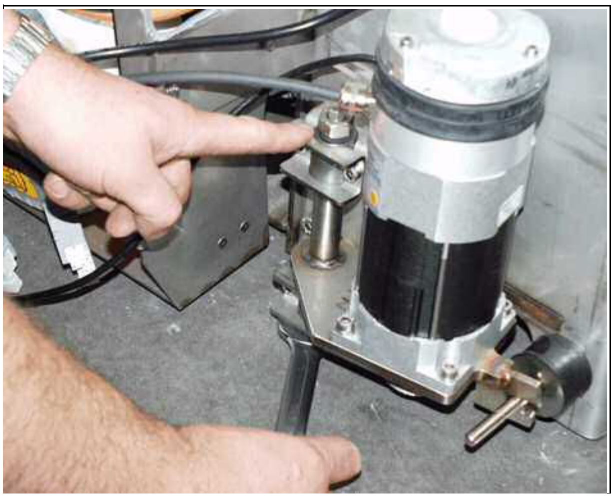
- Remove the ptab electronics plate and disconnect cables belonging to the vertical motor.

Fig. 48: Release toothed belt

- Release toothed belt, press on the motor to remove toothed belt, and replace belt, if necessary.

> **i** The vertical lifting motor consists of magnetic objects. When replacing the motor, use caution around the magnetic field. Maintain a safe distance to the magnetic field during this procedure.

Fig. 49: Release M12 nuts and remove motor



- Remove M12 nuts and turn the motor aside.
- Remove the four Allen screws and **remove the motor immediately from the magnetic field**

3.2.4 Assembly

> i The vertical lifting motor consists of magnetic objects. When replacing the motor, use caution around the magnetic field. Maintain a safe distance to the magnetic field during this procedure.

- Replace motor and safety immediately with the four Allen screws. **When replacing the motor, use caution around the magnetic field.**
- Move the motor back into its original position.
- Reinsert the toothed belt.
- Stretch the toothed belt and tighten the M12 nuts.

3.2.5 Concluding steps

- Reconnect all cables.
- Switch ON F22 (RF-room).
- Install covers.

3.2.6 Adjustments

- n.a.

3.2.7 Final checks

- Switch on patient table and verify proper vertical movement.

3.3 Spindle

3.3.1 General

The table is moved in vertical direction via a special bearing spindle. That spindle consists of a toothed wheel, upper spindle bearing, end stop, spindle nut, lower spindle bearing and lower toothed wheel.

There are two different methods described for replacing the spindle.

If the main bearing is defective it may be necessary to use **method I** with the **table transport device**.

If the vertical movement is still possible, however, the spindle should be replaced due to a damaged brake, use **method II** with a **massive wooden** beam. The wooden beam is part of the spare parts delivery.

The spindle can be replaced by one person.

CAUTION

Strong magnetic field!

Ferromagnetic spindle may cause injury.

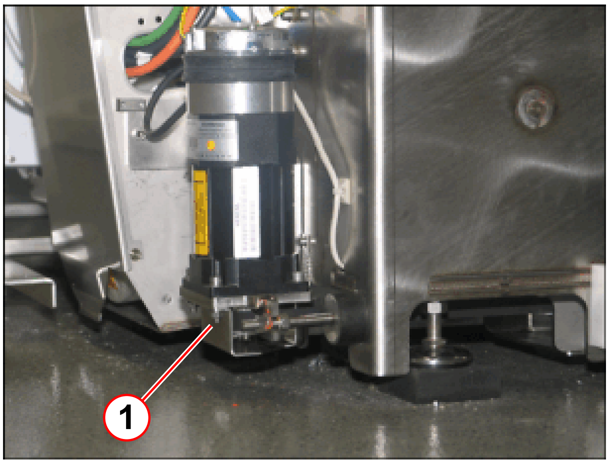
» **Ramp down the magnet**

Two types of spindles are introduced. The newer type has an additional bearing on the spindle brake.

In some pictures, minor differences may occur.

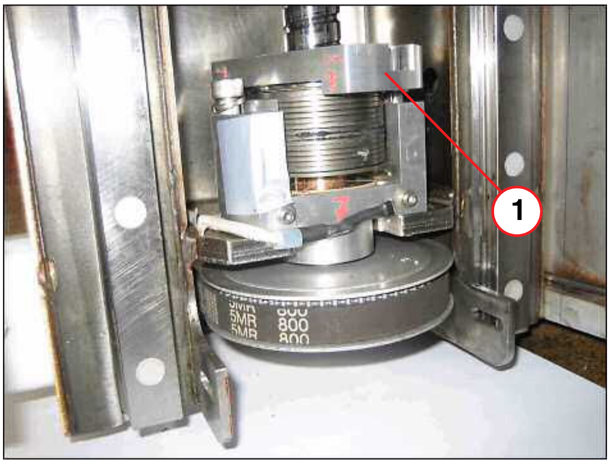
New version of spindle bearing

Fig. 50: Vertical motor



(1) Additional safety cover

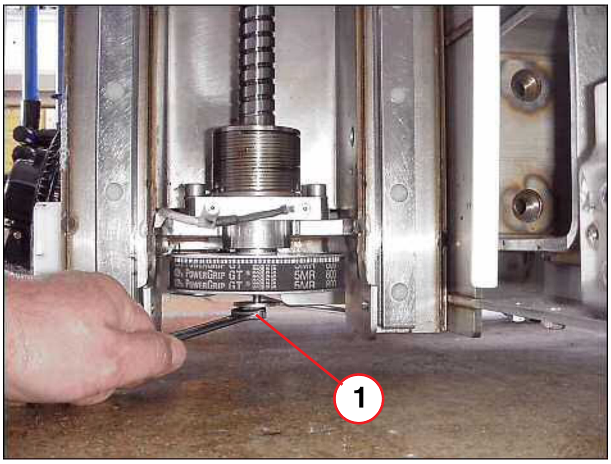
Fig. 51: New version of brake bearing



(1) Additional bearing

Old version of spindle bearing

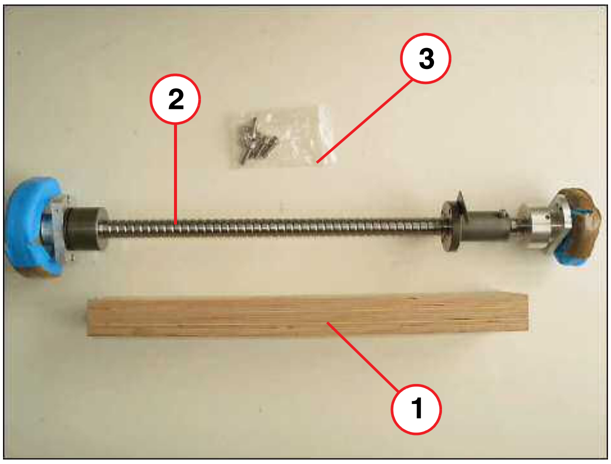
Fig. 52: Toothed disk



(1) Losse mounting screw of toothed wheel 10 mm to12 mm

3.3.2 Spare part items of spindle

Fig. 53: Spindle spare part



(1) Wooden beam
(2) Spindle complete
(3) Spare screws

Fig. 54: Spindle

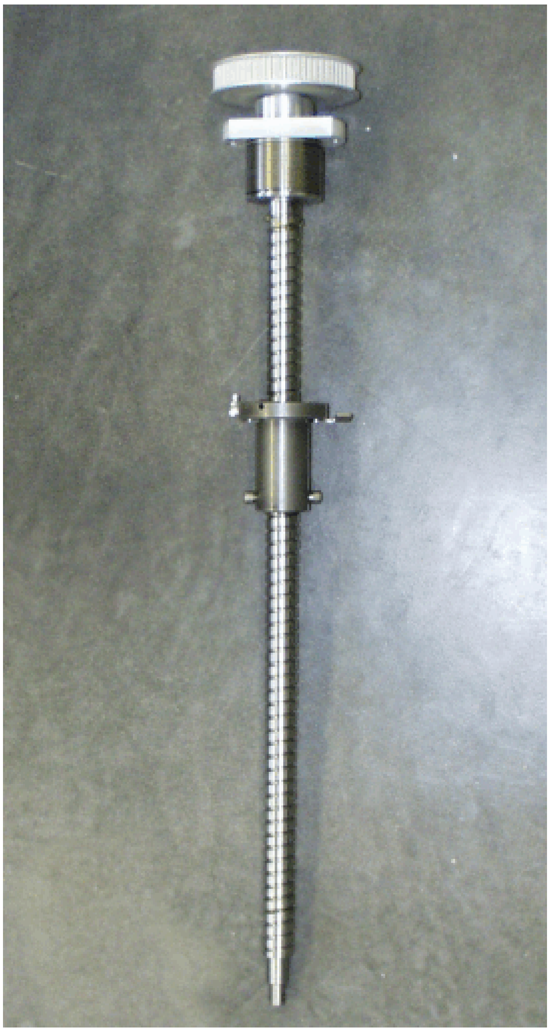
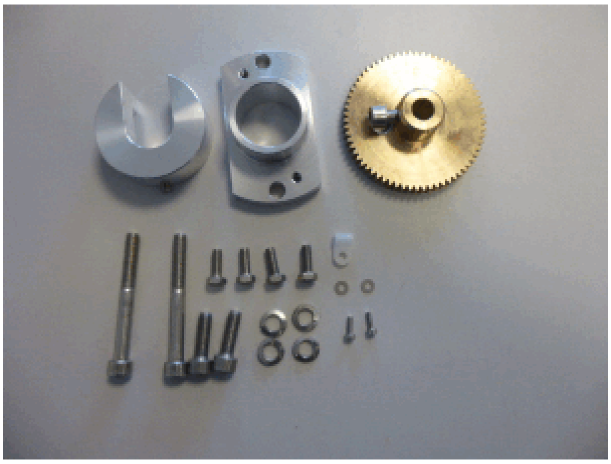
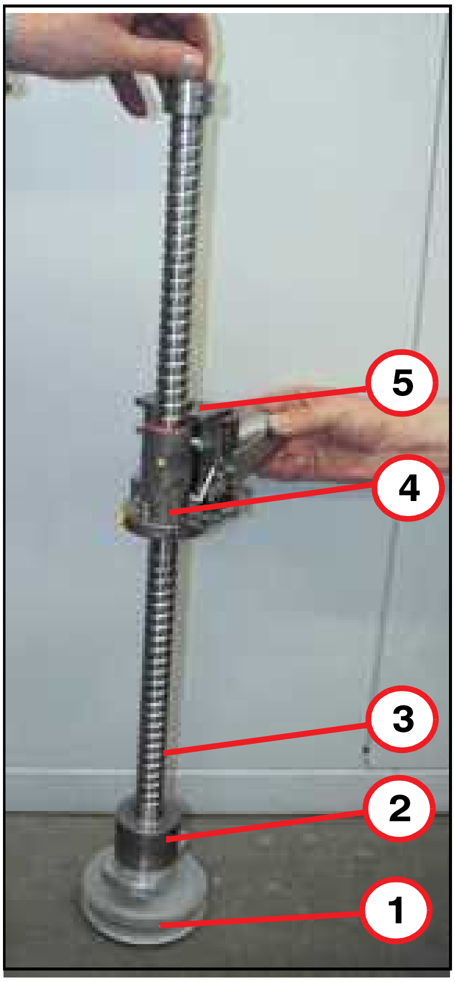


Fig. 55: Spare material



3.3.3 Principle of spindle (old version)

Fig. 56: Principle of spindle



(1) Toothed wheel, lower spindle
(2) Mechanical brake
(3) Spindle:
(4) Spindle nut:
(5) Spindle nut driver

The spindle consists of:

- Lower spindle toothed wheel, driven by a horizontal motor (1).
- Mechanical brake, active during down movement (2).
- Spindle nut with ball bearing circulation spindle (4).
- Spindle nut driver (5).

3.3.4 Tools and time required

3.3.4.1 Time

The estimated time does **not include** ramping the magnet.

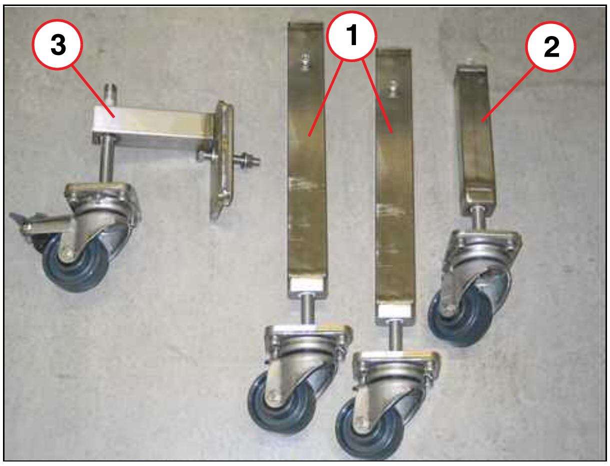
Method I, with table transport device, approx. 120 min

Method II, with wooden beam, approx. 60 min

3.3.4.2 Tools

Table transport device (method I).

Fig. 57: Table transport device



(1) Transport roller 1
(2) Transport roller 2
(3) Transport roller 3

Massive wooden beam as part of the spare parts delivery for the spindle.

Non-magnetic tool kit.

Loctite 221.

Working gloves.

Ratchet and socket kit.

3.3.5 Preliminary

- **Ramp down the magnet**

Any use of grease or oil for the patient table spindle may reduce the brake force and will cause a vertical brake fault.

3.3.6 Disassembly procedurel with table transport device

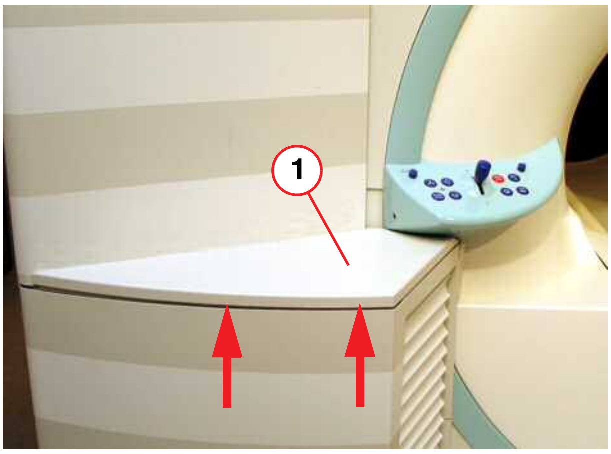
3.3.6.1 Covers

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 58: Service access cover removed

- Remove the service access cover.

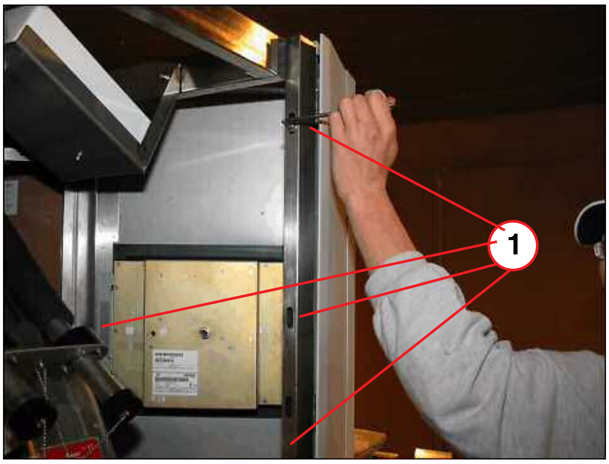
Fig. 59: Lifting column, top cover



(1) Top cover

- Remove top cover of lifting column.

Fig. 60: Top front cover above lifting column



(1) Screws to be removed

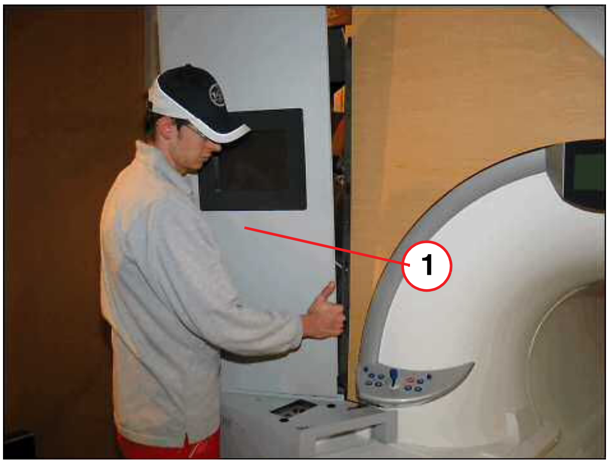
- Remove all screws of top front cover.

Fig. 61: Top front cover above lifting column

(1) Screws to be removed

- Remove all screws inside the top front cover and disconnect cables, if necessary.

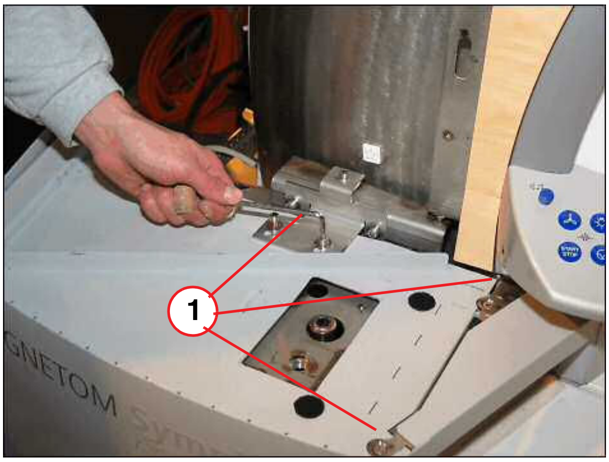
Fig. 62: Top front cover above lifting column



(1) Remove cover

- Remove top front cover.

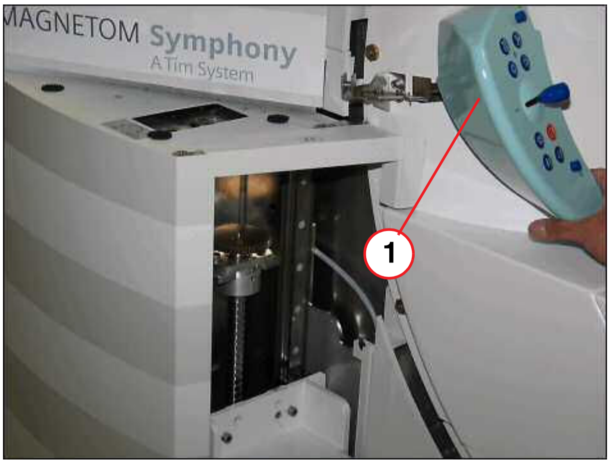
Fig. 63: Lifting column cover



(1) Screws to be removed

- Remove screws from the lifting column and bracket.

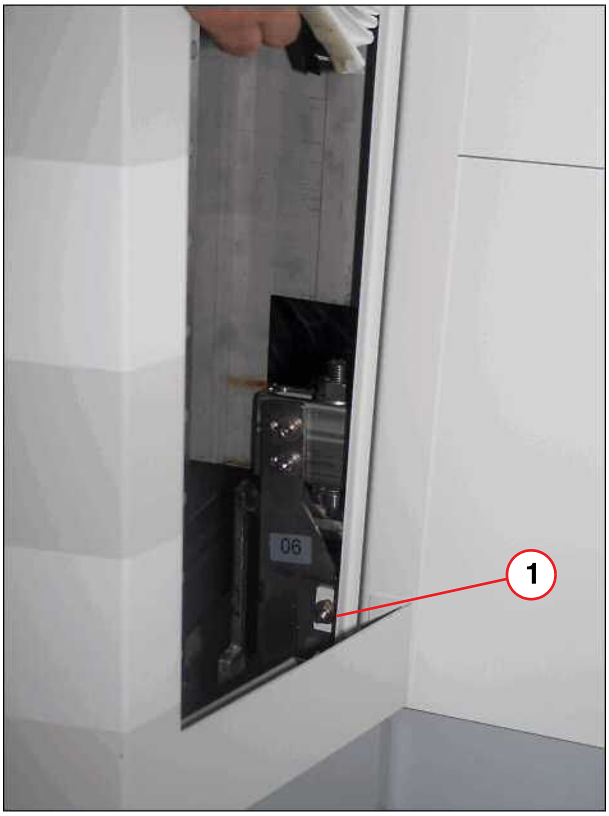
Fig. 64: Control panel



(1) Control panel removed

- Separate the bellows from the adhesive tape and remove upper bellows.
- Loose screw of operating panel, disconnect cable if necessary and remove it.
- Remove the screw behind the control panel to disassembly the lifting column cover.

Fig. 65: Lower part of lifting column



(1) Bellows removed, screw to be removed

- Separate the bellows from the adhesive tape and remove lower bellows.
- Remove screw on the lower part of the lifting column.

Fig. 66: Lifting column cover

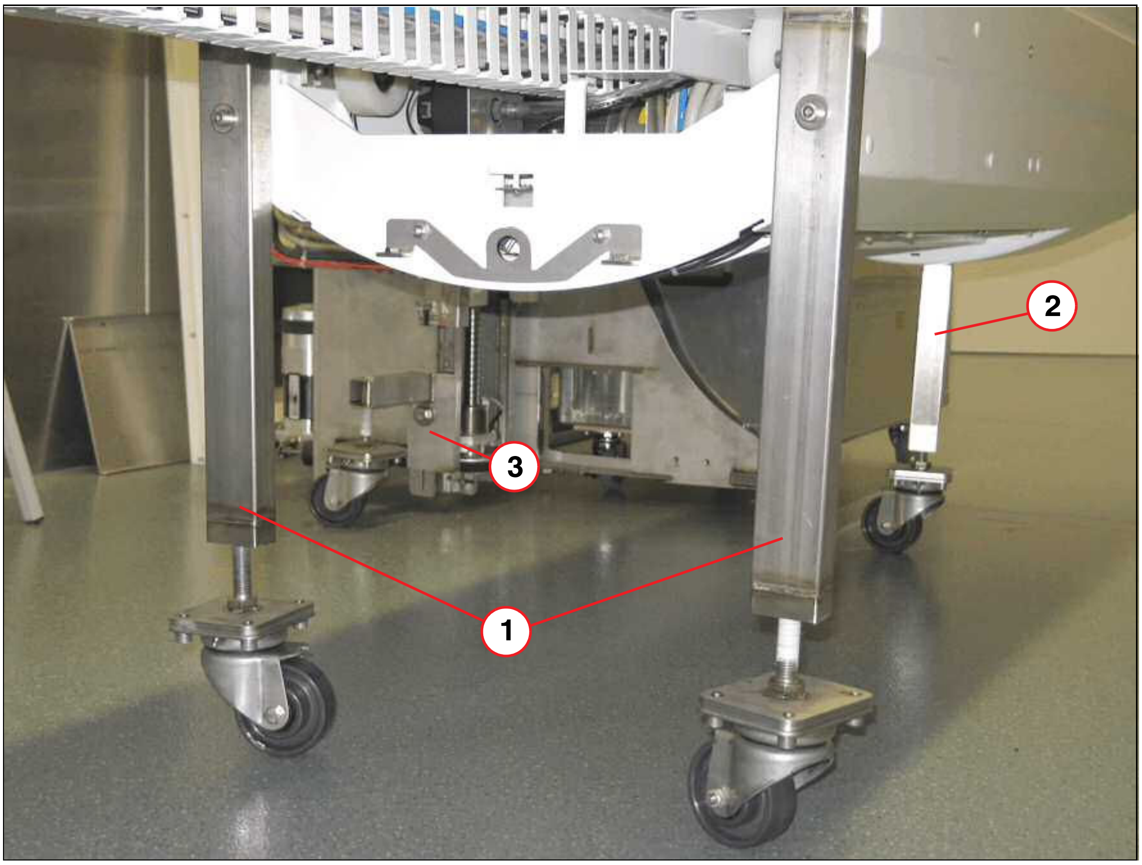
- Remove lifting column cover.

3.3.6.2 Support beam

- Move the table up by hand, use the crank.
- Mount the table service device and lower the table until the lifting column rests on the table service device.

- Switch OFF F22 (RF- room) in the ACC.

Fig. 67: Table transport device installation



(1) Transport roller 1
(2) Transport roller 2
(3) Transport roller 3

Fig. 68: Remove M8 Allen screw

- Remove the four M8 Allen screws, two above and two below.

Fig. 69: Remove two Allen screws inside the support beam

- Remove two Allen screws inside the support beam.

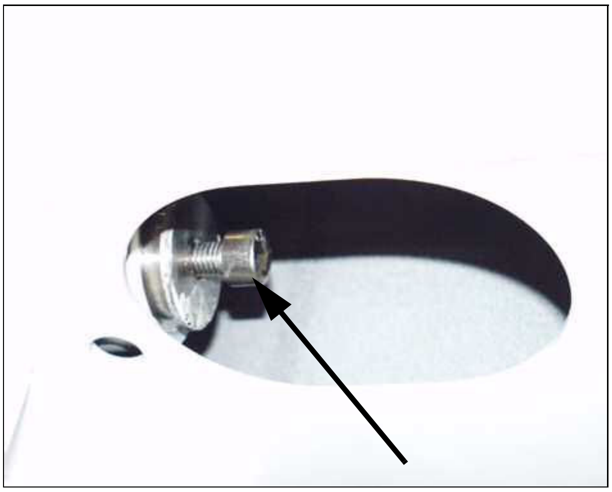
Fig. 70: Energy chain for lifting column

- Dismount the energy chain.

Fig. 71: Remove the M6 Allen screw

- Remove the M6 Allen screw of the spindle nut driver.

Fig. 72: M6 Allen screw, inside the support beam



- Pull back the M6 Allen screw (inside the support beam) to release the spindle nut from the spindle nut driver. The table with the lifting column is now mechanically separated from the spindle.

Fig. 73: M8 Allen screw of support beam

- Remove the bellows support and remove the remaining M8 Allen screws.

Fig. 74: Remove the rest of all (16 in total) other M8 Allen screws

- Remove all other M8 Allen screws and carefully remove the switch panel. Make sure the lifting column rests safely on the table service device.
- To access the spindle, carefully push the table to the side.

3.3.6.3 Toothed belt

Fig. 75: Release toothed belt

- Release toothed belt, press on the motor to remove toothed belt, and replace belt, if necessary.

> If no motor brake is active, the table can **move down** slowly.

3.3.6.4 Cog wheel

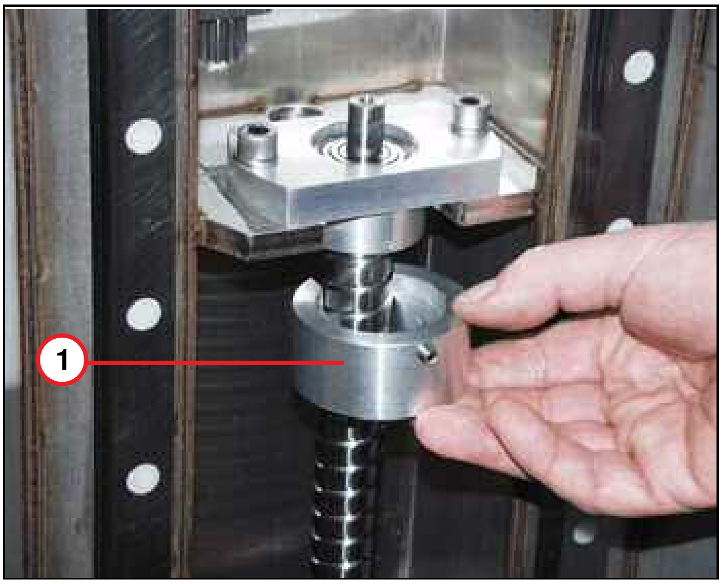
Fig. 76: Upper toothed wheel

(1) Grub screw of upper toothed wheel

- Release the grub screw and remove the cog wheel.

3.3.6.5 End stop

Fig. 77: Upper mechanical end stop



(1) Upper mechanical end stop removed

- Remove the mechanical end stop.

3.3.6.6 Guide bearing

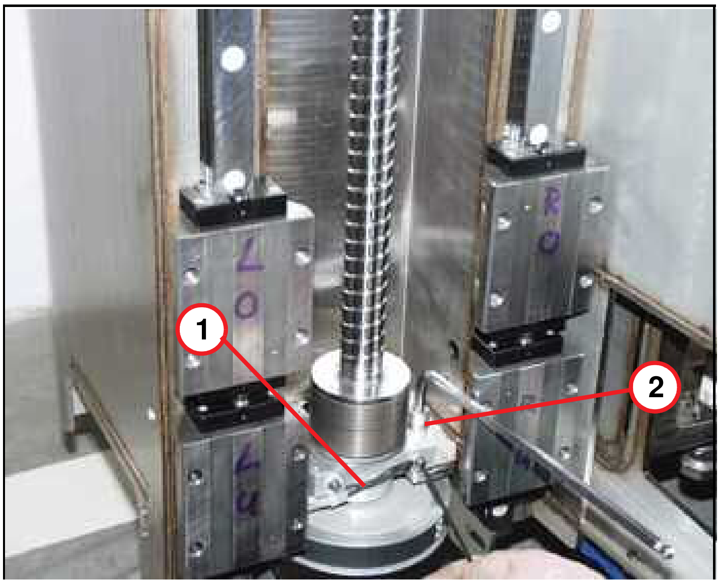
Fig. 78: Spindle bearing

(1) Upper spindle bearing

- Remove both M8 Allen screws from the upper guide bearing.
- If the upper guide bearing cannot be removed slightly, use the longer screws delivered with the spare part.

3.3.6.7 Spindle

Fig. 79: Lower spindle component



(1) Temperature sensor
(2) Lower spindle, bearing block

- Remove the M3 Allen screw for the temperature sensor and both M8 Allen screws for the lower spindle bearing.

- Remove the spindle and replace it. Make sure the thread from the temperature sensor is positioned properly.

> If there are washers bonded to the bottom of the spindle bearing, don’t remove them.

Fig. 80: Remove spindle

3.3.6.8 Disassembly procedure II with a massive wooden beam

Covers

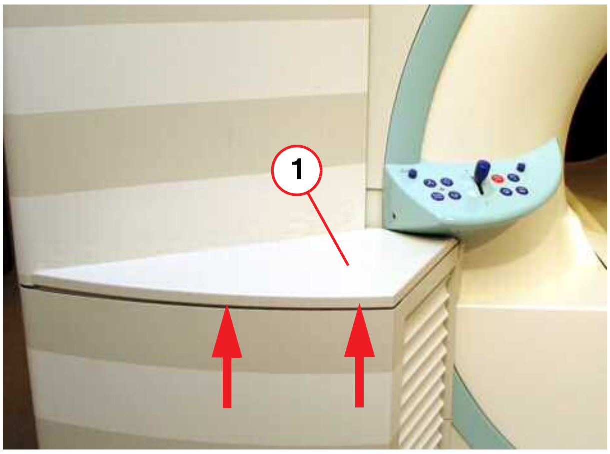
Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

For the vertical movement during exchange procedure II, the patient table can be driven with the normal motor function. Use caution when moving the patient table.

Fig. 81: Service access cover removed

- Remove the service access cover.

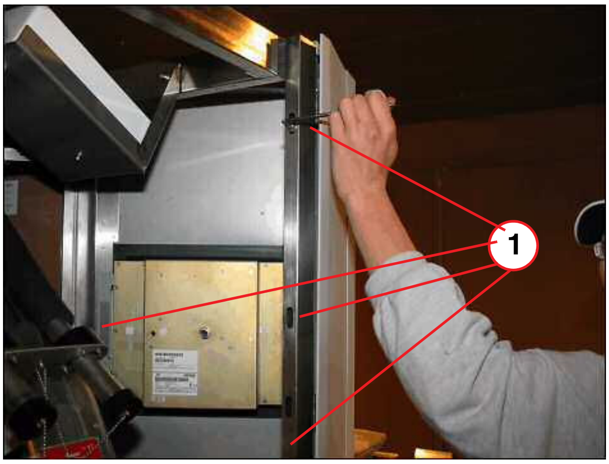
Fig. 82: Lifting column, top cover



(1) Top cover

- Remove top cover of lifting column.

Fig. 83: Top front cover above lifting column



(1) Screws to be removed

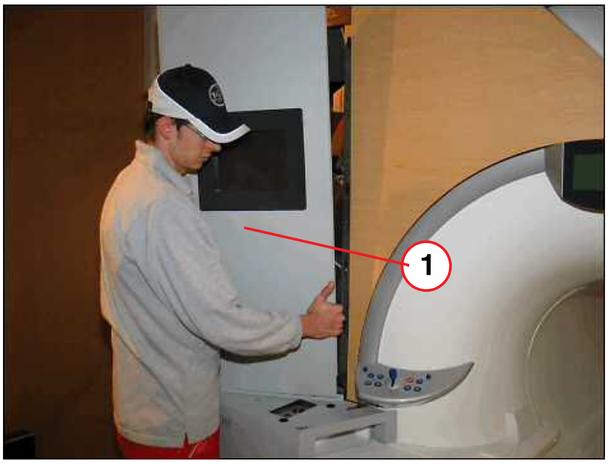
- Remove all screws of top front cove.r

Fig. 84: Top front cover above lifting column

(1) Screws to be removed

- Remove all screws inside the top front cover and disconnect cables, if necessa-ry.

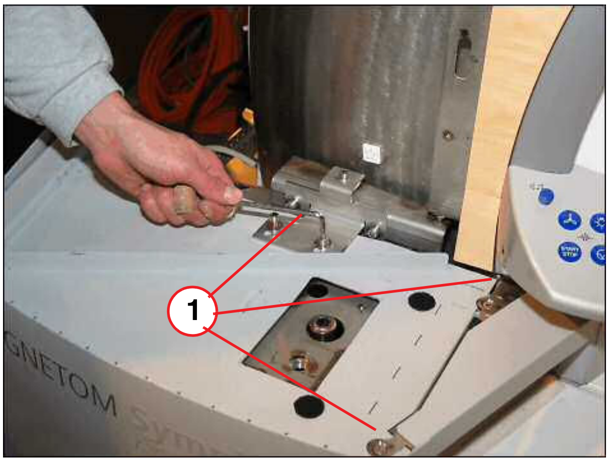
Fig. 85: Top front cover above lifting column



(1) Remove cover

- Remove top front cover.

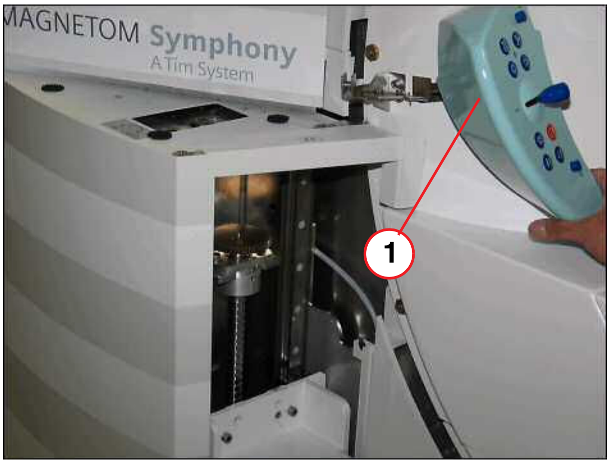
Fig. 86: Lifting column cover



(1) Screws to be removed

- Remove screws from the lifting column and bracket.

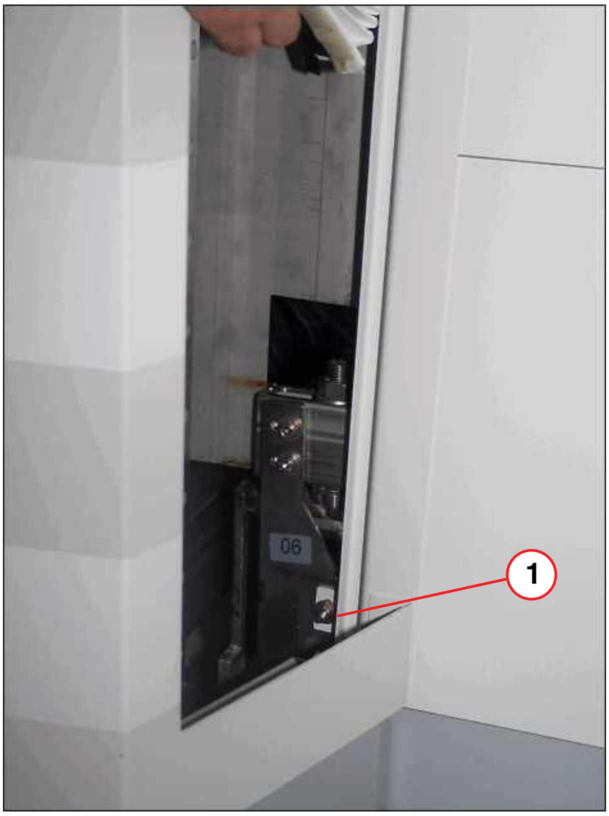
Fig. 87: Control panel



(1) Control panel removed

- Separate the bellows from the adhesive tape and remove upper bellows.
- Loose screw of operating panel, disconnect cable if necessary and remove it.
- Remove the screw behind the control panel to disassembly the lifting column cover.

Fig. 88: Lower part of lifting column



(1) Bellows removed, screw to be removed

- Separate the bellows from the adhesive tape and remove lower bellows.
- Remove screw on the lower part of the lifting column.

Fig. 89: Lifting column cover

- Remove lifting column cover.

Cog wheel

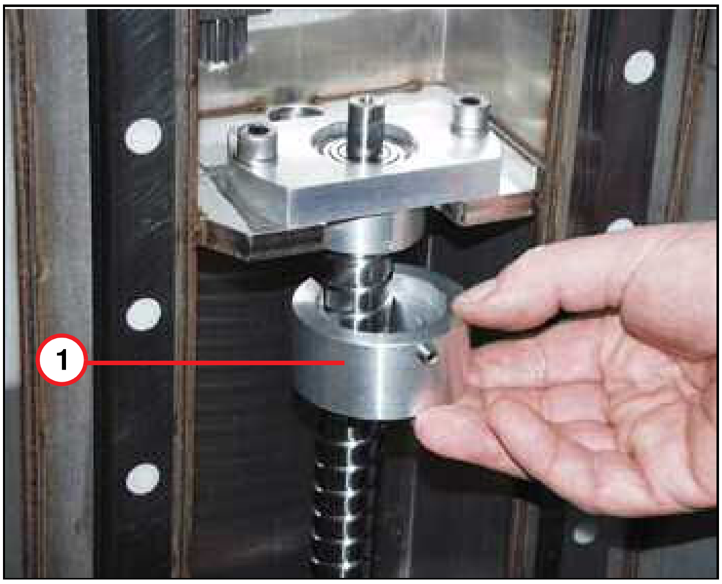
Fig. 90: Upper toothed wheel

(1) Grub screw of upper toothed wheel

- Drive the table down until the upper spindle bearing and the toothed wheel are accessible.
- Release the grub screw and remove the cog wheel.

End stop

Fig. 91: Upper mechanical end stop



(1) Upper mechanical end stop removed

- Remove the end stop.

Guide bearing

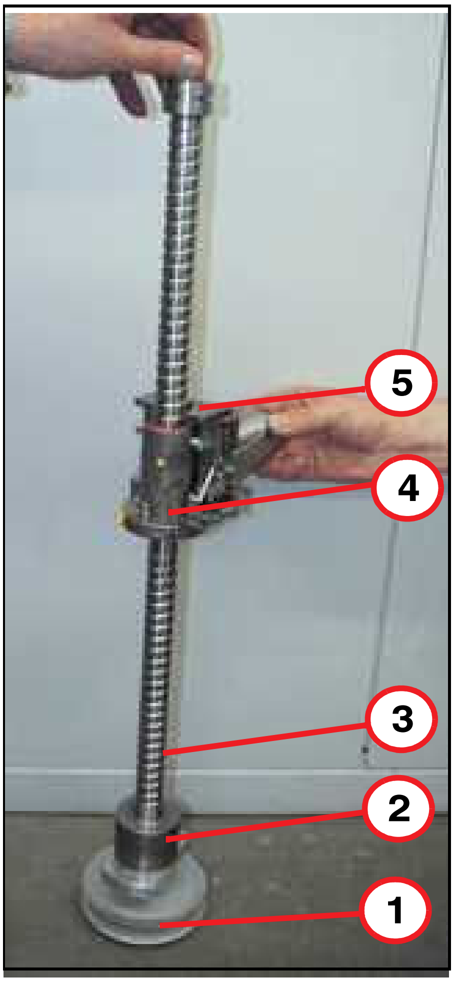
Fig. 92: Spindle bearing

(1) Upper spindle bearing

- Remove both M8 Allen screws from the upper guide bearing.
- If the upper guide bearing cannot be removed slightly, use the longer screws delivered with the spare part.

Spindle

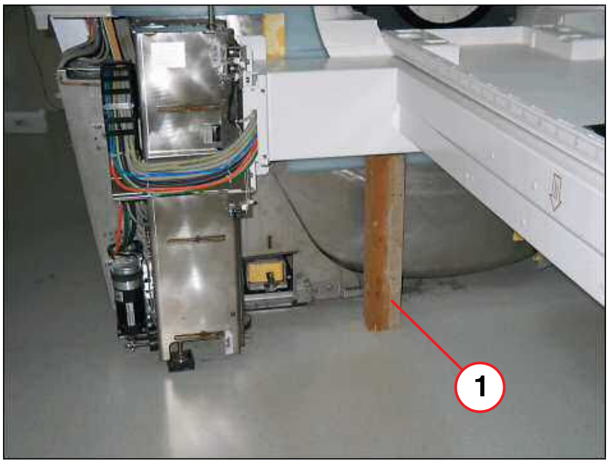
Fig. 93: Principle of spindle



(1) Toothed wheel, lower spindle
(2) Mechanical brake
(3) Spindle:
(4) Spindle nut:
(5) Spindle nut driver

- Move the table up by pressing the table **UP** button. The lower edge of the table (support beam) must reach a height of approx. 600 mm from the floor.

Fig. 94: Support frame



(1) Wooden beam under support frame

- Use a massive wooden beam and place it under the support frame. Wooden beam delivered with the spindle spare part. (Alternative use a frame with a height of 550 mm to carry a load of 220 kg).
- Lower the table by pressing the table **DOWN** button until the lifting column rests on the massive wooden beam or frame. The error (MRI_PXX) ID **63** will show up.
- Switch OFF F22 (RF-room) in the ACC.

Fig. 95: Release toothed belt

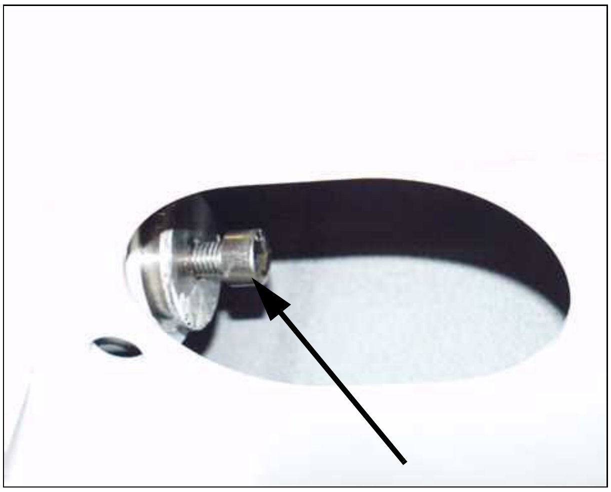
- Release the nut and push the motor to remove the toothed belt.

> **i** If no motor brake is active, the table can **move down** slowly.

Fig. 96: Remove the M6 Allen screw

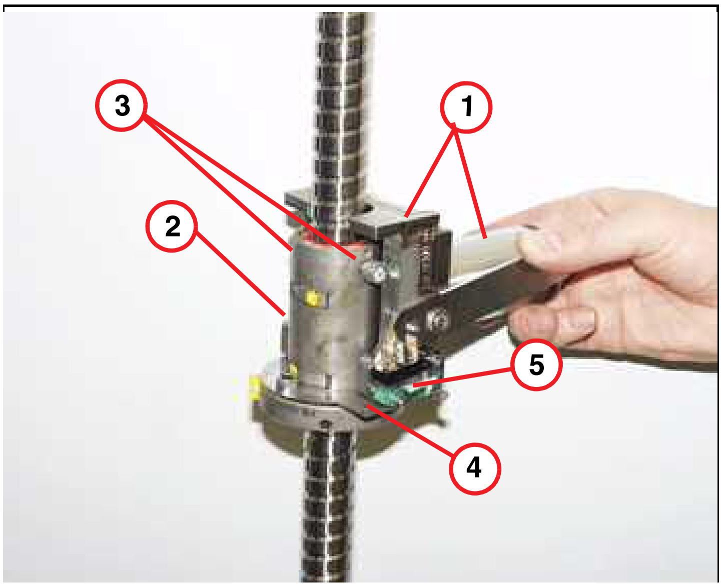
- Remove the M6 Allen screw of the spindle nut driver.

Fig. 97: M6 Allen screw, inside the support beam



- Pull back the M6 Allen screw (inside the support beam) to release the spindle nut from the spindle nut driver. The table with the lifting column is now free mechanically!

Fig. 98: Function of spindle nut

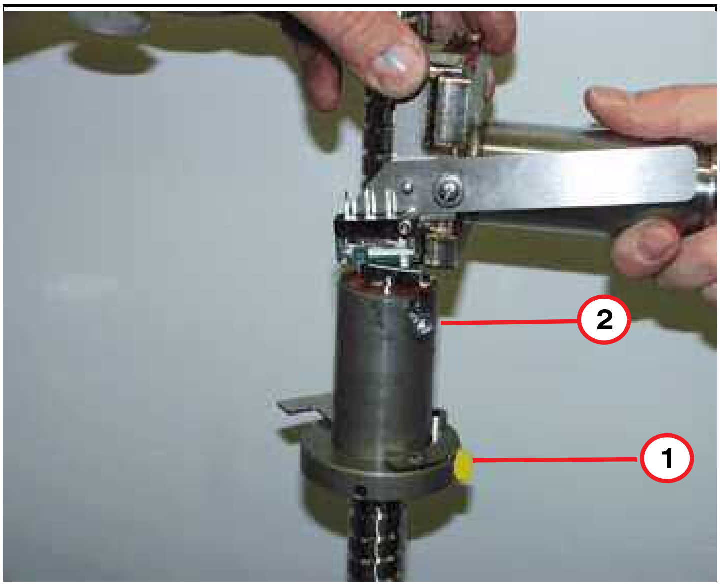


(1) Spindle nut driver
(2) Spindle nut:
(3) Allen screws
(4) Switchplate
(5) Switch S30

- The principle behind disassembly the spindle: Is to turn the spindle nut without demounting the support frame so that the Allen screws are no longer in contact with the spindle nut driver.

  As a result, the spindle nut has to be turned.

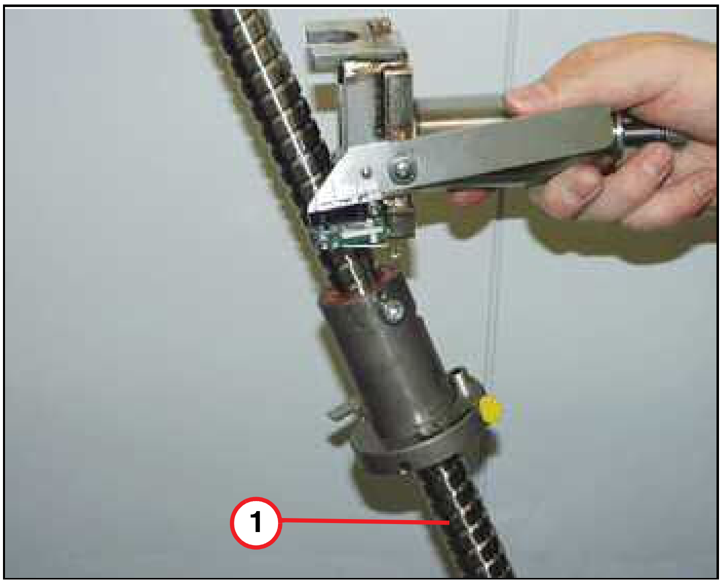
Fig. 99: Spindle nut prepared for spindle replacement



(1) Turned lubrication nipple
(2) Turned Allen screws of spindle nut

- To release the spindle nut from the spindle nut driver: Turn the spindle nut until the lubrication nipple looks towards the support beam, while the spindle nut remains in this position.
- The spindle nut can be driven down by holding the spindle nut and turning the lower toothed wheel by hand.

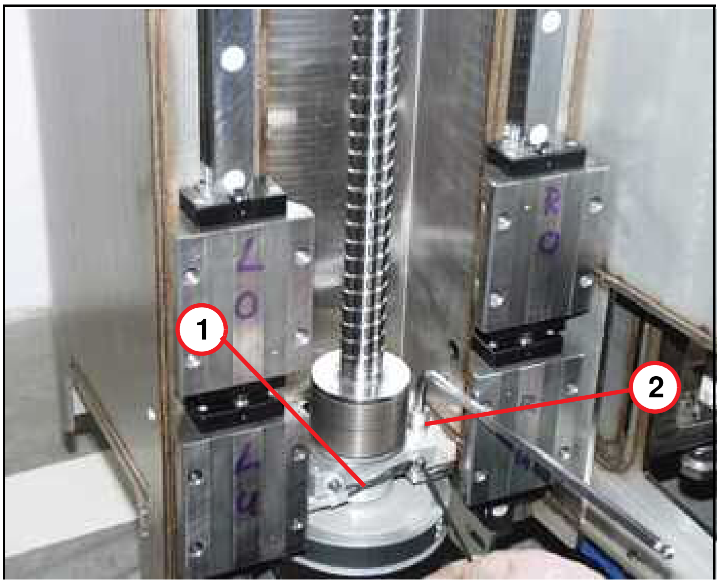
Fig. 100: Principle of spindle nut replacement



(1) Spindle tilt to weave it out

- In case the spindle nut gets stuck due to the spindle nut driver, tilt the top of spindle. In this way, the spindle has more room to pass the spindle nut while driving down.

Fig. 101: Lower spindle component



(1) Temperature sensor
(2) Lower spindle, bearing block

- Remove the M3 Allen screw for the temperature sensor and both M8 Allen screws for the lower spindle bearing.

> If there are washers bonded to the bottom of the spindle bearing, don't remove them.

- Remove the spindle and replace it. Make sure the thread from the temperature sensor is positioned properly.

Fig. 102: Remove spindle

Assembly method I

Spindle

- Mount the new spindle with both M8 Allen screws and safety the lower bearing, re-connect the temperature sensor.
- Position the spindle nut by turning the cog wheel by hand (until the spindle nut rea-ches the same level as the support beam).
- Tighten the M8 Allen screws of the lower and upper spindle bearing . Make sure the thread from the temperature sensor is positioned properly.

i If no motor brake is active, the table can **move down** slowly.

- Insert the upper cog wheel and tighten the grub screw.
- Insert the end stop.

Support beam

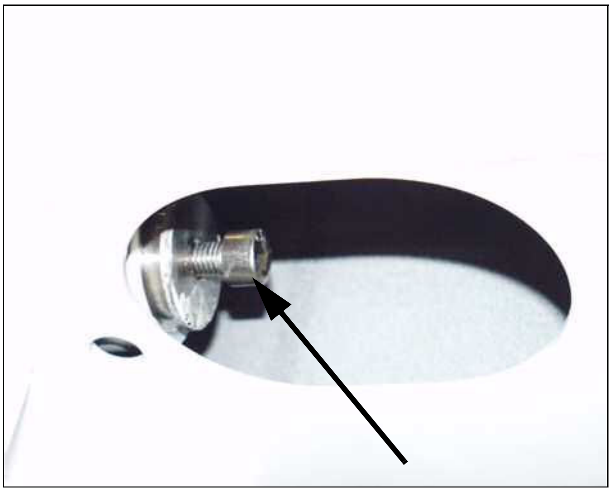
Fig. 103: Move table back

- Carefully move the table back to the lifting column.
- Move up the bearings and install them to the lifting column with one screw at each corner.

Fig. 104: Safety bearings with Allen screws

- Reassemble the switch bracket, insert all Allen screws and tighten them.

Fig. 105: M6 Allen screw, inside the support beam



- Turn the spindle nut into position and push the M6 Allen screw (inside the support beam) back to contact the spindle nut driver.

- Insert the M6 Allen screw (apply Loctite 221) and safety the spindle nut driver.

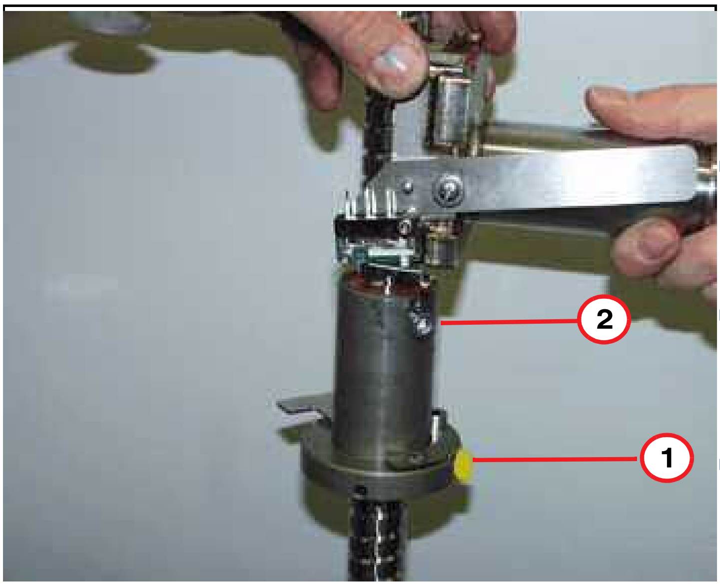
Fig. 106: M6 Allen screw of spindle nut driver

Make sure the M6 Allen screw of the spindle nut driver **cannot** be pulled back.

Assembly method II

Spindle

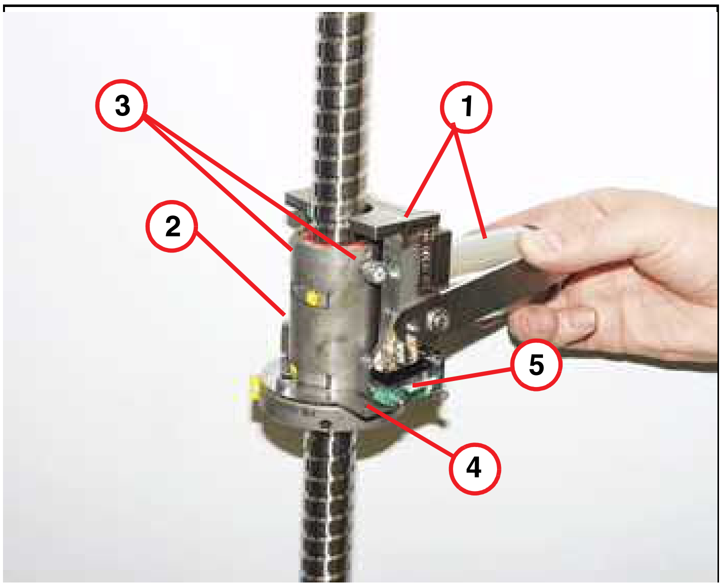
Fig. 107: Spindle nut prepared for spindle replacement



(1) Turned lubrication nipple
(2) Turned Allen screws of spindle nut

- Insert the new spindle and safety the lower spindle bearing with both M8 Allen screws. Reconnect the temperature sensor. Make sure the thread from the temperature sensor is positioned properly.
- Drive the spindle nut up by holding it and turning the lower toothed wheel by hand. Until the spindle nut reaches the same level as the spindle nut driver.
- Turn the spindle nut until the lubrication nipple looks toward the support beam, while the spindle nut remains in this position.
- Drive the spindle nut up into the spindle nut driver: By holding the spindle nut in this position and turning the lower toothed wheel by hand until it stops.

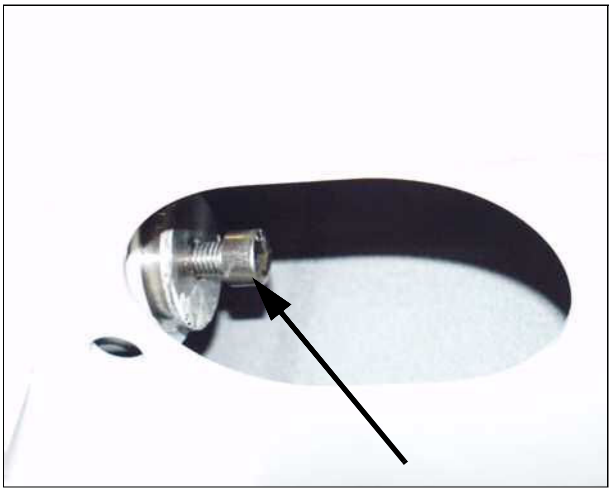
Fig. 108: Function of spindle nut



(1) Spindle nut driver
(2) Spindle nut:
(3) Allen screws
(4) Switchplate
(5) Switch S30

- Turn the spindle nut and lower the spindle nut until both Allen screws are positioned in the spindle nut driver. The switch bracket of S30 is in the correct position.

Fig. 109: M6 Allen screw, inside the support beam



- Turn the spindle nut into position and push the M6 Allen screw (inside the support beam) back to contact the spindle nut driver.

- Insert the M6 Allen screw (apply Loctite 221) and safety the spindle nut driver.

Fig. 110: M6 Allen screw of spindle nut driver

Make sure the M6 Allen screw of the spindle nut driver **cannot** be pulled back.

- Switch on F22 (RF- room) in the ACC.
- Drive the table up, by pressing the table **UP** button and remove the wooden frame, lower the table until the upper spindle part is accessible.

> If no motor brake is active, the table can **move down** slowly.

- Remount the upper spindle bearing and tighten the M8 Allen screws of the lower and upper spindle bearing.
- Insert the upper cog wheel and tighten the grub screw.
- Insert the end stop.

Concluding steps

- Install the toothed belt of the vertical drive and tighten it. If necessary, replace belt.
- Install the lower bellows bracket.
- Install the bellows guide.
- If necessary switch ON F22 (RF-room) ptab and verify proper vertical movement.
- Drive the table up and down (approx. 20 times) until the spindle is getting hot and the error message **(MRI_PXX) 66** shows up. Wait until the spindle is cooled down.
- Install the covers.

Adjustments

- n.a.

Final checks

- Verify proper vertical movement.

3.4 Brake Disk

3.4.1 General

The brake disk is responsible for the brake torque during downward movement of the patient table. The brake disk can be replaced (in accordance with the serial numbers) with the magnet at field as long as anti-magnetic tools are used.

CAUTION

Strong magnetic field.

Magnetic objects may cause injury.

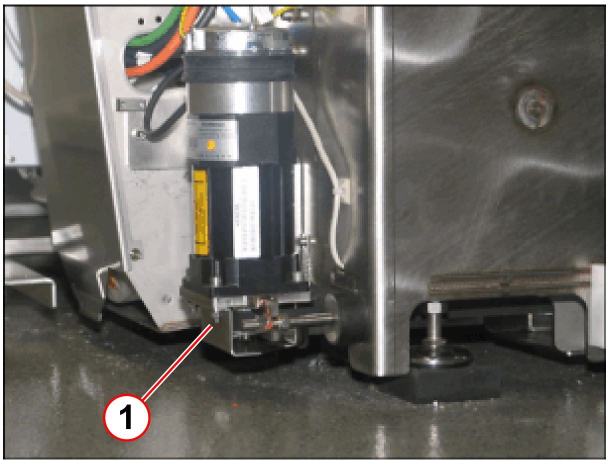
» **Maintain a safe distance to the magnetic field.**

Two types of spindles are introduced. The newer type has an additional bearing on the spindle brake.

In some pictures, minor differences may occur.

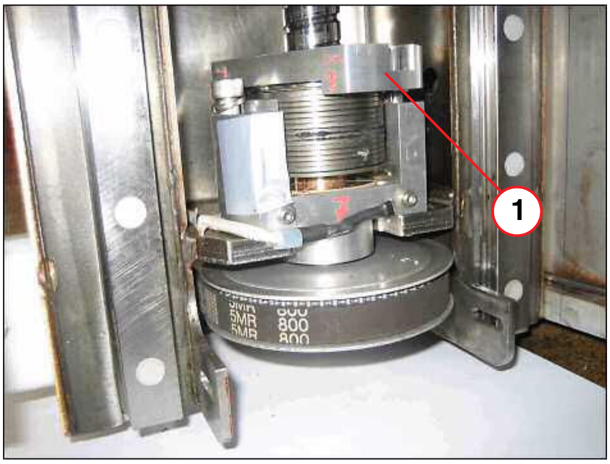
New version of spindle bearing

Fig. 111: Vertical motor



(1) Additional safety cover

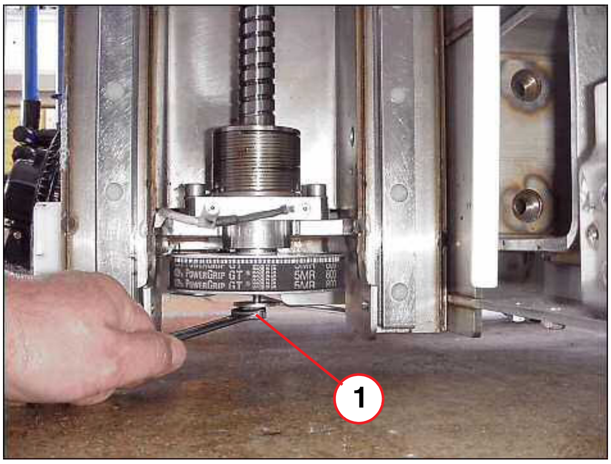
Fig. 112: New version of brake bearing



(1) Additional bearing

Old version of spindle bearing

Fig. 113: Toothed disk



(1) Losse mounting screw of toothed wheel 10 mm to12 mm

3.4.2 Tools and time required

3.4.2.1 Time

Approx. 60 min

3.4.2.2 Tools

Non-magnetic tool kit.

Wooden beam (delivered with the spare part disk).

3.4.3 Preliminary

- **Only for basic tables with serial numbers > 1529, exchangeable tables with serial number > 1094, and all spindles replaced after June 2005**
- **A worn brake disk can be indicated by error 77 after the number of lifts exceeds 100000 movements.**
- Switch OFF F22 (RF-room) in the ACC.

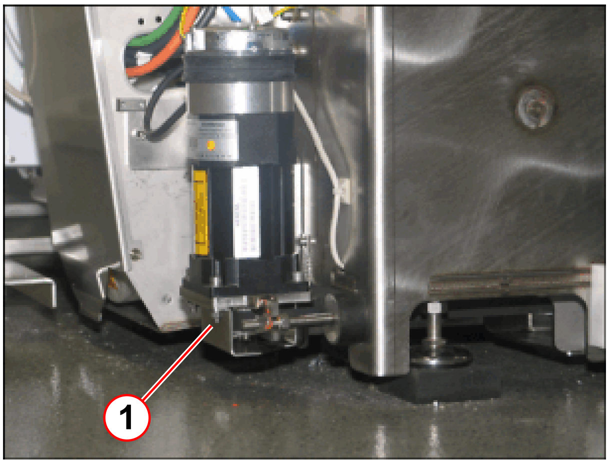
3.4.4 Disassembly

3.4.4.1 Brake disk

Any use of grease or oil for the patient table spindle may reduce the brake force and will cause a vertical brake fault.

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

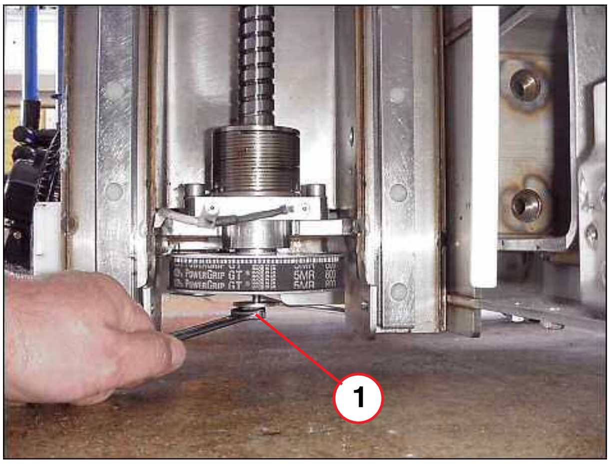
Fig. 114: Vertical motor



(1) Additional safety cover

- On newer system, remove the safety cover.

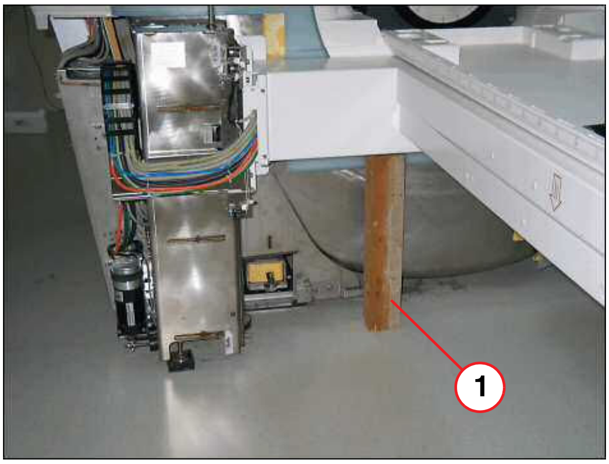
Fig. 115: Toothed disk



(1) Losse mounting screw of toothed wheel 10 mm to12 mm

- Release mounting screw of toothed wheel approx. 10 mm- 12 mm (do not remove the screw).

Fig. 116: Support frame



(1) Wooden beam under support frame

- Place the wooden beam under the support frame of the patient table.

⚠ CAUTION

Possible damage of the spindle nut.

» **After the wooden support beam is supporting the frame and the load is removed from the spindle nut, do not continue to lower the table with the manual crank.**

Fig. 117: Crank for vertical movement

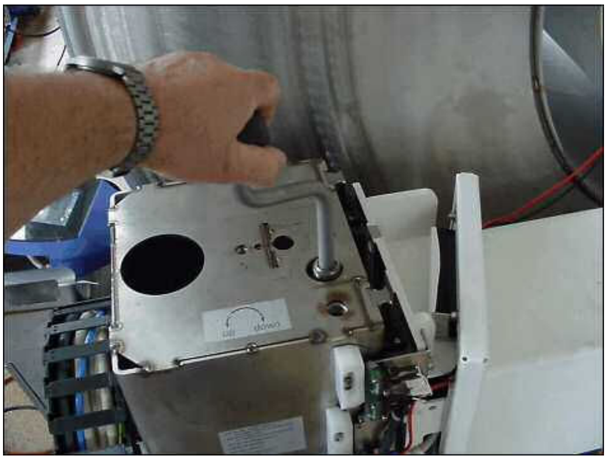
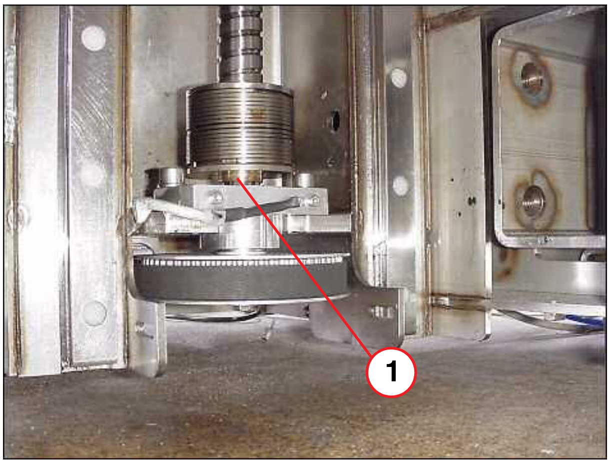


Fig. 118: Brake disk with clearance



(1) Clearance between brake disk and mouting bracket approx.10 mm -12mm.

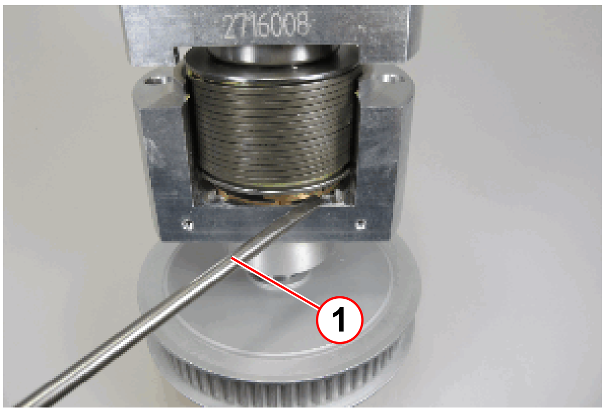
- Crank the patient table down until the complete spring clutch is raised and a clearance of 10 mm- 12 mm is visible above the brake disk.

  The clearance is limited by the released screw on the button of the toothed wheel.

  !If the clearance is not visible, it might be necessary to lift up the spindle manually by using a big screwdriver!

  After the wooden support beam is supporting the frame and load is removed from the spindle nut, do not continue to lower the table with the manual crank.

Fig. 119: Removing the brake disc



(1) Flat-bladed screw driver

- Remove the old brake disk using a small flat bladed screwdriver.

  Clean the clearance after removing the old brake disk.

Fig. 120: Lower surface: copper colored

Fig. 121: Upper surface: grooved

- A worn brake disk can be indicated by Error 77.

  The color of the sliding surface will be primarily bronze or copper.

  Clean the clearance after removing the old brake disk.

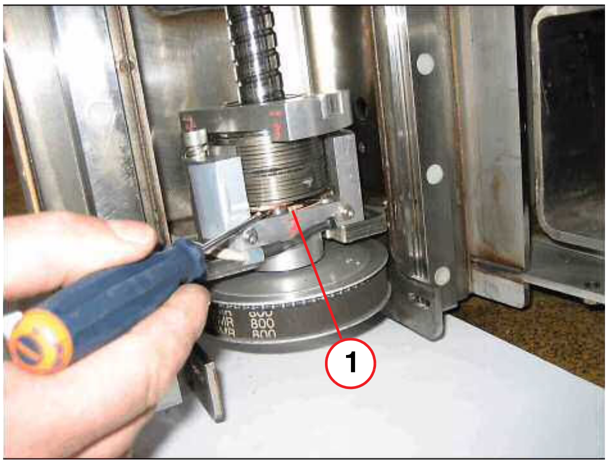
> Make sure that the grooved side of the brake disc faces up.

Fig. 122: Brake disks

Fig. 123: Replace new brake disk

(1) Flad blade screw driver

Fig. 124: Removing the brake disk (newer version)

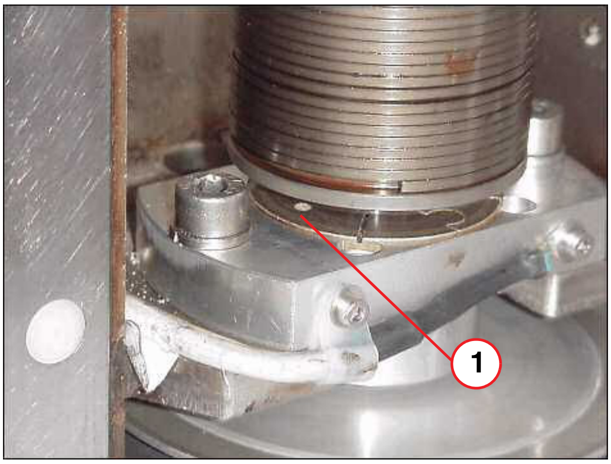


(1) Flat blade screw driver

- The spare part disk halves are positioned below the clutch section.

  Clean the clearance after removing the old brake disk.
- Using a small flat bladed screwdriver, first insert the half with the hole and close it in place with the securing pin.

Fig. 125: Brake disk

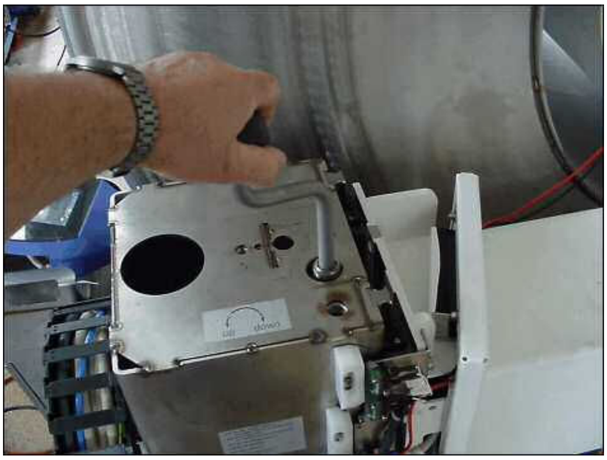


(1) One half of brake disk placed into the securing pin

- Both disk halves lie flush. The half with the hole is closed in place with the securing pin.

  Both halves, one fitted into the other, have to lie flush.

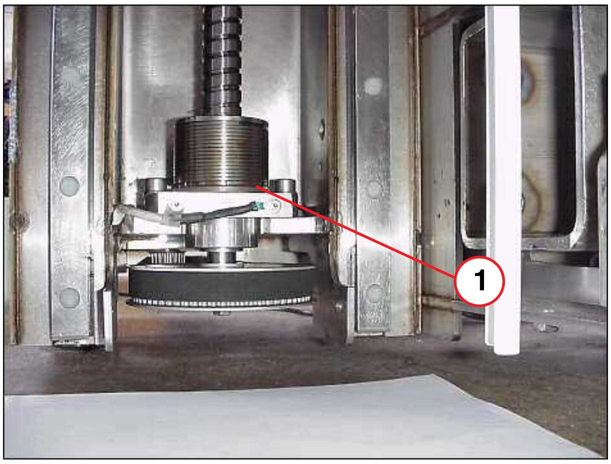
Fig. 126: Crank for vertical movement



- Use the hand crank to move the table up until the spring clutch is in contact with the brake disk.
- Remove the wooden beam under the support frame.

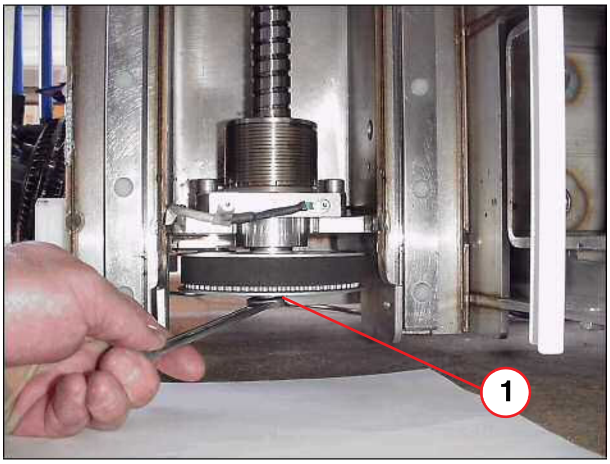
  There should be no clearance between the brake disk and the spring clutch.

Fig. 127: Brake disk in contact with spring clutch



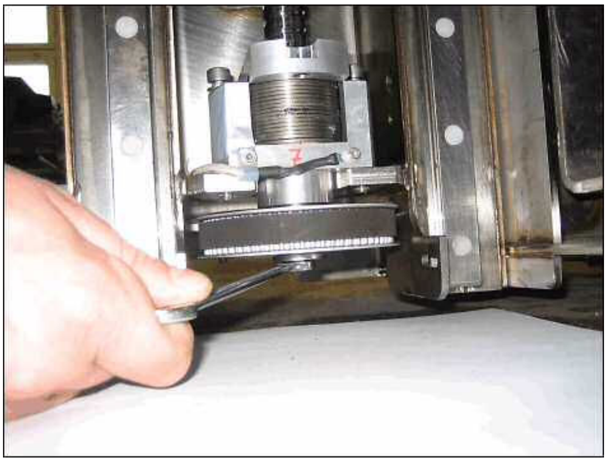
(1) Brake disk seated on the spring clutch

Fig. 128: *Mounting screw of toothed disk*



(1) Apply Loctite and tighten mouting screw

Fig. 129: *Mounting screw of toothed disk (new version)*



- Apply Loctite 221 and tighten the mounting screw of the toothed disk.
- If present, remount safety bracket.

3.4.5 Assembly

- Install the lower bellows bracket.
- Install the bellows guide.

3.4.6 Concluding steps

- Install the covers.

3.4.7 Final checks

- Switch ON F22 (RF-room) for the patient table and verify proper vertical movement.
- Initiate up and down movement several times. The spindle brake needs to be applied several times before full stopping power is attained.

3.5 Tabletop

3.5.1 General

When replacing the tabletop plate, we recommend a second person be used to help.

A minimum distance between the magnet and the wall is required for replacement of the tabletop plate. Depending on the system type, the minimum distances are:

- Avanto **1450 mm**
- Espree **1650 mm**
- Trio a Tim System **850 mm**
- Symphony a Tim System **1300 mm**
- Verio **1300 mm**

3.5.2 Tools and time required

3.5.2.1 Time

Approx. 60 min

3.5.2.2 Tools

Non-magnetic tool kit

3.5.3 Preliminary

- Switch OFF F22 (RF-room) in the ACC

> According to the system/room specifications the movement range of the horizontal movement can be reduced.

With the whole body version of the horizontal drive, the movement range of the horizontal movement may be reduced. If that is the case, remove the end stops. (→ Fig. 306 Page 179)

Before disassembly the patient table covers on a system with a partial drive unit: The toothed belt must be released to push the table top to the service end of the magnet. Refer to (→ Prerequisite for cover removal / partial drive / Page 104)

3.5.4 Disassembly

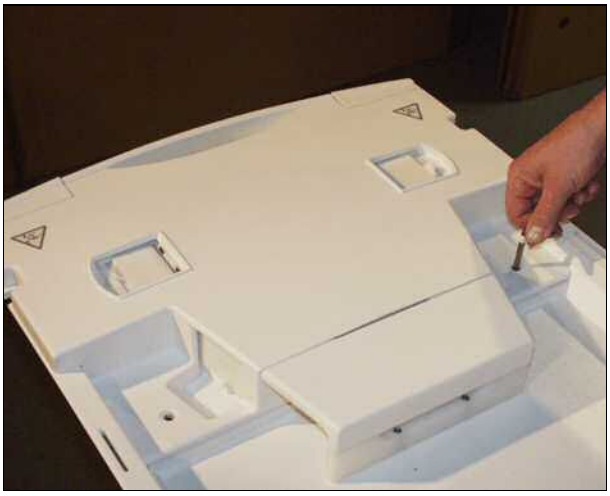
3.5.4.1 Covers

Use caution when moving the tabletop plate > 30 kg. A second person is necessary.

Be careful to avoid damaging any cables when removing the covers of the patient table. Use caution when moving the patient table.

Basic table

Fig. 130: Coil plug cover (foot end)



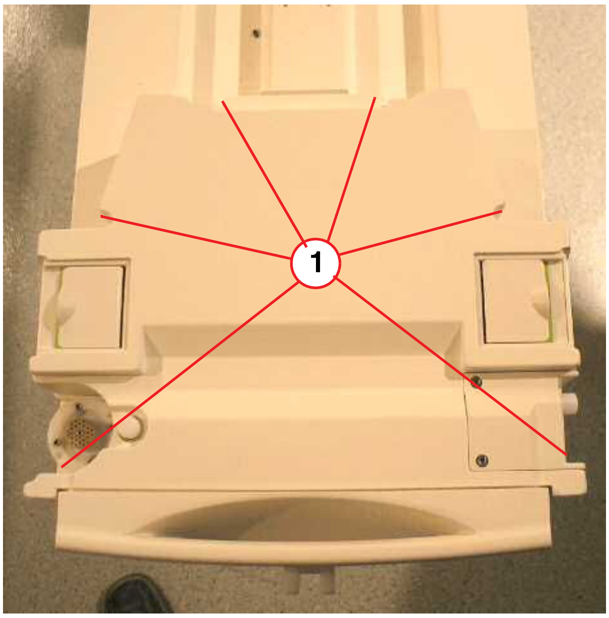
- Remove the cover of the coil plug.

Fig. 131: Remove Allen screws

- Remove the lower Allen screws of the cover from the coil plugs.

Removable table

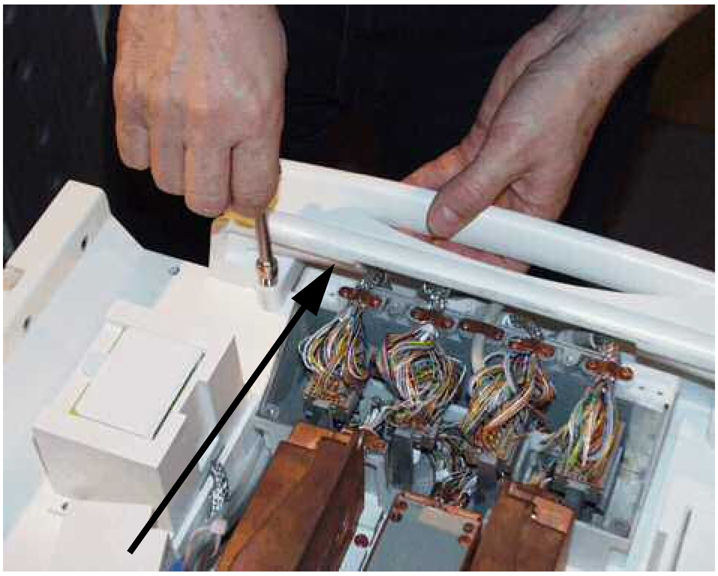
Fig. 132: RF-Cover RTT



(1) Screws to be removed

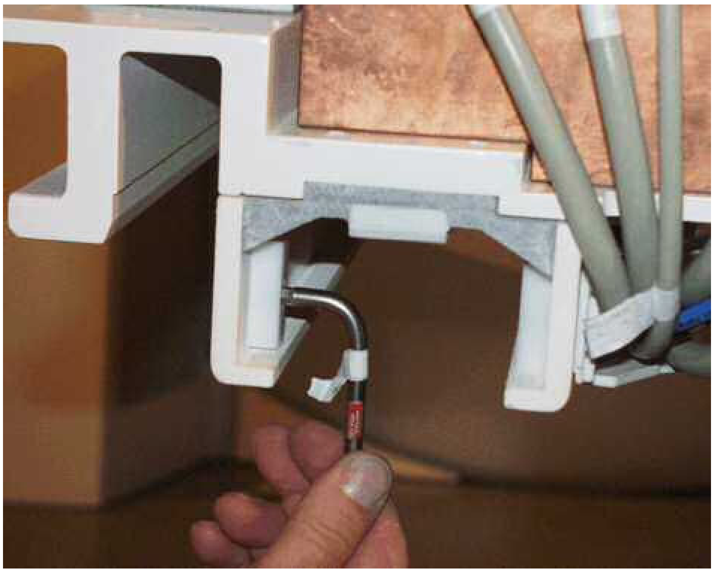
- Remove the carrier plate of the RTT.
- Remove six screws on top of the cover from the coil plugs
- Disconnect cables.

Fig. 133: Foot-end cover



- Remove the two M8 Allen screws, if necessary disconnect cables, and remove foot-end plate.

Fig. 134: Mechanical end stop



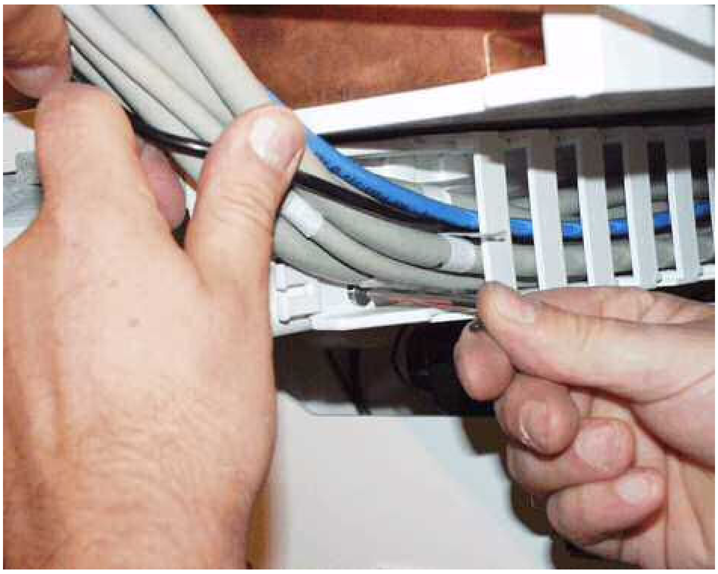
- Remove mechanical end stops.

3.5.4.2 Cables/energy chain

- Disconnect all cables and air hoses. Use a flat screwdriver to open the RF connectors.

Fig. 135: Cable connections

Fig. 136: Energy chain



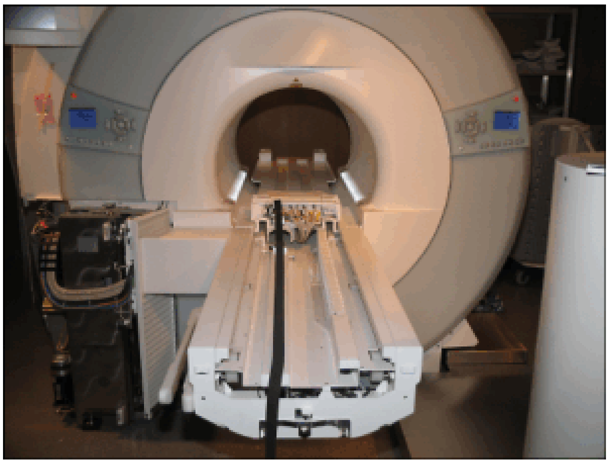
- Dismount the energy chain.

3.5.4.3 Tabletop plate

- Release the emergency clutch and push the tabletop plate manually into the magnet until the cantilever arm is no longer touching the tabletop plate.

Fig. 137: Tabletop plate, pushed into the magnet

Fig. 138: *Tabletop moved into the magnet*



- Carefully push the tabletop plate manually into the magnet; not valid for Trio A Tim Systems.

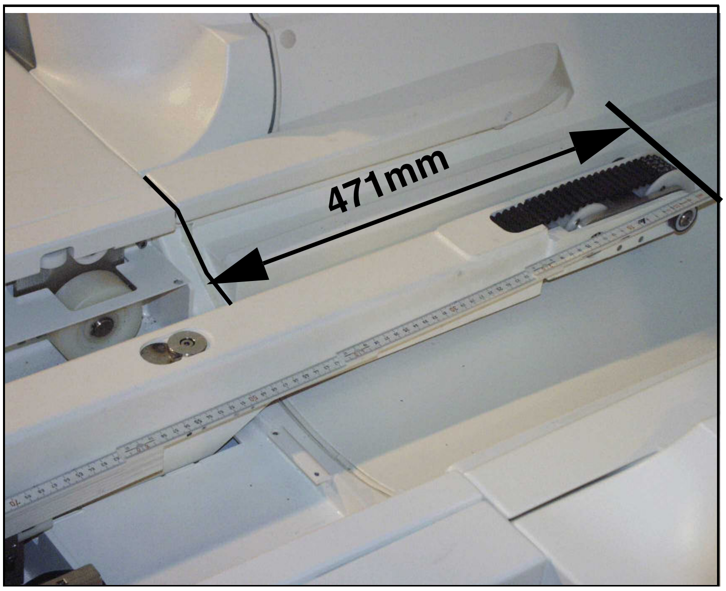
Fig. 139: *Tabletop, removed*

- Lift up the tabletop - a second person is necessary - and carry it to the front of the magnet, replace tabletop plate.

3.5.5 Assembly

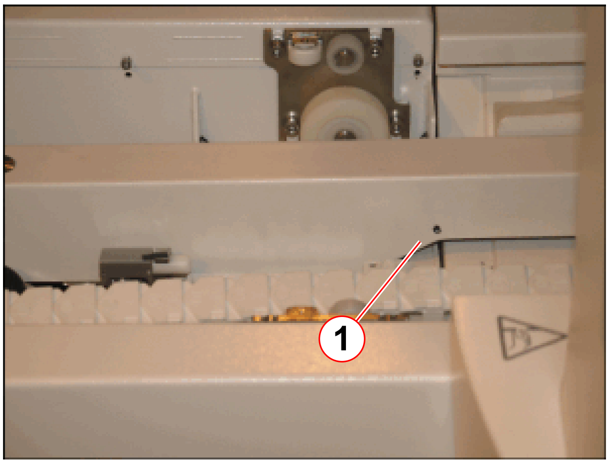
> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 140: Distance of cantilever arm



- Before you insert the new tabletop plate, please do a check of the distance from the end of the cantilever arm to the support frame and adjust to 471 mm.
- Move the new tabletop plate from the front to the back of the magnet.
- Install the energy chain.

Fig. 141: Mark on the telescopic arm



(1) Marking hole in the telescopic arm

- On newer systems the distance mark shows the correct length of the cantilever arm.

> Use the same length of screws when remounting the energy chain

3.5.6 Concluding steps

- Install the end stops to the original position, consider the room length while doing it.
- Reconnect all cables and air hoses.
- Install foot-end plate and the cover of the coil plugs.

3.5.7 Adjustments

For adjustment of the toothed belt, please refer to (→ Adjustments / Page 176)

3.5.8 Final checks

- Switch ON F22 (RF-room) ptab and check for smooth horizontal movement of the tabletop plate.

3.6 Body of the energy chain

3.6.1 General

All cables of the ptab are running in a flexible chain (energy chain). This chain can be replaced.

3.6.2 Tools and time required

3.6.2.1 Time

Approx. 30/45 min

3.6.2.2 Tools

Non-magnetic tool kit

3.6.3 Preliminary

- Switch OFF F22 (RF-room) in the ACC.

> According to the system/room specifications the movement range of the horizontal movement can be reduced

With the whole body version of the horizontal drive, the movement range of the horizontal movement may be reduced. If that is the case, remove the end stops.
(➔ Fig. 306 Page 179)

Before disassembly the patient table covers on a system with a partial drive unit: The toothed belt must be released to push the table top to the service end of the magnet. Therefore refer to. (➔ Prerequisite for cover removal / partial drive / Page 104)

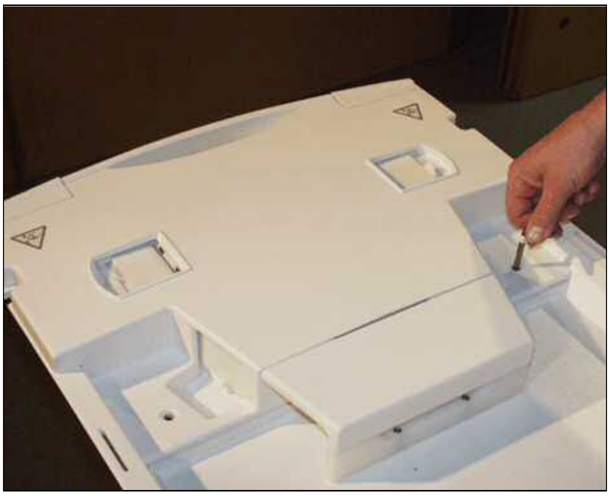
3.6.4 Disassembly

3.6.4.1 Covers

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Basic table

Fig. 142: Coil plug cover (foot end)



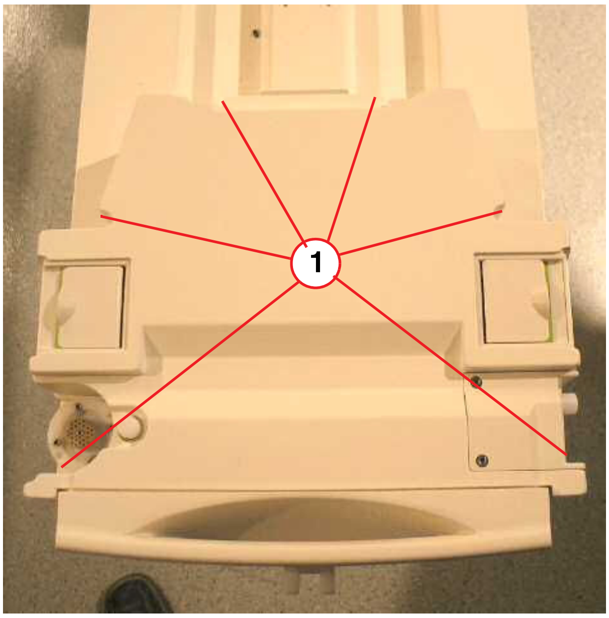
- Remove the Allen screws on top of the cover from the coil plugs.

Fig. 143: Remove Allen screws

- Remove the lower Allen screws of the foot-end cover.

Removable table

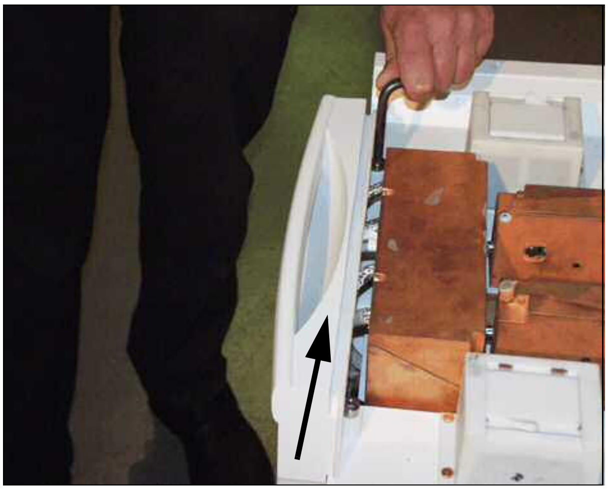
Fig. 144: RF-Cover RTT



(1) Screws to be removed

- Remove the carrier plate of the RTT.
- Remove six screws on top of the cover from the coil plugs.
- Disconnect cables.

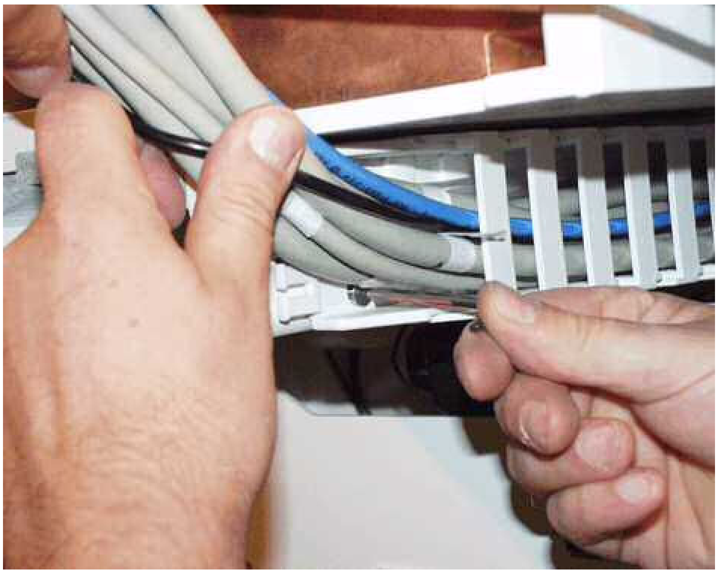
Fig. 145: Remove end plate



- Remove the two M8 Allen screws, if necessary disconnect cables, and remove foot-end plate.

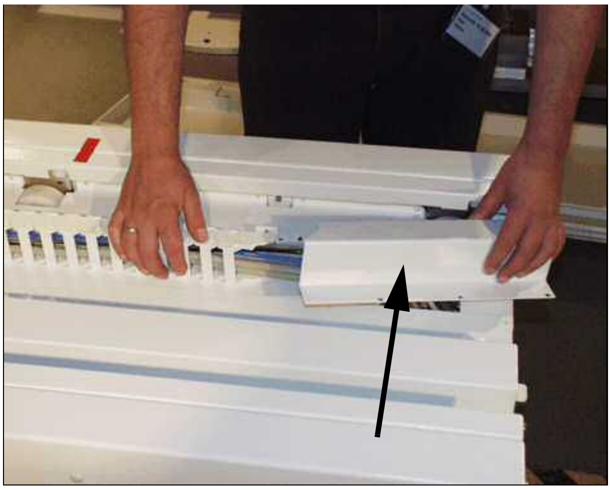
3.6.4.2 Energy chain

Fig. 146: Energy chain



- Push the tabletop plate manually into the magnet to access the energy chain.
- Remove the lower part of the energy chain.

Fig. 147: Remove cover of energy chain



- Remove the upper energy chain cover.

Fig. 148: Remove energy chain

- Remove the upper energy chain of the bracket.

3.6.5 Assembly

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

- Open each chain link by using a screwdriver and replace damaged parts

> Make sure all cables are routed in parallel inside the chain

Fig. 149: Release the chain links

- Install the energy chain.

3.6.6 Concluding steps

- Install covers.

3.6.7 Adjustments

- n.a.

3.6.8 Final checks

- Switch ON F22 (RF-room) and check for smooth movement of the energy chain

3.7 Guide rollers

3.7.1 General

The table is moved in the horizontal direction via guide rollers (special bearings). Each roller can be replaced separately.

3.7.2 Tools and time required

3.7.2.1 Time

Approx. 30/45 min

3.7.2.2 Tools

Non-magnetic tool kit

Loctite 221

3.7.3 Preliminary

- Switch OFF F22 (RF-room) in the ACC

According to the system/room specifications the movement range of the horizontal movement can be reduced.

With the whole body version of the horizontal drive, the movement range of the horizontal movement may be reduced. If that is the case, remove the end stops.
(➔ Fig. 306 Page 179)

Before disassembly the patient table covers on a system with a partial drive unit: The toothed belt must be released to push the table top to the service end of the magnet. Refer to (➔ Prerequisite for cover removal / partial drive / Page 104)

3.7.4 Disassembly

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

3.7.4.1 Covers

- Push the tabletop plate manually into the magnet to access the guide rollers.

- Remove Allen screws and remove the affected side covers.

Fig. 150: Remove cover from the affected side

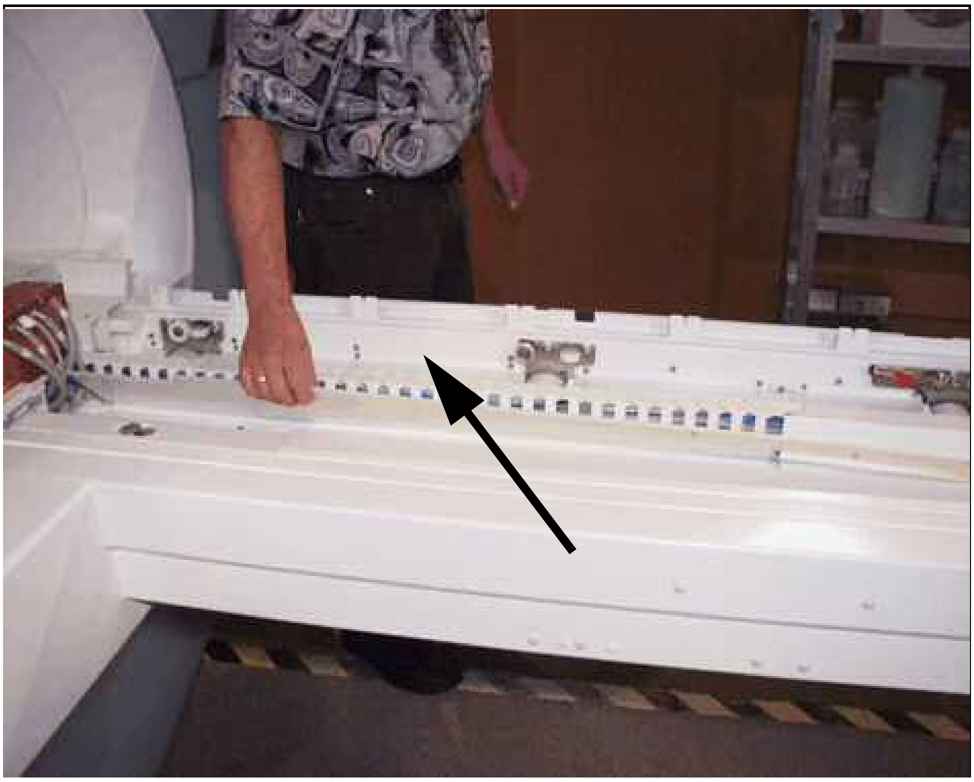


Fig. 151: Remove cover from the affected side

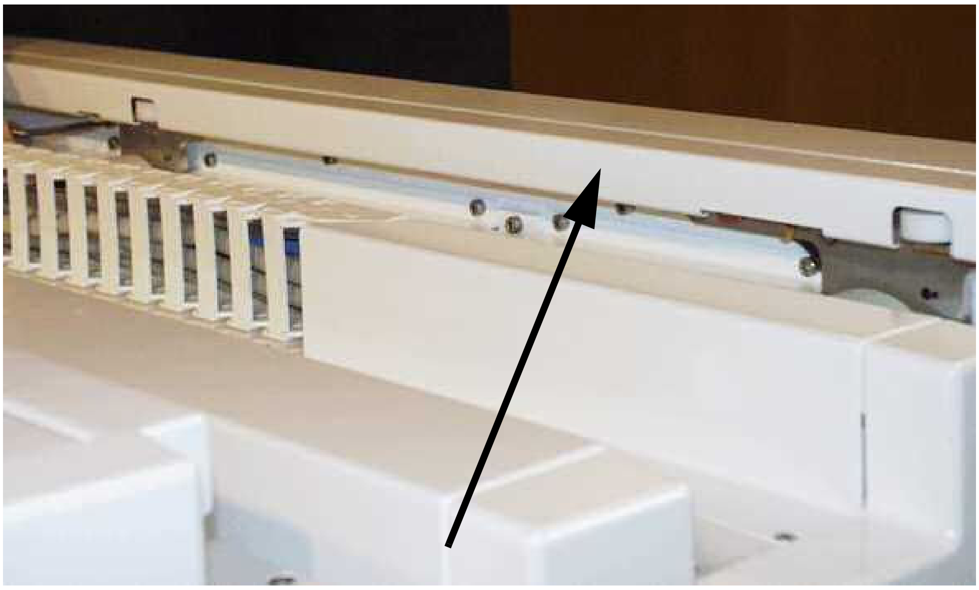
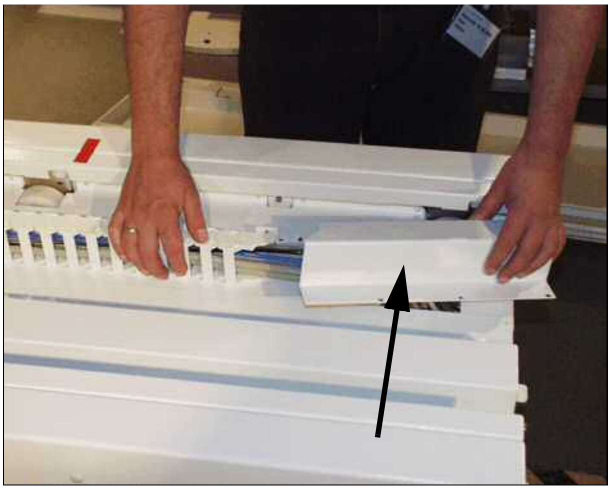


Fig. 152: Remove cover of energy chain

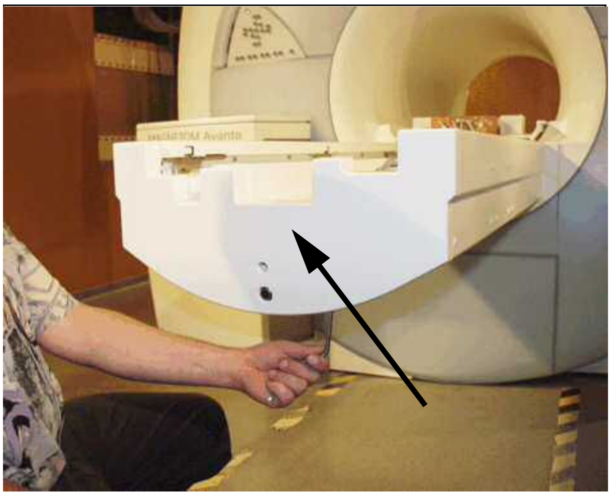


- Remove the upper energy chain cover.

Fig. 153: Remove energy chain

- Remove the upper energy chain of the bracket.

Fig. 154: Foot-end cover



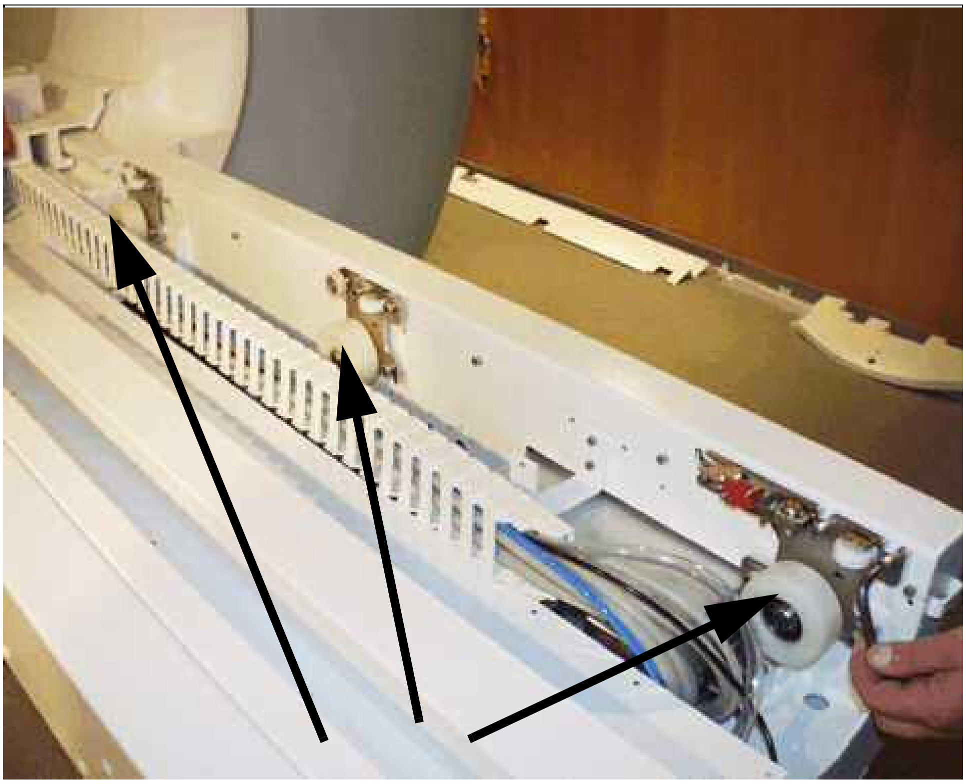
- Remove the foot-end cover of the patient table.

3.7.5 Assembly

For assembling the partial drive, please refer to (➔ Assembly / Page 173) not valid for Trio A Tim Systems.

- Replace the defective guide roller and safety screws by apply Loctite 221.

Fig. 155: Replace affected guide roller



3.7.6 Concluding steps

- Switch ON F22 (RF-room) in the ACC.

3.7.7 Adjustments

For adjustment of the toothed belt, please refer to (➔ Adjustments / Page 176), not valid for Trio A Tim Systems.

3.7.8 Final checks

- Install covers.
- Check smooth horizontal movement.

3.8 MDSD device

3.8.1 General

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

The MDSD device is responsible for driving the horizontal movement of the ptab. Two types of MDSD are introduced.

MDSD for whole body movement.

With revision level 2 and higher, the grounding shield and the external filter must not be used anymore. On older systems, the grounding shield and/or the external filter may be installed. Partial drive is not affected.

MDSD for partial drive, not valid for Trio A Tim Systems.

Each of the MDSD device can be replaced**, they are n o t compatible**.

3.8.2 Tools and time required

3.8.2.1 Time

Approx. 120 min

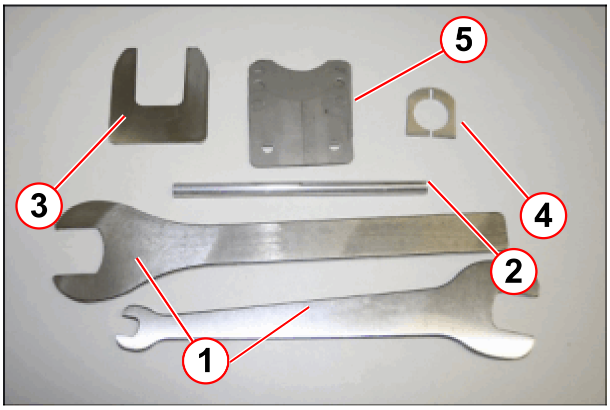
3.8.2.2 Tools

Non-magnetic tool kit

Philips screwdriver

Service aid MDSD (delivered with the MDSD spare part)

Fig. 156: MDSD service aid



(1) Spanners
(2) Feed-in tool for fiber optic cable
(3) Clamp claw
(4) Clamp collar
(5) Shim plates

For patient tables with partial drive MDSD A4330 (p.n. 10277315), the only service tool included in the spare part is the feed-in tool fiber optical cable. Only for the whole body MDSD A4320 (p.n. 8112240) a complete service aid set, as shown in the figure above, is needed and included in the spare part.

3.8.3 Preliminary

- Position patient table at home position
- Switch OFF F22 (RF-room) in the ACC

According to the system/room specifications the movement range of the horizontal movement can be reduced.

With the whole body version of the horizontal drive, the movement range of the horizontal movement may be reduced. If that is the case, remove the end stops. (→ Fig. 306 Page 179)

Before disassembly the patient table covers on a system with a partial drive unit: The toothed belt must be released to push the table top to the service end of the magnet. Therefore refer to (→ Prerequisite for cover removal / partial drive / Page 104), not valid for Trio A Tim Systems

3.8.4 Disassembly

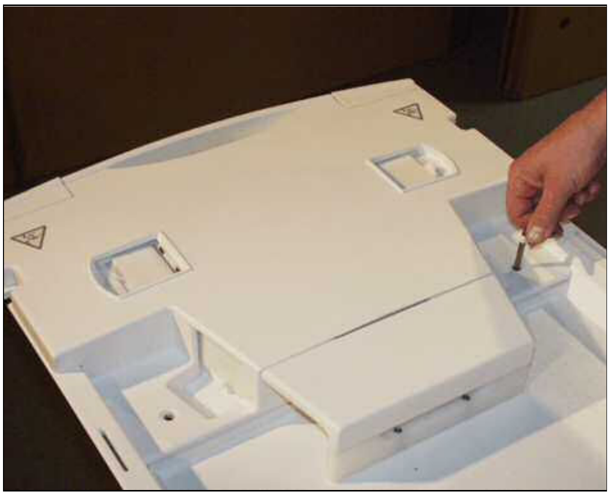
3.8.4.1 Prerequisite for cover removal / partial drive

Be careful and avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

The tabletop plate may fall - support the tabletop plate.

Basic table

Fig. 157: Coil plug cover (foot end)



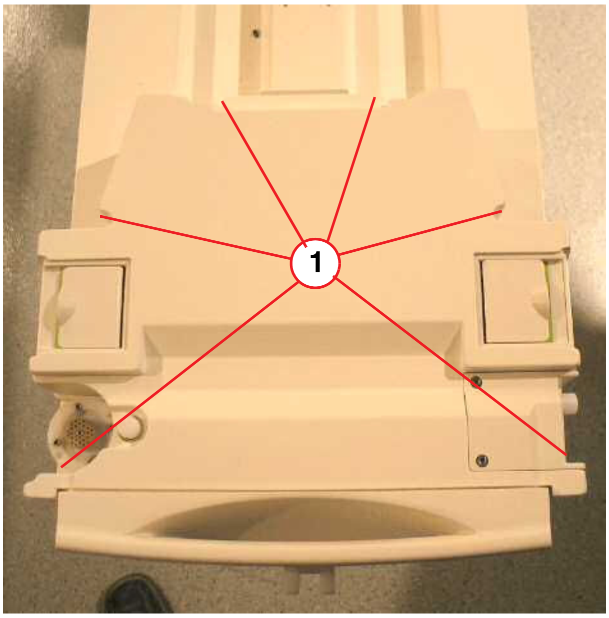
- Remove the Allen screws on top of the cover from the coil plugs.

Fig. 158: Remove Allen screws

- Remove the lower Allen screws of the cover from the coil plugs.

Removable table

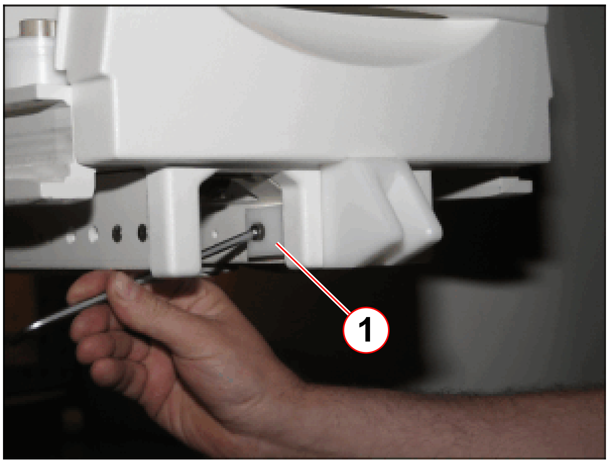
Fig. 159: RF-Cover RTT



(1) Screws to be removed

- Remove the carrier plate of the RTT.
- Remove six screws on top of the cover from the coil plugs.
- Disconnect cables.

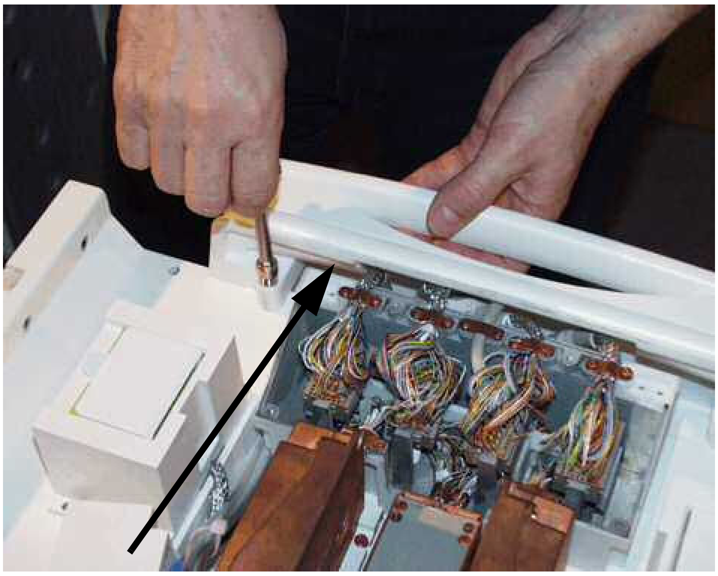
Fig. 160: Additional mounting bracket for foot end cover (newer systems)



(1) Allen screw

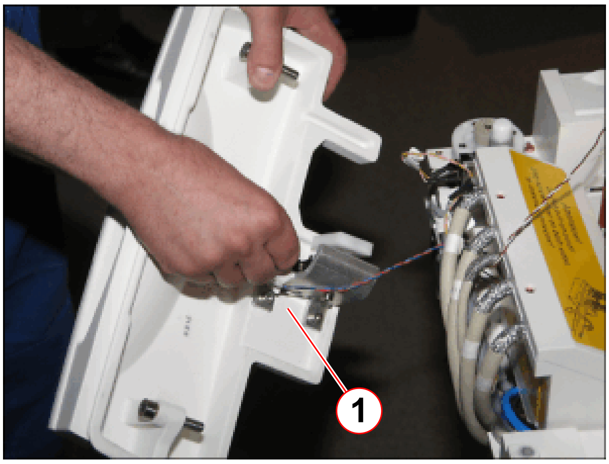
- On newer systems, remove Allen screw of mounting bracket.

Fig. 161: Foot-end cover



- Remove screws of the foot-end cover.

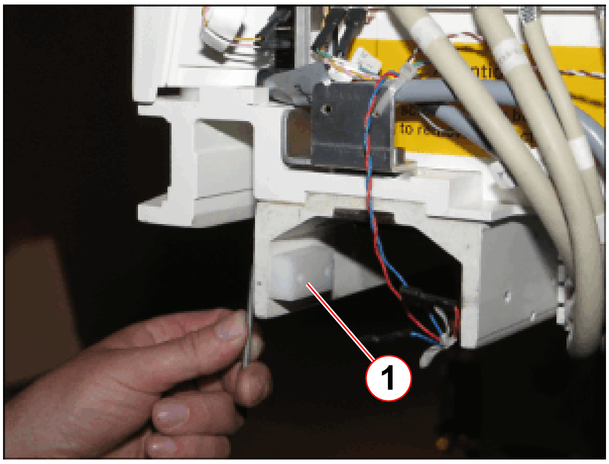
Fig. 162: Foot end cover, removed



(1) Disconnect cables

- Disconnect cable connections and remove the foot-end cover..
- Slide out the lower cover on the bottom of the patient table.

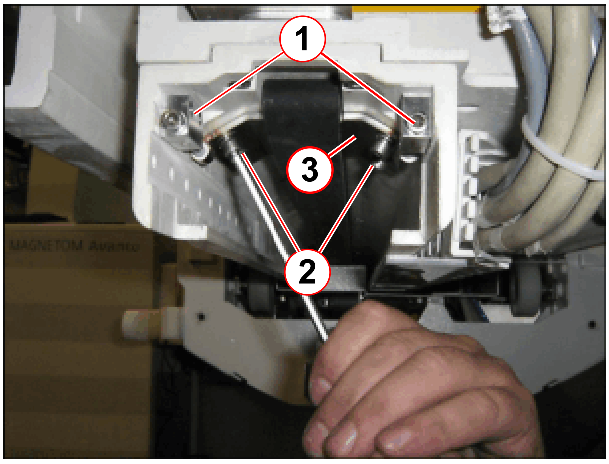
Fig. 163: Mechanical end stop of partial drive unit



(1) End stop to be removed

- Remove mechanical end stop.

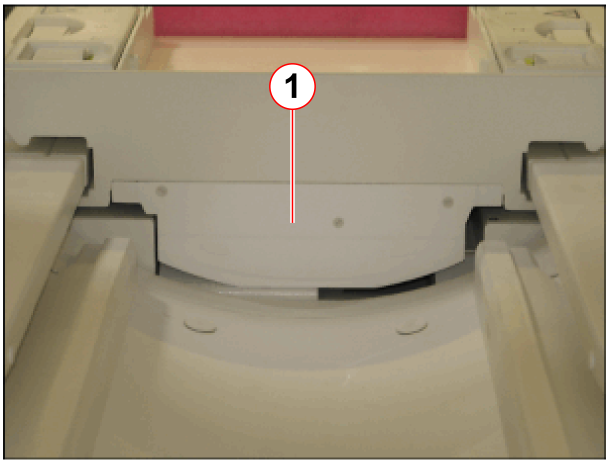
Fig. 164: Tension unit for toothed belt



(1) Allen screws for adjusting belt tension
(2) Attachment screws
(3) clamping plate

- Release tension of toothed belt (1), remove Allen screws (2) and tooth belt tension unit, not valid for Trio A Tim Systems.

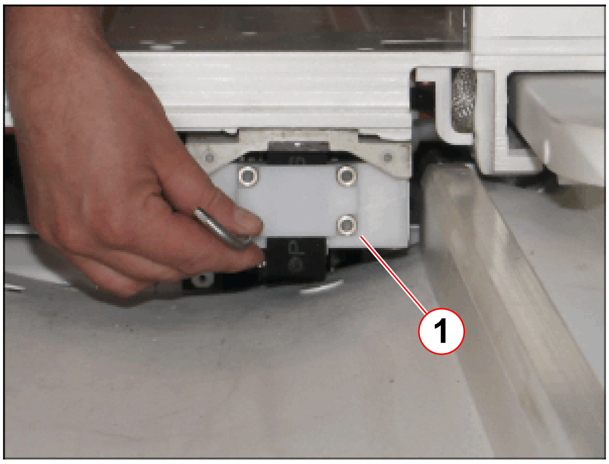
Fig. 165: Cover of toothed belt mount (head end)



(1) Cover of toothed belt mount

- Remove the cover of the toothed belt mount of toothed belt (1) on the head end from the ptab, not valid for Trio A Tim Systems.

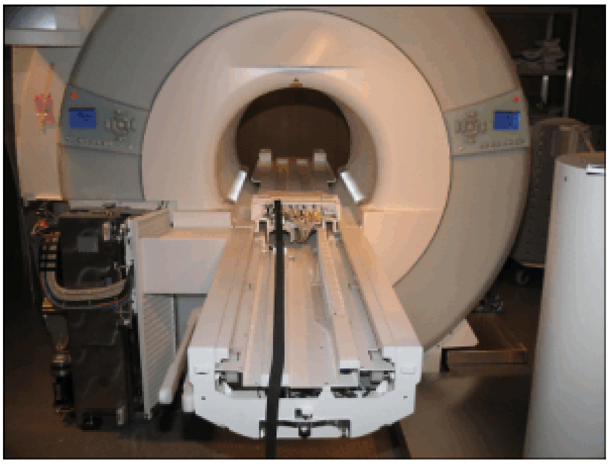
Fig. 166: Toothed belt attachment (head end)



(1) Allen screws of toothed belt mount

- Remove the toothed belt mount (1) on the head end from the ptab, not valid for Trio A Tim Systems.

Fig. 167: Tabletop moved into the magnet



- Carefully push the tabletop plate manually into the magnet.

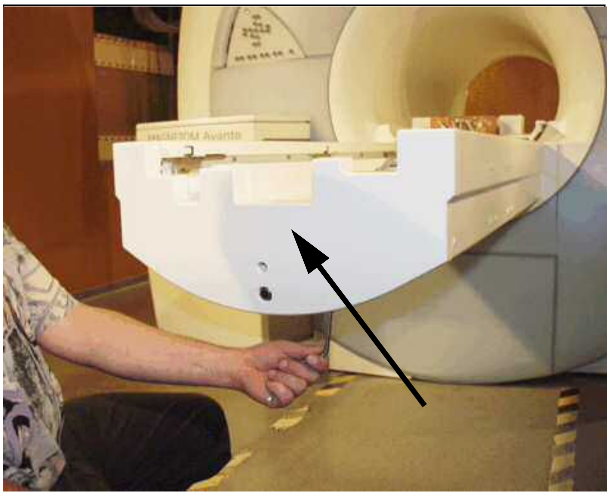
3.8.4.2 Covers

> ℹ Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

> ℹ For disassembling/assembling the patient table covers we recommend a mid-sized Philips screwdriver.

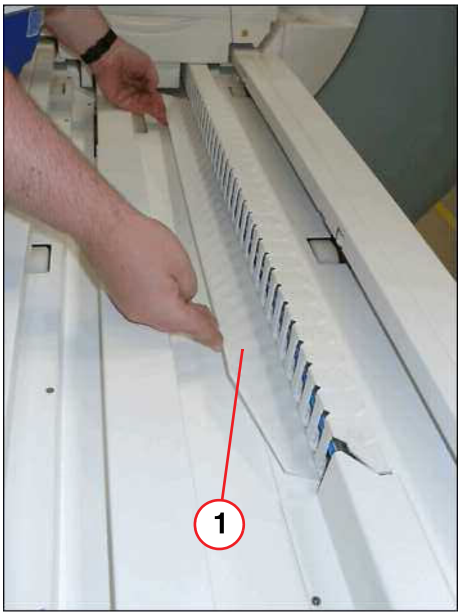
- Push the tabletop plate manually into the magnet (whole body movement).

Fig. 168: Foot-end cover



- Remove the foot-end cover of the patient table.

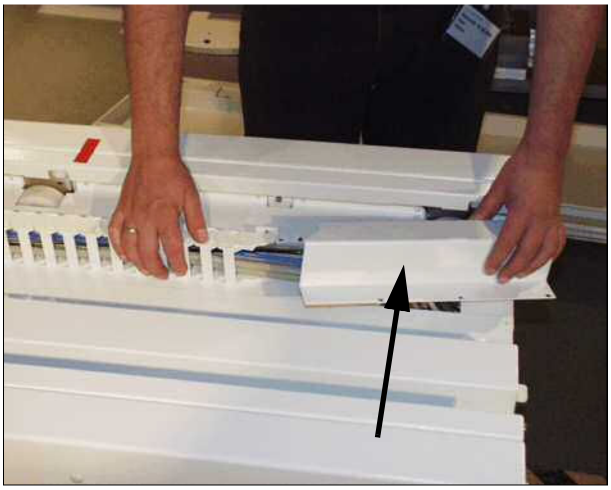
Fig. 169: Plastic inner cover



(1) Plastic inner cover

- Remove inner plastic cover.

Fig. 170: Remove cover of energy chain



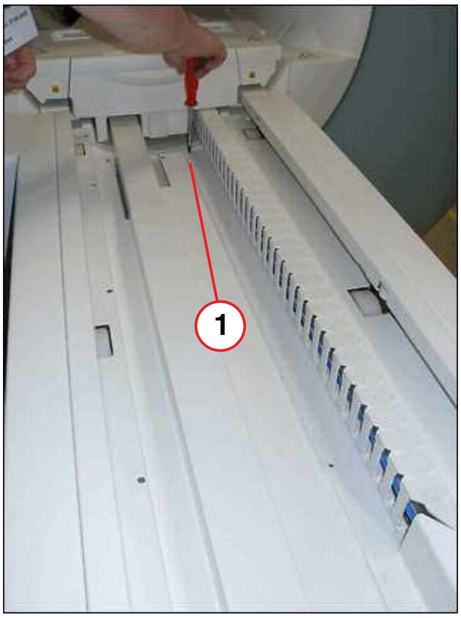
- Remove the upper energy chain cover.

Fig. 171: Remove energy chain

- Remove the upper energy chain of the bracket.

- Remove right guide rail cover (do not remove screws, only loosen).

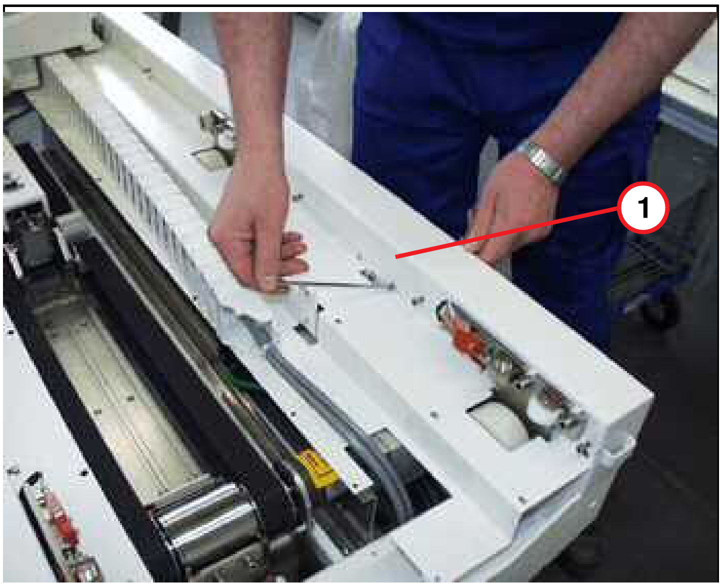
Fig. 172: Inner main cover



(1) Screws to be removed

- Remove screws of main inner cover.

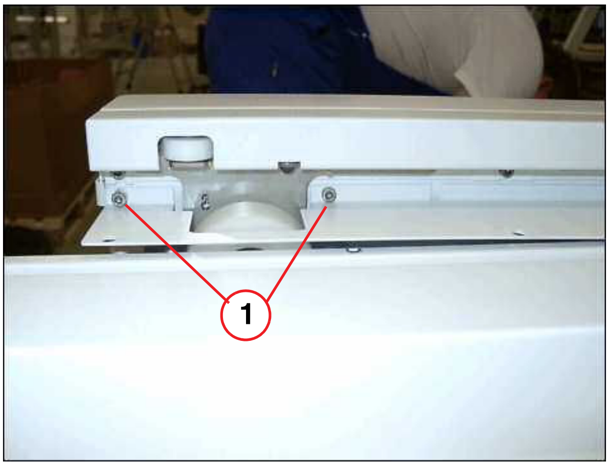
Fig. 173: Inner cover



(1) Top inner cover

- Remove right inner cover.

Fig. 174: Allen screws

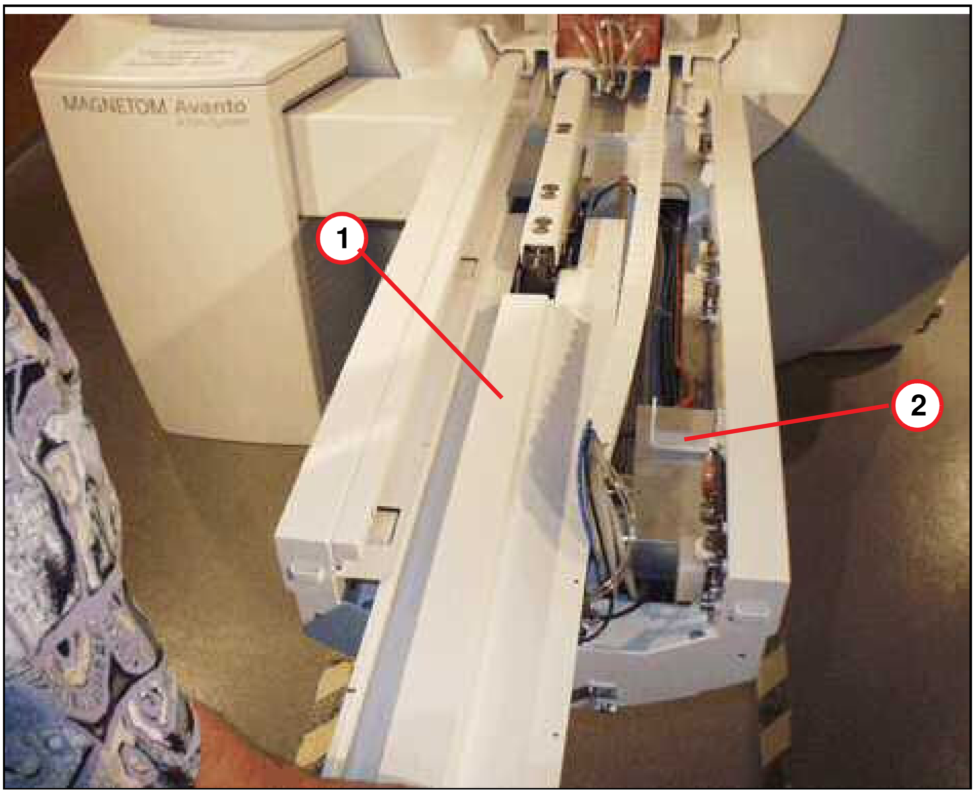


(1) Screws to be removed

- Remove two Allen screws on the left side on the table end (foot) before you remove the main inner cover.

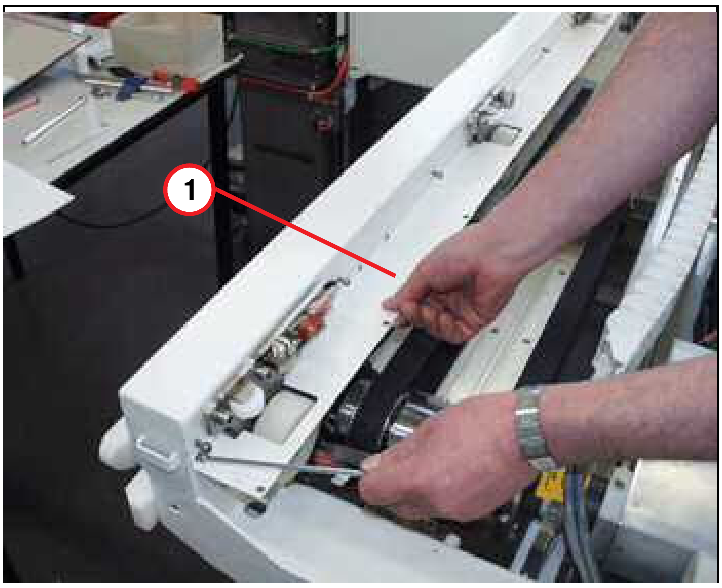
- Remove bracket above the MDSD device.
- Remove main inner cover.

Fig. 175: Disassembling



(1) Main inner cover
(2) Mounting bracket for energy chain

Fig. 176: Inner cover, left side



(1) Inner cover, left side

- Covers removed.

3.8.4.3 MDSD device of whole body drive

Fig. 177: Example - removing FOCs from MDSD

(1) Cam of FOC connector
(2) Screwdriver, flat blade

- The figure below shows the FOC plug connection (principle) inside the MDSD device. Use a flat blade screwdriver to disconnect the FOCs inside the MDSD device.

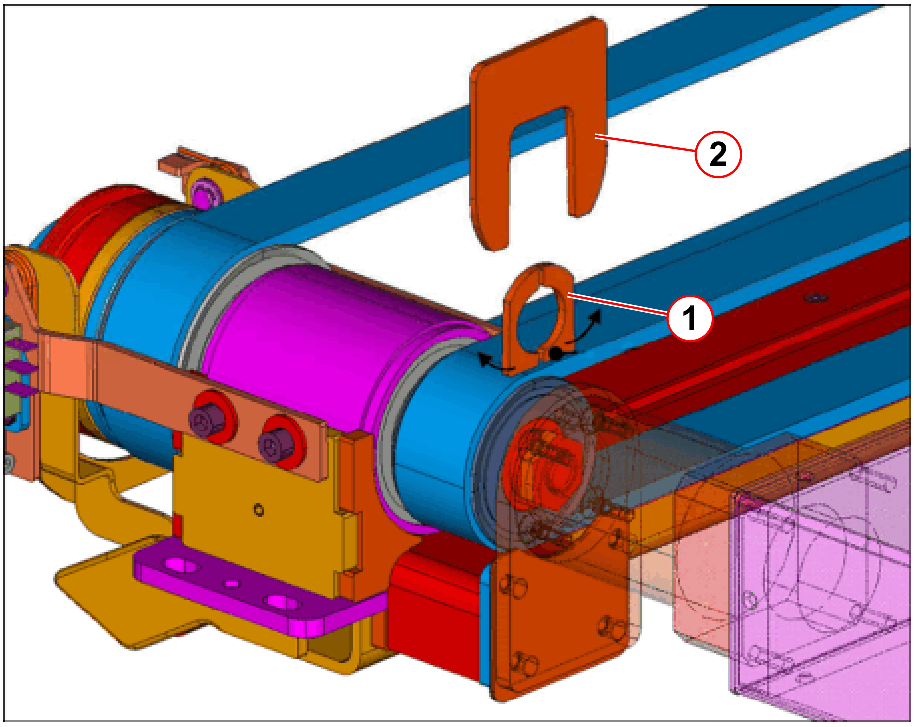
Fig. 178: Fiber-optic cables of MDSD

- Feed the flat blade screwdriver through the hole in the aluminum cover and into the FOC's plug connection. Try to reach the cam of the FOC and carefully push it sideways while at the same time pulling it out from the FOC. Refer to illustration.

- Release the horizontal emergency movement, enabling manual horizontal movement.

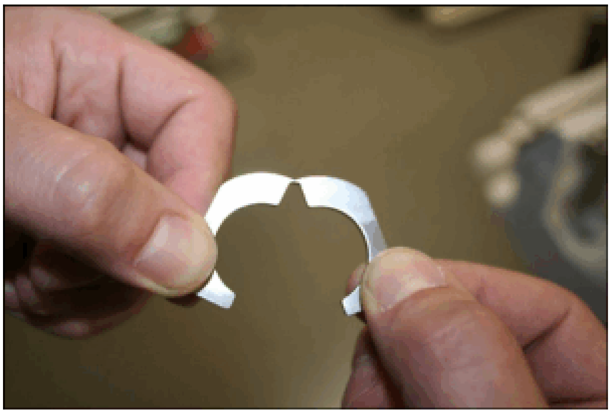
> To avoid the overwinding of the MDSD clamping nut, the clamp collar and the claw fastener (part of the service aid) must be used as described.

Fig. 179: MDSD service aid, principle of use



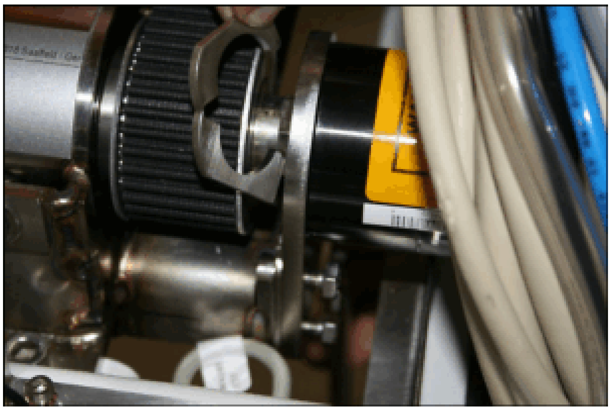
(1) Clamp collar
(2) Claw fastener

Fig. 180: Clamp collar



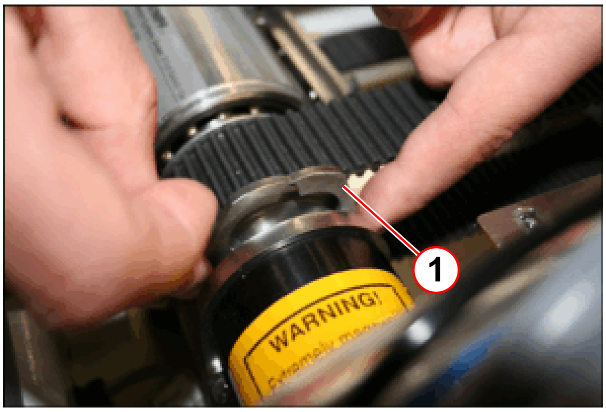
- Carefully bend up the clamp collar, as shown.

Fig. 181: Clamp collar



- Insert the clamp collar.

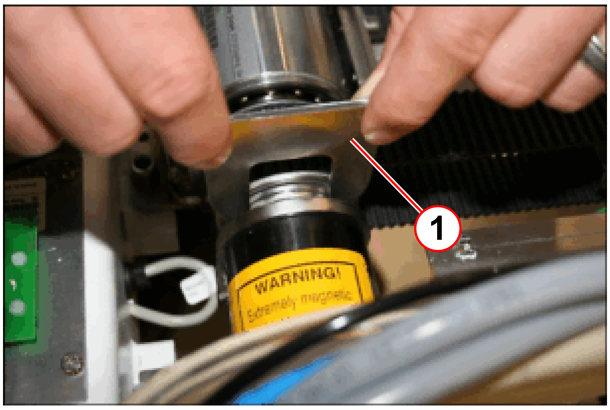
Fig. 182: Clamp collar



(1) Insert clamp collar

- Press the clamp collar together.

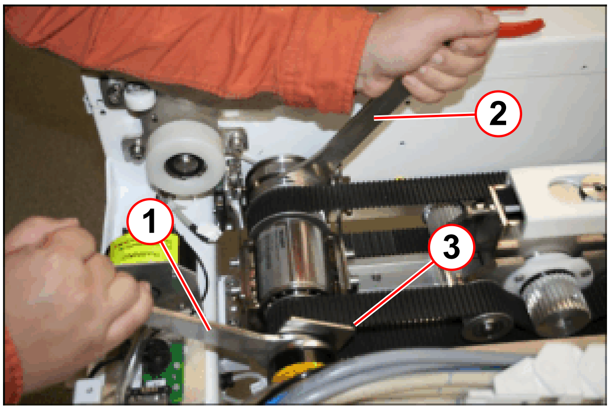
Fig. 183: Claw fastener



(1) Insert claw fastener over the clamp collar.

- Insert the claw fastener (1) over the clamp collar.

Fig. 184: RELEASE or TIGHTENING of the clamping nut



(1) RELEASE, pulling the spanner toward the foot end, TIGHTEN, pushing toward the magnet
(2) Holding the spanner tight
(3) Claw fastener

- Use the open-end spanners of the MDSD service aid to release the MDSD clamping nut.
- **RELEASE,** pulling the open-end spanner (1) **toward the foot end,** while holding the open-end spanner (2).
- The claw fastener should remain.

Fig. 185: RELEASING or TIGHTENING the clamping nut

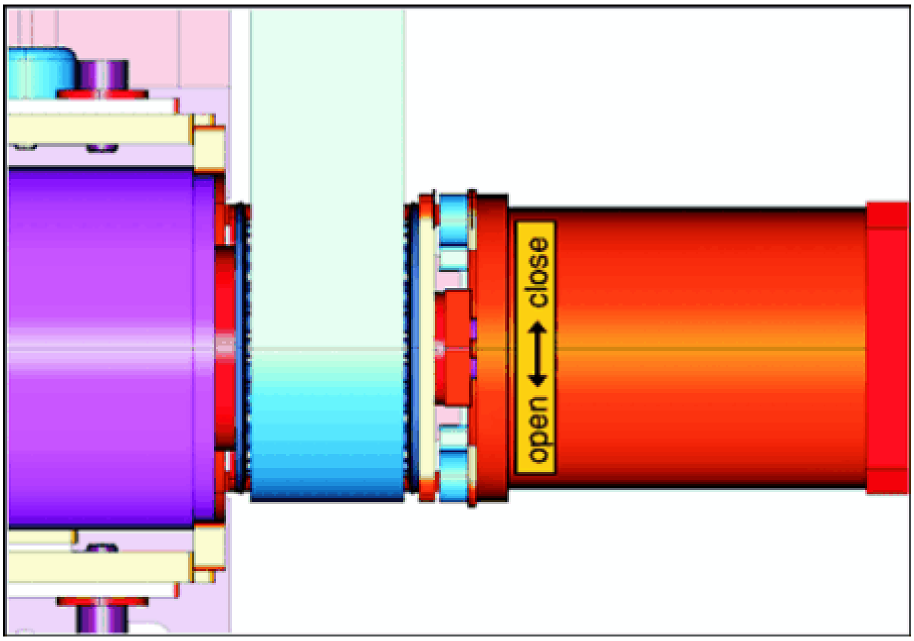
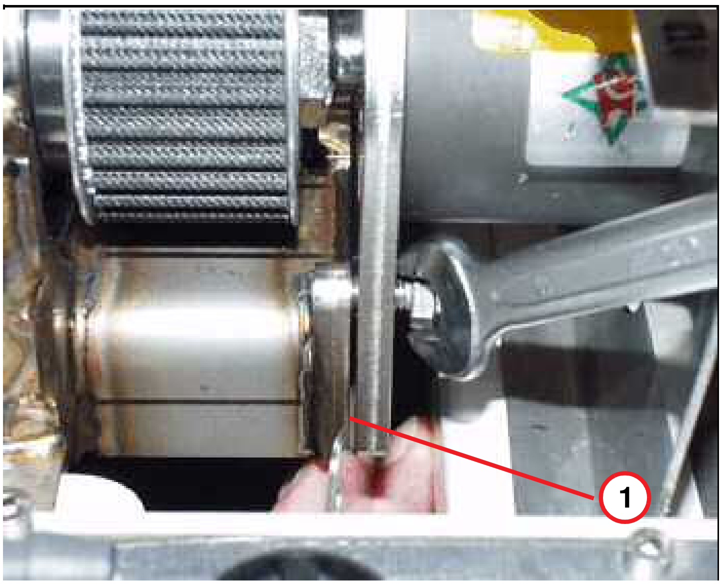


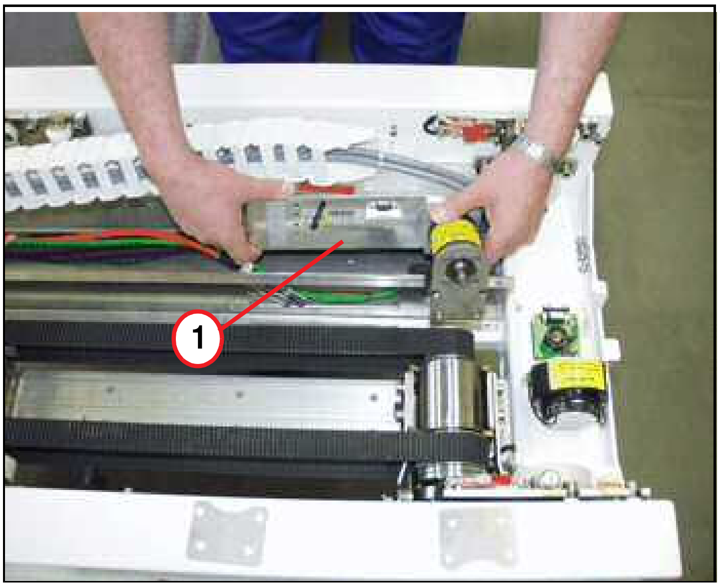
Fig. 186: M6 hex screws of MDSD bracket



(1) Shim gap appears when securing the 22 mm nut

- Remove the four M6 hex nuts.

Fig. 187: MDSD device



(1) Replace MDSD device

- Replace the MDSD drive, take care of the correct version.
- Remove the MDSD service aid (clamp collar and claw fastener).

CAUTION

The MDSD drive consists of magnetic objects!

Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field!**

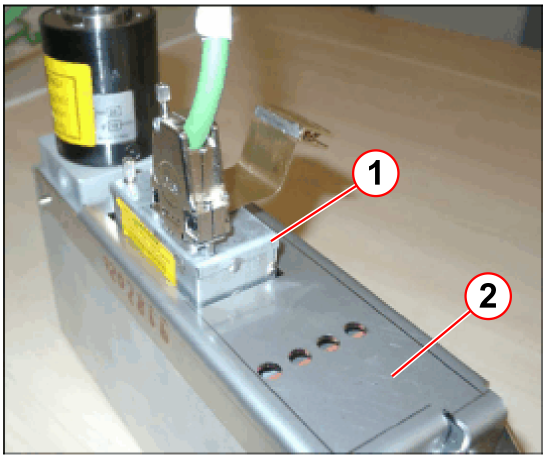
CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

Fig. 188: MDSD with external filter and grounding shield



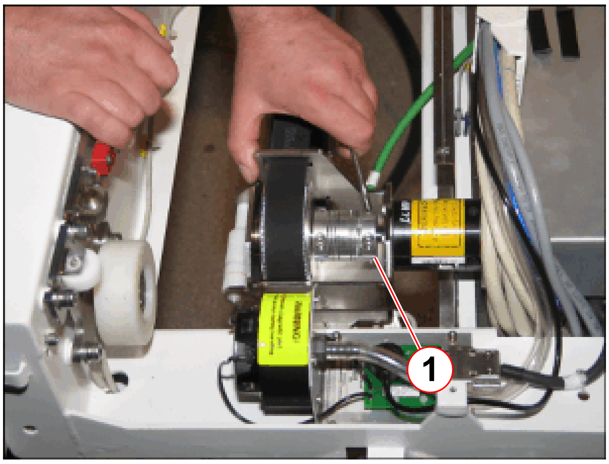
(1) External filter
(2) Grounding shield

Note:

If a grounding shield and/or an external filter is installed**, do not use them anymore for the new MDSD** (applies to revision level 02 and higher).

3.8.4.4 MDSD device of partial drive, not valid for Trio A Tim Systems

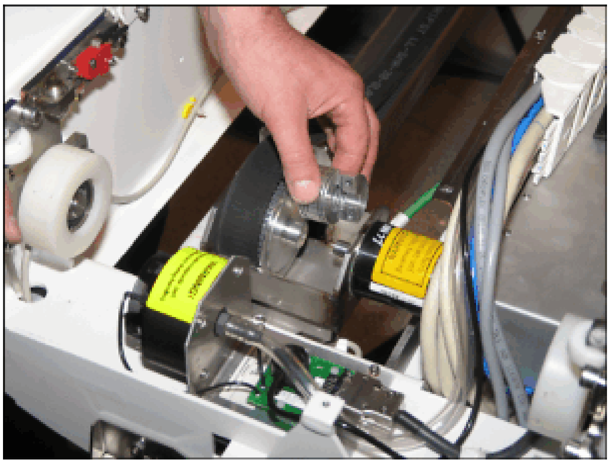
Fig. 189: Clutch of MDSD, partial drive



(1) Allen screws of MDSD clutch

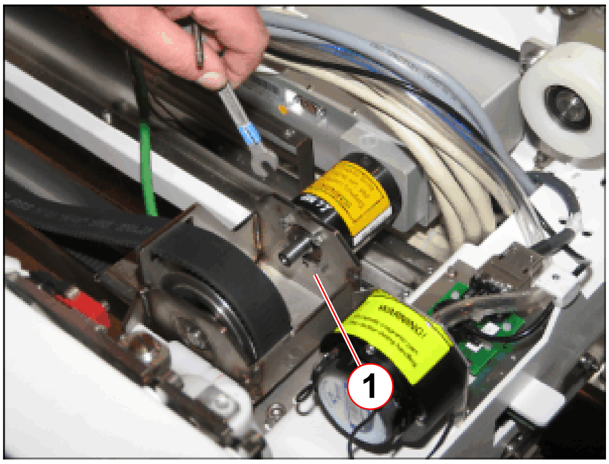
- Remove Allen screws of MDSD clutch (1).

Fig. 190: Clutch of MDSD removed



- Remove MDSD clutch.

Fig. 191: MDSD clutch removed



(1) Mounting screws of MDSD

- Remove the hex screws (1) of the mounting bracket from the MDSD.

Fig. 192: Removing the MDSD

- Replace the MDSD drive.
- Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!

CAUTION

The MDSD drive consists of magnetic objects!

Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field.**

CAUTION

Strong magnetic field!

Magnetic objects may cause injury.

» **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

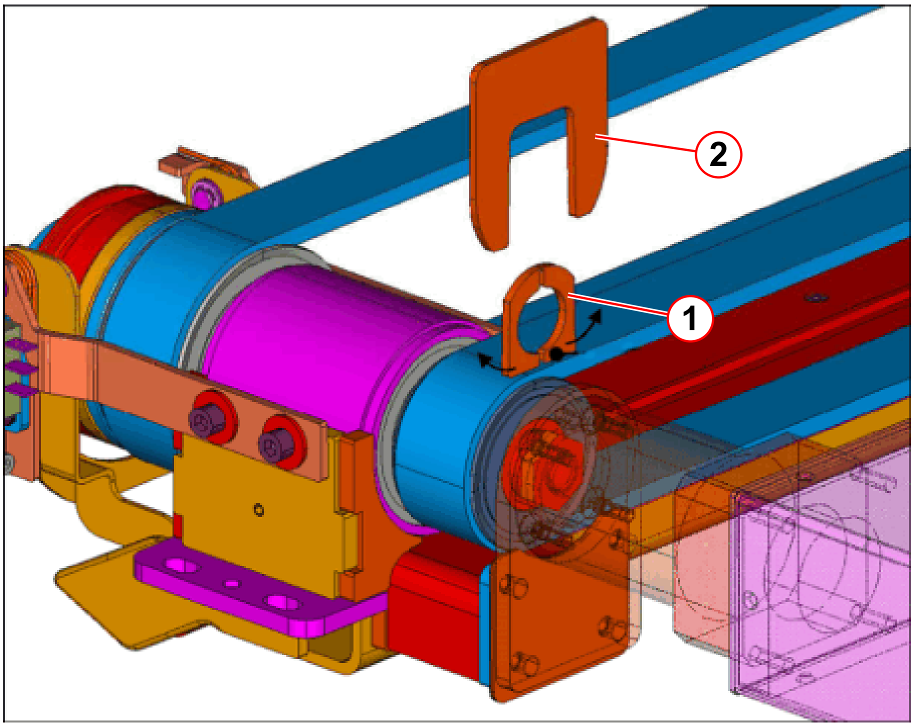
3.8.5 Assembly

3.8.5.1 MDSD drive of whole body drive

The service aid for the MDSD is a prerequisite for assemly.

To avoid the overwinding of the MDSD clamping nut, the clamp collar and the claw fastener (part of the service aid) must be used as described.

Fig. 193: MDSD service aid, principle of use

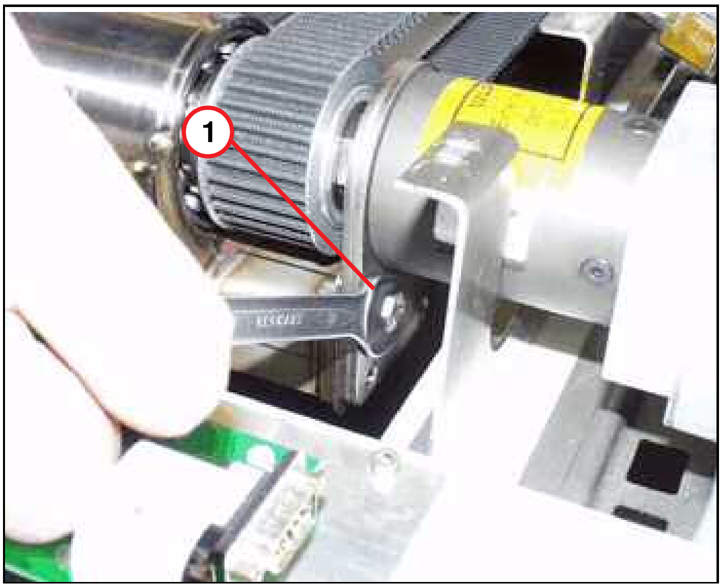


(1) Clamp collar
(2) Claw fastener

Fig. 194: Set of shim plates MDSD drive

- Shim plates must be used when assembling the MDSD device.

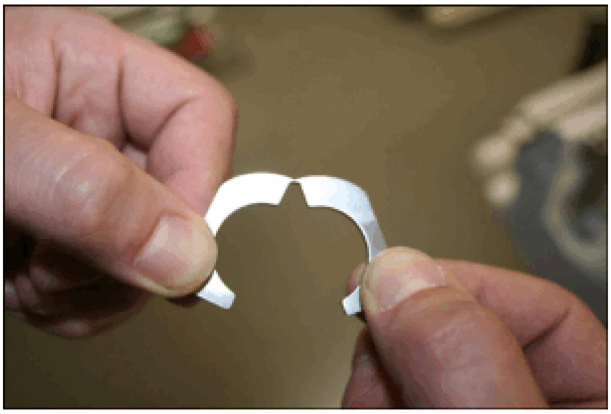
Fig. 195: Hex screws of MDSD



(1) M6 hex screws

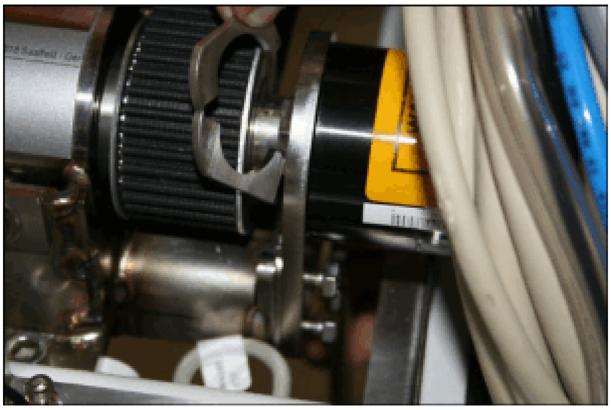
- Insert only the upper two M6 hex nuts and don't tighten them, use them only to hold (as a torque bracket).

Fig. 196: Clamp collar



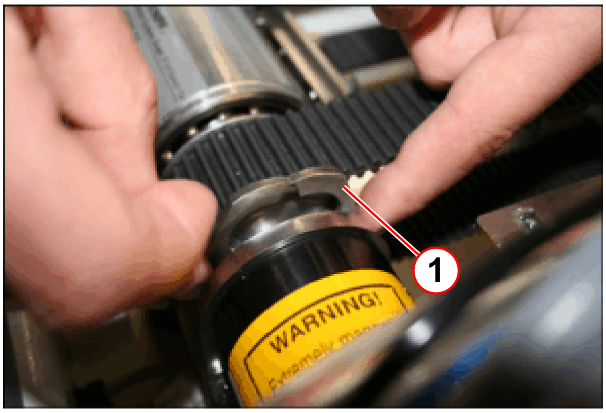
- Carefully bend up the clamp collar, as shown.

Fig. 197: Clamp collar



- Insert the clamp collar.

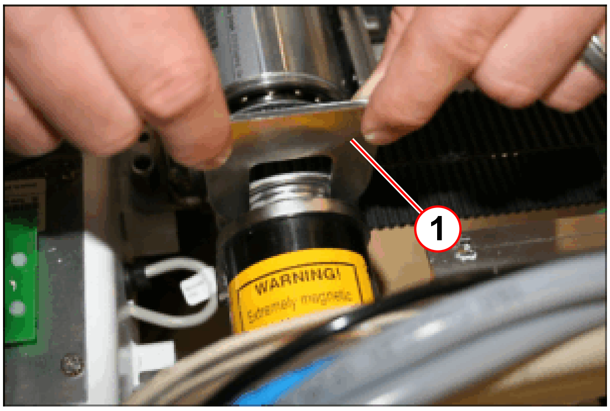
Fig. 198: Clamp collar



(1) Insert clamp collar

- Press the clamp collar together.

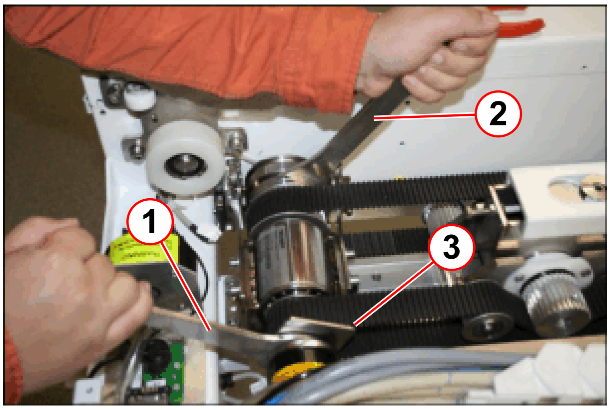
Fig. 199: Claw fastener



(1) Insert claw fastener over the clamp collar.

- Insert the claw fastener (1) over the clamp collar.

Fig. 200: RELEASE or TIGHTENING of the clamping nut



(1) RELEASE, pulling the spanner toward the foot end, TIGHTEN, pushing toward the magnet
(2) Holding the spanner tight
(3) Claw fastener

- Use the open-end spanners of the MDSD service aid to tighten the MDSD clamping nut.
- **TIGHTEN,** pushing the open-end spanner (1) **to the magnet,** while holding the open-end spanner (2).
- The claw fastener should remain.

Fig. 201: RELEASING or TIGHTENING the clamping nut

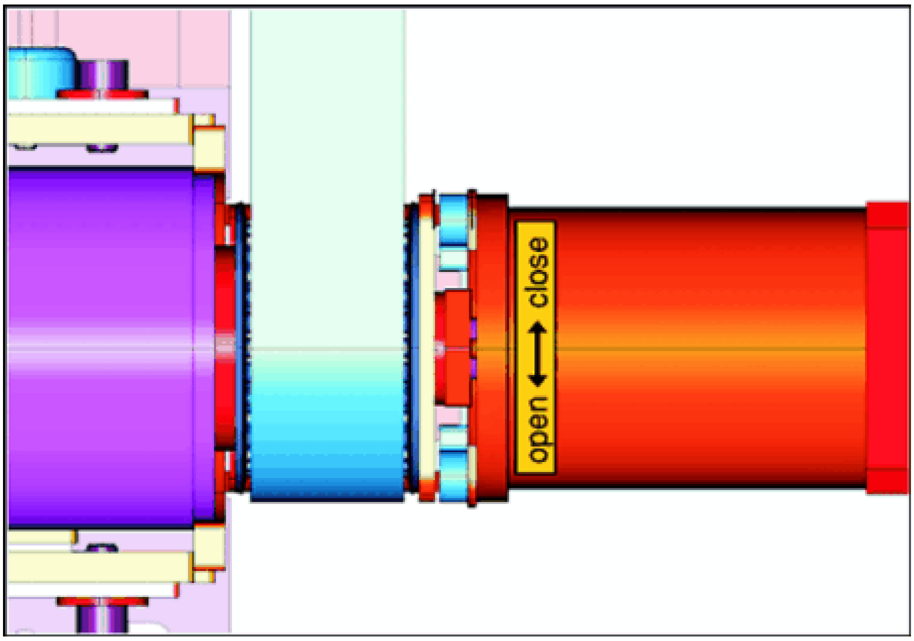
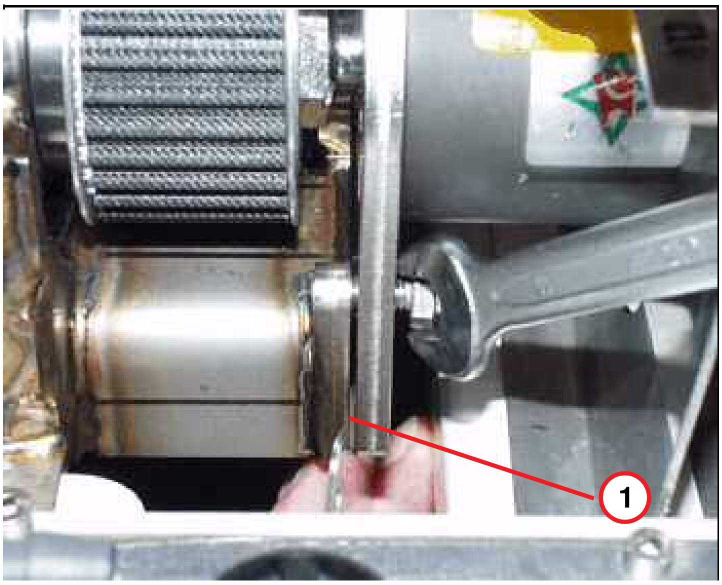


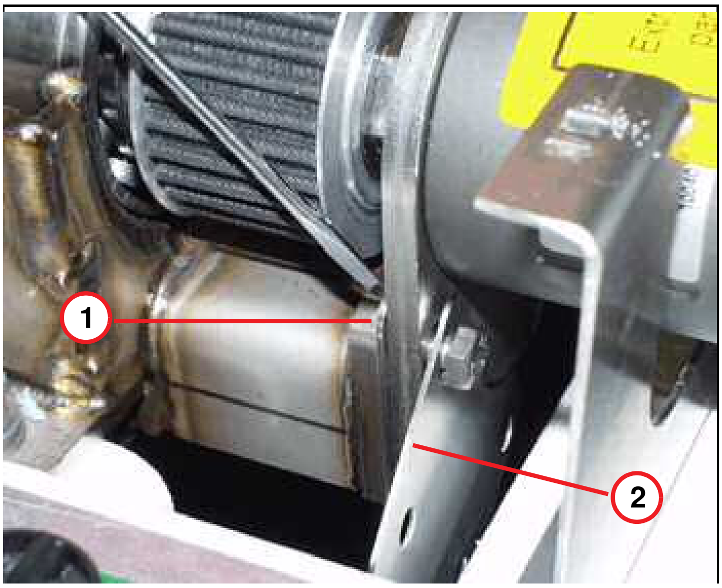
Fig. 202: M6 hex screws of MDSD bracket



(1) Shim gap appears when securing the 22 mm nut

- While torquing the 22 mm nut, a small clearance appears.

Fig. 203: Insert shim plates



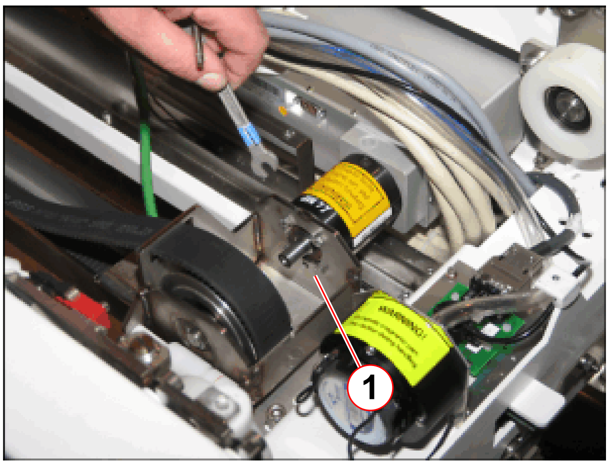
(1) Shim plates to be added
(2) Spare shim plates

- Use the shim plates to fill the clearance between the motor carrier and the axle (delivered with the spare unit, or component of the original part). Tighten all four hex screws.

3.8.5.2 MDSD drive of partial drive, not valid for Trio A Tim Systems

> **CAUTION**
>
> **Strong magnetic field!**
>
> **Magnetic objects may cause injury.**
>
> » **Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!**

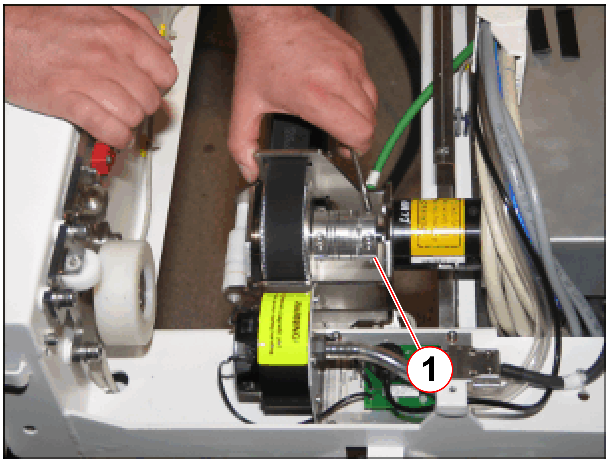
Fig. 204: MDSD clutch removed



(1) Mounting screws of MDSD

- Remove the hex screws (1) of the mounting bracket from the MDSD.

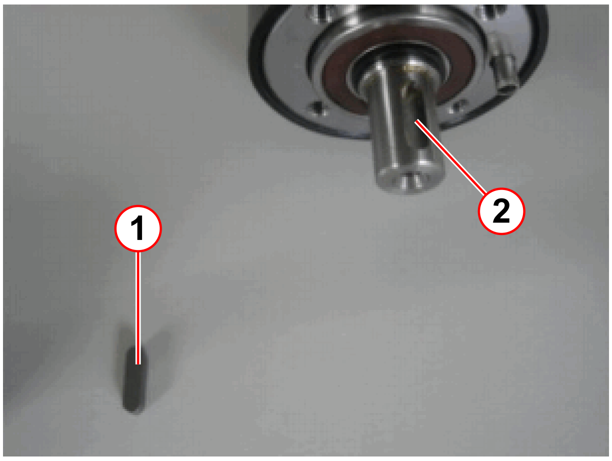
Fig. 205: Clutch of MDSD, partial drive



(1) Allen screws of MDSD clutch

- Remove Allen screws of MDSD clutch (1).
- Hold the ferromagnetic part with both hands while moving it to the correct position and to remove the unit directly out of the field!

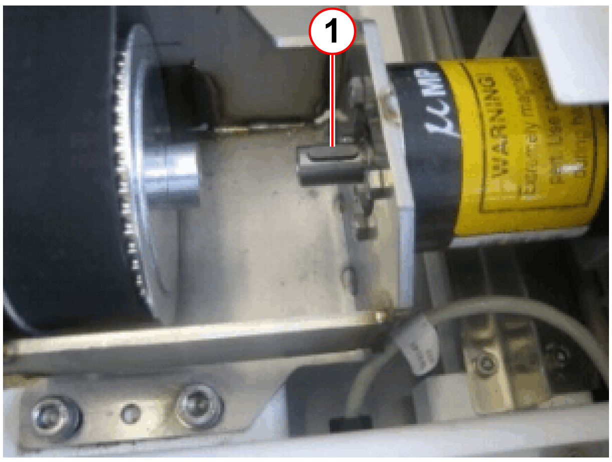
Fig. 206: Assembling the partial drive MDSD



(1) Key
(2) Notch of the MDSD axis

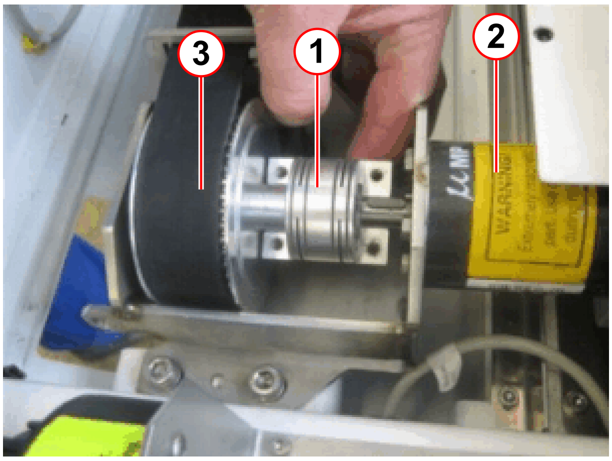
- For partial drive tables insert the key (1), delivered with the spare part, into the notch of the MDSD axis.

Fig. 207: Key inserted in MDSD axis



(1) Key

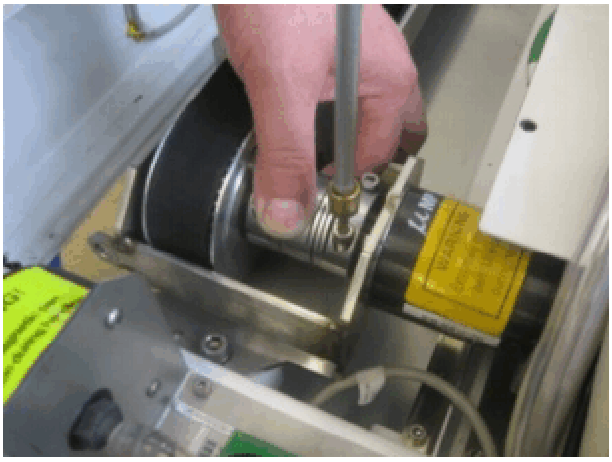
Fig. 208: Fixation of the clutch



(1) Clutch
(2) MDSD for partial drive
(3) Drive unit TK

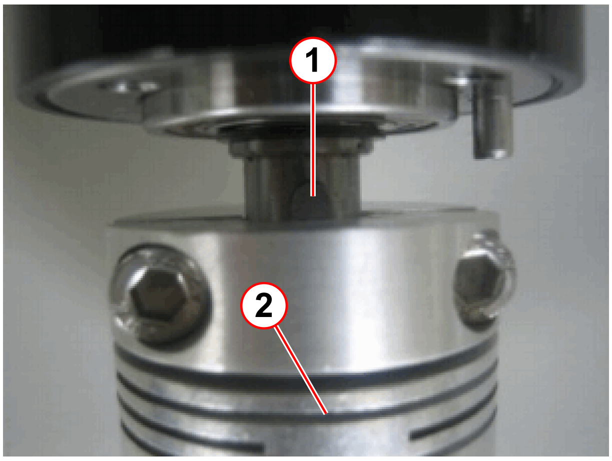
- Place the clutch as shown in pos. (1).

Fig. 209: Fixation of the clutch



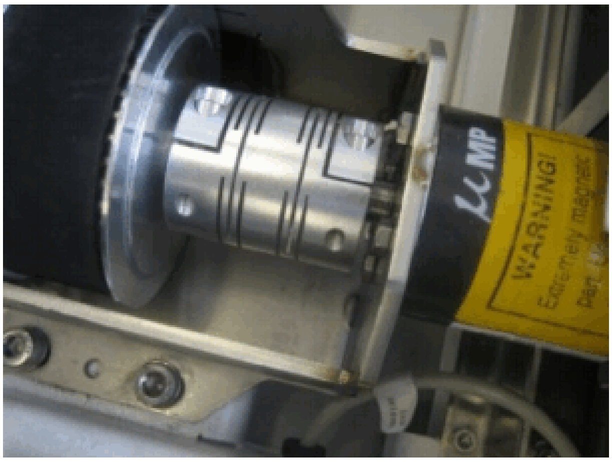
- Position the other half of the clutch so that the key fits into the notch of the clutch and tighten the screws of the clutch.

Fig. 210: Key position with clutch



(1) Key
(2) Clutch

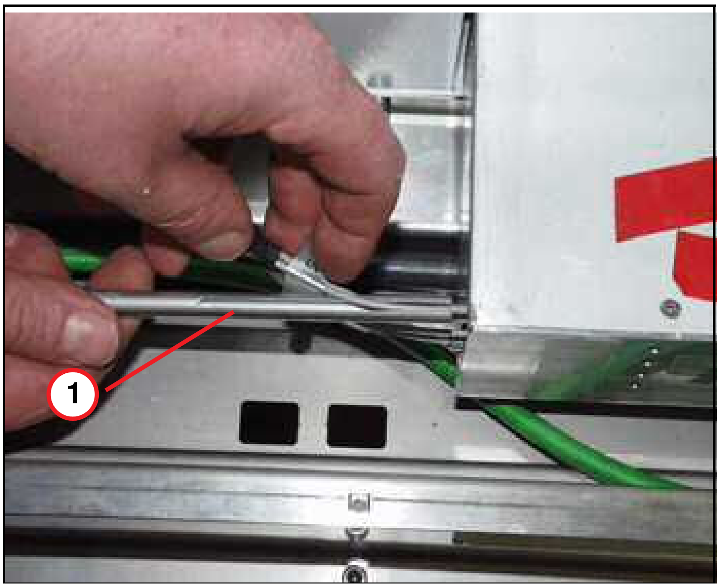
Fig. 211: Clutch installed



For further assembling of the partial drive, please refer to (→ Assembly / Page 173), not valid for Trio A Tim Systems.

3.8.6 Concluding steps

Fig. 212: MDSD FOC connections



(1) FOC feed-in tool

- Reconnect the cables and FOCs on the MDSD drive and use the feed-in tool (service aid) delivered with the spare part.

- Reinstall the covers in reverse order, tighten all screws at the end of the assembling procedure.
- Switch ON F22 (RF-room).

3.8.7 Adjustments

For adjustment of the toothed belt, please refer to (→ Adjustments / Page 176), not valid for Trio A Tim Systems.

3.8.8 Final checks

- Please refer to final checks horizontal drive unit (→ Final checks / Page 163).

3.9 Horizontal drive unit

3.9.1 General

The horizontal drive unit is responsible for the horizontal movement of the ptab. The unit can only be replaced in full.

There are two types of horizontal drives unit in the patient table.

1. "Telescopic Drive" that translates a relatively small movement range into a wide range to support whole body applications.
2. "Partial drive" allows the horizontal table movement without whole body application and a limited movement range (not valid for Trio A Tim Systems).

3.9.2 Tools and time required

3.9.2.1 Time

Approx. 120 min

3.9.2.2 Tools

Non-magnetic tool kit

Philips screwdriver

3.9.3 Preliminary

- Switch OFF F22 (RF-room) in the ACC

> i According to the system/room specifications the movement range of the horizontal movement can be reduced.

- If the movement range of the horizontal movement is reduced it may be necessary to remove the end stops. (→ Fig. 306 Page 179)

3.9.4 Disassembly

» **Please follow the procedure if the horizontal drive unit needs to be replaced.**

CAUTION

The horizontal drive units consist of magnetic objects!

Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field!**

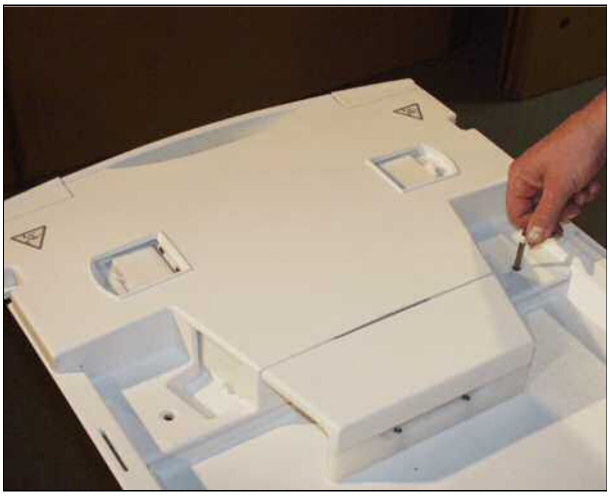
3.9.4.1 Tabletop plate

Be careful and avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

The tabletop plate may fall - support the tabletop plate.

Basic table

Fig. 213: Coil plug cover (foot end)



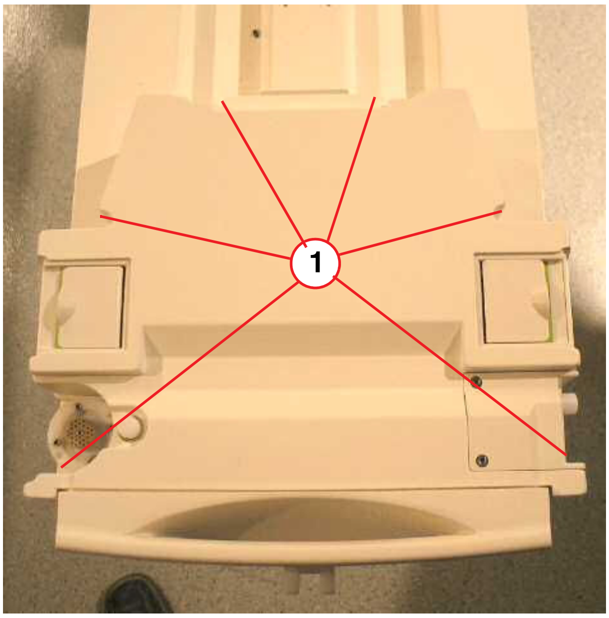
- Remove the Allen screws on top of the cover from the coil plugs.

Fig. 214: Remove Allen screws

- Remove the lower Allen screws of the cover from the coil plugs.

Removable table

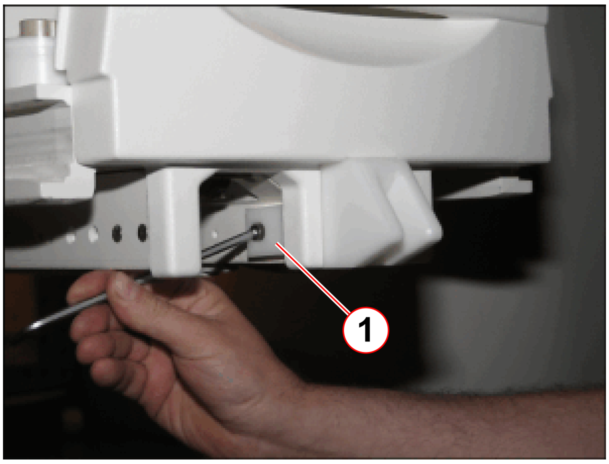
Fig. 215: RF-Cover RTT



(1) Screws to be removed

- Remove the carrier plate of the RTT.
- Remove six screws on top of the cover from the coil plugs.
- Disconnect cables.

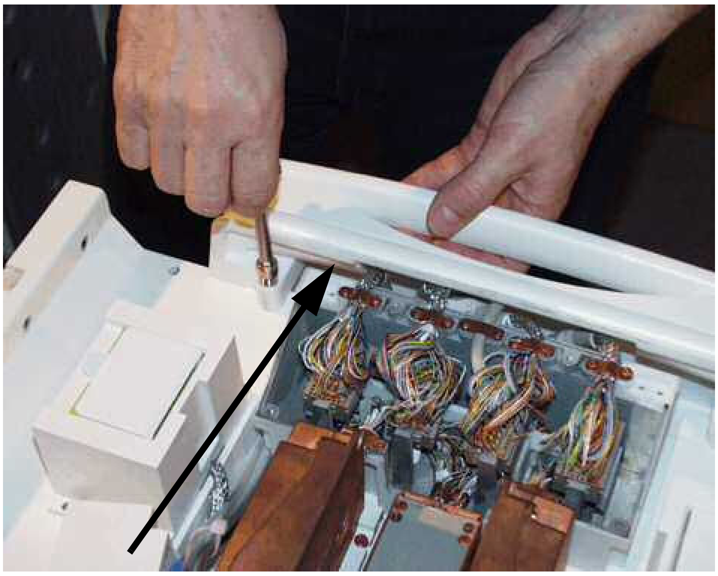
Fig. 216: Additional mounting bracket for foot end cover (newer systems)



(1) Allen screw

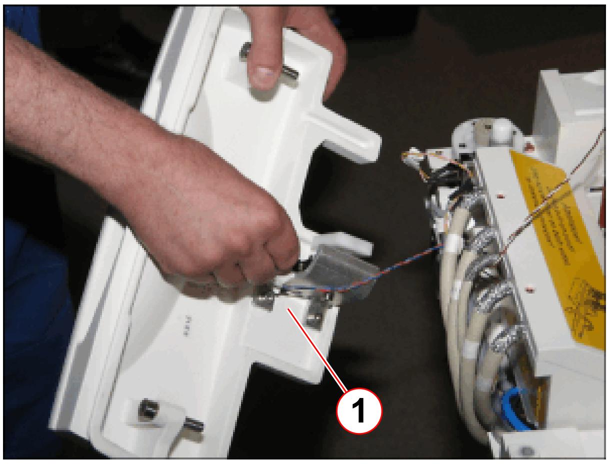
- On newer systems, remove Allen screw of mounting bracket.

Fig. 217: Foot-end cover



- Remove screws of the foot-end cover.

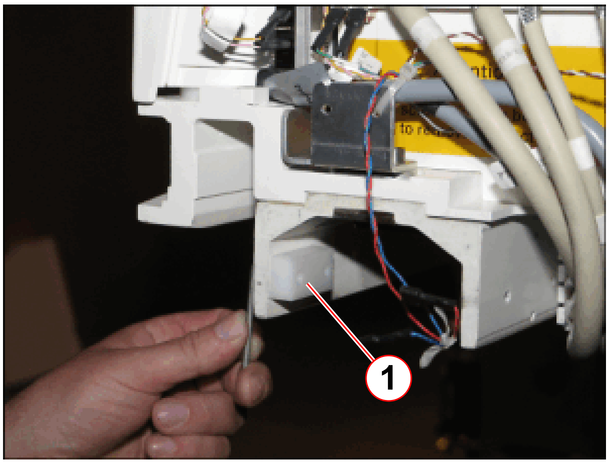
Fig. 218: Foot end cover, removed



(1) Disconnect cables

- Disconnect cable connections and remove the foot-end cover.
- Slide out the lower cover on the bottom of the patient table.

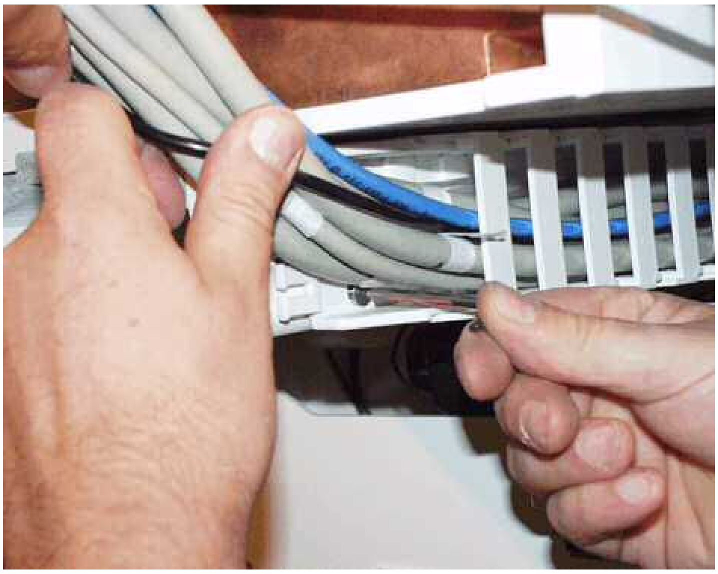
Fig. 219: Mechanical end stop of partial drive unit



(1) End stop to be removed

- Remove mechanical end stop.

Fig. 220: Energy chain

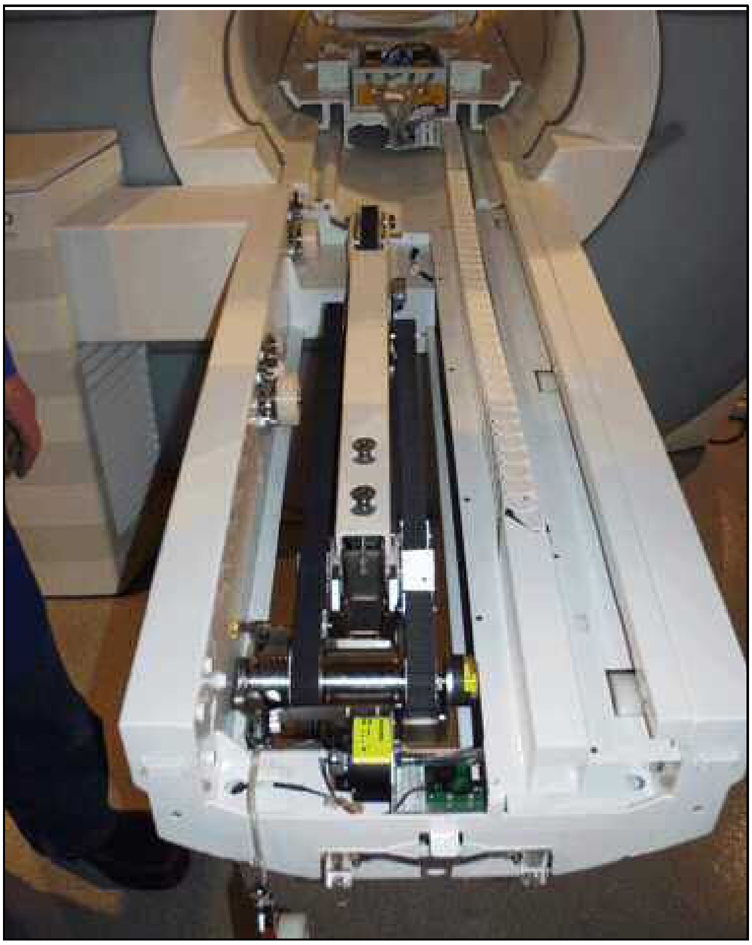


- Remove the lower part of the energy chain

Whole Body / telescopic drive

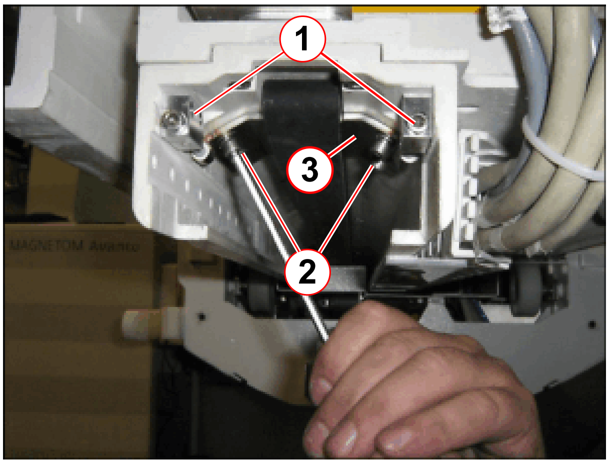
- For the telescopic drive: Release the emergency clutch and carefully push the tabletop plate manually into the magnet until the cantilever arm is no longer touching the tabletop plate. Support the table to ensure it does not fall.

Fig. 221: Tabletop, magnet end



Partial drive, not valid for Trio A Tim Systems

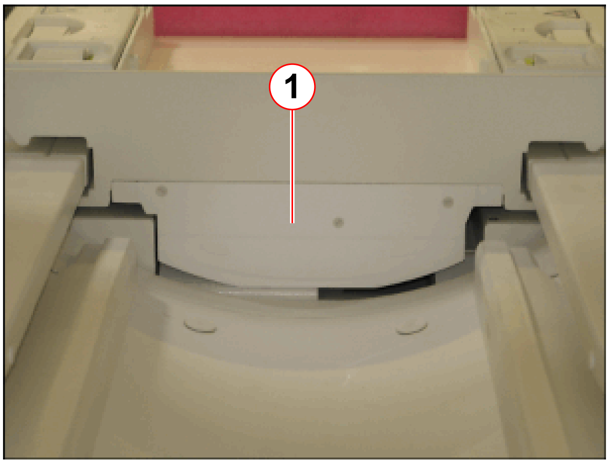
Fig. 222: Tension unit for toothed belt



(1) Allen screws for adjusting belt tension
(2) Attachment screws
(3) clamping plate

- Release tension of toothed belt (1), remove Allen screws (2) and tooth belt tension unit.

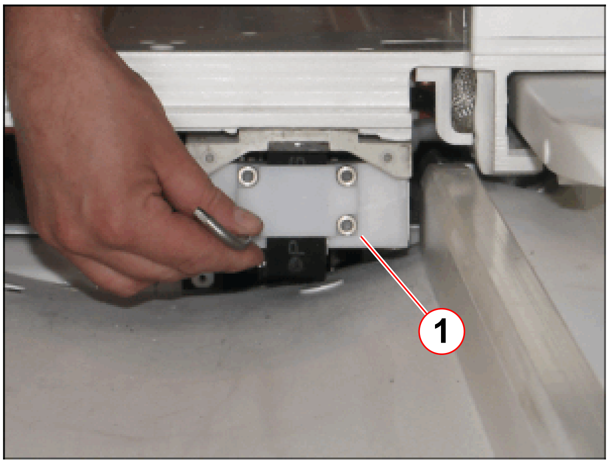
Fig. 223: Cover of toothed belt mount (head end)



(1) Cover of toothed belt mount

- Remove the cover of the toothed belt mount of toothed belt (1) on the head end from the ptab.

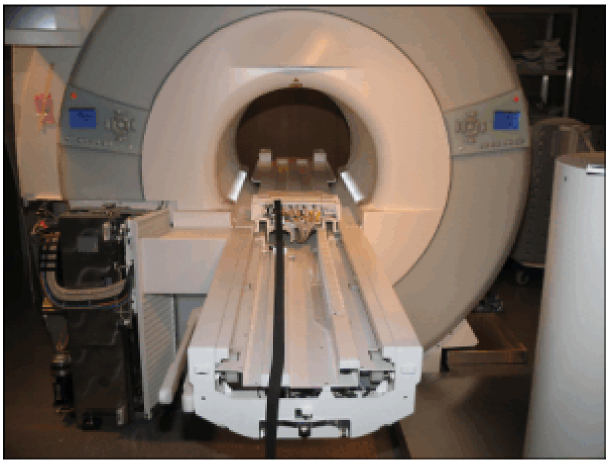
Fig. 224: *Toothed belt attachment (head end)*



(1) Allen screws of toothed belt mount

- Remove the toothed belt mount (1) on the head end from the ptab.

Fig. 225: *Tabletop moved into the magnet*

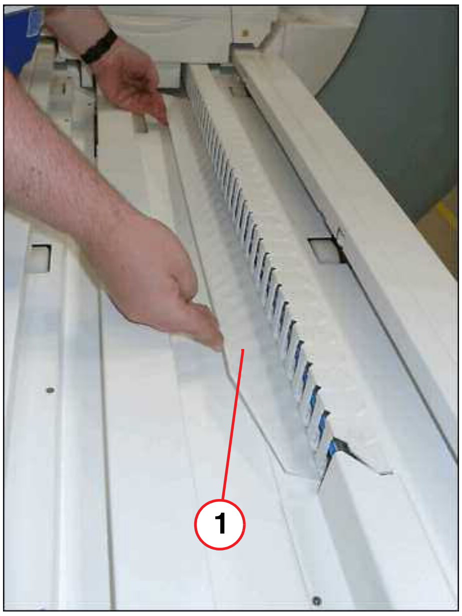


- Carefully push the tabletop plate manually into the magnet.

3.9.4.2 Covers / cables / energy chain

> Be careful and avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

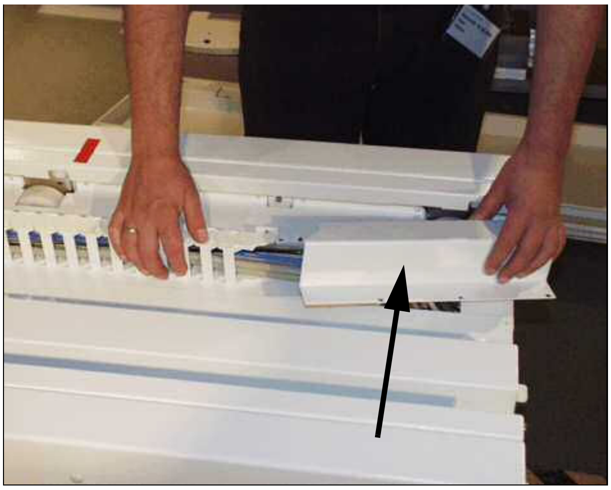
Fig. 226: Plastic inner cover



(1) Plastic inner cover

- Remove inner plastic cover.

Fig. 227: Remove cover of energy chain

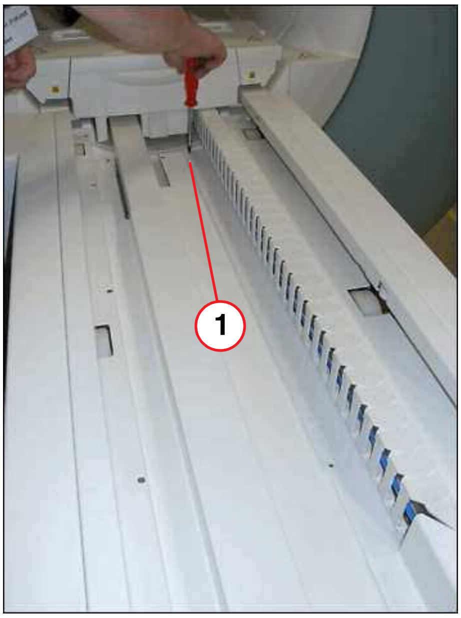


- Remove the upper energy chain cover.

Fig. 228: Remove energy chain

- Remove the upper energy chain of the bracket.

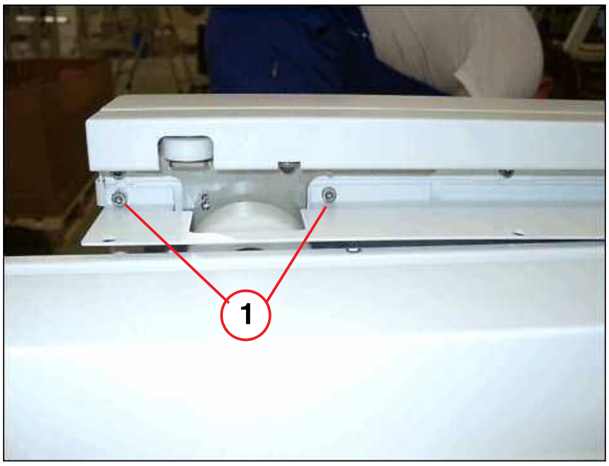
Fig. 229: Inner main cover



(1) Screws to be removed

- Remove screws of main inner cover.

Fig. 230: Allen screws

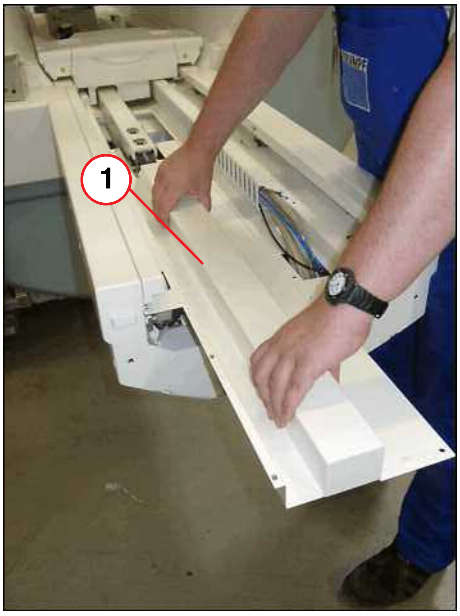


(1) Screws to be removed

- Remove two Allen screws at the left side of the table end (foot) before you remove the main inner cover.

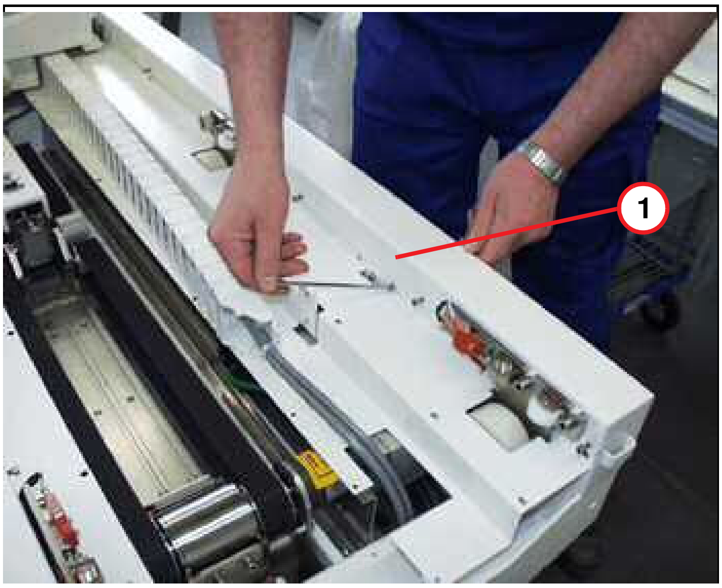
- Remove the inner main cover.

Fig. 231: Inner main cover



(1) Inner main cover

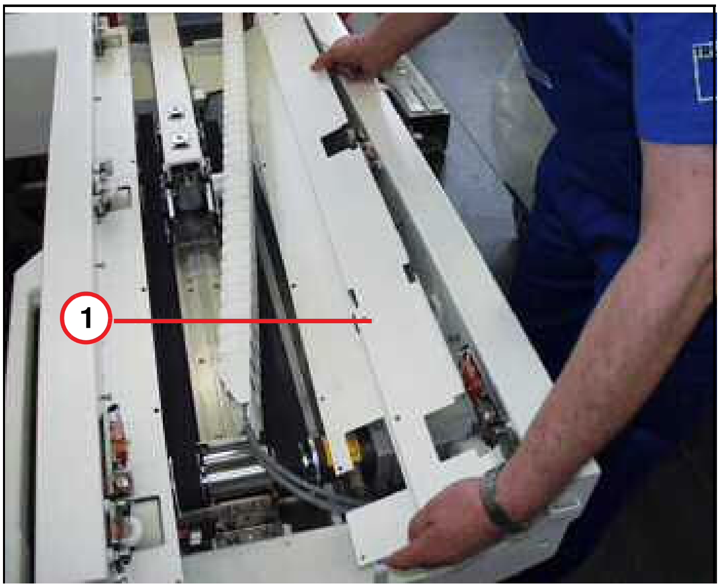
Fig. 232: Inner cover



(1) Top inner cover

- Remove the right guide rail cover and the inner top cover.

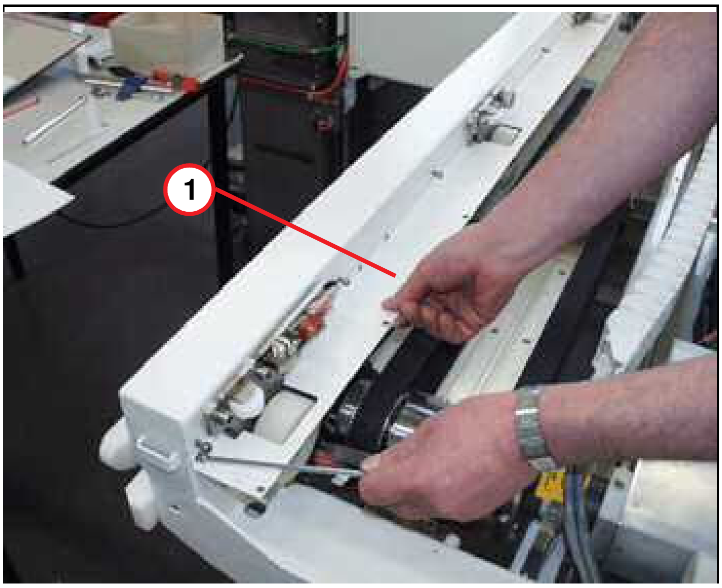
Fig. 233: Inner cover



(1) Inner cover, right side

- Remove the inner right cover.

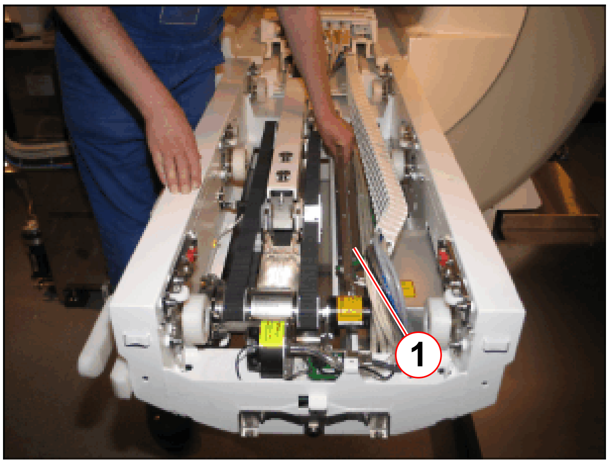
Fig. 234: Inner cover, left side



(1) Inner cover, left side

- Remove the left guide rail and the inner cover, left side.

Fig. 235: Covers removed



(1) Support plate

- Remove support plate.

3.9.4.3 Removal of horizontal drive unit / telescopic drive

CAUTION

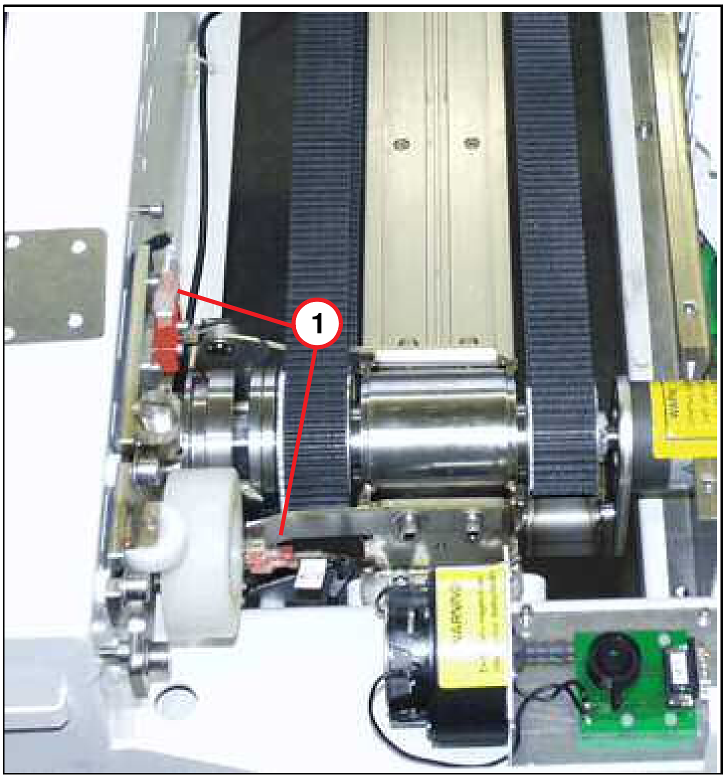
The horizontal drive units consist of magnetic objects!

Magnetic objects may cause injury.

» **Maintain a safe distance from the magnetic field.**

» **Fasten the horizontal drive unit with the arresting cable.**

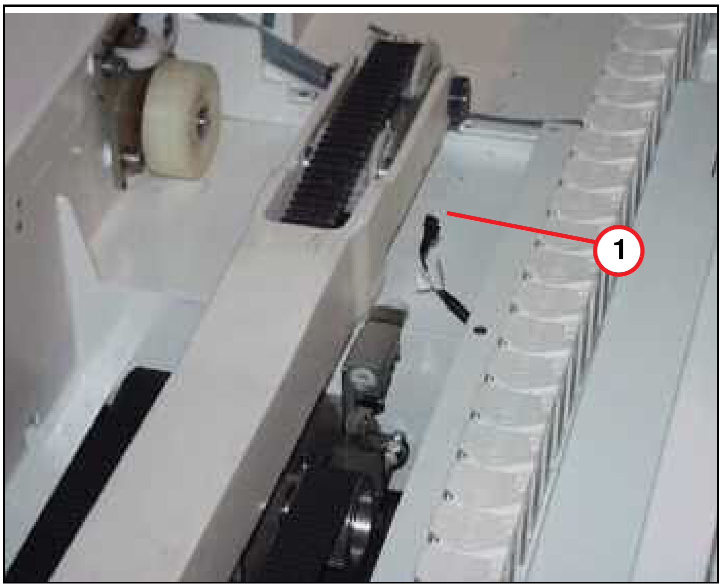
Fig. 236: Control switches



(1) Switch S51 and S32

- Disconnect the switches, S51 (emergency release), S32.

Fig. 237: Plug S10

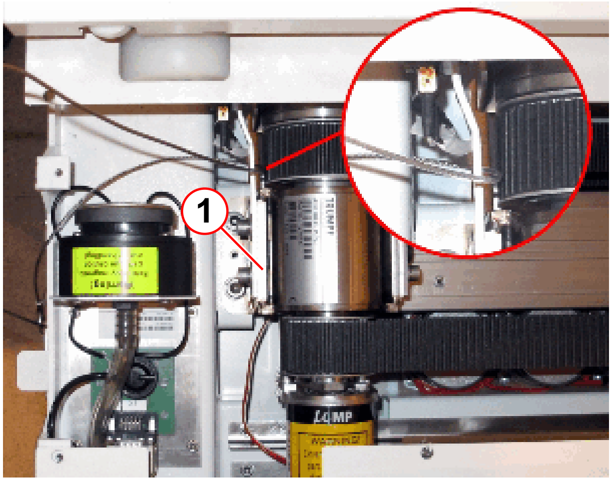


(1) S10 plug, disconnected

- Disconnect the cables of S10 (reference switch).

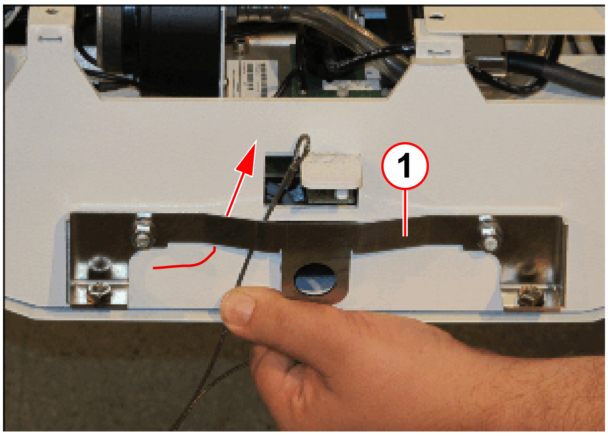
- For removing and mounting, the horizontal drive unit (p/n 8112257) must be secured with the arresting cable.
- Secure the horizontal drive with the arresting cable as shown below

Fig. 238: Arresting cable secured on the horizontal drive



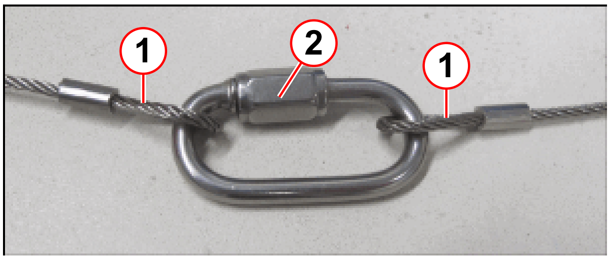
(1) Horizontal drive bracket

Fig. 239: Arresting cable secured on the mounting angle



(1) Mounting angle

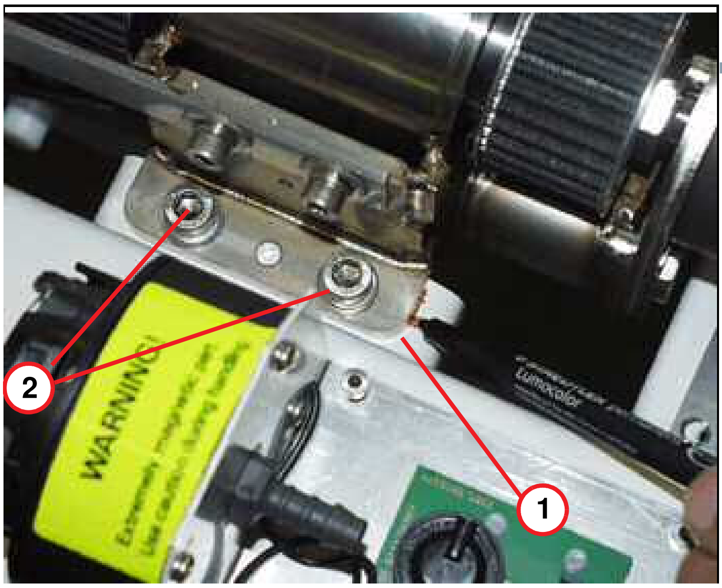
Fig. 240: Hook



(1) Loops
(2) Hook

- Secure the arresting cable loops (pos.1) by closing the hook (pos.2).

Fig. 241: Position of drive bracket



(1) Original position of drive bracket
(2) Fastener of horiz. drive unit

- Mark the position of the carrier plate with respect to the drive bracket.
- Remove the two M8 Allen screws of the horizontal drive bracket.

Fig. 242: M8 Allen screw, horizontal drive unit

- Remove the M8 Allen screw underneath the patient table.

Fig. 243: Example - removing FOCs from MDSD

(1) Cam of FOC connector
(2) Screwdriver, flat blade

- The figure below shows the FOC plug connection (principle) inside the MDSD device. Use a flat blade screwdriver to disconnect the FOCs inside the MDSD device.

Fig. 244: Fiber-optic cables of MDSD

- Feed the flat blade screwdriver through the hole in the aluminum cover and into the FOC's plug connection. Try to reach the cam of the FOC and carefully push it sideways. At the same time, pull it out from the FOC. Refer to the illustration.

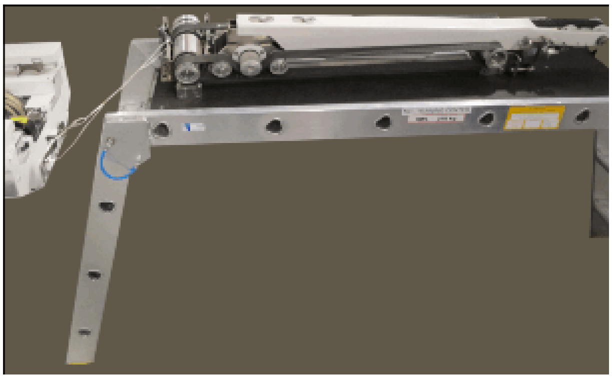
CAUTION

Due to the strong magnetic parts of the horizontal drive, it is very important to move the horizontal drive always away from the magnet!

Do not move the drive closer to the magnet than absolutely necessary.

» Maintain a safe distance from the magnetic field.

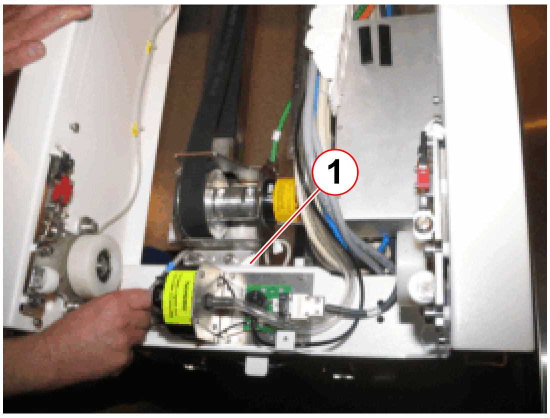
Fig. 245: Horizontal drive removed on the service platform



- Store the secured horizontal drive at the service platform at the patient end of the table.
- Replace the horizontal drive with the new one. Replace the original MDSD device as described in the MDSD chapter. (→ MDSD drive of whole body drive / Page 122)
- For the replacement, the arresting cable must be used.

3.9.4.4 Removal of partial drive unit, not valid for Trio A Tim Systems

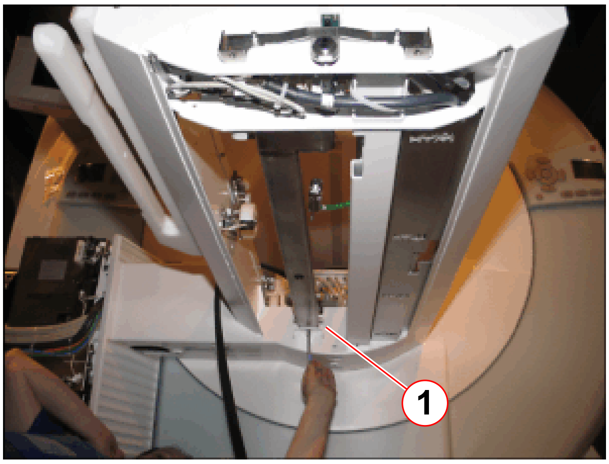
Fig. 246: Partial drive assembly



(1) Allen screws of mounting bracket

- Remove Allen screws of partial drive unit, foot end side.

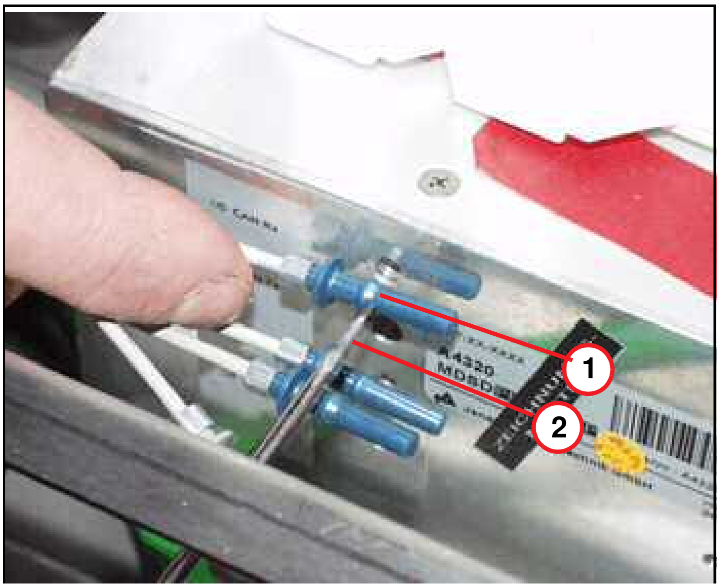
Fig. 247: Partial drive unit, button view



(1) Allen screw for fastening the partial drive unit

- Remove Allen screw on the bottom of the partial drive unit.

Fig. 248: Example - removing FOCs from MDSD



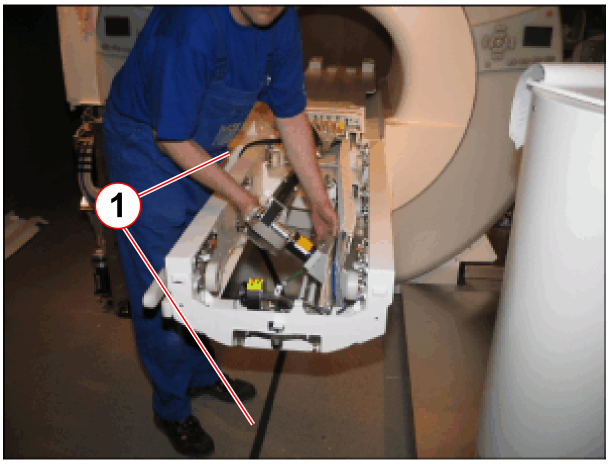
(1) Cam of FOC connector
(2) Screwdriver, flat blade

- The figure below shows the FOC plug connection (principle) inside the MDSD device. Use a flat blade screwdriver to disconnect the FOCs inside the MDSD device.

Fig. 249: Fiber-optic cables of MDSD

- Feed the flat blade screwdriver through the hole in the aluminum cover and into the FOC's plug connection. Try to reach the cam of the FOC and carefully push it sideways while at the same time pulling it out from the FOC. Refer to illustration.

Fig. 250: Partial drive assembly, removed



(1) Position of toothed belt

- Remove the partial drive unit and store it on a clean workbench.
- Replace the original MDSD device as described in the MDSD chapter. (→ MDSD drive of whole body drive / Page 122)

3.9.5 Assembly

3.9.5.1 Horizontal drive unit / telescopic drive

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

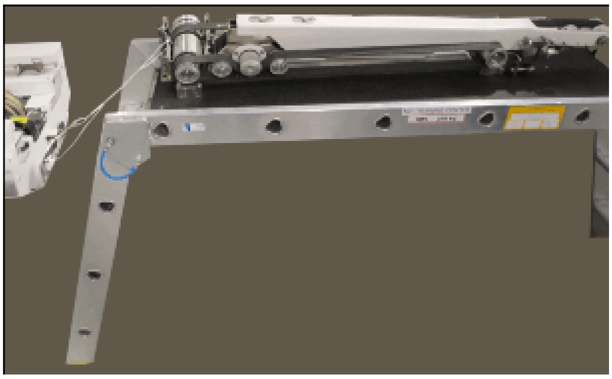
CAUTION

Due to the strong magnetic parts of the horizontal drive, it is very important to move the horizontal drive always away from the magnet!

Do not move the drive closer to the magnet than absolutely necessary.

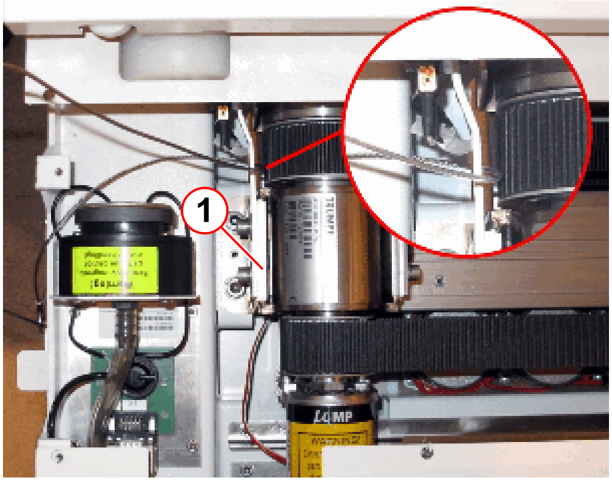
» **Maintain a safe distance from the magnetic field.**

Fig. 251: Horizontal drive removed on the service platform



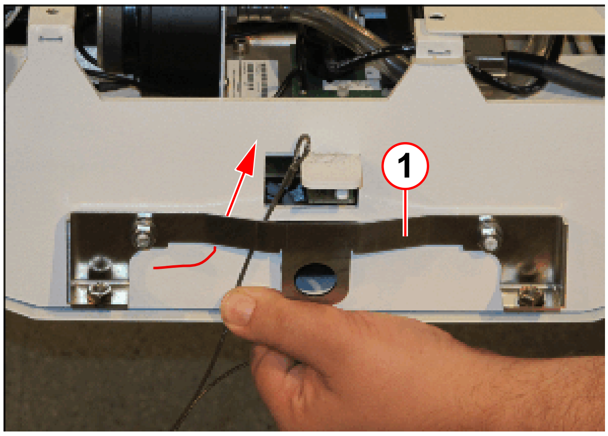
- Store the secured horizontal drive at the service platform at the patient end of the table.
- Secure the horizontal drive with the arresting cable as shown in the pictures below.
- Replace the horizontal drive with the new one. Replace the original MDSD device as described in the MDSD chapter. (→ MDSD drive of whole body drive / Page 122)

Fig. 252: Arresting cable secured on the horizontal drive



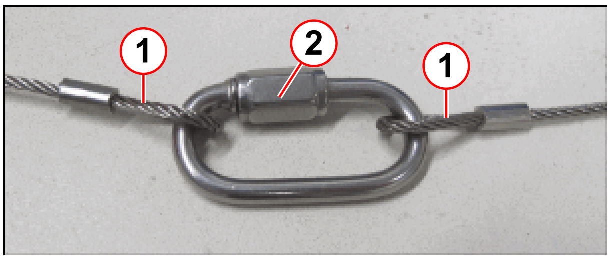
(1) Horizontal drive bracket

Fig. 253: Arresting cable secured on the mounting angle



(1) Mounting angle

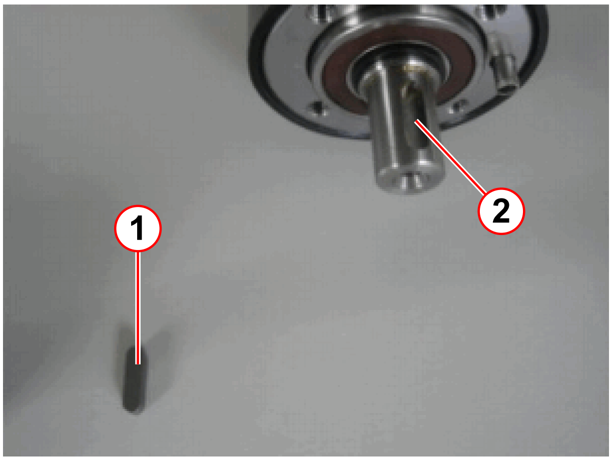
Fig. 254: Hook



(1) Loops
(2) Hook

- Secure the arresting cable loops (pos.1) by closing the hook (pos.2).

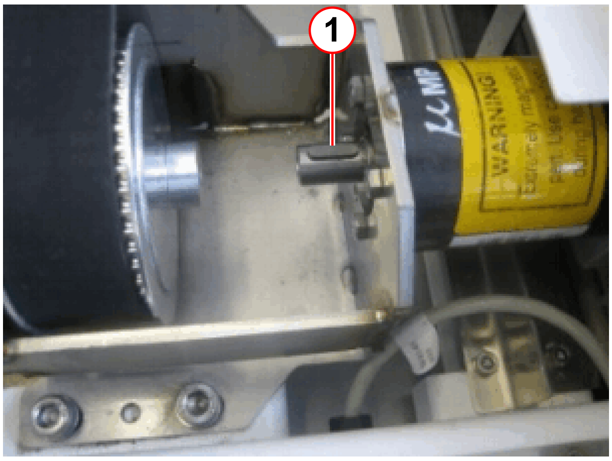
Fig. 255: Assembling the partial drive MDSD



(1) Key
(2) Notch of the MDSD axis

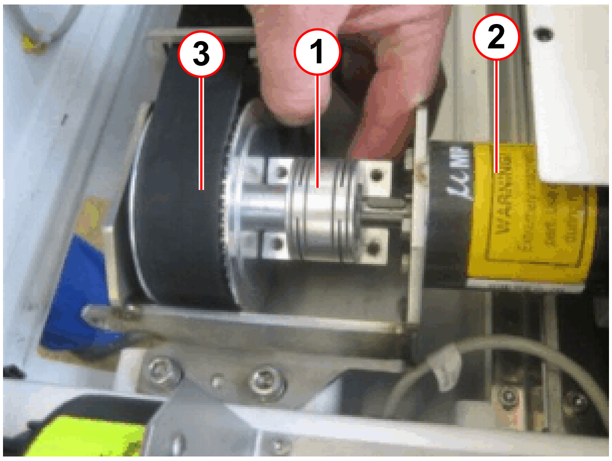
- For partial drive tables insert the key (1), delivered with the spare part, into the notch of the MDSD axis.

Fig. 256: Key inserted in MDSD axis



(1) Key

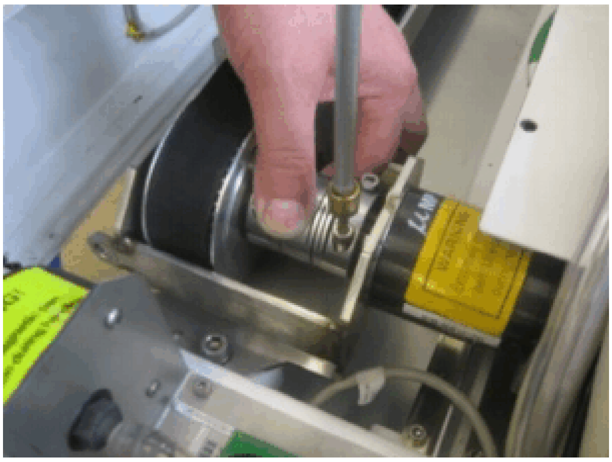
Fig. 257: Fixation of the clutch



(1) Clutch
(2) MDSD for partial drive
(3) Drive unit TK

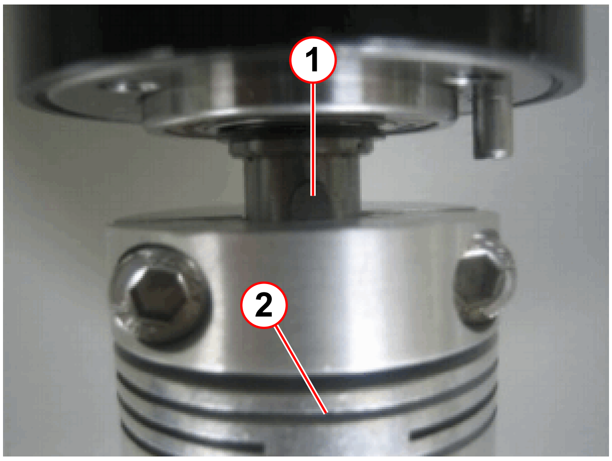
- Place the clutch as shown in pos. (1).

Fig. 258: Fixation of the clutch



- Position the other half of the clutch so that the key fits into the notch of the clutch and tighten the screws of the clutch.

Fig. 259: Key position with clutch



(1) Key
(2) Clutch

Fig. 260: Clutch installed

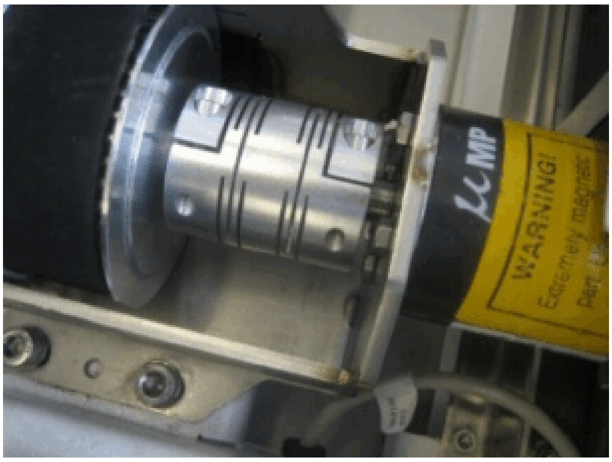
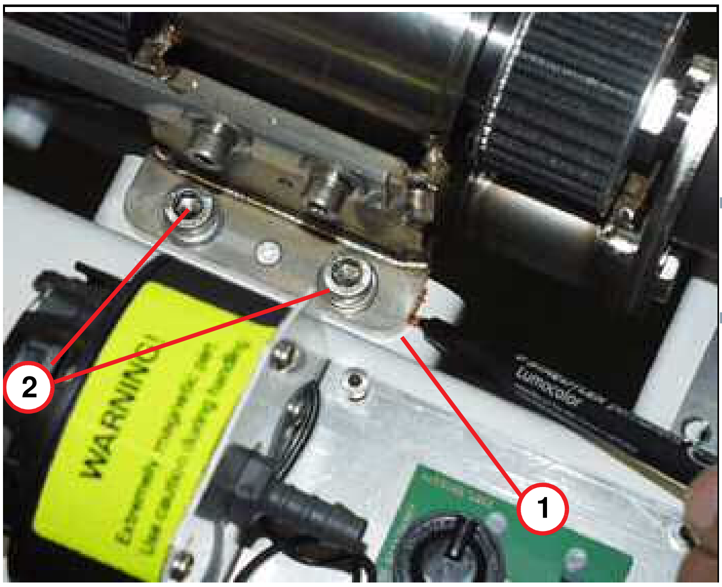


Fig. 261: Position of drive bracket



(1) Original position of drive bracket
(2) Fastener of horiz. drive unit

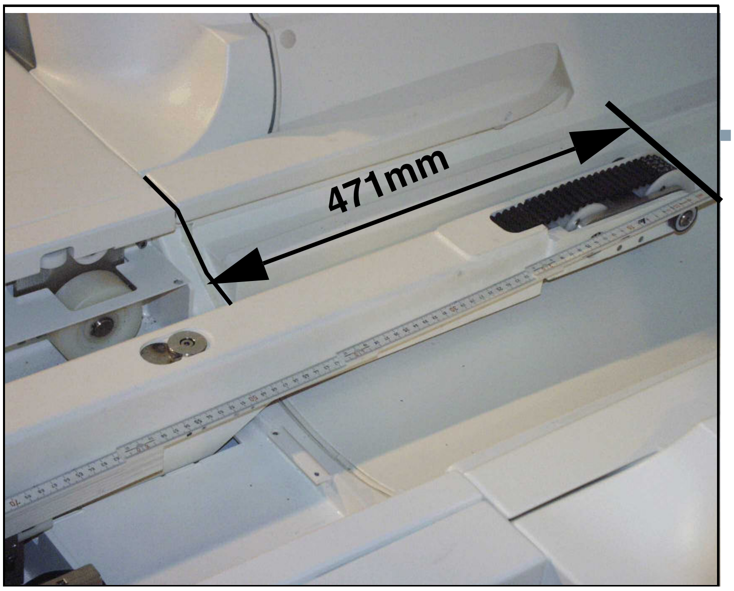
- Replace the horizontal drive with the new one and use the key delivered with the spare part as described before. Replace the original MDSD device as described in the MDSD chapter. (➔ MDSD drive of whole body drive / Page 122)
- Reinstall the horizontal drive unit and adjust the drive bracket to the marked position.
- Reinsert the Allen screws of the horizontal drive mechanism, apply Loctite 221.

Fig. 262: M8 Allen screw, horizontal drive unit

- Reinsert the Allen screw of the horizontal drive underneath the table, apply Loctite 221.

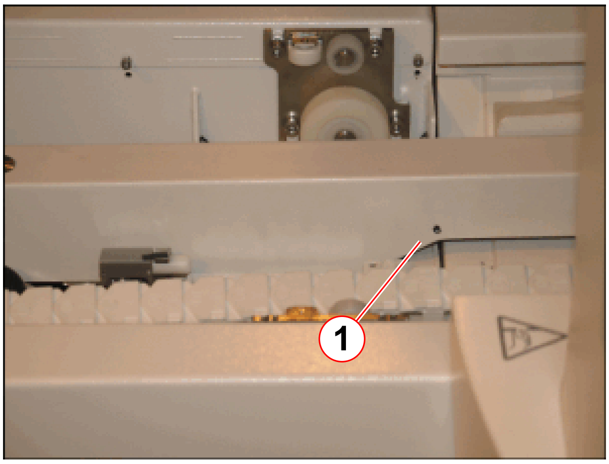
- Removal the arresting cable.
- Reconnect switch S10 and S51.
- Reconnect all cables to the MDSD device, use the feed-in tool as described in the MDSD chapter.
- Reinstall the assembly of the guide roller, apply Loctite 221.
- Reconnect switch S32.
- Reinstall all inner table covers.

Fig. 263: Distance of cantilever arm



- Before you insert the new tabletop plate, please do a check of the distance from the end of the cantilever arm to the support frame. Adjust to 471 mm.
- Carefully connect the tabletop plate to the toothed belt of the horizontal drive.

Fig. 264: Mark on the telescopic arm



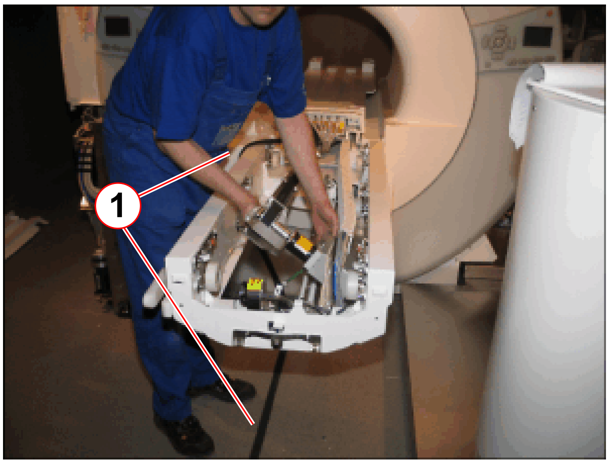
(1) Marking hole in the telescopic arm

- On newer systems the distance mark shows the correct length of the cantilever arm.

3.9.5.2 Partial drive unit, not valid for Trio A Tim Systems

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

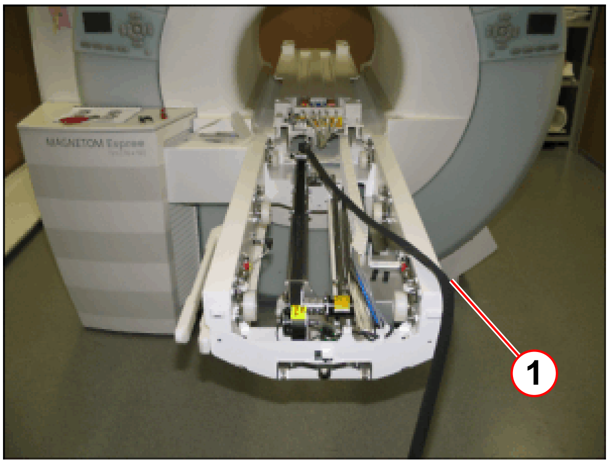
Fig. 265: Partial drive assembly, removed



(1) Position of toothed belt

- Reinsert the horizontal drive unit.

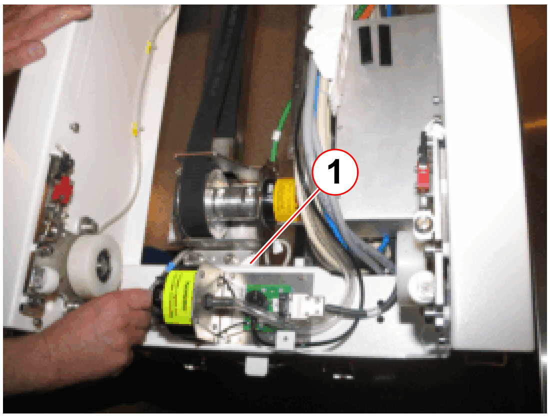
Fig. 266: Tabletop into the magnet; toothed belt route



(1) To avoid misassembling, lay toothed belt over side

- Before you insert the the cover, ensure the correct toothed belt route.

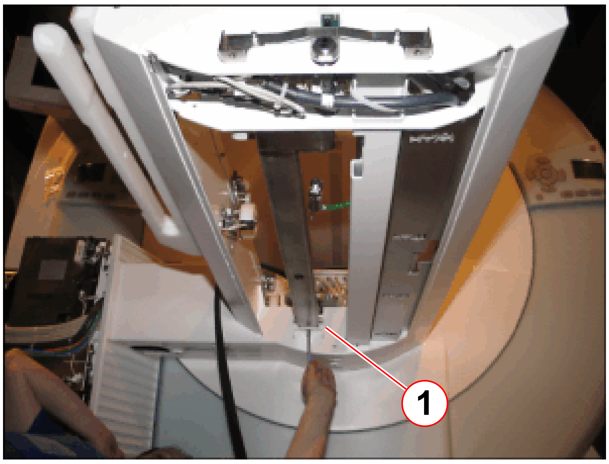
Fig. 267: Partial drive assembly



(1) Allen screws of mounting bracket

- Reinstall the drive unit.
- Reinsert Allen screws of partial drive unit, foot end side, apply Loctite 221.

Fig. 268: Partial drive unit, button view

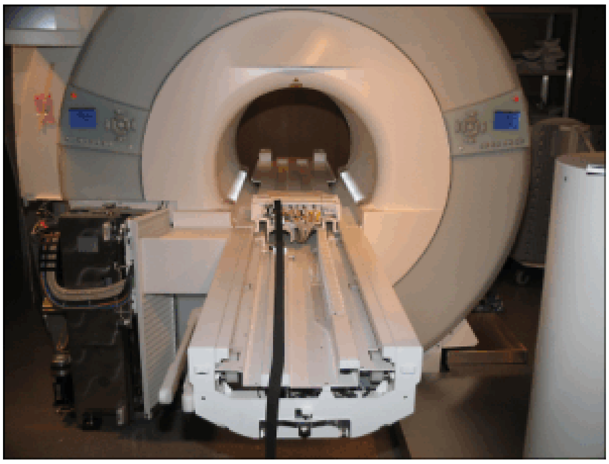


(1) Allen screw for fastening the partial drive unit

- Reinsert Allen screw on the bottom of the partial drive unit, apply Loctite 221.

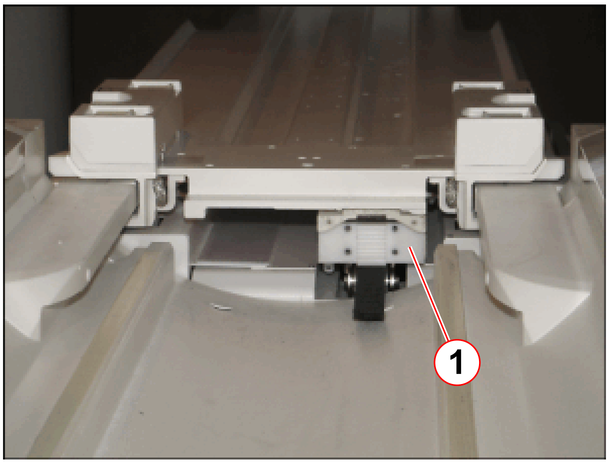
- Reconnect switches.
- Reconnect all cables to the MDSD device, use the feed-in tool as described in the MDSD chapter. (→ MDSD drive of whole body drive / Page 122)
- Reinstall all inner table covers.

Fig. 269: Tabletop moved into the magnet



- Make sure that the toothed belt runs correct under the table top.

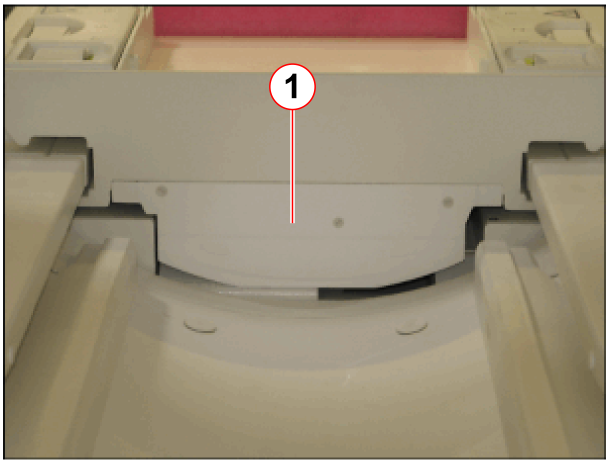
Fig. 270: Toothed belt attachment (head end)



(1) Toothed belt mount (head end)

- Carefully pull the tabletop plate manually back to the food end side (patient end).
- Install the mount of toothed belt on the head end from the ptab.

Fig. 271: Cover of toothed belt mount (head end)



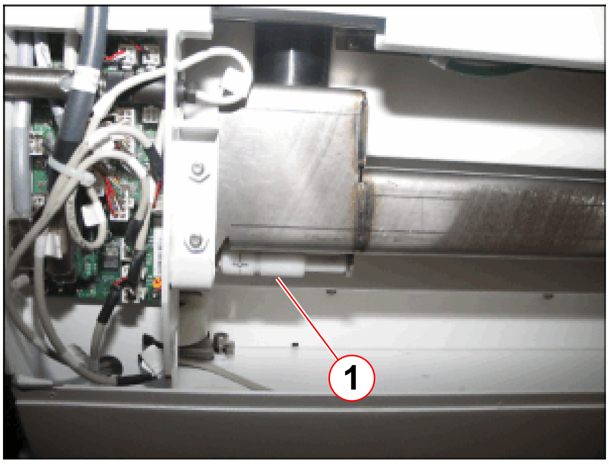
(1) Cover of toothed belt mount

- Install the cover of the toothed belt mount(1) on the head end from the ptab.

Fig. 272: Mounting the toothed belt (foot end)

- Principle of toothed belt tension unit, located on the foot end side of the ptab.

Fig. 273: Service tool for toothed belt tension



(1) Storage position of service tool for toothed belt tension

- The service aid for adjustment of the toothed belt tension is located underneath the table top.
- Snap out the service aid for adjustment of the toothed belt tension.

3.9.6 Concluding steps

- Install the energy chain.

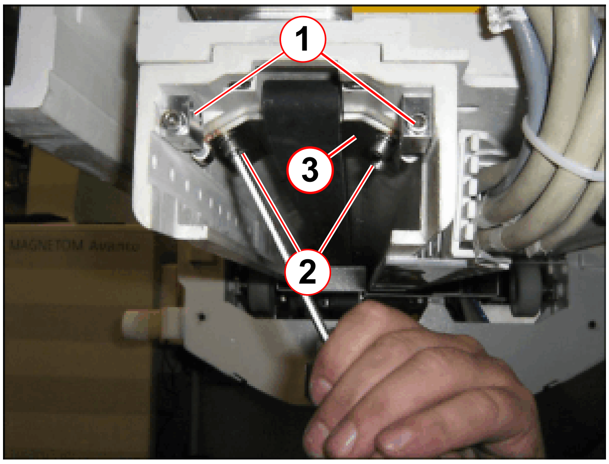
> ℹ Use the same length of screws when reinstalling the energy chain

- Make sure you replace the covers in the proper sequence and tighten all screws at the end of the assembling procedure.
- Install the end stops.
- Install the remaining covers.
- Install foot-end plate and cover from the coil plug.

3.9.7 Adjustments, not valid for Trio A Tim Systems

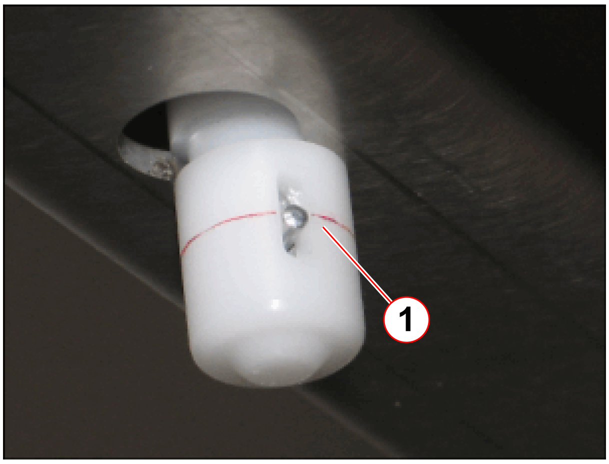
Fig. 274: Service tool, toothed belt tension

Fig. 275: Tension unit for toothed belt



(1) Allen screws for adjusting belt tension
(2) Attachment screws
(3) clamping plate

Fig. 276: Service tool for toothed belt tension in use



(1) Correct tension of toothed belt (pin aligned with mark)

- Connect service aid to measure the tension of the toothed belt.
- Tighten the Allen screws (→ 1/Fig. 275 Page 162) until the pin is aligned with the mark (→ 1/Fig. 276 Page 162).
- When the toothed belt is adjusted to the proper tension, tighten the Allen screws (→ 2/Fig. 275 Page 162).
- Remove the service aid and place it on the store position (→ 1/Fig. 273 Page 161).

3.9.8 Final checks

- Switch ON F22 (RF-room) ptab and check for smooth horizontal movement of the tabletop plate.

The system should be on and SYNGO should be running.

- Move the table all the way out.
  - » The table should move constant and without any extraordinary noise.
  - » The table stops in the horizontal end position.
- Lower the table all the way.
  - » The table stops in the vertical end position.
  - » The table should move without any extraordinary noise.
- Move the table all the way up again.
  - » The table should move without any extraordinary noise.
  - » The table stops in the vertical end position.
- Move the table all the way in.
  - » The table should move constant and without any extraordinary noise.
  - » The table stops in the horizontal end position.
- Move the table to the Home position.

3.10 Toothed belt of horizontal drive

3.10.1 General

The "Partial drive" allows the horizontal table movement without whole body application in a limited movement range. A toothed belt is responsible for transferring the engine power into a horizontal table movement.

3.10.2 Tools and time required

3.10.2.1 Time

Approx. 60 min

3.10.2.2 Tools

Non-magnetic tool kit

Philips screwdriver

3.10.3 Preliminary

- Switch OFF F22 (RF-room) in the ACC.

> ℹ According to the system/room specifications the movement range of the horizontal movement can be reduced.

- If the movement range of the horizontal movement is reduced it may be necessary to remove the end stops. (➔ Fig. 306 Page 179)

3.10.4 Disassembly

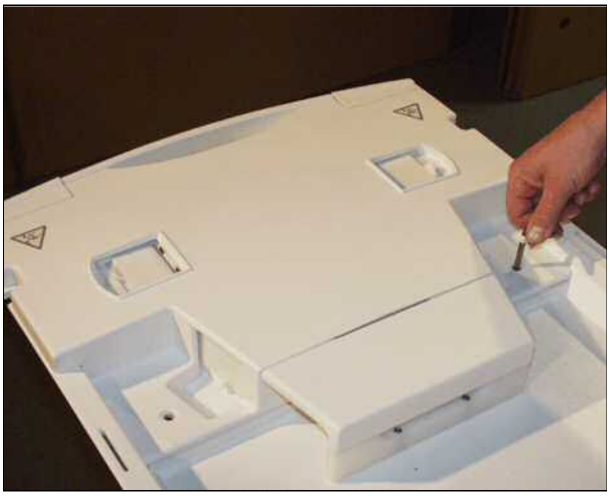
3.10.4.1 Tabletop plate

> ℹ Be careful and avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

> ℹ The tabletop plate may fall - support the tabletop plate

Basic table

Fig. 277: Coil plug cover (foot end)



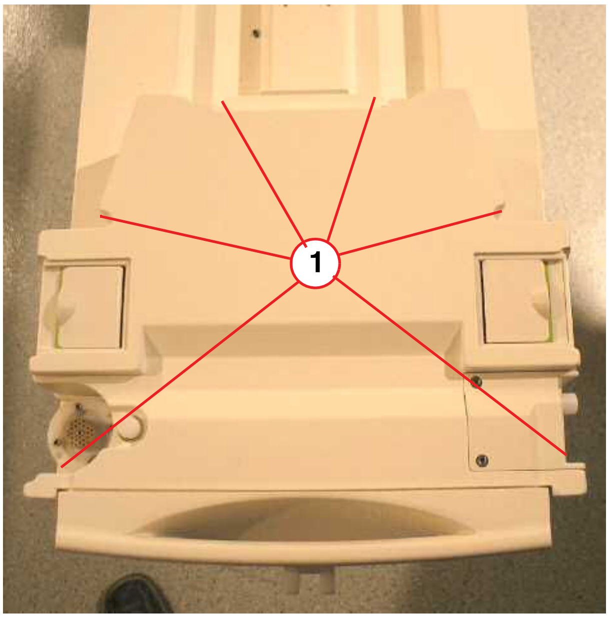
- Remove the Allen screws on top of the cover from the coil plugs.

Fig. 278: Remove Allen screws

- Remove the lower Allen screws of the cover from the coil plugs.

Removable table

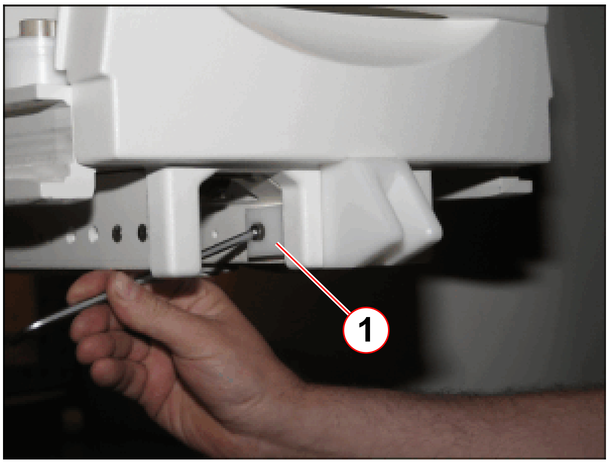
Fig. 279: RF-Cover RTT



(1) Screws to be removed

- Remove the carrier plate of the RTT.
- Remove six screws on top of the cover from the coil plugs.
- Disconnect cables.

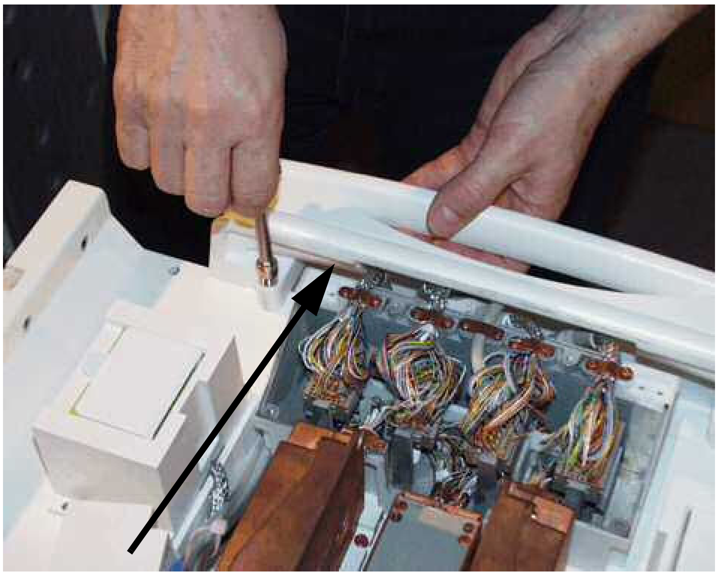
Fig. 280: Additional mounting bracket for foot end cover (newer systems)



(1) Allen screw

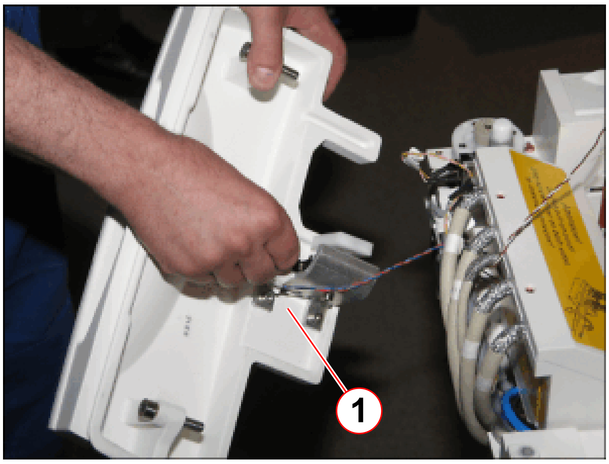
- On newer systems, remove Allen screw of mounting bracket.

Fig. 281: Foot-end cover



- Remove screws of the foot-end cover.

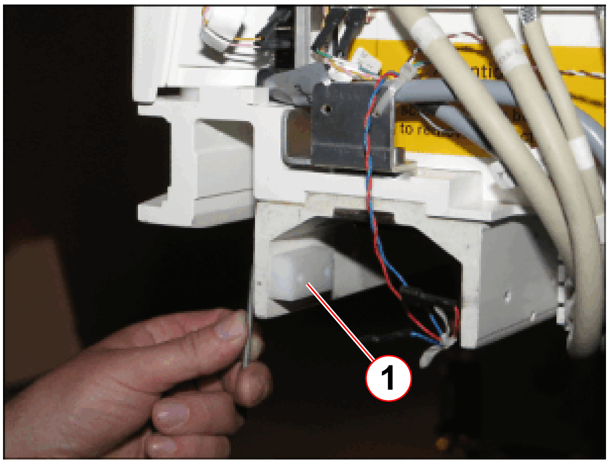
Fig. 282: Foot end cover, removed



(1) Disconnect cables

- Disconnect cable connections and remove the foot-end cover.
- Slide out the lower cover on the bottom of the patient table.

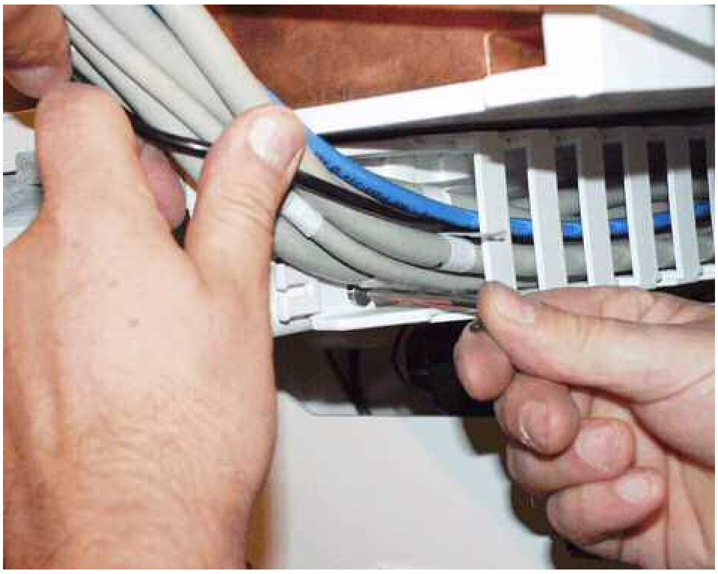
Fig. 283: Mechanical end stop of partial drive unit



(1) End stop to be removed

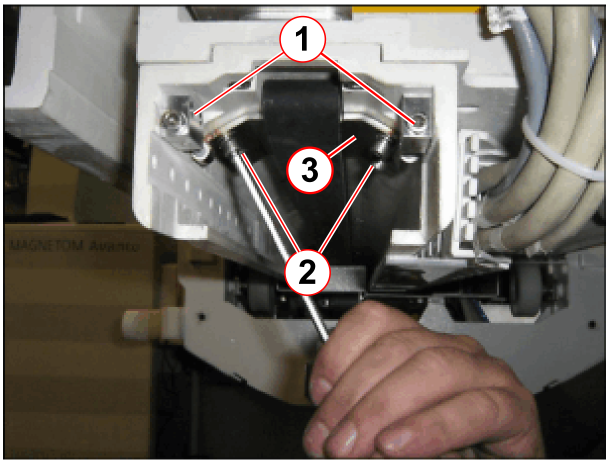
- Remove mechanical end stop.

Fig. 284: Energy chain



- Remove the lower part of the energy chain.

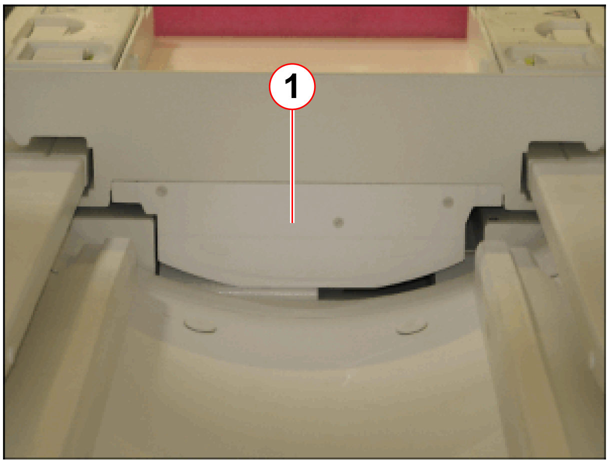
Fig. 285: Tension unit for toothed belt



(1) Allen screws for adjusting belt tension
(2) Attachment screws
(3) clamping plate

- Release tension of toothed belt, remove Allen screws and tooth belt tension unit.

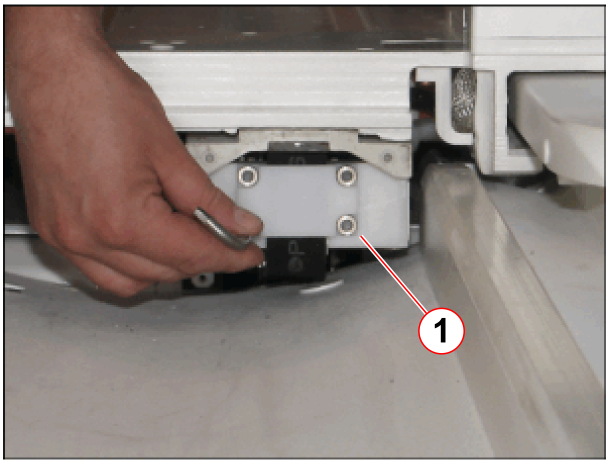
Fig. 286: Cover of toothed belt mount (head end)



(1) Cover of toothed belt mount

- Remove the cover of the toothed belt mount of toothed belt (1) on the head end from the ptab.

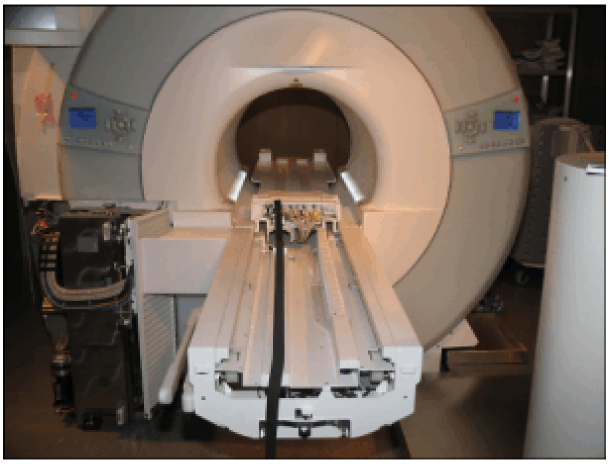
Fig. 287: Fixation of tooth belt - service end



(1) Allen screws of toothed belt mount

- Remove the mount of toothed belt on the head end from the ptab.

Fig. 288: Tabletop moved into the magnet

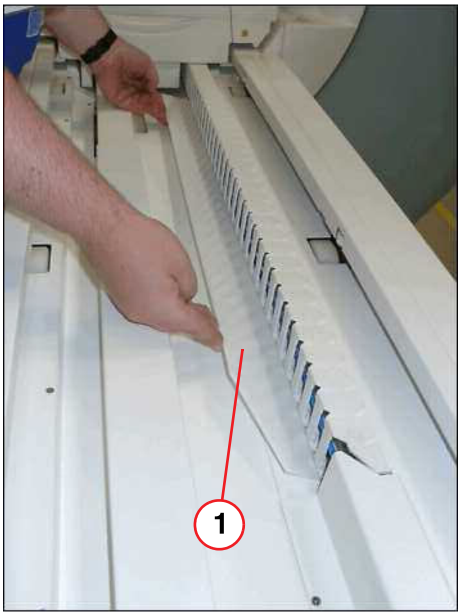


- Carefully push the tabletop plate manually into the magnet.

3.10.4.2 Covers / cables / energy chain

> Be careful and avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

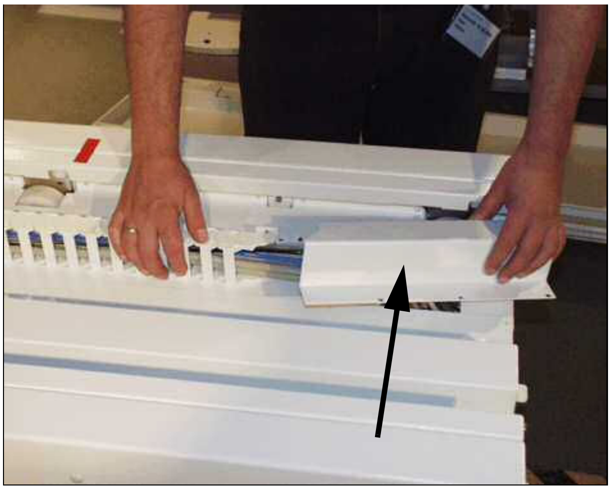
Fig. 289: Plastic inner cover



(1) Plastic inner cover

- Remove inner plastic cover.

Fig. 290: *Remove cover of energy chain*

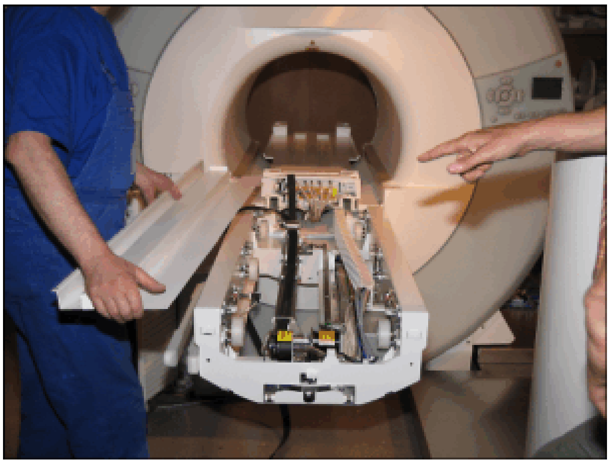


- Remove the upper energy chain cover.

Fig. 291: *Remove energy chain*

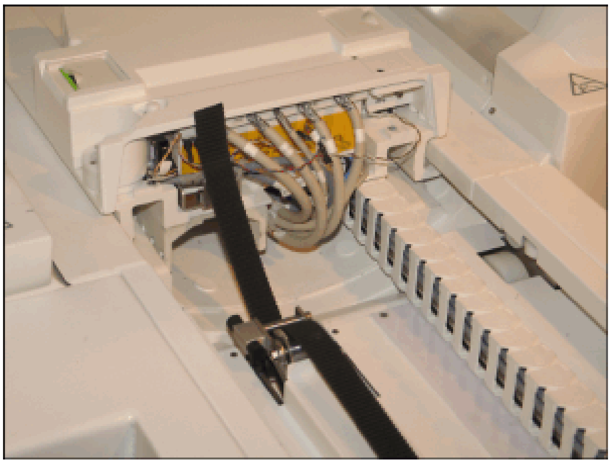
- Remove the upper energy chain of the bracket.

Fig. 292: *Covers removed*



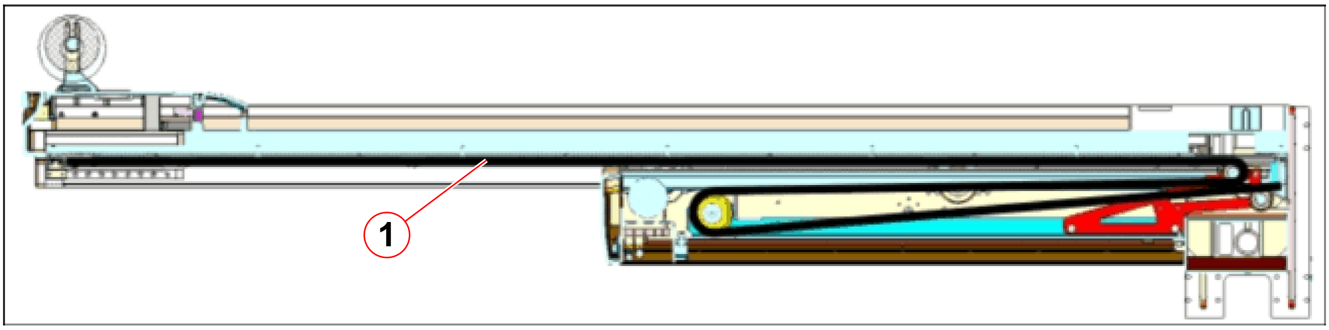
- Remove covers until the toothed belt is reachable.

Fig. 293: Toothed belt rooting



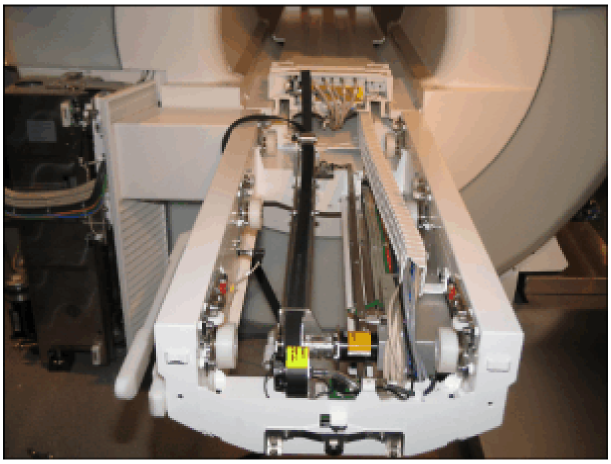
- Principle of toothed belt rooting, head end table carrier.

Fig. 294: Schematic of the toothed belt routing



(1) Toothed belt

Fig. 295: Covers removed, ready for toothed belt exchange

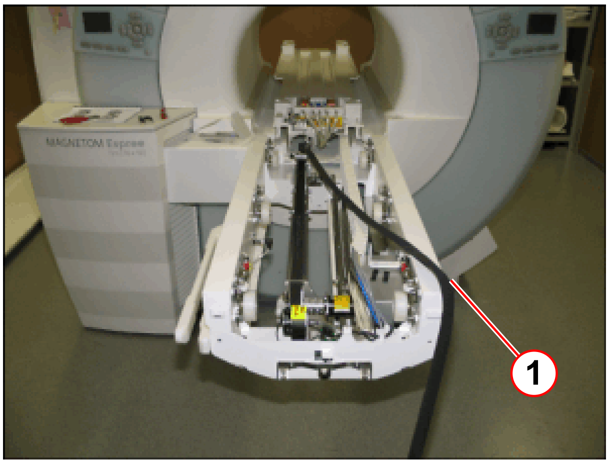


- Remove inner coves to get access to the toothed belt unit.
- Replace toothed belt.

> ℹ Use the same length of screws when reinstalling the energy chain.

3.10.5 Assembly

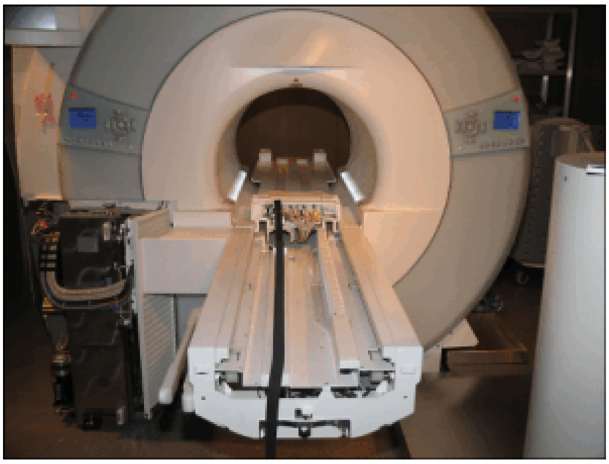
Fig. 296: Tabletop into the magnet; toothed belt route



(1) To avoid misassembling, lay toothed belt over side

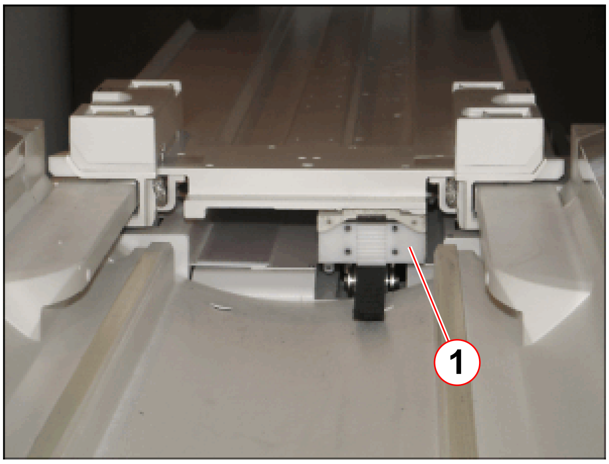
- Before you insert the the cover, ensure the correct toothed belt route.

Fig. 297: Tabletop moved into the magnet



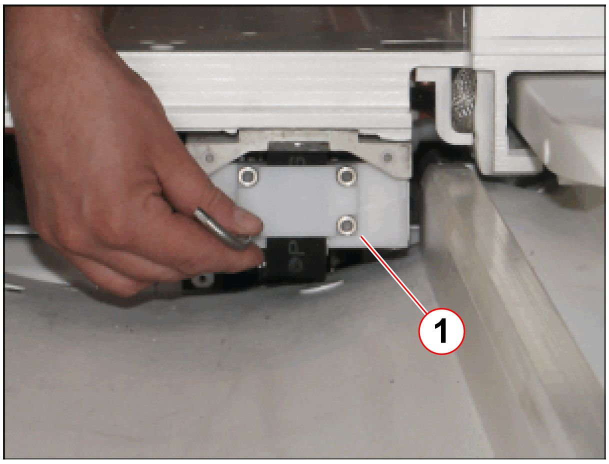
- Make sure you remount the covers in the proper sequence and tighten all screws at the end of the assembly procedure.
- Install the energy chain.

Fig. 298: Toothed belt attachment (head end)



(1) Toothed belt mount (head end)

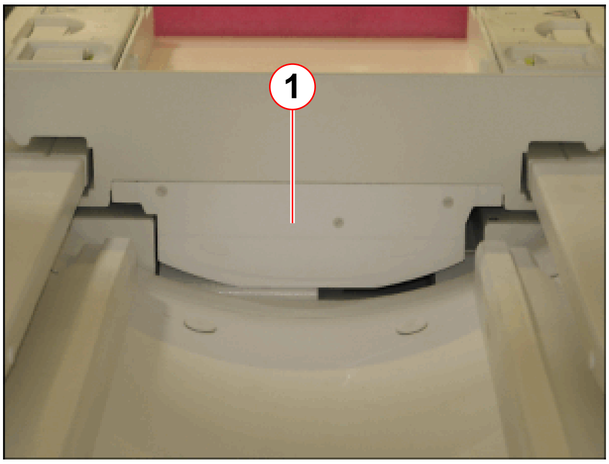
Fig. 299: Fixation of tooth belt - service end



(1) Allen screws of toothed belt mount

- Carefully pull the tabletop plate manually back to the food end side (patient end).
- Remount the mount of toothed belt on the head end from the ptab.

Fig. 300: Cover of toothed belt mount (head end)



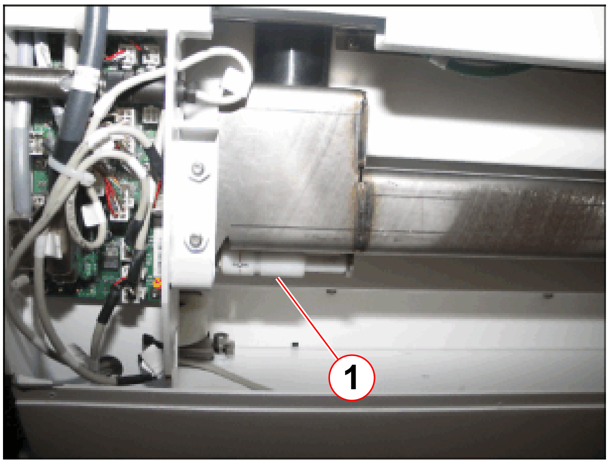
(1) Cover of toothed belt mount

- Install the cover of the toothed belt mount (1) on the head end from the ptab.

Fig. 301: Mounting the toothed belt (foot end)

- Principle of toothed belt tension unit, located on the foot end side of the ptab.

Fig. 302: Service tool for toothed belt tension



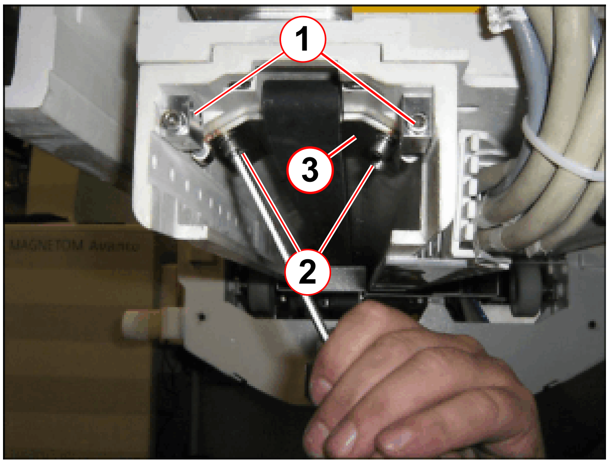
(1) Storage position of service tool for toothed belt tension

- The service aid for adjustment of the toothed belt tension is located underneath the table top.
- Snap out the service aid for adjustment of the toothed belt tension.

3.10.6 Adjustments

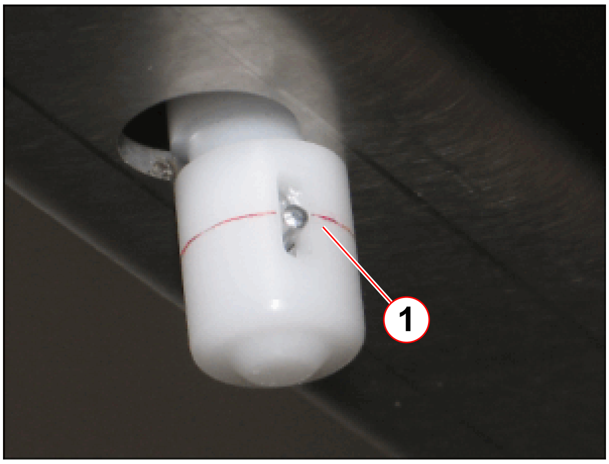
Fig. 303: Service tool, toothed belt tension

Fig. 304: Tension unit for toothed belt



(1) Allen screws for adjusting belt tension
(2) Attachment screws
(3) clamping plate

Fig. 305: Service tool for toothed belt tension in use



(1) Correct tension of toothed belt (pin aligned with mark)

- Connect service aid to measure the tension of the toothed belt.
- Tighten the Allen screws (➔ 1/Fig. 304 Page 176) until the pin is aligned with the mark. (➔ 1/Fig. 305 Page 176)
- When the toothed belt is adjusted to the proper tension, tighten the Allen screws. (➔ 2/Fig. 304 Page 176)
- Remove the service aid and place it on the store position. (➔ 1/Fig. 302 Page 175)

3.10.7 Concluding steps

- Install the end stops if necessary.
- Install the remaining covers.
- Install foot end plate and the cover of the coil plugs.

3.10.8 Final checks

- Please refer to final checks horizontal drive unit. (→ Final checks / Page 163)

3.11 Sound transducer

3.11.1 General

The sound transducer is responsible for patient communication. The sound transducer can be replaced. Consider the cable run during assembling.

CAUTION

The sound transducer consists of magnetic objects!

Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field!**

3.11.2 Tools and time required

3.11.2.1 Time

Approx. 60 min

3.11.2.2 Tools

Non-magnetic tool kit

Philips screwdriver, midsize

3.11.3 Preliminary

- Switch OFF F22 (RF-room) in the ACC.

According to the system/room specifications the movement range of the horizontal movement can be reduced.

With the whole body version of the horizontal drive, the movement range of the horizontal movement may be reduced. If that is the case, remove the end stops.
(→ Fig. 306 Page 179)

Before disassembly the patient table covers on a system with a partial drive unit: The toothed belt must be released to push the table top to the service end of the magnet. Therefore refer to (→ Prerequisite for cover removal / partial drive / Page 104).

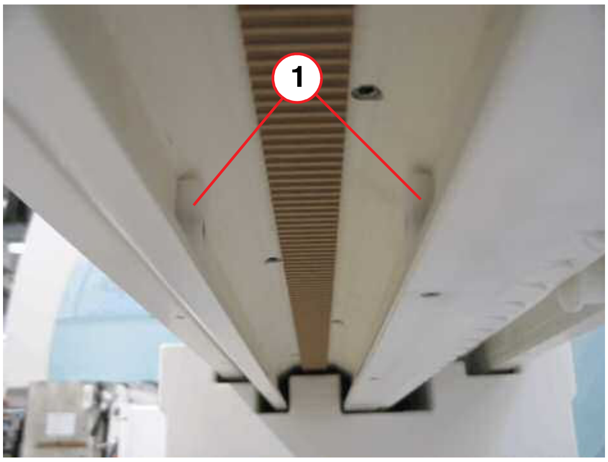
3.11.4 Disassembly

3.11.4.1 Covers

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

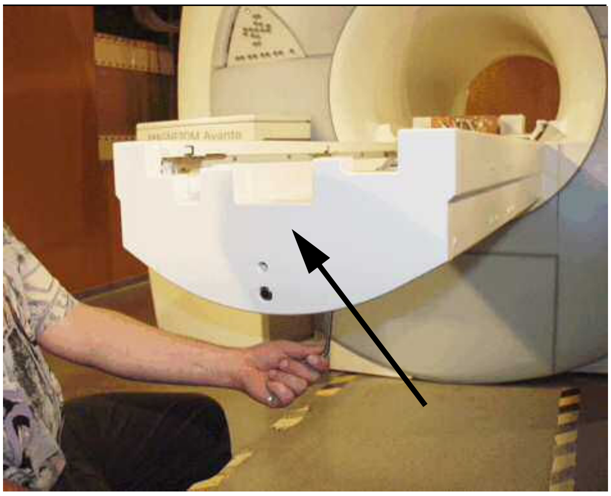
- For disassembling/assembling the patient table covers we recommend a mid-sized Philips screwdriver.
- Push the tabletop plate manually into the magnet
- If necessary, remove the mechanical end stops

Fig. 306: Table carrier underside (whole body, limited movement range)



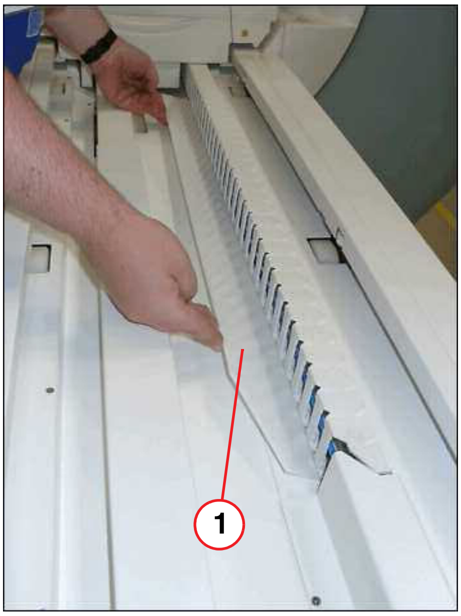
(1) Mechanical end stops

Fig. 307: Foot-end cover



- Remove the foot-end cover of the patient table.

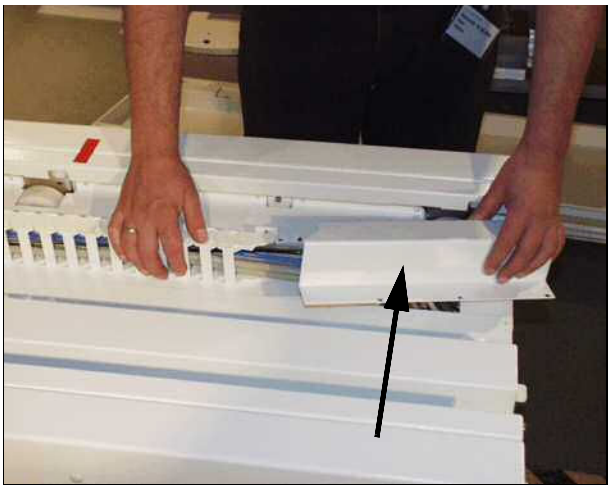
Fig. 308: Plastic inner cover



(1) Plastic inner cover

- Remove inner plastic cover.

Fig. 309: Remove cover of energy chain

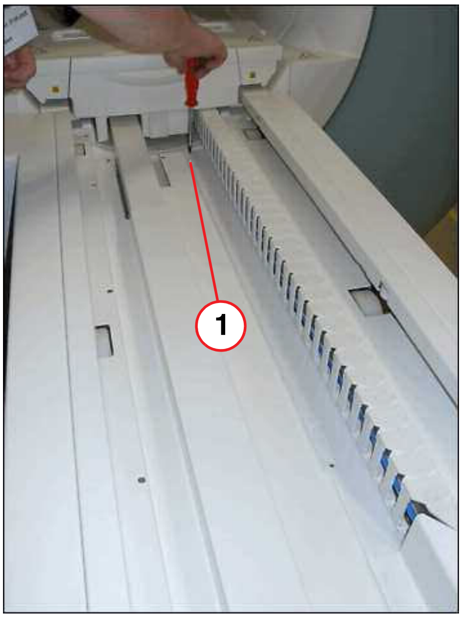


- Remove the upper energy chain cover.

Fig. 310: Remove energy chain

- Remove the upper energy chain of the bracket.

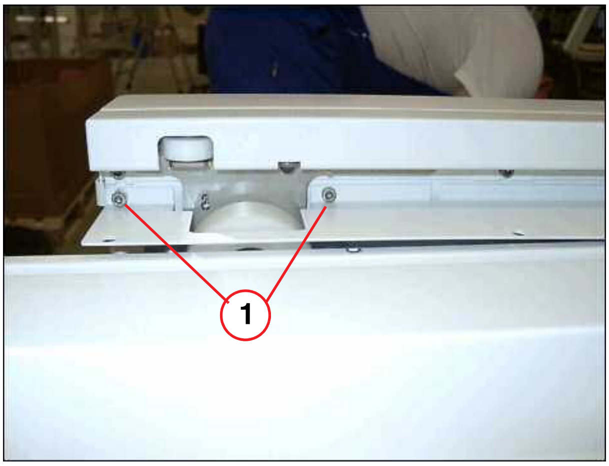
Fig. 311: Inner main cover



(1) Screws to be removed

- Remove screws of main inner cover.

Fig. 312: Allen screws

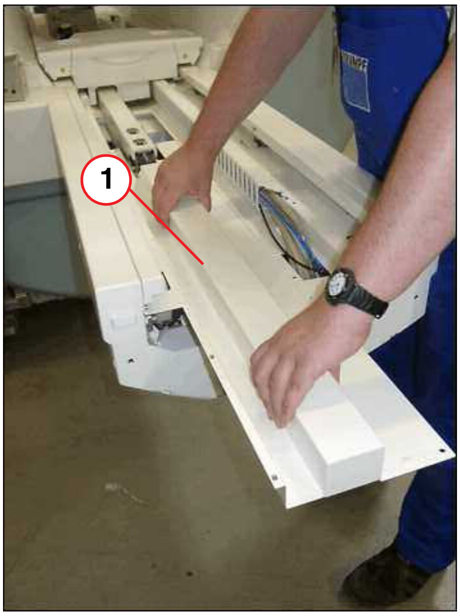


(1) Screws to be removed

- Remove two Allen screws at the left side of the table end (foot) before you remove the main inner cover.

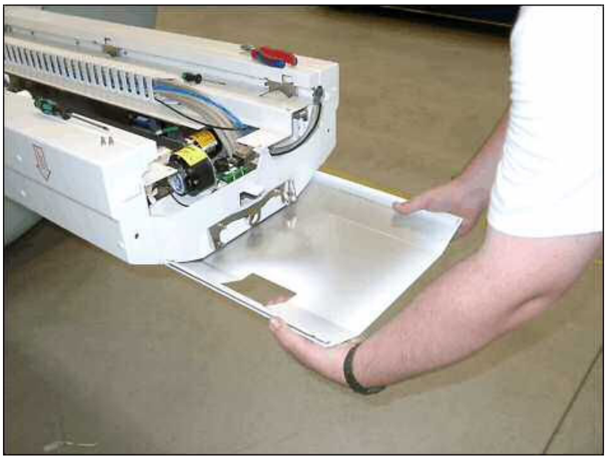
- Remove main inner cover.

Fig. 313: Inner main cover



(1) Inner main cover

Fig. 314: Button cover



- Remove bottom cover.

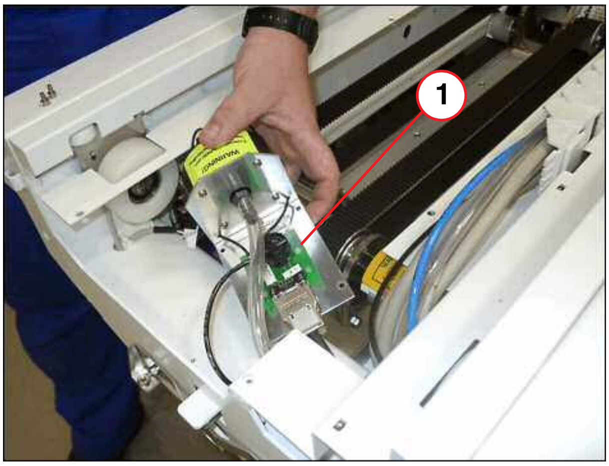
3.11.4.2 Sound transducer

Fig. 315: Cut air hose

- Cut black air hose approx. 5 cm.

- Remove four screws, air hose and sound transducer.

Fig. 316: Sound transducer



(1) Sound transducer to be removed

CAUTION

The sound transducer consists of magnetic objects!

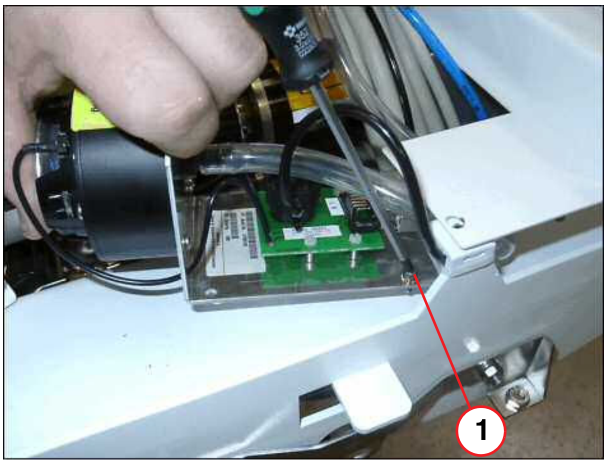
Magnetic objects may cause injury.

» **Maintain a safe distance to the magnetic field!**

3.11.5 Assembly

3.11.5.1 Sound transducer

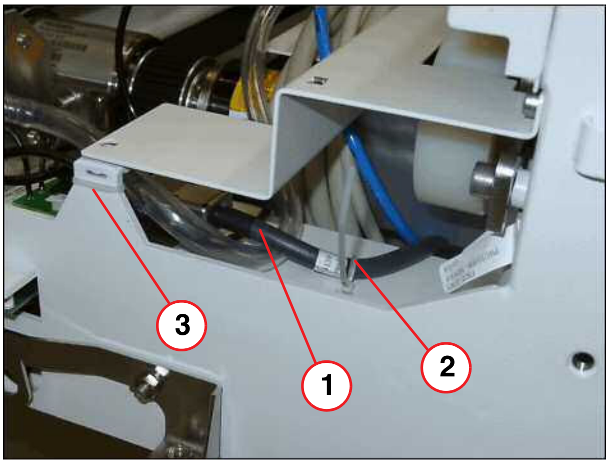
Fig. 317: Sound transducer



(1) Insert first Allen screw

- Insert first one Allen screw as shown.

Fig. 318: Cable run



(1) Run the black cable above the air hose
(2) Secure black cable with tie wrap
(3) Fix small black air hose with tie wrap

- To prevent the cable from being crushed, pay attention and route the cables as shown, e.g. the black cable x 23 above the big air hose. Safety small black air hose and black cable x 23 as shown.

3.11.6 Concluding steps

For assembling the partial drive, please refer to (➔ Assembly / Page 173), not valid for Trio A Tim Systems.

- Reinstall the covers in reverse order, tighten all screws at the end of the assembling procedure.
- If necessary, reinstall the mechanical end stops.
- Switch ON F22 (RF-room).

3.11.7 Adjustments

For adjustment of the toothed belt, please refer to (➔ Adjustments / Page 176), not valid for Trio A Tim Systems.

3.11.8 Final checks

- Check for smooth patient table movement.
- Do a check of the function of the patient communication.

3.12 RF coil plugs

3.12.1 General

The coil plugs at the foot (magnet front) and on the head (magnet end) positions of the patient table can be replaced.

The coil plugs are covered with caps, each of which can be replaced in case of damage.

3.12.2 Tools and time required

3.12.2.1 Time

Approx. 30 min/ 30 min

3.12.2.2 Tools

Non-magnetic tool kit

Service plug

3.12.3 Preliminary

- Switch OFF F24 (RFIS)
- Switch OFF F22 (RF-room) in the ACC

3.12.4 Disassembly

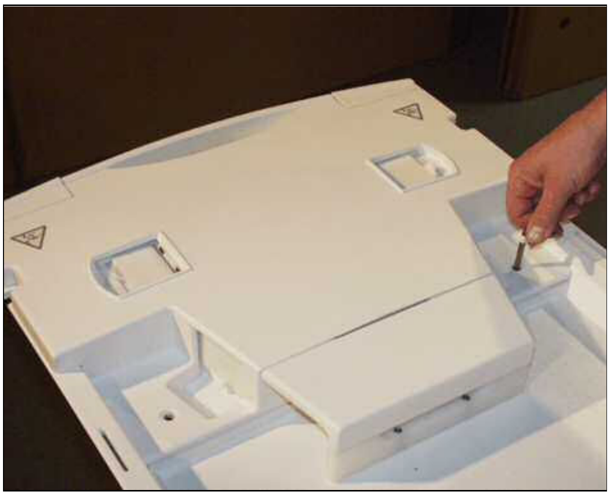
3.12.4.1 Coil plugs at the foot end side (magnet front)

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

- If the movement range of the horizontal movement is reduced it may be necessary to remove the end stops. (➔ Fig. 306 Page 179)

Basic table

Fig. 319: Coil plug cover (foot end)



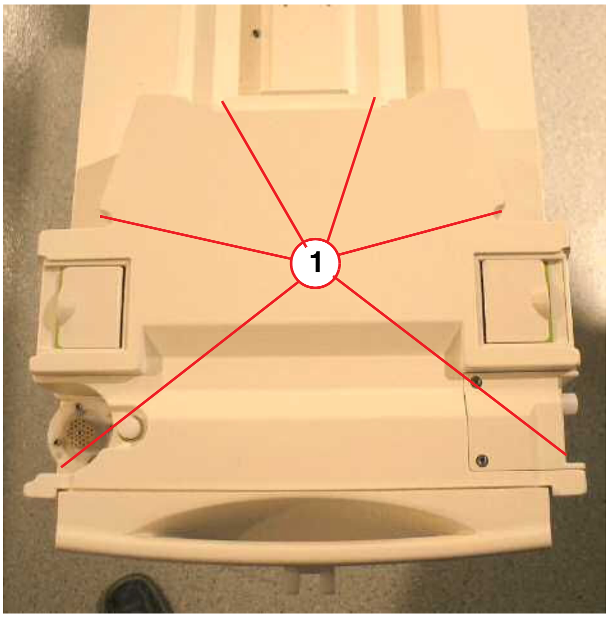
- Remove the Allen screws on top of the cover from the coil plugs.

Fig. 320: Remove Allen screws

- Remove the lower Allen screws of the cover from the coil plugs.

Removable table

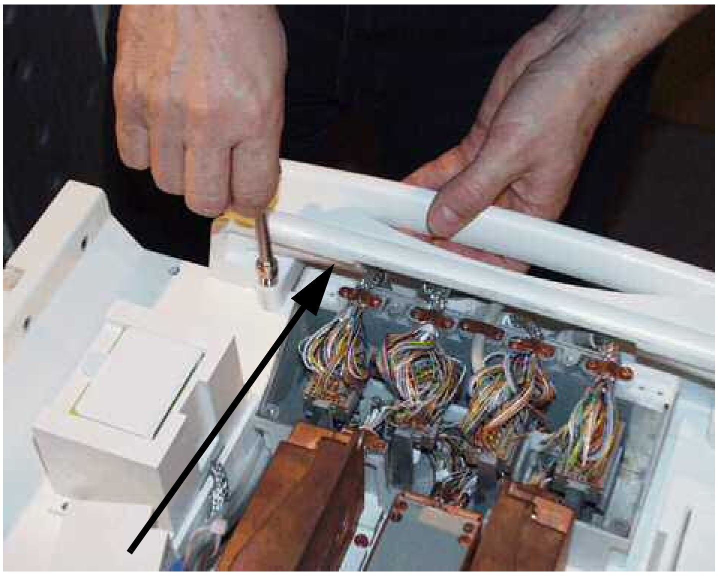
Fig. 321: RF-Cover RTT



(1) Screws to be removed

- Remove the carrier plate of the RTT.
- Remove six screws on top of the cover from the coil plugs.
- Disconnect cables.

Fig. 322: Foot-end cover



- Remove the two M8 Allen screws, if necessary disconnect cables, and remove foot end plate.

3.12.4.2 RF coil plugs

Observe the information in the note when disassembly the coil plugs.

Do not pull on cord, use a flat screwdriver on both sides to remove connector.

Fig. 323: RF cover from interconnect box

- Remove the RF cover from the interconnect box.

- Loosen the screws of defective RF coil plugs.
- Coil plug 7-8 belongs to X101.
- Coil plug 9-10 belongs to X104.
- Release the corresponding strain relief from the coil plug (X101 or X104).

- Disconnect RF cables, use a flat screwdriver to open the RF connectors.

Fig. 324: Cable connections

- Install the corresponding covers.

3.12.5 Assembly

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

- Replace the defective RF coil plugs including the RF wave trap.
- Install the new RF coil plug assembly.
- Reconnect RF cables.

> Be careful to avoid pinching any cables when assembling the cover of the coil plug connectors.

- Make sure the RF cable connections are installed properly.
- Reinstall the strain relief from the replaced coil plug.

3.12.6 Concluding steps

- Reinstall the RF cover of the interconnect box.

- Install the corresponding covers.

3.12.7 Final checks

- Switch ON F24 (RFIS).
- Start QA coil check and do a check of the replaced coil plug.

3.12.8 Disassembly

3.12.8.1 Coil plugs at the head end (back of the magnet) / Coil plug caps

Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Do not pull on cord, use a flat screwdriver on both sides to remove the connector

- Manually push the tabletop plate into the magnet until the head end of the tabletop plate is accessible.

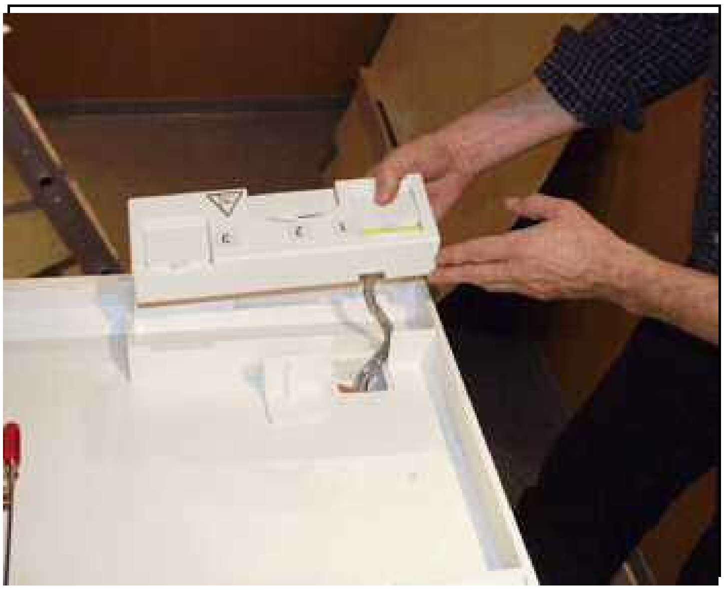
- Remove the screws of the affected coil plug.

Fig. 325: Screws of coil plugs (magnet end, back view)

- Remove the cable strain relief of the affected coil plug.

Fig. 326: Strain relief of coil plug cable (magnet end, back view)

Fig. 327: Coil plug assembly (head end)



- Remove the affected coil plug assembly.

Fig. 328: Plug fastener of coil plugs (magnet end)

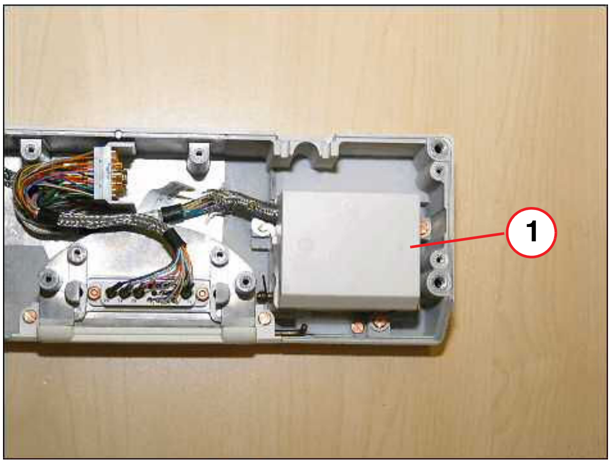
- Remove the RF cover of the affected coil plug assembly by loosening the screws.

3.12.8.2 Replacing the coil plug caps

The coil plugs are covered with caps, each of which can be replaced in case of damage.

- Remove inner cover of defective coil plug.

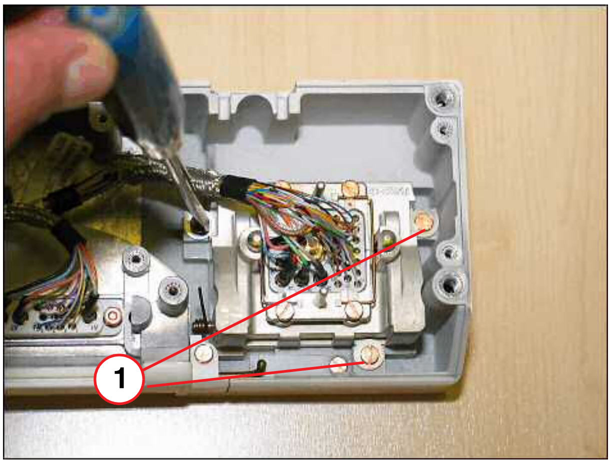
Fig. 329: Remove cover



(1) Cover of coil plug assembly

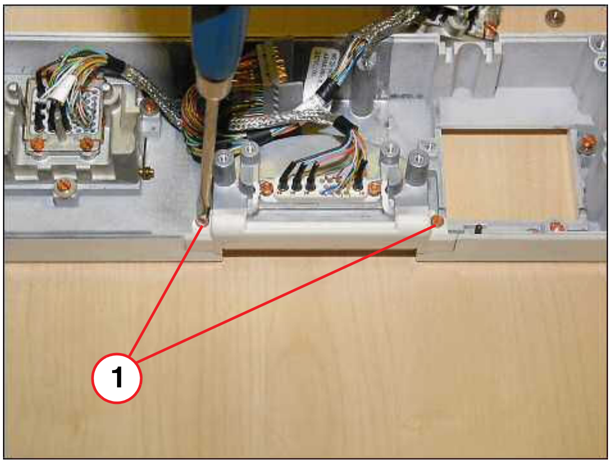
- Remove the screws of the defective coil plug cap.

Fig. 330: Coil plug housing



(1) Screws to be removed

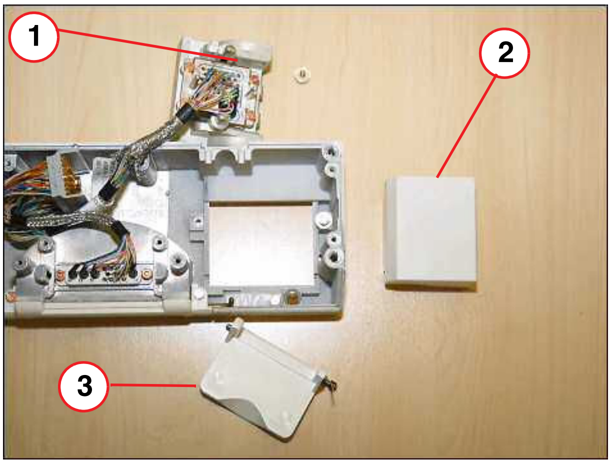
Fig. 331: Coil plug



(1) Screws to be removed

- Replace the defective coil plug cap.

Fig. 332: Coil plug



(1) Coil plug housing
(2) Cover of coil plug housing
(3) Coil plug cap

- Remove the coil plug fastener.
- Disconnect RF cables; use a flat screwdriver to open the RF connectors.

Fig. 333: Cable connections (magnet end)

3.12.9 Assembly

- Replace the defective coil plug cap.
- Replace the defective coil plug assembly.
- Replace protective film (part of the coil plug assembly delivery) if necessary. For additional information, refer to UI 26/04.
- Connect the RF cables and install the RF connector.
- Install the coil plug fastener.

> **CAUTION**
>
> **Bend the cable as shown in the picture below.**
>
> **Any contact between the shield of the cable and the housing of the RF trap must be avoided.**
>
> **» Otherwise an overheating could happen in the components.**

Fig. 334: RF traps

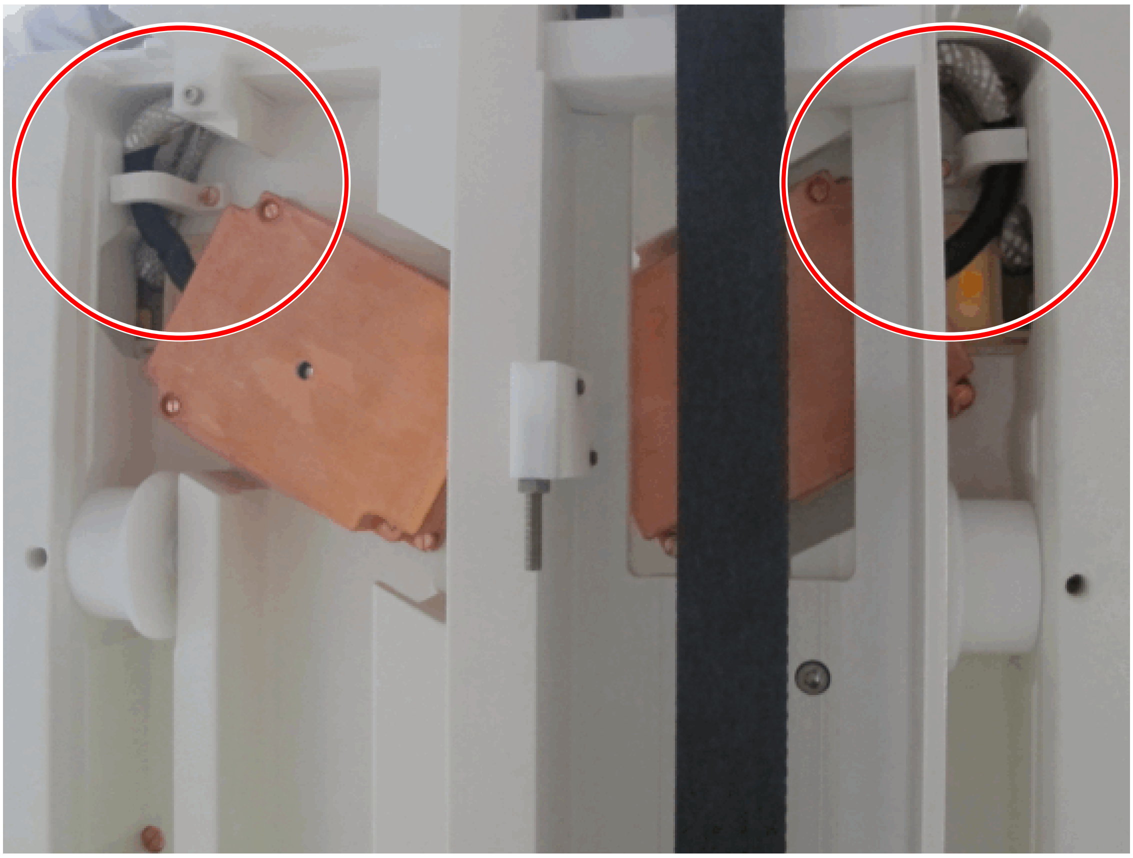


Fig. 335: Correct cable routing, left RF trap

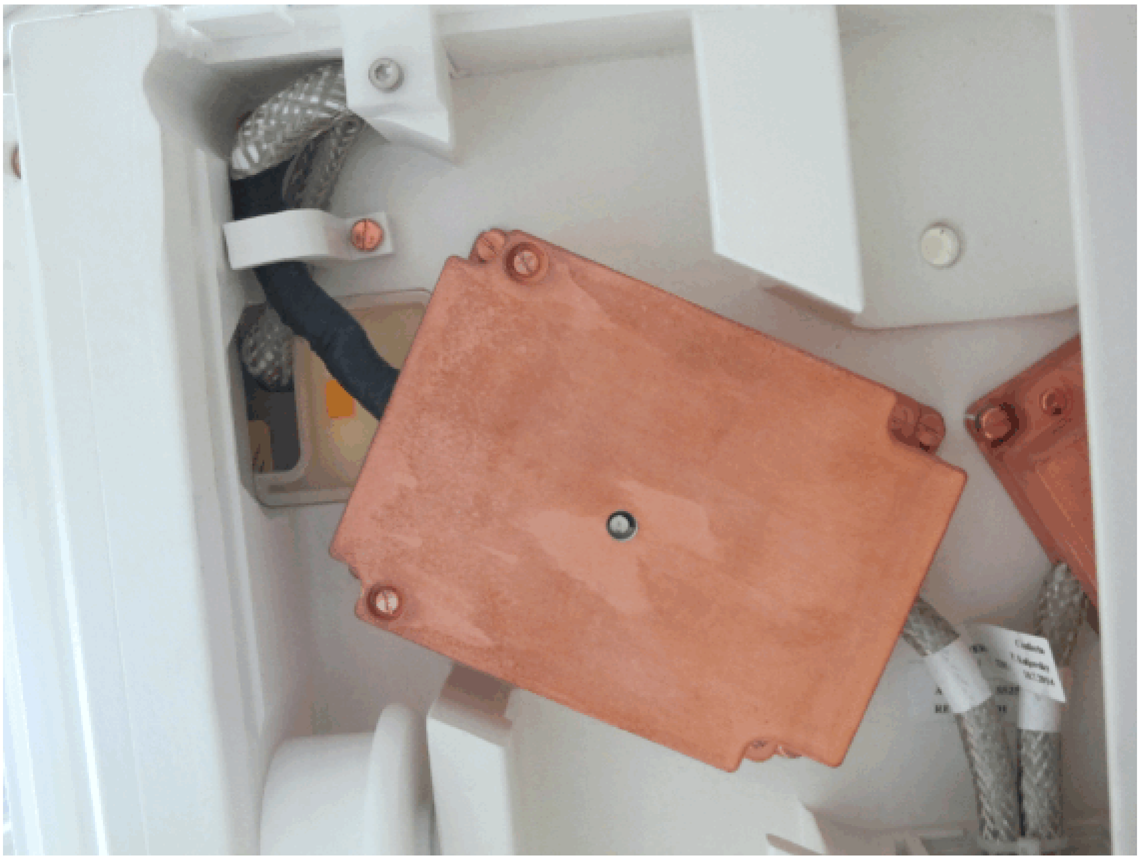


Fig. 336: Correct cable routing, right RF trap

3.12.10 Concluding steps

- Reinstall the RF cover.

> **i** Be careful to avoid pinching any cables when assembling the cover of the coil plug connectors.

- Reassemble the coil plug assembly.
- Install the coil plug assembly to the tabletop plate.

3.12.11 Final checks

- Switch ON F24, F22.
- Start QA coil check .

3.13 Control switch S30

3.13.1 General

The S30 maintains the vertical collision identification. If the S30 detects uncoupling of the spindle nut, the switching contact is open. In case the S30 has to be replaced, replacement is described in this part.

3.13.2 Tools and time required

3.13.2.1 Time

Approx. 60 min

3.13.2.2 Tools

Non-magnetic tool kit

3.13.3 Preliminary

- Switch OFF F22 in the ACC.

3.13.3.1 Location and principle of S30

- The switch location of the S30 is shown in the following figures.

Fig. 337: Princip of upper spindle part

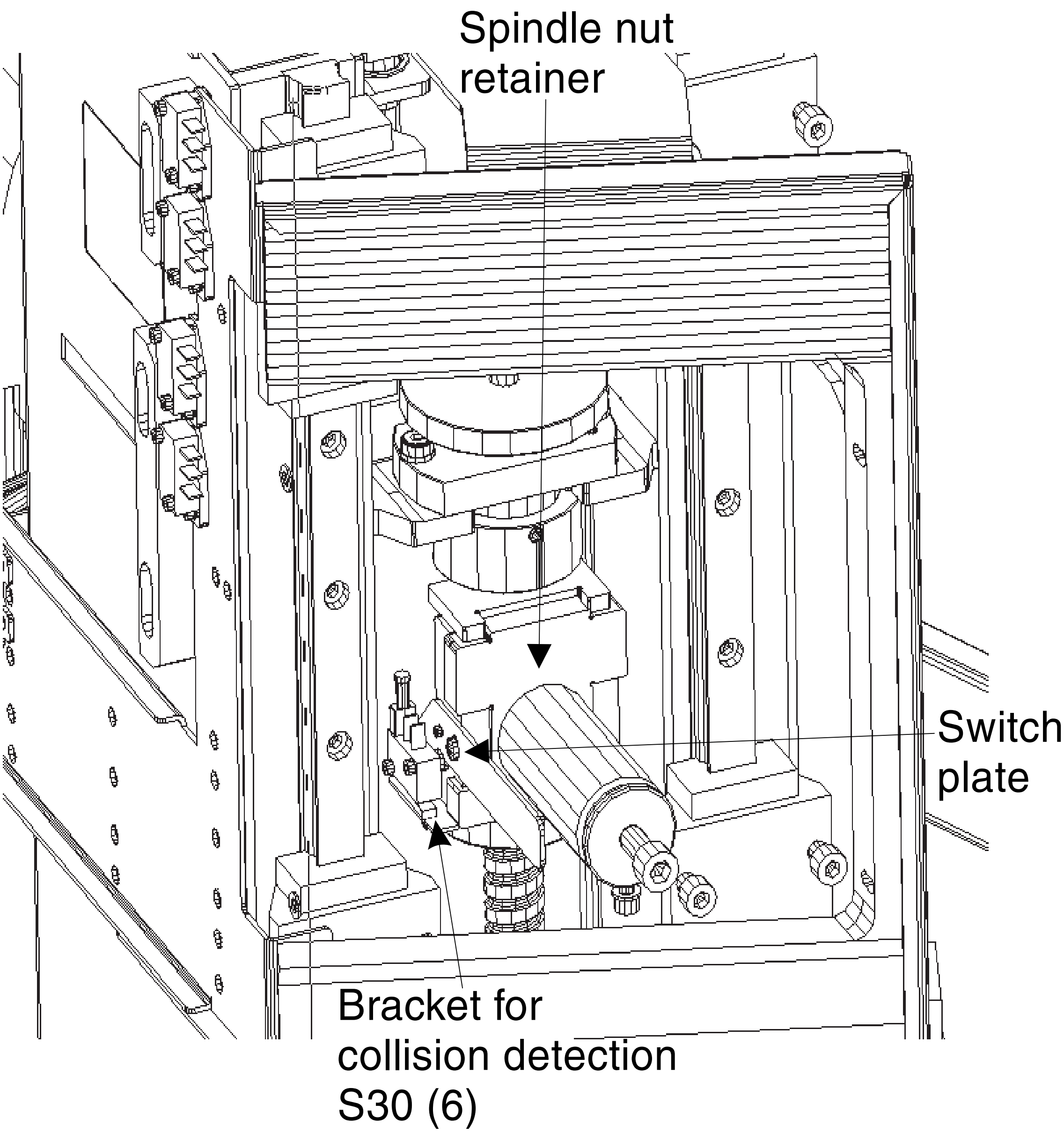
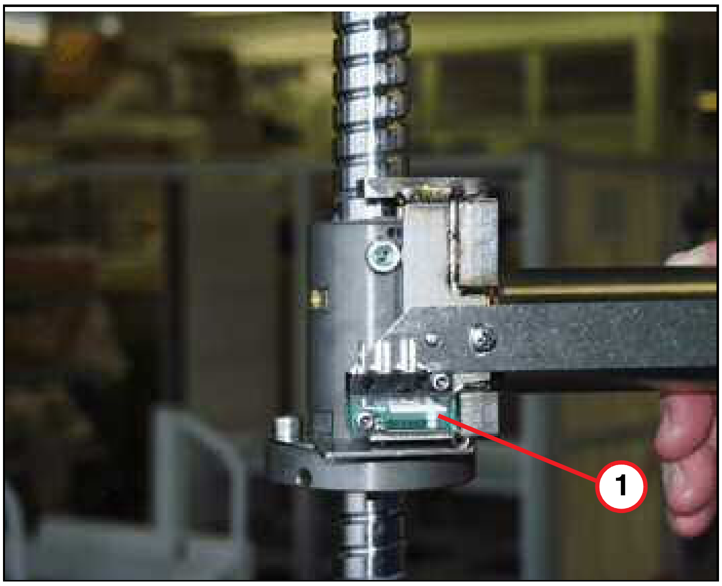


Fig. 338: Spindle with S30



(1) Principle of S30, normally closed

- The principle of S30, normal state= switch is actuated, switching contact is closed, rotate mounting bracket for switch adjustment.

Fig. 339: S30 assembly

- S30 assembly.

3.13.4 Disassembly

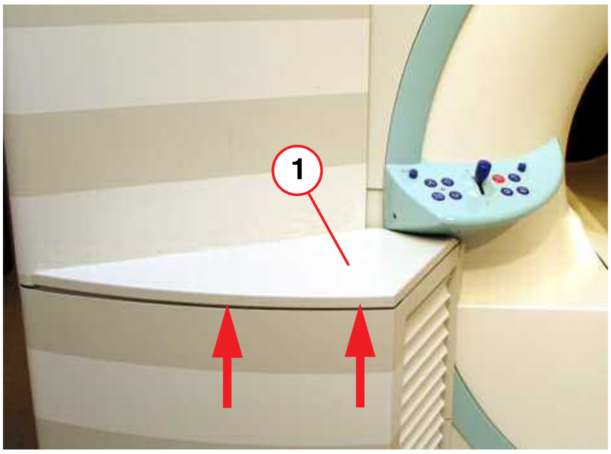
3.13.4.1 Covers

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 340: Service access cover removed

- Remove the service access cover.

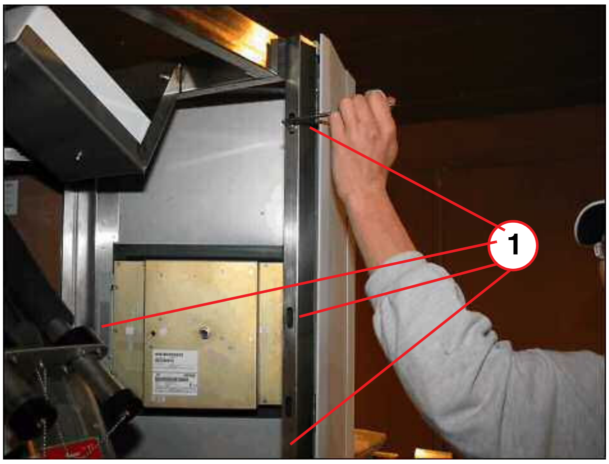
Fig. 341: Lifting column, top cover



(1) Top cover

- Remove top cover of lifting column.

Fig. 342: Top front cover above lifting column



(1) Screws to be removed

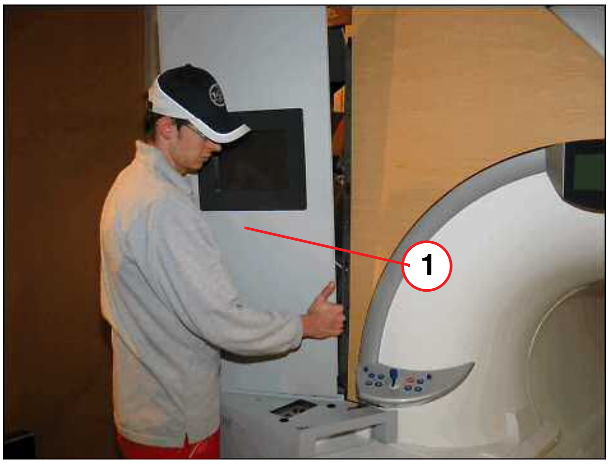
- Remove all screws of top front cover.

Fig. 343: Top front cover above lifting column

(1) Screws to be removed

- Remove all screws inside the top front cover and disconnect cables, if necessary.

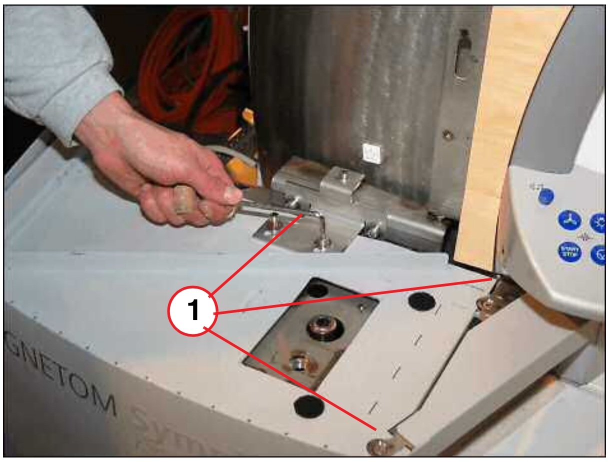
Fig. 344: Top front cover above lifting column



(1) Remove cover

- Remove top front cover.

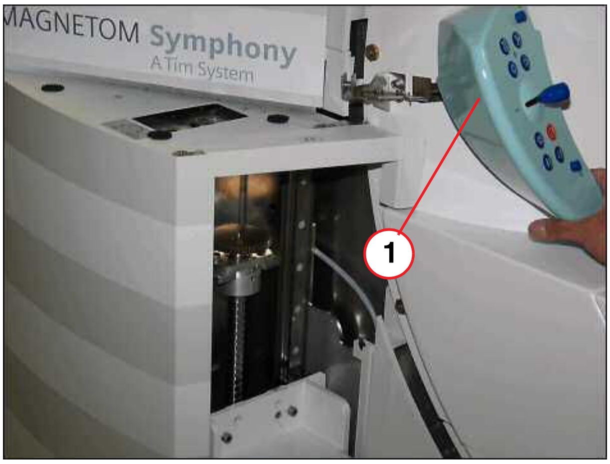
Fig. 345: Lifting column cover



(1) Screws to be removed

- Remove screws from the lifting column and bracket.

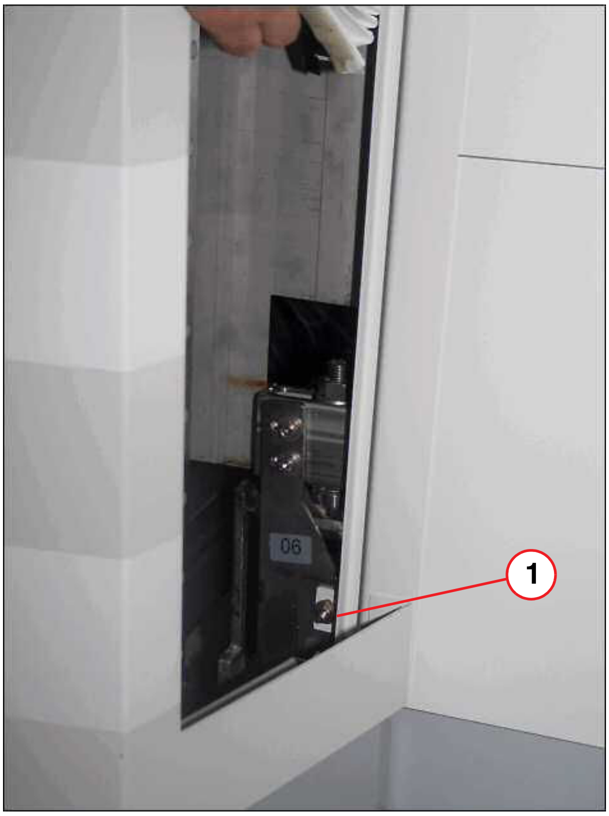
Fig. 346: Control panel



(1) Control panel removed

- Separate the bellows from the adhesive tape and remove upper bellows.
- Loose screw of operating panel, disconnect cable if necessary and remove it.
- Remove the screw behind the control panel to disassembly the lifting column cover.

Fig. 347: Lower part of lifting column



(1) Bellows removed, screw to be removed

- Separate the bellows from the adhesive tape and remove lower bellows.
- Remove screw on the lower part of the lifting column.

Fig. 348: Lifting column cover

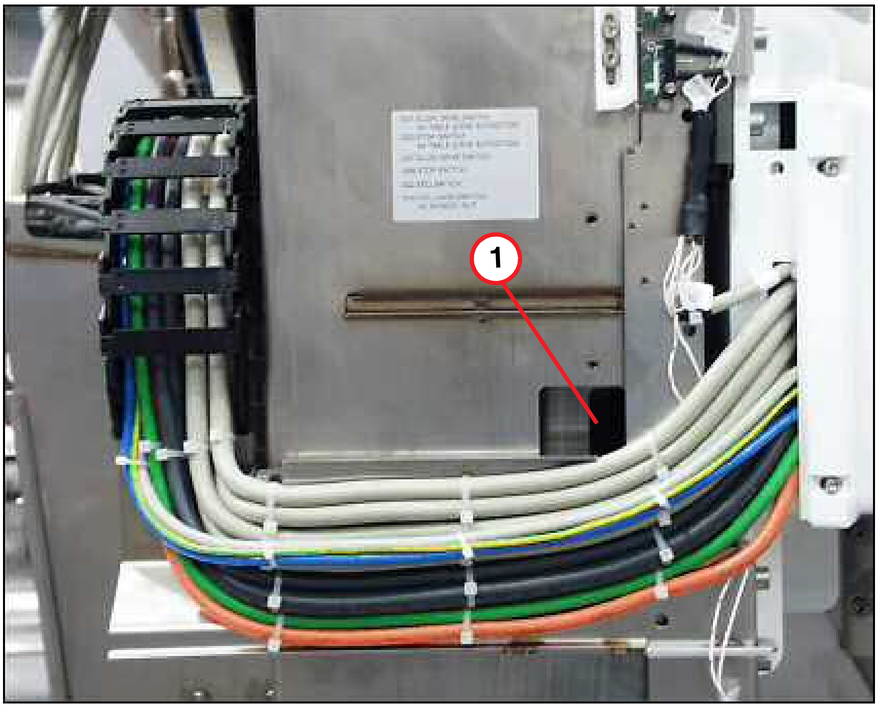
- Remove lifting column cover.

3.13.4.2 Switch assembly S30

- If necessary cut off tie wraps from the cables of the lifting column.

- Beginning with patient table serial number 1178, the lifting column is equipped with a service cut-out to access the S30 switch.

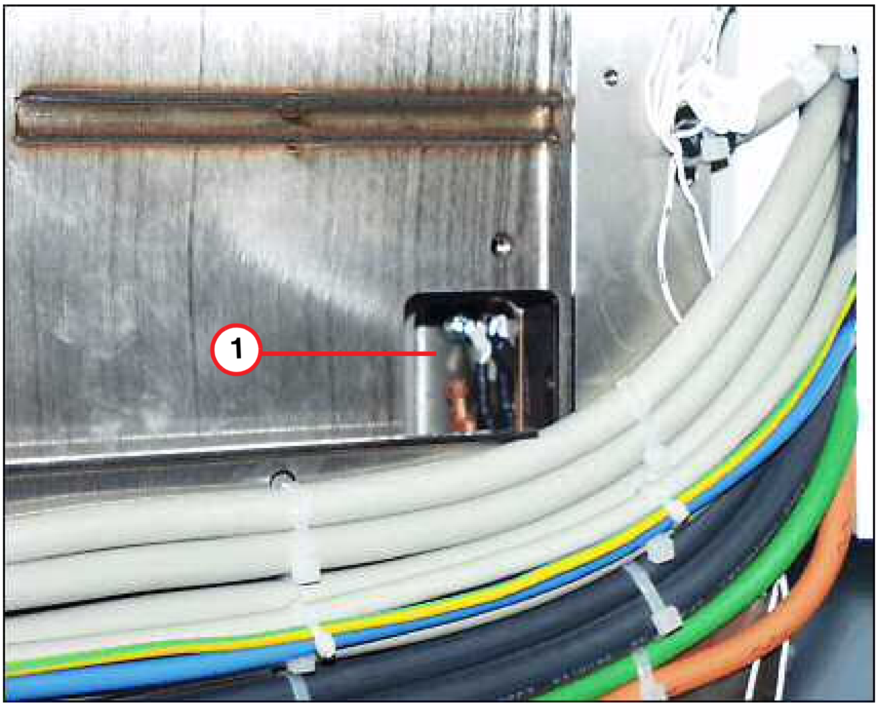
Fig. 349: Cut-out on lifting column



(1) Access to S30

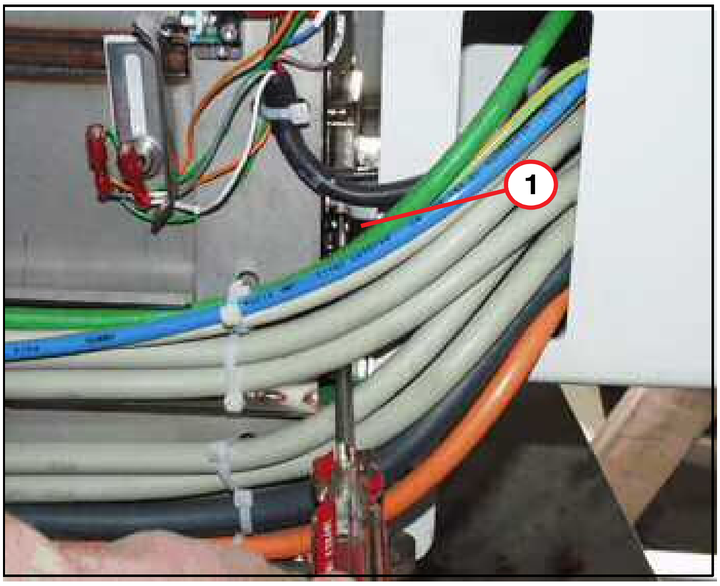
- Use a Phillips screwdriver to remove the S30 assembly.

Fig. 350: Lifting column, location of S30



(1) S30

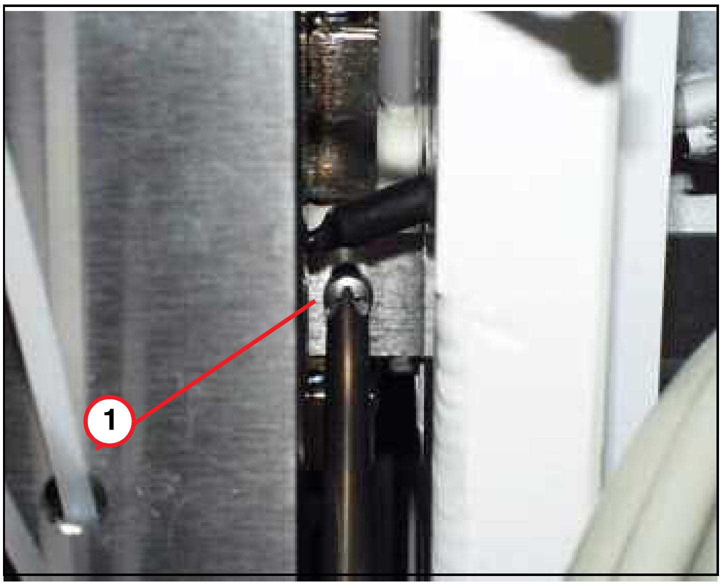
Fig. 351: S30 attachment



(1) Feed the Phillips screwdriver through the cables

- For older systems use a Philips screwdriver to remove the S30 assembly.

Fig. 352: S30 mounting bracket



(1) Tighten screw after adjustment

Fig. 353: S30 switch assembly

- Catch the S30 switch assembly from underneath the support beam and pull it out.

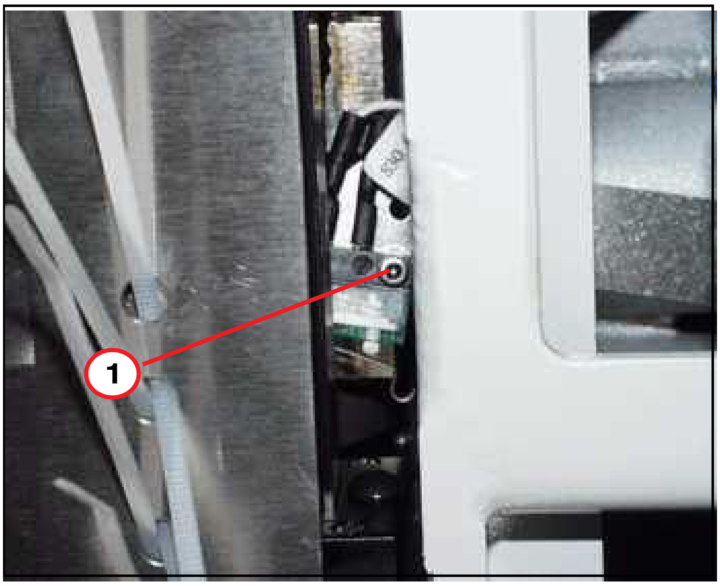
3.13.5 Assembly

> Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 354: S30 switch assembly

- Disconnect cables and replace the defective S30 switch assembly.
- Feed in the new S30 switch assembly.

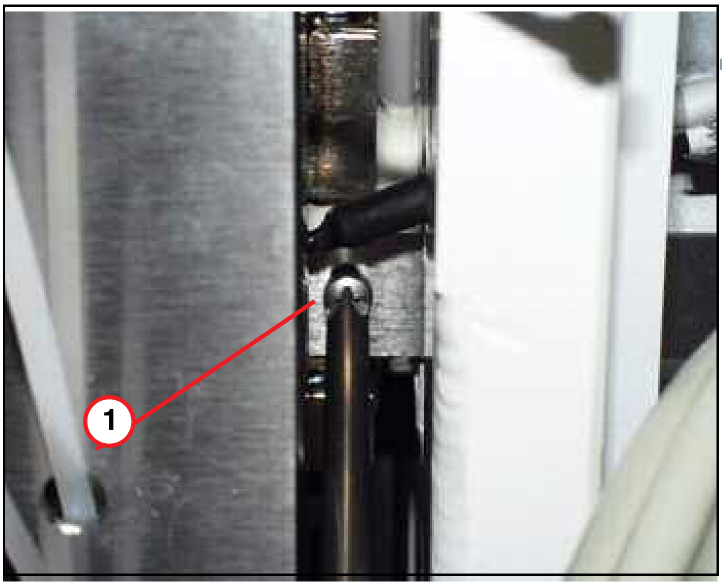
Fig. 355: Location of S30



(1) Mounting screw and rotation point of S30

- Slide the S30 switch assembly through the support beam to the spindle nut for fastening.

Fig. 356: S30 mounting bracket



(1) Tighten screw after adjustment

- Safety the S30 switch assembly with the Philips screw and adjust the S30 switch.
- If the switch assembly is properly adjusted, S30 is always closed.

3.13.6 Concluding steps

- If necessary reconnect cables and install cables to the lifting column.
- Reinstall the covers.

3.13.7 Final checks

- Switch ON F24 (RFIS).

- Start patient table movement and check for smooth operation.
- Start patient table service menu for switch test and check S30 HI status.
- Start to move the table down until it reaches the end stop.
- Insert hand crank and lower the table some more until the switch status of the S30 changes from HI to LOW in the Service mode.

3.14 Cable replacement

3.14.1 General

The replacement of a defective cable, running in the energy chain through the patient table, is described in this part.

3.14.2 Tools and time required

3.14.2.1 Time

Approx. 180 min

3.14.2.2 Tools

Non-magnetic tool kit

Tie wraps

3.14.3 Preliminary

- Switch OFF the corresponding circuit breaker.
- Switch OFF F22 (RF-room) in the ACC.

> i According to the system/room specifications the movement range of the horizontal movement can be reduced.

With the whole body version of the horizontal drive, the movement range of the horizontal movement may be reduced. If that is the case, remove the end stops.
(→ Fig. 306 Page 179)

Before disassembly the patient table covers on a system with a partial drive unit: The toothed belt must be released to push the table top to the service end of the magnet. Therefore refer to (→ Prerequisite for cover removal / partial drive / Page 104).

3.14.4 Disassembly

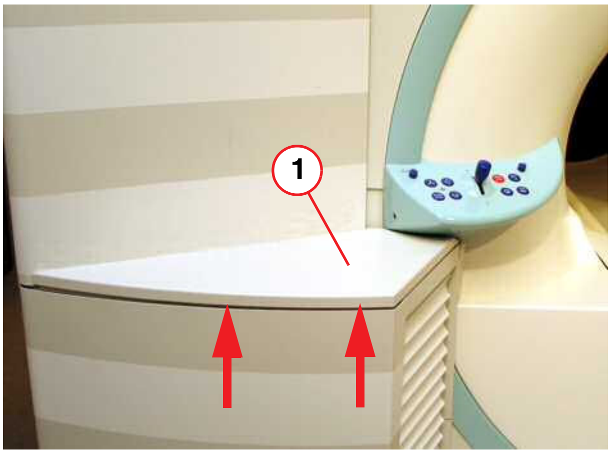
3.14.4.1 Covers

> i Be careful to avoid damaging cables while removing the covers of the patient table. Use caution when moving the patient table.

Fig. 357: *Service access cover removed*

- Remove the service access cover.

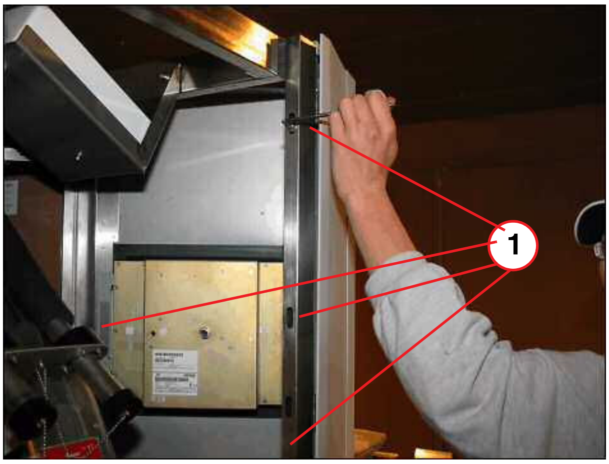
Fig. 358: *Lifting column, top cover*



(1) Top cover

- Remove top cover of lifting column.

Fig. 359: Top front cover above lifting column



(1) Screws to be removed

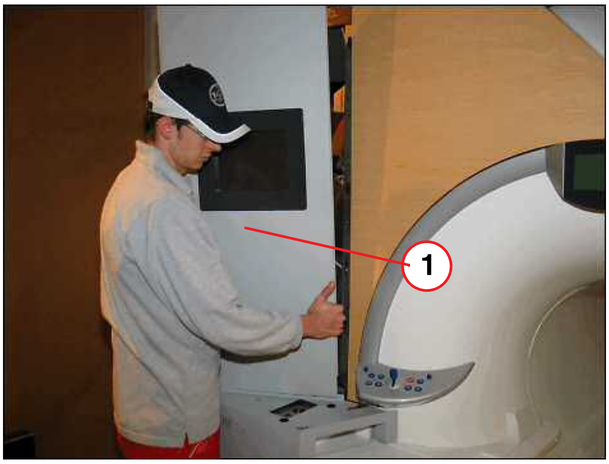
- Remove all screws of top front cover.

Fig. 360: Top front cover above lifting column

(1) Screws to be removed

- Remove all screws inside the top front cover and disconnect cables, if necessary.

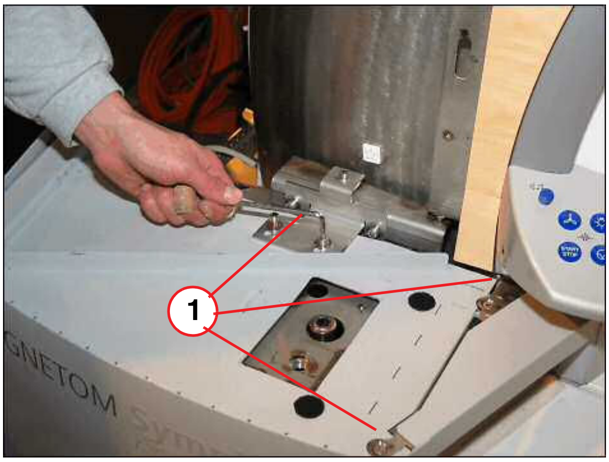
Fig. 361: Top front cover above lifting column



(1) Remove cover

- Remove top front cover.

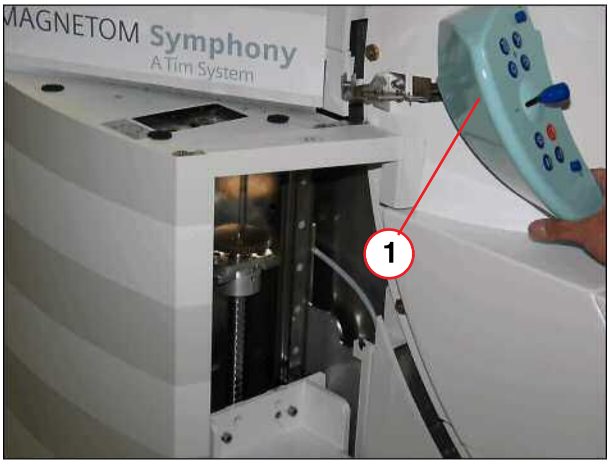
Fig. 362: Lifting column cover



(1) Screws to be removed

- Remove screws from the lifting column and bracket.

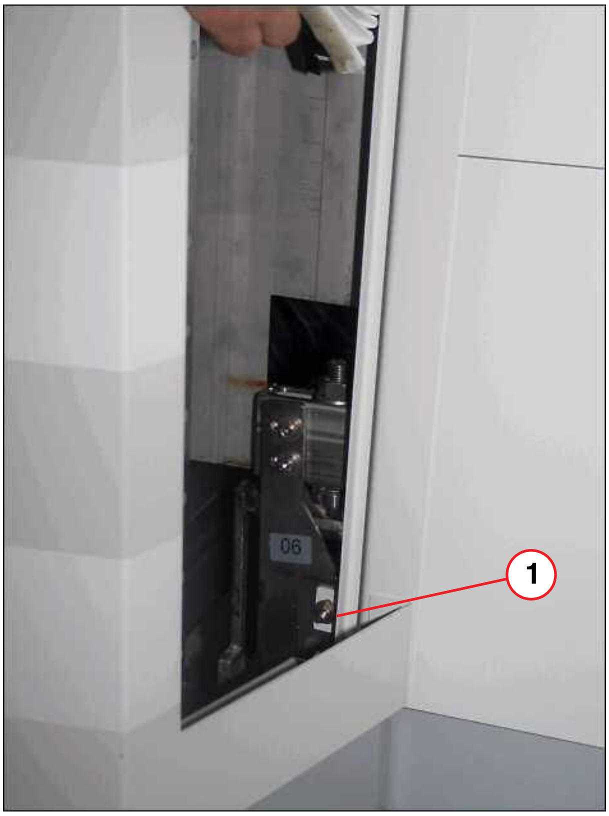
Fig. 363: Control panel



(1) Control panel removed

- Separate the bellows from the adhesive tape and remove upper bellows.
- Loose screw of operating panel, discon-nect cable if necessary and remove it.
- Remove the screw behind the control panel to disassembly the lifting column cove.r

Fig. 364: Lower part of lifting column



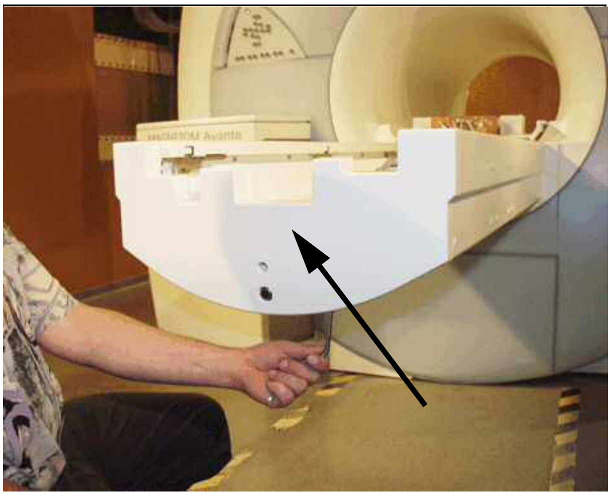
(1) Bellows removed, screw to be removed

- Separate the bellows from the adhesive tape and remove lower bellows.
- Remove screw on the lower part of the lifting column.

Fig. 365: Lifting column cover

- Remove lifting column cover.

Fig. 366: Foot-end cover

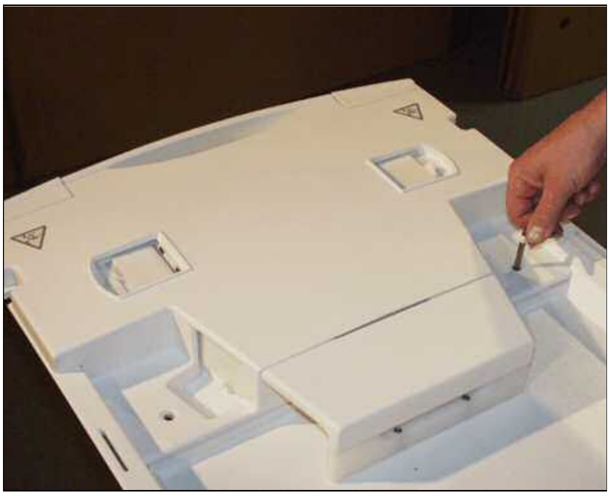


- Remove the foot-end cover of the patient table.

Fig. 367: Remove Allen screws

- Remove the lower Allen screws of the cover from the coil plugs.

Fig. 368: Coil plug cover (foot end)



- Remove the Allen screws on top of the cover from the coil plugs.

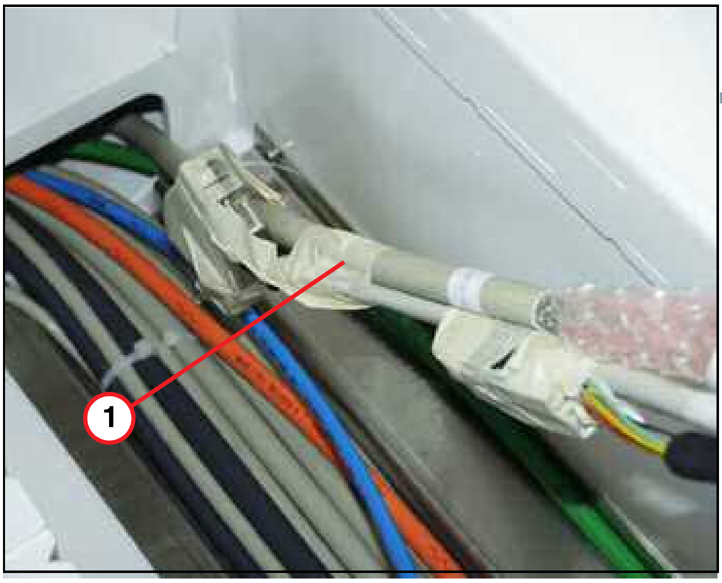
3.14.5 Assembly

For assembling the partial drive, please refer to (→ Assembly / Page 173), not valid for Trio A Tim Systems.

The replaced cable must be the same length as the old one. Therefore, measure it before replacing it. Measure from end to end; i.e., from the end of the energy chain to the end of the affected cable at the side of the table.

To avoid damage on the cables in the moving parts (energy chain), the replaced cable should run parallel with the other cables.

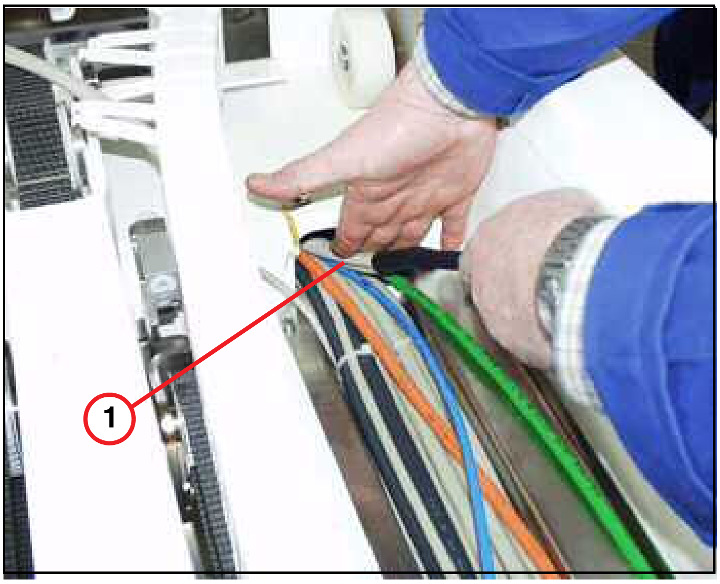
Fig. 369: Cable replacement



(1) Tape the new cable to the defective one

- Tape the new cable to the defective one at table side. If necessary cut off connector of defective cable.
- From the magnet side, pull the new cable through the tabletop plate support beam.

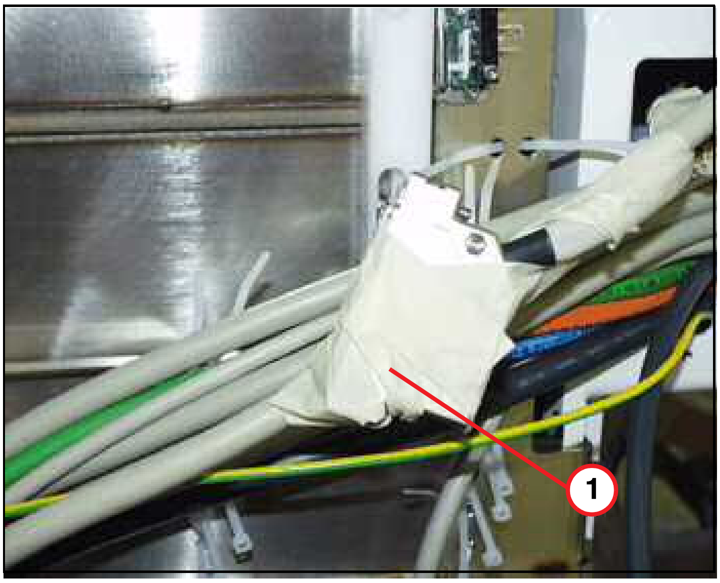
Fig. 370: Cable replacement



(1) Push the new cable

- At the same time as you pull the cable from the magnet side: Push the new cable from the table side through the tabletop plate support beam.

Fig. 371: Replace defective cable



(1) Pull the new cable by using the defective one

- Feed the cable through the support beam. Pull the cable out until it is of the same length as the old one. Measured from end to end, that is, from the end of the energy chain to the end of the affected cable at table side.

Fig. 372: Energy chain for lifting column

- The replaced cable has to run parallel without crossing any other cable in the energy chain.
- Run the cable through the energy chain on the lifting column. Please ensure that the cables in the energy chain run in parallel.

- Align the new cable and install it with tie wraps.
- Connect the cable to the affected component.
- Close the chain links of the energy chains.

3.14.6 Adjustments

For adjustment of the toothed belt, please refer to (→ Adjustments / Page 176).

3.14.7 Concluding steps

- Reinstall mechanical components.
- Reinstall the covers of the patient table.
- Install the corresponding covers.

3.14.8 Final checks

- Switch ON F24 (RFIS).
- Switch ON the corresponding circuit breaker.
- Do a check of the proper table movement, without any grinding or noise.
- Do a check of the proper function of the replaced cable.

3.15 Control panel

3.15.1 General

This section describes the replacement of the control panels.

3.15.2 Tools and time required

3.15.2.1 Time

Approx. 20 min.

3.15.2.2 Tools

Non-magnetic tool kit

3.15.3 Disassembly

3.15.3.1 Control panel

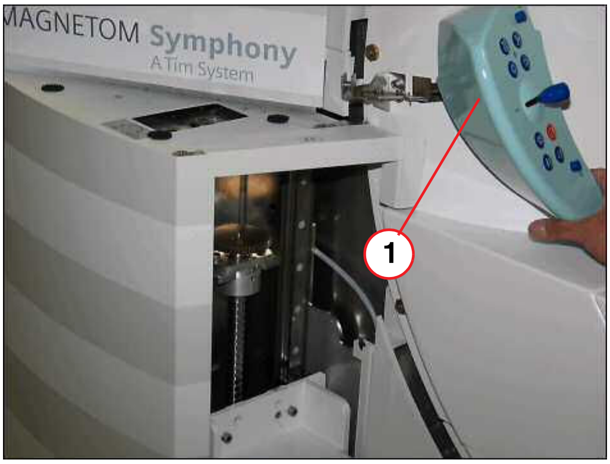
There are two control panels on the left and right front side of the magnet. As an option, a third control panel can be placed in back of the magnet.

This section describes how to disassembly the control panel on the left front side as well as the one in back.

Fig. 373: Control panel

- Remove screw on the other side of the control panel, disconnect the cables and remove the control panel.

Fig. 374: Control panel



(1) Control panel removed

3.15.4 Preliminary

- Switch OFF F22 (RF-room).

3.15.5 Assembly

- Replace control panel.
- Reconnect the cables..
- Remount the control panel.

3.15.6 Concluding steps

- n.a.

3.15.7 Adjustments

- n.a.

3.15.8 Final checks

- Switch ON F22 (RF-room).
- Test function of control panel.

3.16 Light localizer/Display

3.16.1 General

This section describes the replacement of the components light localizer and the display. The light localizer is located in the display unit.

3.16.2 Tools and time required

3.16.2.1 Time

Approx. 25 min./20 min.

3.16.2.2 Tools

Non-magnetic tool kit.

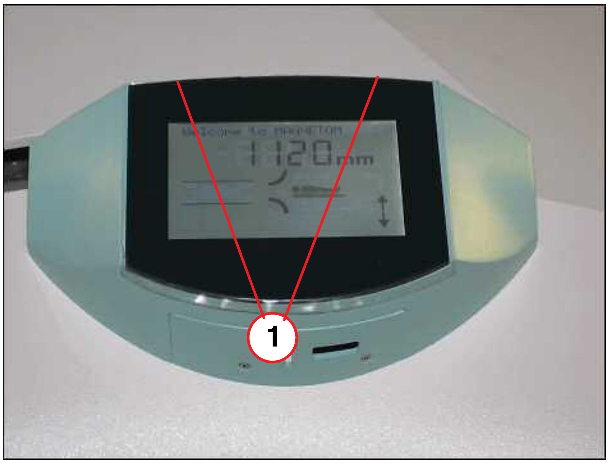
3.16.3 Preliminary

- Switch OFF F22 (RF-room).

3.16.4 Disassembly

3.16.4.1 Display/ light localizer

Fig. 375: Display



(1) Screws to be removed

- Remove screws on the top of the display.

Fig. 376: Display unit with light localizer

- Disconnect cables from the defective part and remove it if necessary.

3.16.5 Assembly

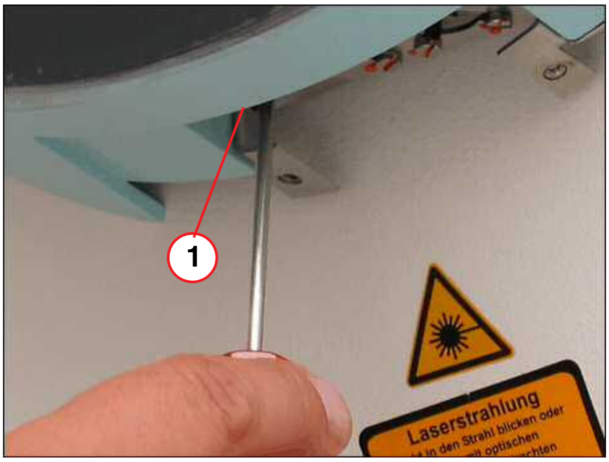
- Replace defective part.
- Reassemble the display unit.
- Reconnect the cables.

3.16.6 Concluding steps

- Remount display unit to the front cover.

3.16.7 Adjustments

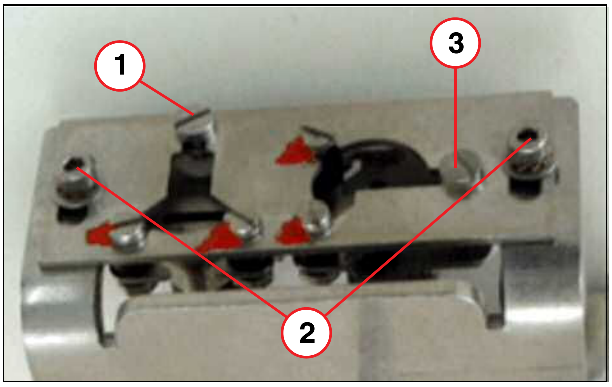
Fig. 377: Light localizer



(1) Adjustment of light localizer

- Star with the adjustment procedure described below.

Fig. 378: Light localizer



(1) Setting screw A
(2) Attachment B
(3) Setting screw C

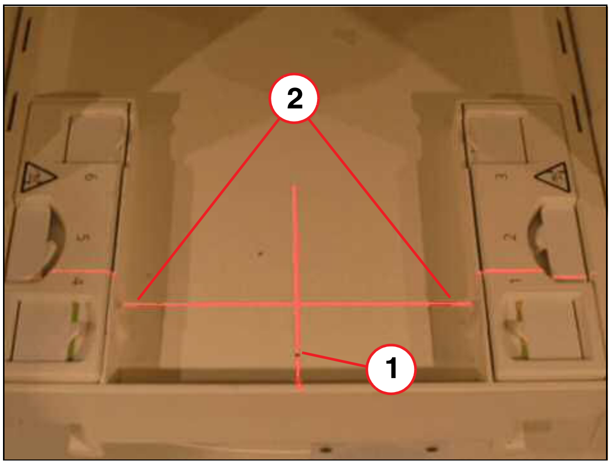
The principle for adjusting the light localizer is the same for both the Avanto and the Espree.

Adjustment screw A (1), for laser adjustment A horizontal.

Mounting B (2), for laser adjustment B horizontal.

Adjustment screw C (3), for laser adjustment B vertical.

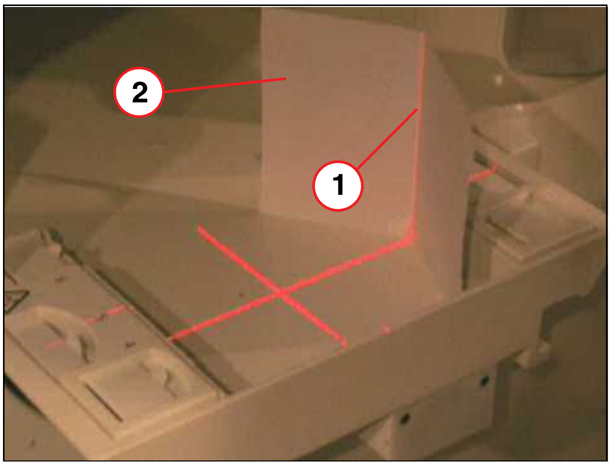
Fig. 379: Laser 1



(1) Laser A horizontal
(2) Laser B horizontal

- Release the two screws and adjust laser B (2) horizontally to the tabletop plate, the left and right distance are equal.
- Use screw A and adjust laser A (1) horizontally to the reference hole in the tabletop plate.

Fig. 380: Laser and auxiliary tools



(1) Laser B vertical
(2) Auxiliary tool, folded paper

- Crumple up a white piece of paper and place it on the tabletop plate.
- Use screw C and adjust laser B (2) vertically along the white paper.

3.16.8 Final checks

- Switch ON F22 (RF-room).
- Do a test of the function of the replaced part.

4.1 Locking Hooks (Mechanical)

4.1.1 General

The Trolley consists of several components. Due to the mechanical complexity of the trolley, the replacement of FRUs is limited to locking mechanics, PCBs and keypads described in this chapter. In case the hydraulic components need to be replaced, the trolley has to be returned to the factory.

This chapter describes how to replace locking hooks. In addition, this procedure replaces several other mechanical procedures, e.g. raster element, spring element, Bowden wire....

4.1.2 Tools and Time required

4.1.2.1 Time

Approx. 120 min

4.1.2.2 Tools

Non-magnetic tool kit

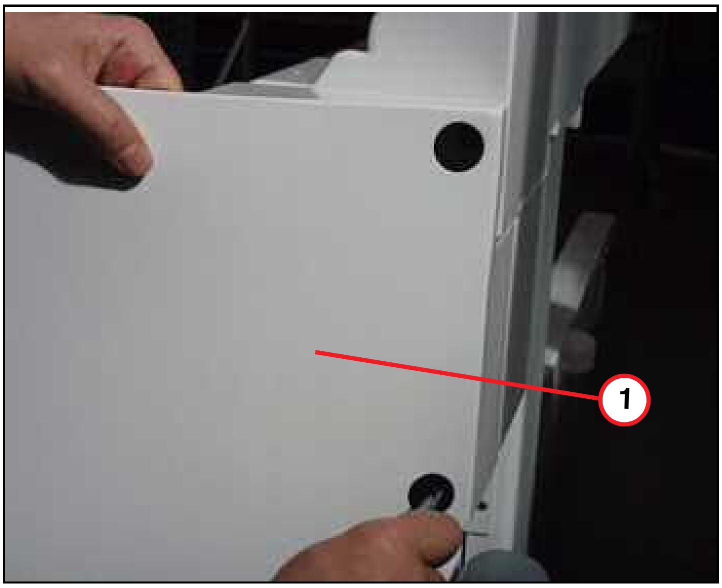
4.1.3 Preliminary

- Move the trolley out of the examination room.

4.1.4 Disassembly

4.1.4.1 Covers

Fig. 381: Trolley, front cover



(1) Front cover to be removed

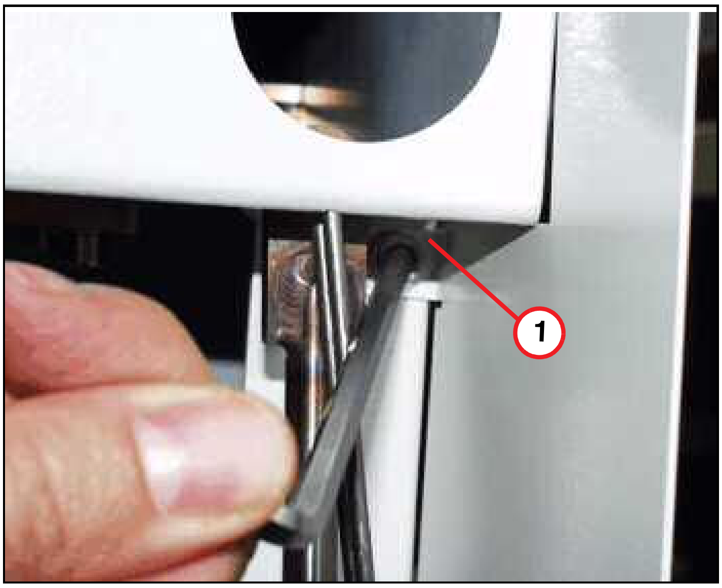
- Remove the 4 plastic caps and remove the front trolley front cover by taking off the four Allen screws.

Fig. 382: Lower cover

(1) Lower cover to be removed

- Remove lower cover by taking off the Allen screws.

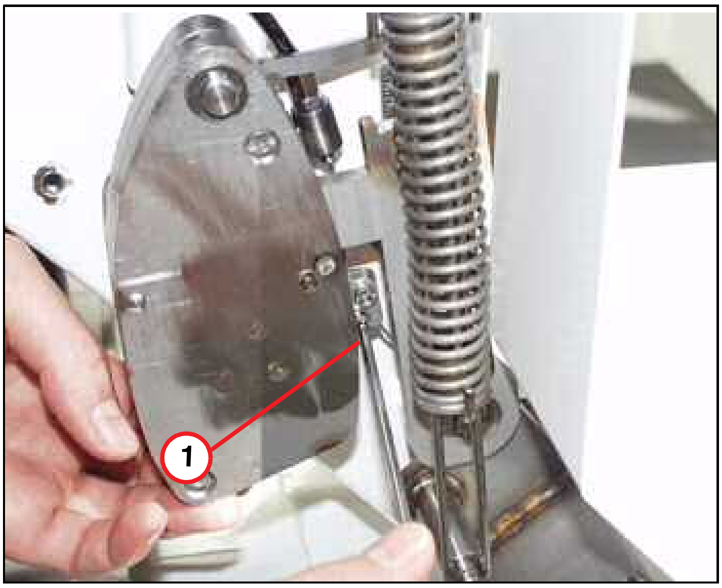
Fig. 383: Allen screw of front cover



(1) Allen screw to be removed

4.1.4.2 Locking mechanics

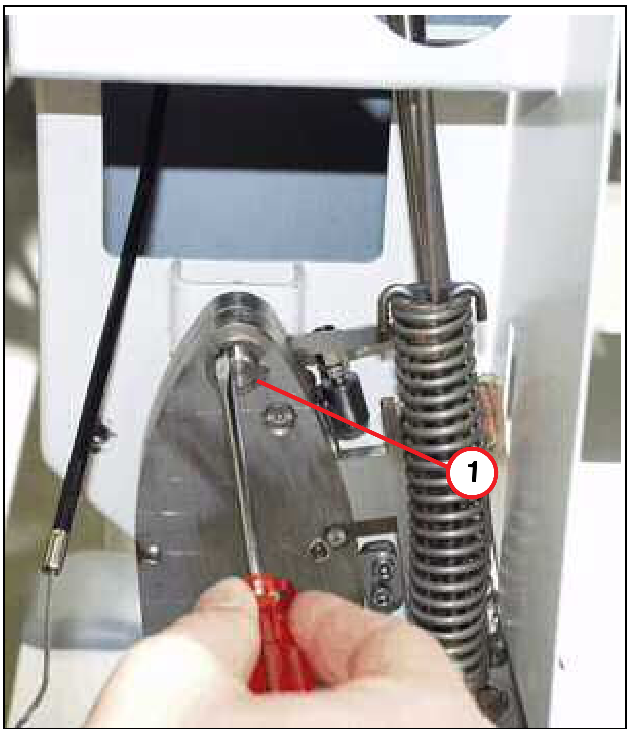
Fig. 384: Clamping plate



(1) Clamping plate of Bowden wire

- Release clamping plate from the Bowden wire and remove the wire.

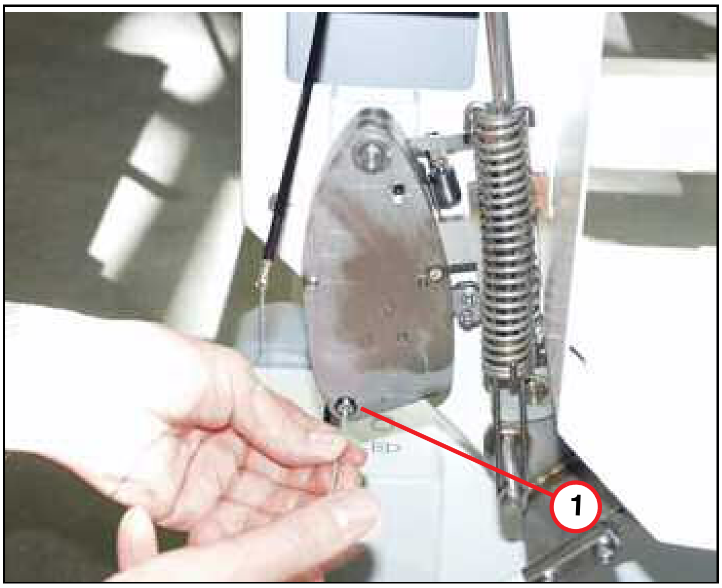
Fig. 385: Locking element



(1) Split ring of locking element

- Remove split ring of rast element.

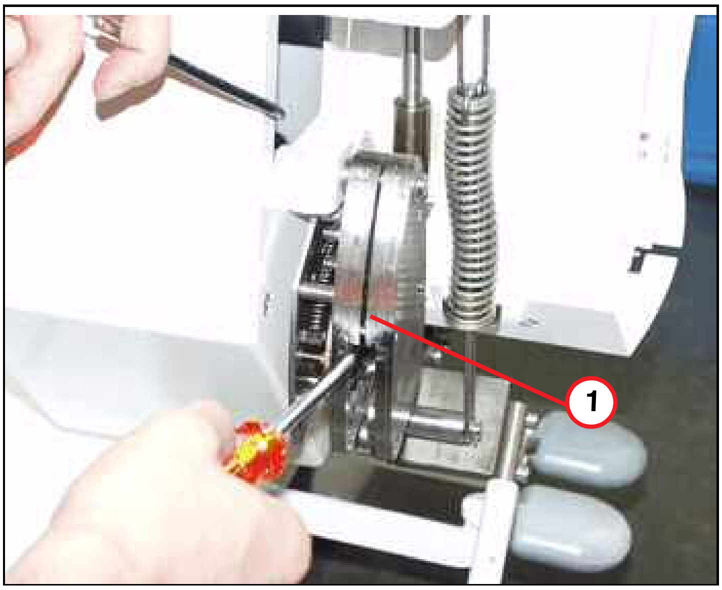
Fig. 386: Rast element



(1) Allen screws of rast element

- Remove the two Allen screws.

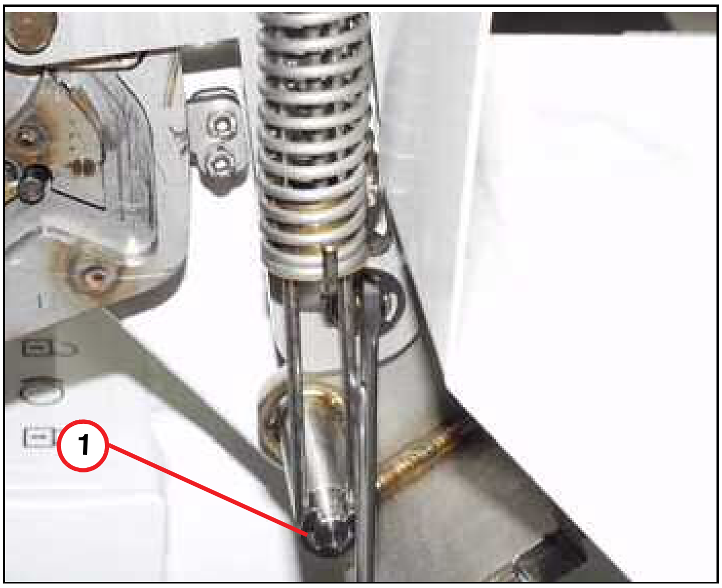
Fig. 387: Rast element



(1) Rast element

- Remove the front plate of the lock-in element by using a flat head screwdriver.

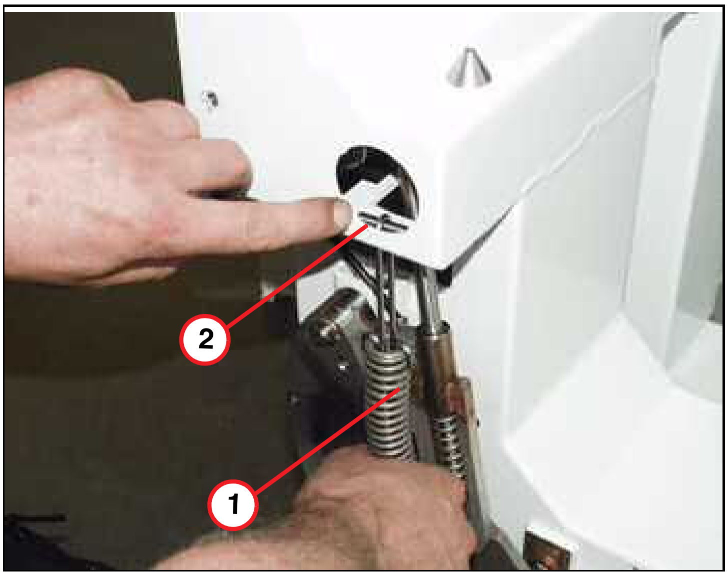
Fig. 388: Spring element



(1) Lower split ring of spring element

- Remove the split ring.

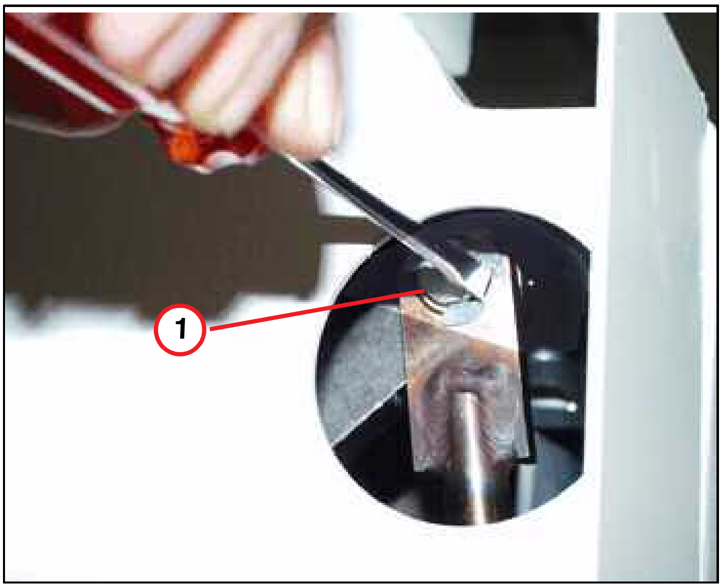
Fig. 389: Actuating spring



(1) Spring element
(2) Bolt to be removed

- Remove bolt and spring element.

Fig. 390: Split ring



(1) Upper split ring

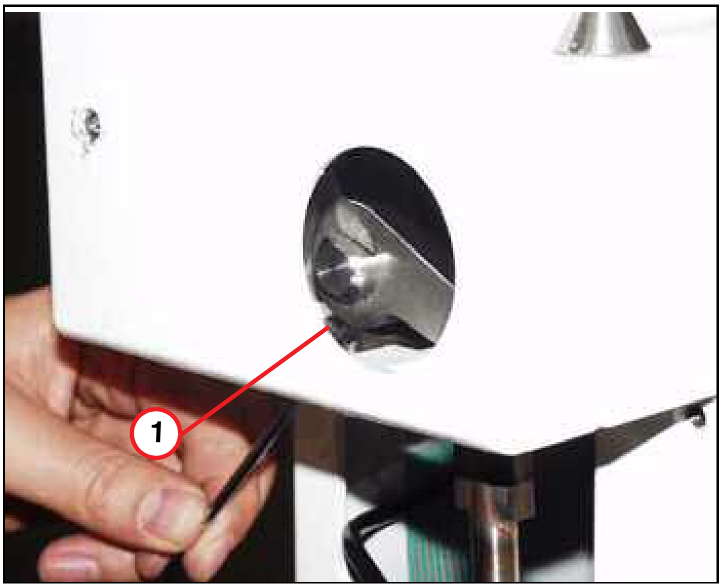
- Remove upper split ring.

Fig. 391: Spring rod

(1) Spring rod

- Remove lower split ring and the spring rod.

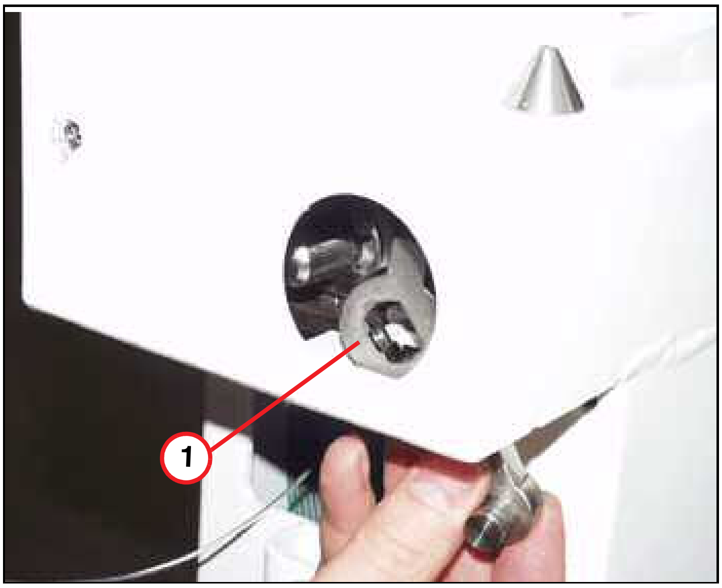
Fig. 392: Grub screw



(1) Grub screw of actuating lever

- Remove the grub screw of the actuating lever.

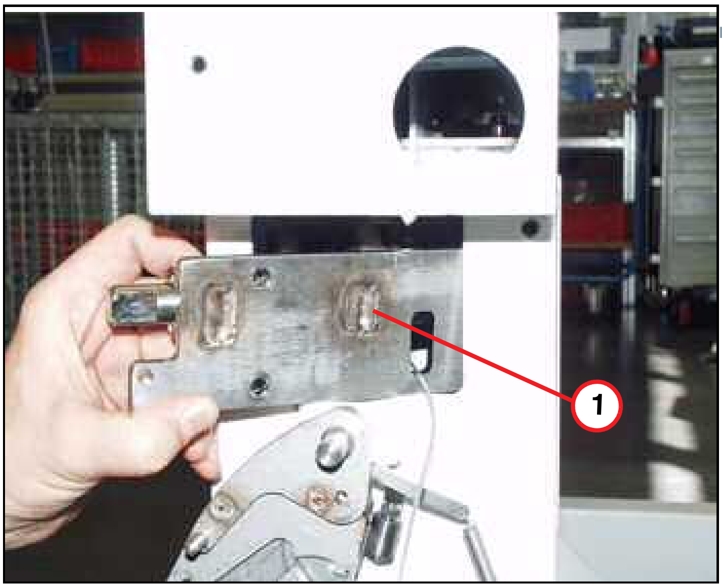
Fig. 393: Actuating lever



(1) Actuating lever removed

- Remove actuating lever.

Fig. 394: Trolley locking mechanics



(1) Trolley locking mechanics removed

- Remove the four Allen screws.
- Remove locking mechanics and replace them.

4.1.5 Assembly

4.1.5.1 Locking mechanics

Fig. 395: Spring rod

(1) Spring rod

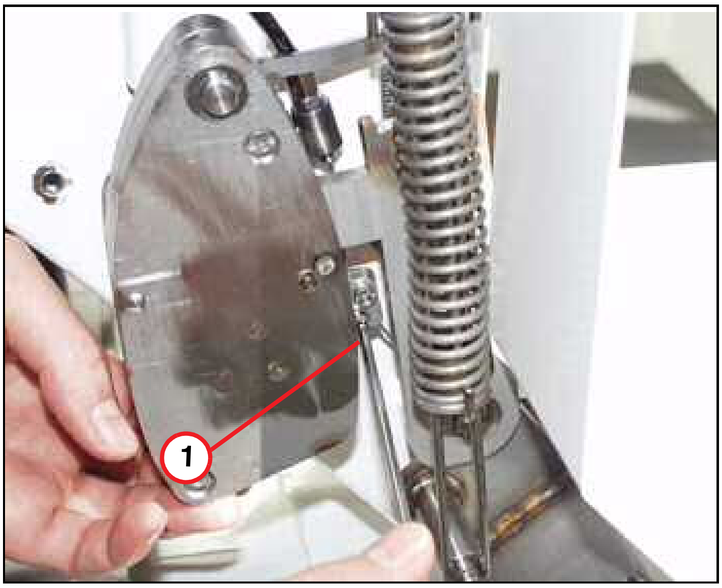
- Remount the actuating lever and tighten the grub screw.
- Remount spring rod.

Fig. 396: Bowden wire for tabletop locking

(1) Insert Bowden wire bush

- Reassemble the bushing of the Bowden wire and do a check of the function of the locking mechanism.

Fig. 397: Clamping plate



(1) Clamping plate of Bowden wire

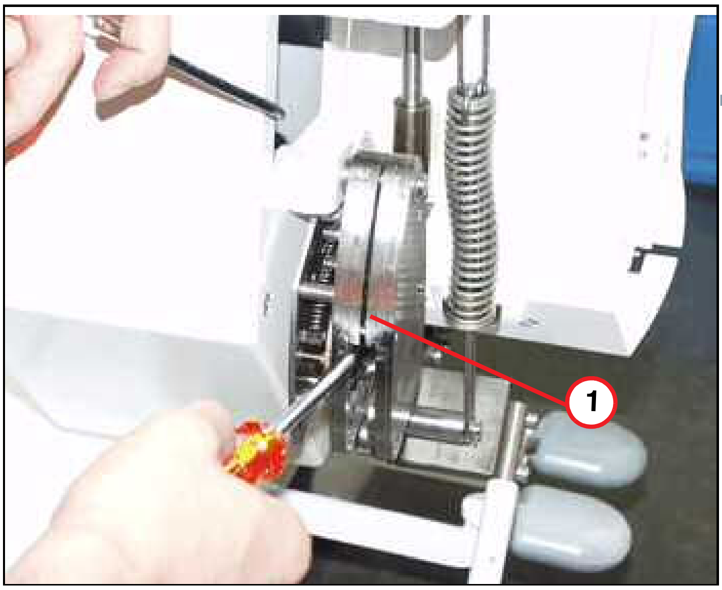
- Insert the Bowden wire into the clamping plate, slowly pull Bowden wire until the plastic hook reacts. Release it a little bit and remain in this position, tighten the clamping plate.

Fig. 398: Trolley hook

(1) Trolley hook assembly

- Insert the split rings from the spring rod
- A properly adjusted Bowden wire leads to the planned hook assembly (plastic and metal hook are level). Tighten the counter nut after the adjustment.

Fig. 399: Rast element



(1) Rast element

- Reassemble the front plate of the lock-in element and insert the Allen screws, the standoff and the split ring.
- Install the spring element, with bolt and split ring.

4.1.6 Final Steps

- Reinstall the covers.

4.1.7 Final Checks

- Position the trolley at the patient table.
- Start trolley function and check for smooth and safe operation.

4.2 Keypads / PCBs

4.2.1 General

This chapter describes how to replace locking hooks. In addition, this procedure replaces several other mechanical procedures, e.g. lock-in element, spring element, Bowden wire....

4.2.2 Tools and time required

4.2.2.1 Time

Approx. 20 min.

4.2.2.2 Tools

Non-magnetic tool kit

4.2.3 Preliminary

- Move the trolley out of the examination room.

4.2.4 Disassembly

4.2.4.1 Keypads / PCBs

Fig. 400: Cover of keypad

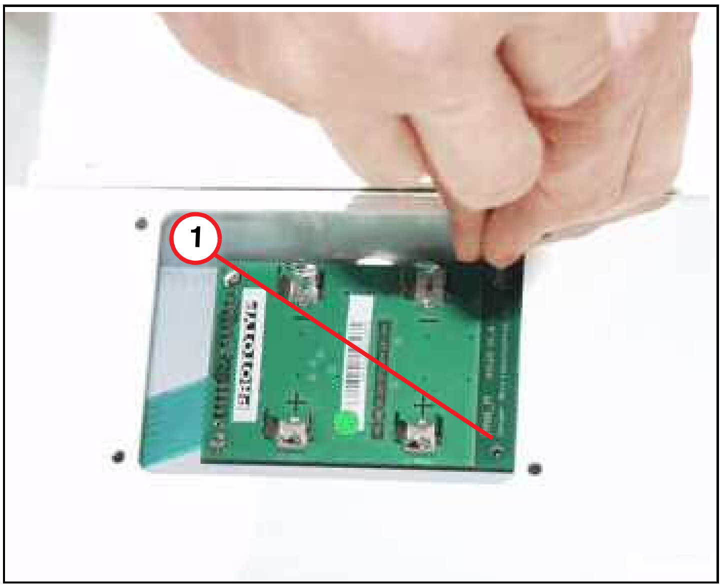
(1) Cover to be removed

- Remove the small cover of the left or right side.

Fig. 401: Battery cover, trolley remote control

- Remove the cover of the PCB.

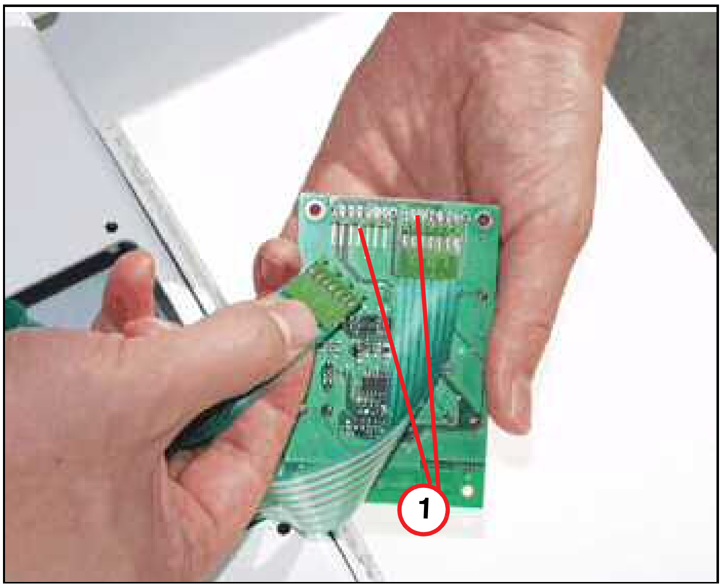
Fig. 402: Circuit board



(1) Screws of PCB removed

- Remove batteries and PCB.

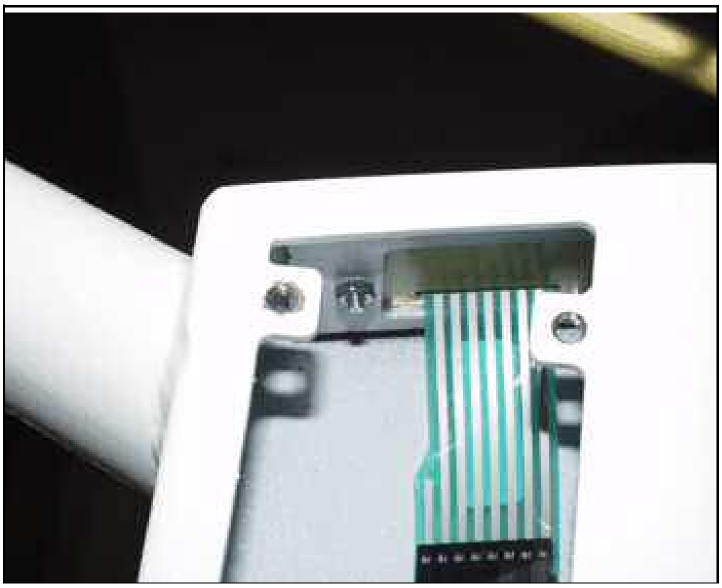
Fig. 403: Cable connections



(1) Cable connections on PCB

- Disconnect cables and replace PCB, if necessary.

Fig. 404: Keypad



- Replace the defective keypad.

4.2.5 Assembly

- Reassemble in reverse order.

4.2.6 Concluding steps

- Reinstall the covers.

4.2.7 Final checks

- Position the trolley at the patient table.
- Start trolley function and check remote control buttons

4.3 Wheel Reinforcement Kit

4.3.1 General

This chapter describes how to install the reinforcement kit (material number 10092160). By installing a wheel reinforcement set in the trolley, the mechanical stress resistance of the trolley wheels improves.

4.3.2 Tools and Time required

4.3.2.1 Time

60 min

2 Persons required

4.3.2.2 Tools

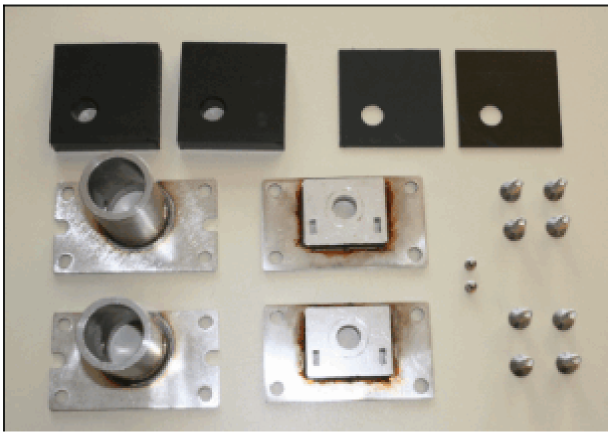
- Heat gun
- Set of Allen keys
- Flat wrench 13 mm
- Loctite 577 (material number 03443348)
- Safety gloves
- Permanent waterproof pen
- Isolation tape
- Protect the trolley surface when putting it on its side (e.g. with a blanket, weight of the trolley is approx. 150 kg.

4.3.3 Preliminary

- Move the trolley out of the examination room.

4.3.4 Installing the Wheel Reinforcement Kit

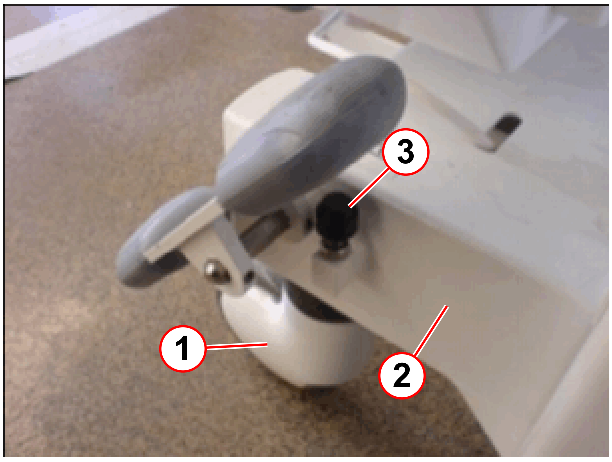
Fig. 405: Wheel Reinforcement Kit



- Check the scope of delivery.

4.3.4.1 Rear Wheels.

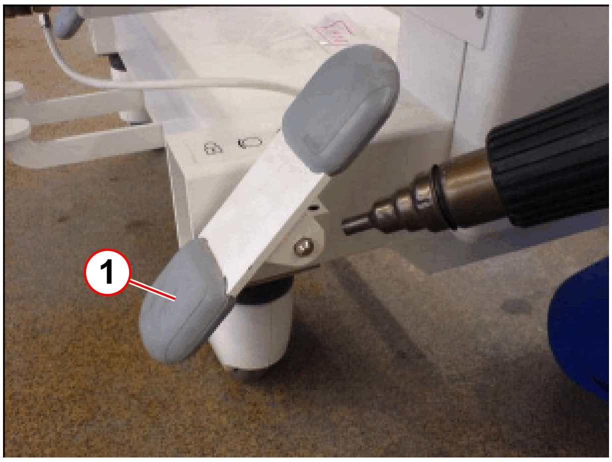
Fig. 406: Removing end stop



(1) Rear wheel
(2) Base frame
(3) End stop

- Move the trolley forward (approx. 0.5 m) so that the rear wheels (1) are parallel to the chassis (2).
- Activate the brakes.
- Loosen the counter nut at the end stop (3)
- Remove the end stop (3).

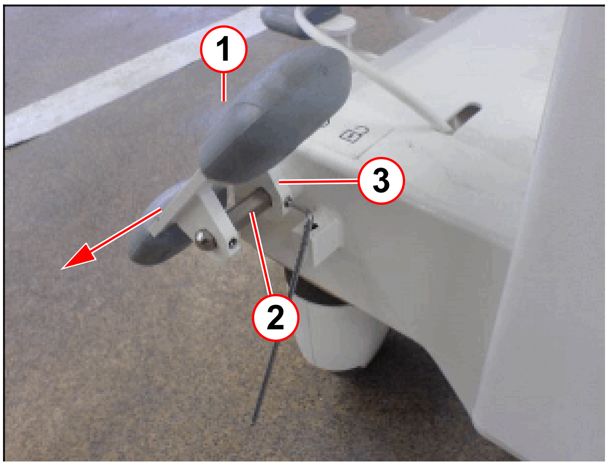
Fig. 407: Brake pedal



(1) Brake pedal

- Heat up the threaded bolts of the brake pedal (1) with a heat gun.
- Be careful not to damage the heat-sensitive surface.

Fig. 408: Brake

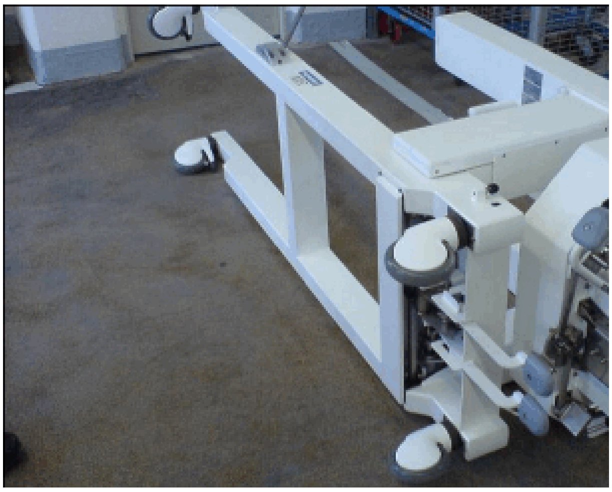


(1) Pedal
(2) Hexagonal rod
(3) Spacer

- Take an Allen key and remove the threaded bolts of the pedal (1)
- Pull out pedal (1) and spacer (3) (white plastic)

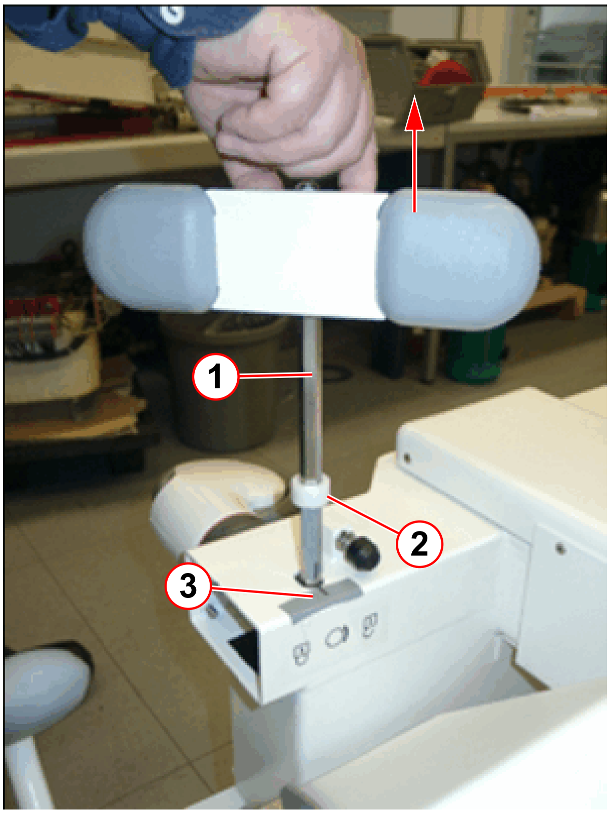
> The rear wheels must be parallel with the base frame and the brake must still be engaged.

Fig. 409: Trolley on the side



- Put the trolley on the side, where the pedal has been removed (cover floor to protect the surface of the trolley, trolley weight approx. 150 kg).
- The following describes the replacement of the wheel take-ups for the front and rear wheels.

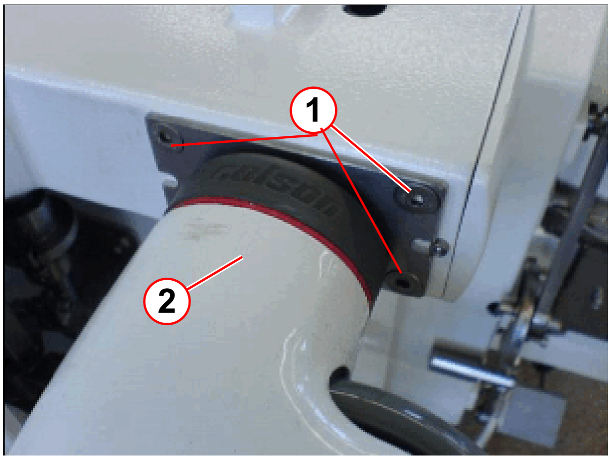
Fig. 410: Pulling out the hexagonal rod



(1) Pedal with hexagonal rod
(2) Spacer
(3) Marking the pedal angle position

- Mark the position of the pedal angle at the rod (3) (e.g. on a strip of isolation tape) – for correct reinstallation.
- Pull the pedal out completely together with the hexagonal rod (1) and spacer (2)

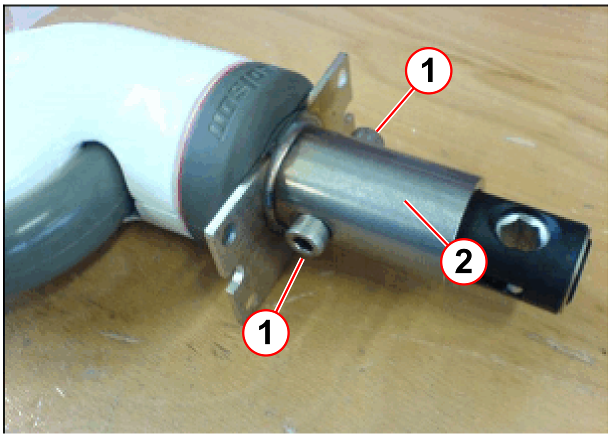
Fig. 411: Screws on the rear wheel



(1) Screws
(2) Wheel with wheel take-up

- If necessary, carefully heat up the screws (1) of a rear wheel
- Take off the rear wheel (2) with wheel take-up

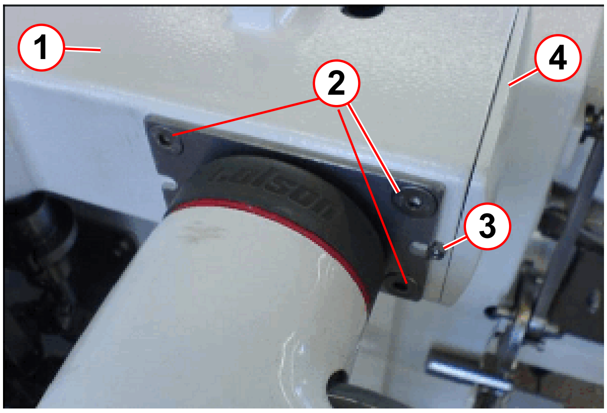
Fig. 412: Wheel tale-up



(1) Screws
(2) Wheel take-up

- The wheel take-up is not symmetrical.
- Fix the wheel take-up (2) with the 2 screws. (1)

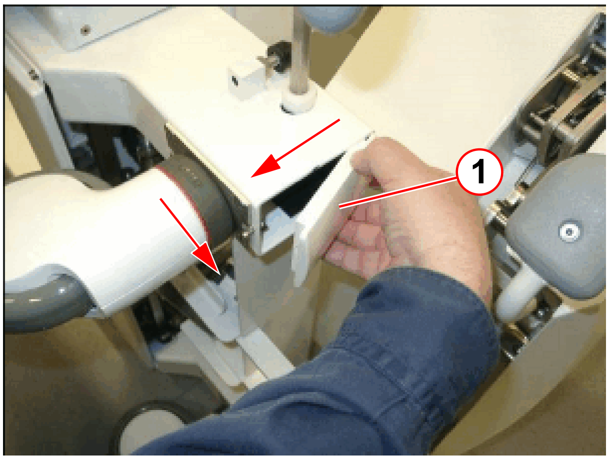
Fig. 413: Reinstall the wheel



(1) Base frame
(2) Screws (4)
(3) Screw
(4) Cover

- Install the modified rear wheel on the base frame (1) with the leveling plates.
- Apply Loctite 577, P.N. 03443348, to screws (2) and tighten them
- Loosen the screw (3) for the cover (4) approx. 8 mm
- Repeat the procedure for the other rear wheel

Fig. 414: Taking off the cover

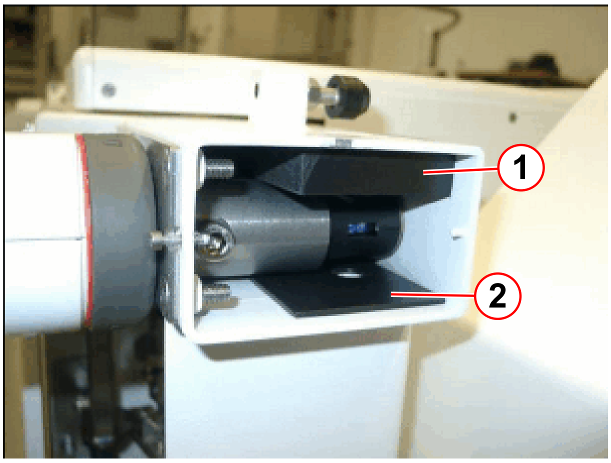


(1) Cover

Remove the cover (1) as follows:

- Shift the cover toward the base frame
- Pull out cover

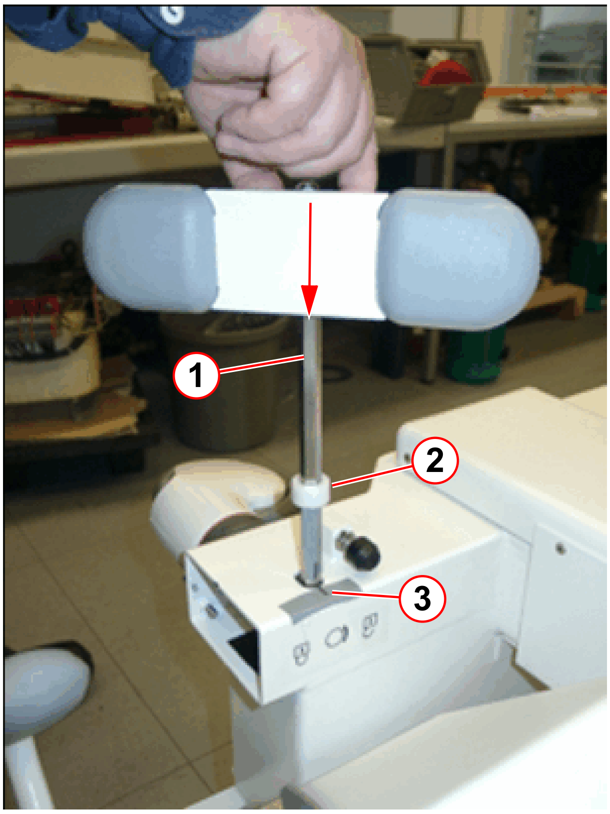
Fig. 415: Installing distance plate



(1) Distance plate 15 mm
(2) Distance plate 2 mm

- Install distance plates, 15 mm (1) and 2 mm (2).
- Repeat the procedure for the other rear wheel.

Fig. 416: Pulling in the hexagonal rod

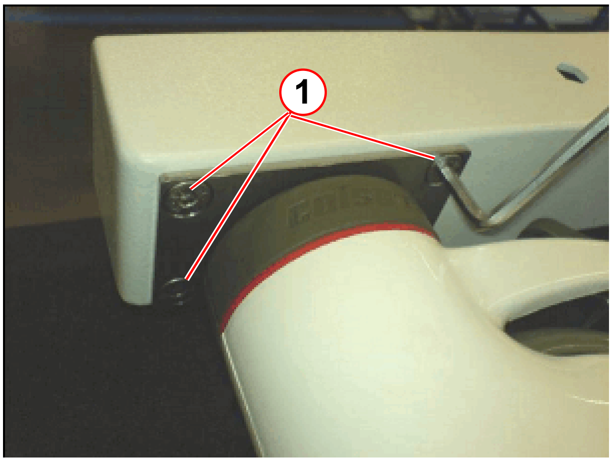


(1) Pedal with hexagonal rod
(2) Spacer
(3) Marking the pedal angle position

- Slide the hexagonal rod with pedal (1) and spacer (2) back in. Make sure that the rod is sliding through the holes of the distance plates.
- After correctly inserting the rod, remove marks (3)
- Install the cover and secure it with the screw

4.3.4.2 Front Wheels

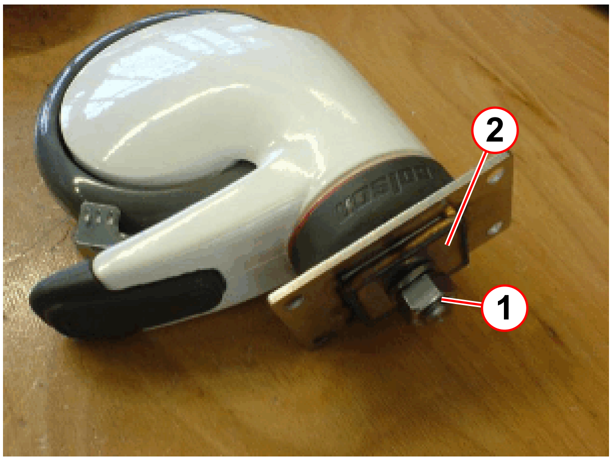
Fig. 417: Remove the front wheels



(1) Screws

- If necessary, carefully heat up the screws (1) of both front wheels
- Take off the front wheels

Fig. 418: *Replace wheel mount*



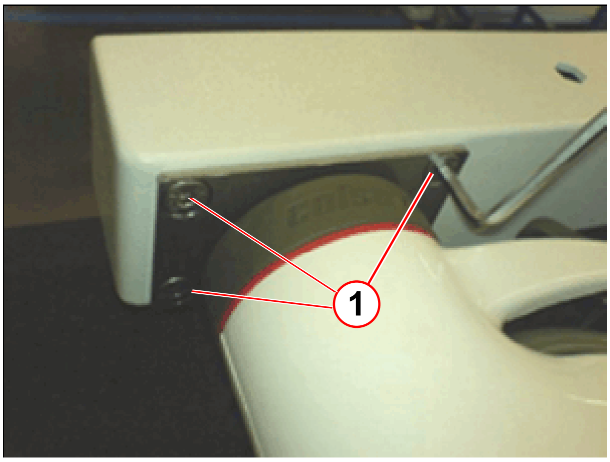
(1) Nut
(2) Mount

- Open the nut (1) and remove the wheel mount.
- Install the new wheel mount in the same way as the old one (material is thicker); use Loctite 577.
- Tighten the nut (1) securely.
- Repeat the procedure for the other front wheel.

Fig. 419: *Front wheel*

- Tighten/loosen the screw on the front wheel plates shown in the picture.

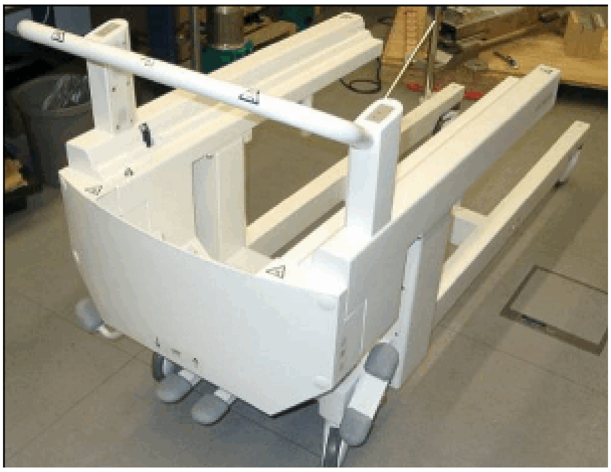
Fig. 420: Install the front wheel



(1) Screws

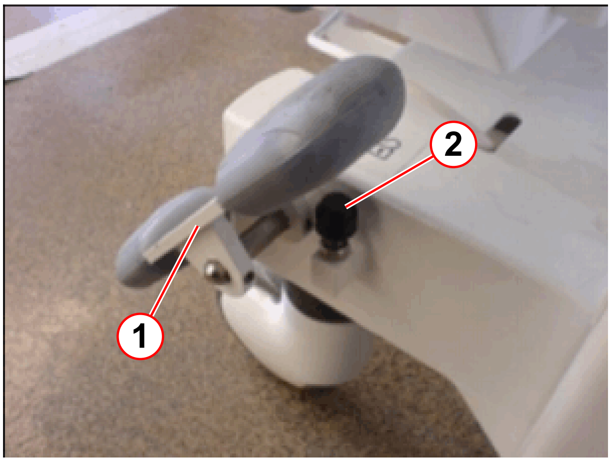
- Install the modified front wheel with the leveling plates.
- Tighten screws (1) securely, use Loctite 577.

Fig. 421: Lift up trolley



- Lift the trolley carefully. Be sure that the brake is still active.

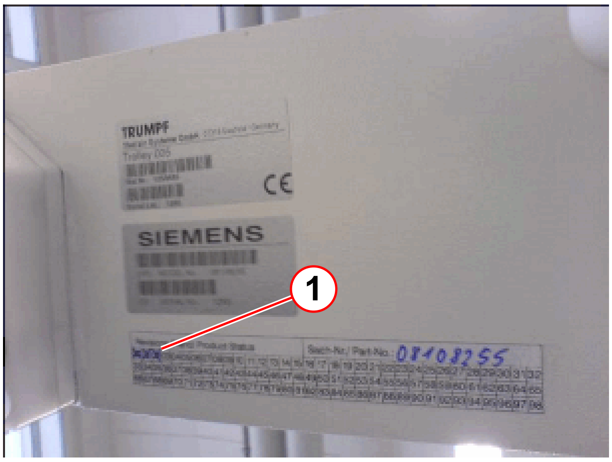
Fig. 422: Installing pedal and end stop



(1) Pedal
(2) End stop of brake pedal

- Reinstall the pedal (1) using Loctite 577 for the threaded bolts.
- The 2 pedals must be parallel
- Insert and align the end stop (2) of the brake pedal. Both pedals must be horizontal when not braked.

Fig. 423: Revision level



(1) Revision level label

- Mark the new revision level (refer to table below) on label (1) using a permanent, waterproof pen

Trolley System designation	Material number	Revision level
Trolley 005	08108255	03
Trolley 094	10018482	03
Trolley 092	10018397	02
Trolley 005/ 022 - 100 mm increased	10018902	01

4.3.5 Final Checks

- Position the trolley near the patient table.
- Start trolley function and check remote control buttons.

5.1 PDAU

5.1.1 General

The PMU consists of several components. However, only the exchange of PDAU needs to be described.

> With software version VB13, a new PDAU has been introduced.
>
> Older software versions will not support the new PDAU, material number 10096522, refer to spare part list.

5.1.2 Tools and time required

5.1.2.1 Time

Approx. 20 min.

5.1.2.2 Tools

Non-magnetic tool kit

Non-magnetic 2.5 mm Allen key

5.1.3 Preliminary steps

- Switch OFF F22 (RF-room).

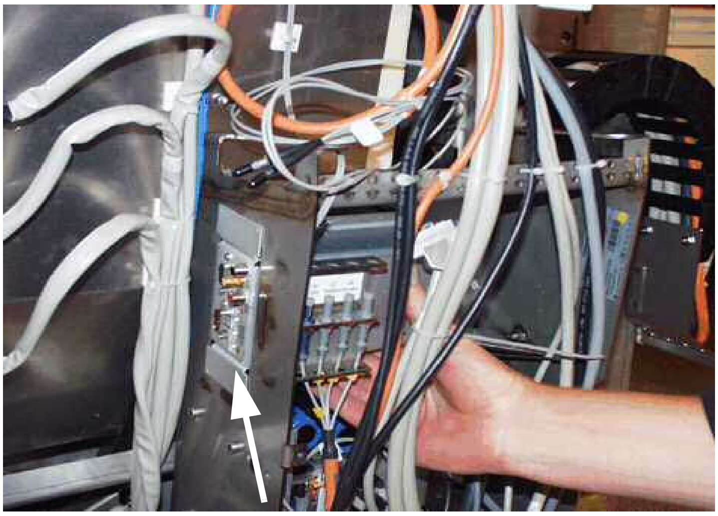
5.1.4 Disassembly

5.1.4.1 Covers

- Remove the magnet side cover and all covers from the lifting column.
- Remove the cover for the patient table electronics.

5.1.4.2 PDAU

Fig. 424: PDAU, top of ptab-electronics

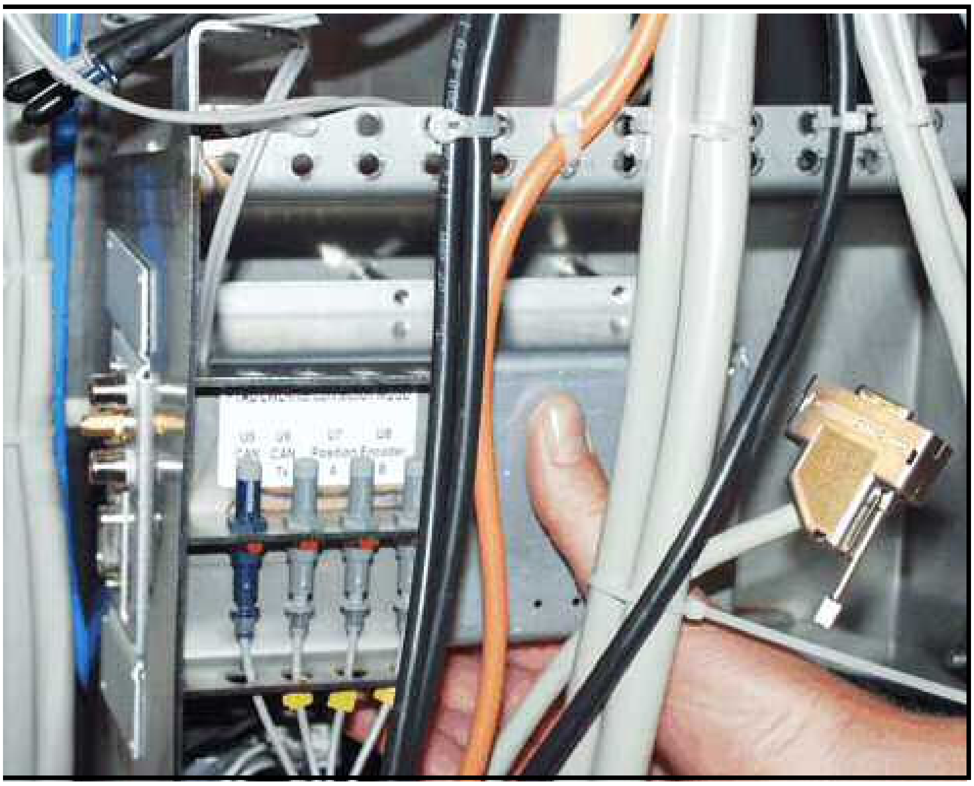


- Disconnect all cables from the PDAU.

- Remove the Allen screws and replace the PDAU.

> The PDAU consists of small magnetic objects. When replacing the PDAU, observe caution. Maintain a safe distance to the magnetic field during this procedure.

Fig. 425: Remove Allen screws of PDAU



5.1.5 Assembly

- Insert the new PDAU and tighten the screws.

5.1.6 Concluding steps

- Reconnect the cables.
- Reassemble the covers.

5.1.7 Adjustments

- n.a.

5.1.8 Final checks

- Switch ON F22 (RF-room).
- Perform, reboot scanner.
- Check PDAU for proper function.

5.2 PMU antenna / microphone

5.2.1 General

This section describes the replacement of the PMU antenna and microphone, located behind the front funnel.

5.2.2 Tools and time required

5.2.2.1 Time

Approx. 25 min./ 25 min.

5.2.2.2 Tools

Non-magnetic tool kit

5.2.3 Preliminary

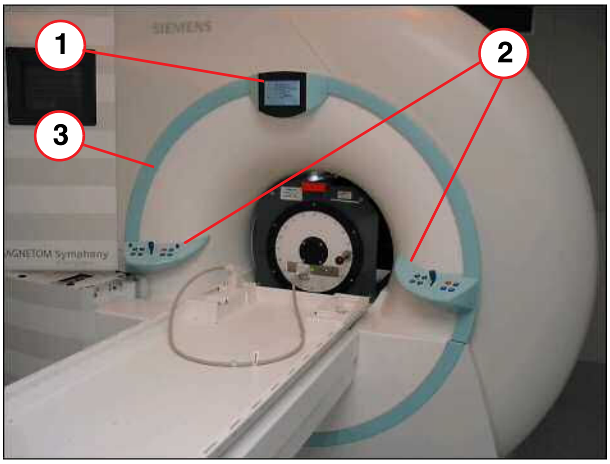
- Switch OFF F22 (RF-room).

5.2.4 Disassembly

Take care of lose cables while removing the front funnel.

5.2.4.1 Front funnel

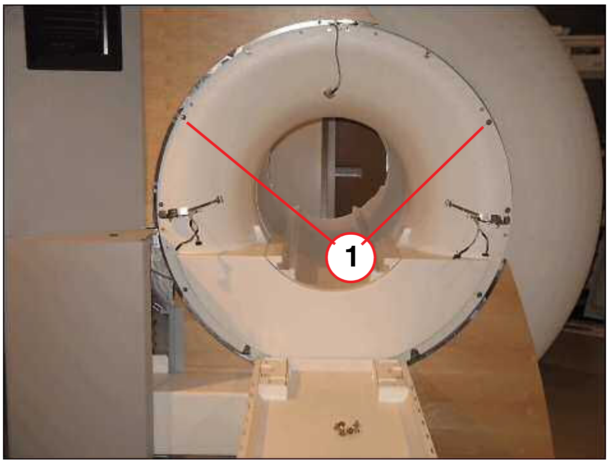
Fig. 426: Parts to be removed



(1) Remove display unit
(2) Remove control units
(3) Remove ring segment

- Remove parts in the order shown in the picture.
- Remove display and disconnect cables
- Disassembly operating panels, disconnect cables and remove them.
- Remove ring of all segments by turning it clockwise.

Fig. 427: Front funnel



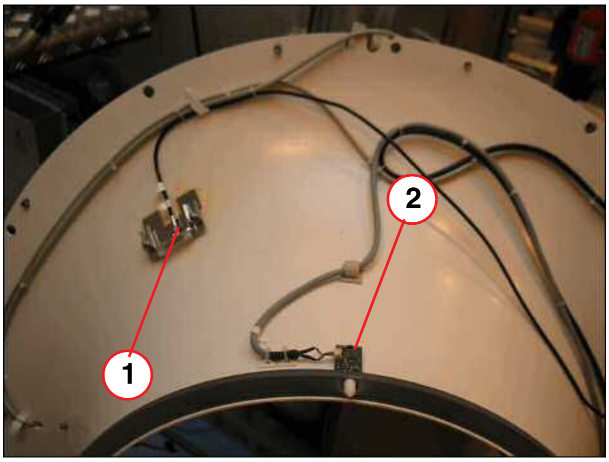
(1) Screws to be removed

- Lower the table by using the hand crank.
- Remove all screws of the front funnel.

Fig. 428: *Remove front funnel*

- Carefully remove the front funnel.

Fig. 429: *Front funnel*



(1) PMU antenna
(2) Microphone

- Disconnect the cable and remove the defective part.

5.2.5 Assembly

- Replace defective part.

5.2.6 Concluding steps

- Reconnect the cables.
- Reassemble parts in the reverse order.

5.2.7 Adjustments

- n.a.

5.2.8 Final checks

- Switch ON F22 (RF-room).
- Test affected component (Intercom, PMU Test).

6.1 CCD camera

6.1.1 General

This section describes the replacement of the CCD patient monitoring camera.

6.1.2 Tools and time required

6.1.2.1 Time

Approx. 15 min.

6.1.2.2 Tools

Non-magnetic tool kit

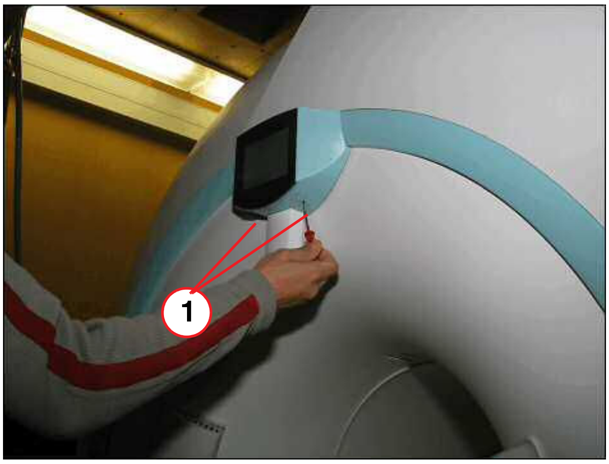
6.1.3 Preliminary

- Switch OFF F22 in the ACC.

6.1.4 Disassembly

6.1.4.1 Display/CCD Camera

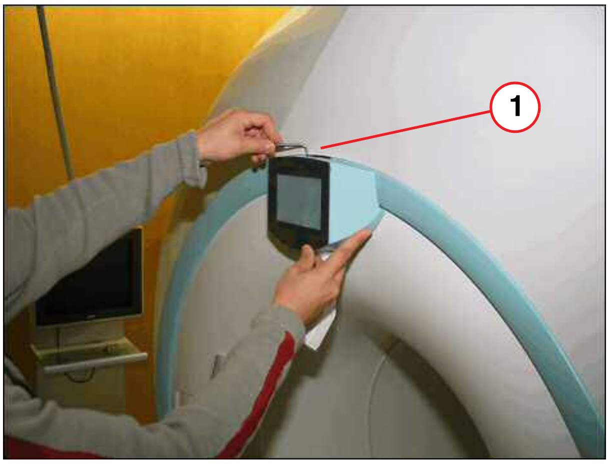
Fig. 430: Display/LCD camera



(1) Screws to be removed

- Remove the two flat blade screws from the cover plate of the CCD camera.

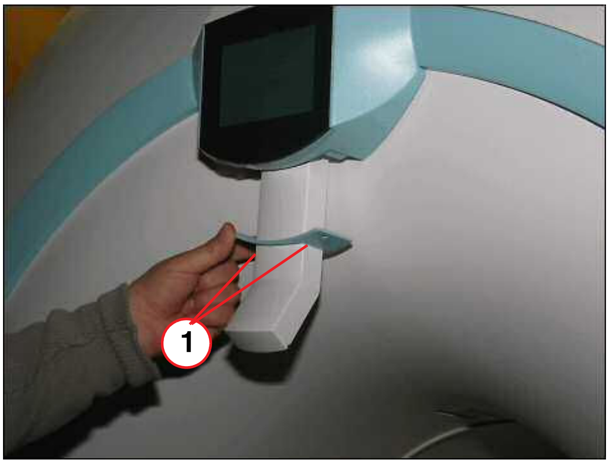
Fig. 431: Display/LCD camera



(1) Screws to be removed

- Remove the two Allen screws inside the outer camera cover.

Fig. 432: Display/LCD camera



(1) Cover to be removed

- Carefully slide the camera cover up and remove them.

6.1.5 Assembly

Fig. 433: Display/LCD camera

(1) Display to be removed

- Disconnect all cables from the CCD camera and replace it.

6.1.6 Concluding steps

- Reconnect the cables.
- Reassemble the covers.

6.1.7 Adjustments

- n.a.

6.1.8 Final checks

- Switch ON F22 (RF-room).
- Visual test of patient monitoring.

- The print number (M6-015.841.15) from the Replacement of Parts is revised with controlled language.
- Editorial changes.

Version 07

- Hint: "Protective earth connection test" was added.
- Caution: "Hold the ferromagnetic part with both hands" was added.
- The assembly procedures for the partial drive MDSD and the Drive unit TK have been updated with a new mechanical key installation.
- Caution: "Possible damage of the spindle nut" was added.

Version 08

- Arresting cable added, for exchange and mount the horizontal drive unit (→ Removal of horizontal drive unit / telescopic drive / Page 145)

Version 09

- Insert the wheel reinforcement kit procedure (→ Wheel Reinforcement Kit / Page 242)

Version 10

- Tools, guide rails: Allen key added. (→ Tools / Page 15)
- Final safety test, test spindle nut safety switch S30 added. (→ Final checks / Page 33)

Version 11

- Add new graphic (→ Fig. 54 Page 44) (→ Fig. 55 Page 44)
- Minor editorial changes (Charm MR 00447613)
- Add in the FRU: Transformer A4114 (→ FRU Overview / Page 9)

Version 12

- RF Coil Plugs (head end): (→ Assembly / Page 197)

  Caution: "Any contact between the shield of the cable and the housing of the trap must be avoided."

  and graphics (→ Fig. 334 Page 197),(→ Fig. 335 Page 198) and (→ Fig. 336 Page 199) added. (MR Charm 441285)

Version 13

- Brake disc: New photo added. Note added that the grooved side must face up. (→ Brake Disk / Page 75)

There are no Hazard IDs in this document.

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